Bubbles in constant-proportion wealth and the unidentified volatility tail

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Abstract

Let a discounted price follow a driftless local-volatility diffusion $dS = \sigma(S)\,dW$ under the pricing measure, absorbed at zero, and let V^π be the wealth of an investor who keeps a constant fraction $\pi > 0$ of wealth in the asset. A natural conjecture is that V^π has a bubble (is a strict local martingale) exactly when S has one and $\pi > 1/2$. The conjecture is false. With V^π taken to be the stochastic exponential of $\pi \int dS/S$, stopped when S reaches zero, we show that on every finite horizon V^π is a strict local martingale if and only if either $\pi > 1/2$ and $\int^\infty x/\sigma^2(x)\,dx < \infty$, or $0 < \pi < 1/2$ and $\int_0 x/\sigma^2(x)\,dx < \infty$. Equivalently, V^π has a bubble exactly when $\pi > 1/2$ and S has a bubble, or $0 < \pi < 1/2$ and S can reach zero. The exposure $\pi = 1/2$ never produces a bubble, and the conjectured threshold holds exactly for prices that cannot reach zero. Brownian motion absorbed at zero is a counterexample at $\pi = 1/4$. A second result concerns identification. If the volatility function is known exactly on a bounded range $(0, R]$, then $\int_c^\infty x/\sigma^2(x)\,dx$ can take every value above its known part $\int_c^R x/\sigma^2(x)\,dx$, and the value infinity, so whether the price has a bubble is not identified. This is an exact version of the extrapolation problem of Jarrow, Kchia and Protter; it is a statement about the coefficient, and it is proved in Lean; it persists when σ is known only on $[r, R]$, where neither endpoint integral is identified. For price data we show, by stopping the price above a high level, that any test of a finite number of price paths on a finite horizon has, at every alternative, power at most its size taken uniformly over a broad null class, and it persists when every null is a driftless local-volatility diffusion, positive and continuous if the alternative is, under any equivalent physical law, including the physical law equal to the pricing law; a test with power must exclude the linear-tail completions of the volatility function used to build those nulls. We characterize the identified set of the bubble $s_0 - E_Q[S_T]$ as a law-dependent interval and, in Sin's stochastic-volatility model, exhibit one physical price law compatible with both a bubble and no bubble. For funds rebalanced at discrete dates, as exchange-traded leveraged funds are, the zero-endpoint effect disappears and, for exposures in $(0, 1]$, a fund has a bubble exactly when its underlying does, and an inverse fund's gap is signed and needs a liquidation rule to describe a nonnegative price. Toward a usable test from prices alone, the visited part of the JKP integral is identified exactly through local time, and we give identification under a declared power tail, pooling across names that share an exponent, calibrated bands and e-values in a finite catalogue of exact price likelihoods, and a list of bubbly names with false-discovery control; none of these yet has useful power on realistic samples. In simulation, the score with a continuous Gaussian null, combined with that list, is the only procedure compared that keeps false discoveries controlled on realistic daily panels, but it fails under repeated volatility regimes, and a declared intraday clock gives high power on a year of exact five-minute prices. On real prices it flags no calm control and almost none of the episodes commonly called bubbles, while four existing tests flag calm controls at the rates we measure. Results that use option prices, namely tests from fund options, the decomposition of a basket's bubble across names, and upper bounds from American calls, rest on conventions about how option quotes relate to the bubble that are not derived from how options are priced, and we state them only as conditional theory. The first result rests on the martingale criterion of Mijatović and Urusov, which we apply rather than re-prove; it is conditional on that criterion and on the stated identification of wealth. Its deterministic part, the classification of the criterion's endpoint conditions, is proved in Lean without conditions, and the conditional classification of wealth is proved there from the cited criteria stated as hypotheses. The results on discretely rebalanced funds and on price data rest on bounded stopping and the supermartingale property of nonnegative local martingales, and on Sin's model, and are proved in Lean from them stated as hypotheses (Appendix A).

1. Introduction

In the strict-local-martingale theory of asset price bubbles [44], a discounted price has a bubble when it is a local martingale but not a martingale under the pricing measure Q. For a driftless one-dimensional diffusion $dS = \sigma(S)\,dW$ on $(0,\infty)$ the question has a complete answer: S is a strict local martingale if and only if the JKP integral $\int^{\infty} x/\sigma^2(x)\,dx$ is finite ([60], Corollary 4.3).

This paper asks whether a rebalancing strategy inherits the bubble. An investor holds the fraction $\pi > 0$ of wealth in S and the rest in the numéraire. The wealth process V^π is then a stochastic exponential of $\pi \int dS/S$, and the question is when it is a martingale. A tempting answer, suggested by the case $\pi = 1$, where V^1 is proportional to S, is that V^π is strict exactly when S is strict and $\pi > 1/2$. We show that the upper threshold $1/2$ is right, but that the statement misses the lower endpoint: when S can reach zero, every exposure strictly between 0 and $1/2$ gives strict wealth, whether or not S has a bubble (Theorem 1, Corollary 2).

The result is an application of a known general theorem, not a new theorem about diffusions. Mijatović and Urusov [60] characterize when a stochastic exponential of a diffusion is a martingale: two endpoint conditions, each decided by Feller's test for explosions applied to an auxiliary diffusion. We apply their Theorem 2.1 with integrand $b_\pi(x) = \pi\sigma(x)/x$, check its hypotheses, and compute the two endpoint tests. The Feller test function for this auxiliary drift family is already written down by Karatzas and Ruf ([45], equation (7.4)); what we add is the classification it implies for every $\pi > 0$, the role of the zero endpoint, and the reading in terms of default.

Theorem 1, like the JKP criterion, turns on integrals of x/σ^2 near the endpoints of the state space, and these are what price data do not reveal. Jarrow, Kchia and Protter ([41], §1 and §5) observe that σ can be estimated only at levels the price has visited, so the tail of the volatility function, and with it the finiteness of the JKP integral, must be extrapolated; they call this the extrapolation problem. Proposition 7 is its exact, set-valued version. Knowing σ exactly on $(0, R]$ leaves every value of the JKP integral above its known part attainable, and infinity too. The observation there is the coefficient on a bounded range, not price data, and Proposition 7 is not a theorem about tests.

Section 3.4 turns to price data. Theorem 18 says that, without structural assumptions, no test based on finitely many price paths on a finite horizon has power above its size at any alternative. The mechanism is elementary: stopping a strict price when it first exceeds a high level gives a true martingale whose paths agree with the original ones except on an event of small probability. The theorem is an application of bounded stopping [38], not a new general impossibility theorem. Proposition 19 characterizes what the law of prices identifies about the size of the bubble: an interval that depends on that law, with no universal answer. Proposition 20 applies Sin's stochastic-volatility model [75] to exhibit a single physical price law with one pricing measure under which the price is a martingale and another under which it is strict. The null class of Theorem 18 contains prices that are frozen above a level. Theorem 22 removes them: the bound holds with nulls that are driftless local-volatility diffusions, under any equivalent physical law including the pricing law itself, so local volatility, positivity and continuity alone do not make a bubble testable.

Section 2 states the setting and the cited inputs. Section 3 gives the theorem, its corollary, a counterexample to the conjecture, the identification result for the coefficient, the discretely rebalanced and inverse funds of exchange-traded practice, what price data cannot identify, and the ingredients of a usable test from prices alone (Section 3.5). Section 4 collects results that rest on conventions about option prices, as conditional theory. Section 5 reports simulations with known answers, including a comparison with four

existing tests, and Section 6 applies the tests to real prices. Section 7 discusses related work and Section 8 lists what is not done.

2. Setting and standing assumptions

Standing assumptions. Throughout:

1. $\sigma : (0,\infty) \to \mathbb{R}$ is Borel, $\sigma(x) \neq 0$ for every $x > 0$, and $1/\sigma^2$ is integrable on every compact subinterval of $(0,\infty)$.
2. On a filtered probability space under Q satisfying the usual conditions, W is a Brownian motion and S is a nonnegative weak solution of $dS = \sigma(S)\,dW$ started at a deterministic $s_0 > 0$, stopped and absorbed at its first hitting time of zero. The equation is interpreted before that time, so σ need not be defined at zero.
3. $\pi > 0$ and $v_0 > 0$ are deterministic constants. Set $b_\pi(x) = \pi\sigma(x)/x$, and let Z^π be the stochastic exponential of $\int b_\pi(S)\,dW$ in the sense of [60] (its equation 7, p. 4), including its convention at the exit time ζ of S from $(0,\infty)$: Z^π keeps its value at ζ if $\int_0^\zeta b_\pi^2(S_t)\,dt < \infty$ and is set to zero otherwise.
4. **Wealth representation (W).** The wealth process under study is $V^\pi = v_0 Z^\pi$ at all times.

Assumption (W) is the identification of the constant-proportion strategy with this stopped exponential. Before ζ it is the usual self-financing equation $dV^\pi = \pi V^\pi\,dS/S$. We do not prove the identification as a statement about admissible trading strategies, and we do not define holdings after S has reached zero; every statement about wealth below is conditional on (W). The exit convention is not a free modelling choice in practice: on $\{\zeta < \infty\}$ the integral $\int_0^\zeta b_\pi^2(S_t)\,dt = \pi^2 \int_0^\zeta \sigma^2(S_t)/S_t^2\,dt$ is almost surely infinite, since otherwise $\log S = \int \sigma(S)/S\,dW - \frac{1}{2}\int \sigma^2(S)/S^2\,dt$ would have a finite limit at ζ, contradicting $S_\zeta = 0$. So V^π is absorbed at zero when S is.

For a fixed $c > 0$ define the extended integrals

$$I_0 = \int_0^c \frac{x}{\sigma^2(x)}\,dx, \qquad I_\infty = \int_c^\infty \frac{x}{\sigma^2(x)}\,dx. \tag{1}$$

Their finiteness does not depend on c, because x/σ^2 is integrable on compact subintervals of $(0,\infty)$ by assumption 1. I_∞ is the JKP integral.

Cited inputs. We use three published results, stated here in our notation; we apply them and do not re-prove them. For a diffusion with drift μ and coefficient σ on $(0,\infty)$ and the reference point c, write $\rho(x) = \exp(-\int_c^x 2\mu/\sigma^2)$ for its scale density and $s(x) = \int_c^x \rho$ for its scale function.

- *Martingale criterion* ([60], Theorem 2.1 and the remark following it). Under the Engelbert–Schmidt conditions on μ and σ and local integrability of b^2/σ^2, the exponential Z of [60] (its equation 7, p. 4), is a martingale on $[0,T]$, for any fixed $T \in (0,\infty)$, if and only if, at each endpoint, either the auxiliary diffusion with drift $\mu + b\sigma$ and coefficient σ does not exit there, or the endpoint is good. Infinity is good if $s(\infty) < \infty$ and $(s(\infty) - s)\,b^2/(\rho\sigma^2)$ is integrable near infinity; zero is good if $s(0) > -\infty$ and $(s - s(0))\,b^2/(\rho\sigma^2)$ is integrable near zero. Goodness is computed with the original scale.
- *Feller's test for explosions*, in the endpoint form of [60], equations (20)–(23), and the global form of [45], Appendix A. A diffusion as above exits at infinity with positive probability if and only if $s(\infty) < \infty$ and

$(s(\infty)-s)/(\rho\sigma^2)$ is integrable near infinity, and exits at zero if and only if $s(0) > -\infty$ and $(s-s(0))/(\rho\sigma^2)$ is integrable near zero.

- *Existence and uniqueness in law* ([60], Section 2, p. 4; [45], Proposition 9.1). Under standing assumption 1 the driftless equation has a weak solution absorbed at zero, and two such equations whose coefficients agree on an interval $(0, n)$ containing s_0 have the same law when stopped at the first exit from $(0, n)$.
- *Share probability* ([47], Definition 2.5 and Theorem 2.12). For a strictly positive continuous local martingale X on a right-continuous, uncompleted filtration of a standard system, there is a probability $\tilde{Q}$ with $\tilde{Q}(A \cap \{t < \tau\}) = E_Q[(X_t/X_0)1_A]$ for $A \in \mathcal{F}_t$, where τ is the explosion time of X under $\tilde{Q}$.
- *Bounded stopping* ([38], Section 2 and Lemma 2.1). If S is a continuous local martingale on a right-continuous filtration with $S_0 = s_0$, and $\tau_n = \inf\{t \geq 0 : |S_t| > n\}$ for $n > s_0$, then the stopped process $S^n = S_{\cdot\wedge\tau_n}$ is a uniformly integrable martingale bounded by n.
- *Supermartingale property* ([38], Section 2, after the proof of Lemma 2.1; [60], p. 5). A nonnegative local martingale with integrable initial value is a supermartingale; in particular $E_Q[S_t]$ is nonincreasing in t.
- *Sin's model* ([75], Theorems 3.2 and 3.6). Under a probability Q^s let $dS_t = S_t v_t^{\alpha} \gamma \cdot dW_t$ and $dv_t = v_t\, a \cdot dW_t + \kappa(L - v_t)\, dt$, with $S_0 = s_0 > 0$, $v_0 = 1$, constants $\alpha, \kappa, L > 0$, nonzero $\gamma, a \in \mathbb{R}^2$ and a two-dimensional Brownian motion W. Then S is a martingale on all finite horizons if and only if $a \cdot \gamma \leq 0$. If $a \cdot \gamma > 0$ and the vectors are not parallel, then for each $T > 0$ there is Q^m_T, equivalent to Q^s on $\mathcal{F}_T$, under which S is a martingale on $[0, T]$; under it the drift of v becomes $\kappa L - (a \cdot \gamma) v^{1+\alpha}$. We use the 1995 report; the journal version [76] has not been read.
- *Finite trading* ([26], Theorem 7.11 and Proposition 7.14). If X is a continuous local martingale, $0 \leq t_0 < \cdots < t_N$ a deterministic grid and h_i finite $\mathcal{F}_{t_i}$-measurable random variables, then $\sum_i h_i(X_{\cdot\wedge t_{i+1}} - X_{\cdot\wedge t_i})$ is a continuous local martingale.
- *JKP criterion* ([60], Corollary 4.3 and the remark following it). Under assumptions 1–2, S is a strict local martingale on every finite horizon if and only if $I_\infty < \infty$.

3. Main results

3.1 Constant-proportion wealth

Theorem 1 (constant-proportion wealth). Assume the standing assumptions and (W). For every finite $T > 0$, V^π is a nonnegative continuous local martingale, and

$$V^\pi \text{ is a strict local martingale on } [0,T] \iff \Big(0 < \pi < \tfrac{1}{2} \text{ and } I_0 < \infty\Big) \text{ or } \Big(\pi > \tfrac{1}{2} \text{ and } I_\infty < \infty\Big). \tag{2}$$

Equivalently, V^π is a martingale on every finite horizon exactly in the following cases.

Exposure	**V^π is a martingale exactly when**
$0 < \pi < 1/2$	$I_0 = \infty$
$\pi = 1/2$	always
$\pi > 1/2$	$I_\infty = \infty$

Status. Conditional on the martingale criterion and on (W); the proof below also uses Feller's test, which the formal proof avoids. Machine-checked: the equivalence between the criterion's endpoint conditions and the negation of the right-hand side of (2) is proved in Lean without conditions, and (2) and the table are proved there from the martingale criterion stated as a hypothesis (Appendix A).

Proof. The integrand b_π is Borel and $b_\pi^2/\sigma^2 = \pi^2/x^2$ is integrable on compact subintervals of $(0,\infty)$; the original drift is $\mu = 0$. So the hypotheses of the martingale criterion hold, and it gives the local-martingale property of Z^π, hence of V^π. The auxiliary diffusion has drift $\mu_\pi(x) = b_\pi(x)\sigma(x) = \pi\sigma^2(x)/x$, coefficient σ and start s_0; its ratio $\mu_\pi/\sigma^2 = \pi/x$ is locally integrable, so Feller's test applies to it.

Both endpoints are bad. With $\mu = 0$ the original scale is $\rho = 1$, $s(x) = x - c$. At infinity $s(\infty) = \infty$. At zero $s(0) = -c$ is finite, but $(s(x) - s(0))\, b_\pi^2(x)/(\rho(x)\sigma^2(x)) = \pi^2/x$ is not integrable near zero. Neither fact depends on σ.

Auxiliary scale. The auxiliary scale density is $\tilde\rho_\pi(x) = (x/c)^{-2\pi}$. With $q = 1 - 2\pi$, the scale function normalized by $\tilde s_\pi(c) = 0$ is

$$\tilde s_\pi(x) = \frac{c^{2\pi}(x^q - c^q)}{q} \quad (\pi \neq \tfrac{1}{2}), \qquad \tilde s_\pi(x) = c\log(x/c) \quad (\pi = \tfrac{1}{2}). \tag{3}$$

Right endpoint. If $\pi \le 1/2$, then $\tilde s_\pi(\infty) = \infty$ and the auxiliary diffusion does not exit at infinity. If $\pi > 1/2$, then $\tilde s_\pi(\infty) = c/(2\pi - 1)$ and

$$\frac{\tilde s_\pi(\infty) - \tilde s_\pi(x)}{\tilde\rho_\pi(x)\sigma^2(x)} = \frac{1}{2\pi - 1}\,\frac{x}{\sigma^2(x)}, \tag{4}$$

so by Feller's test the auxiliary diffusion exits at infinity if and only if $\pi > 1/2$ and $I_\infty < \infty$.

Left endpoint. If $\pi \ge 1/2$, then $\tilde s_\pi(0) = -\infty$ and there is no exit at zero. If $0 < \pi < 1/2$, then $\tilde s_\pi(0) = -c/(1 - 2\pi)$ and

$$\frac{\tilde s_\pi(x) - \tilde s_\pi(0)}{\tilde\rho_\pi(x)\sigma^2(x)} = \frac{1}{1 - 2\pi}\,\frac{x}{\sigma^2(x)}, \tag{5}$$

so the auxiliary diffusion exits at zero if and only if $0 < \pi < 1/2$ and $I_0 < \infty$.

Since both endpoints are bad, the martingale criterion says that Z^π is a martingale on $[0,T]$ exactly when the auxiliary diffusion exits at neither endpoint. Substituting the two exit conditions gives (2) and the table, and multiplication by $v_0 > 0$ changes nothing. □

The finite-cutoff Feller test function of the auxiliary diffusion is already in [45], equation (7.4). In that source the diffusion coefficient is s and the drift is $s\,b$; with $s = \sigma$ and $b(x) = a\,s(x)/x$, $a = \pi$, its drift is our μ_π. For $\pi \neq 1/2$ the kernel $(\tilde s_\pi(x) - \tilde s_\pi(z))/(\tilde\rho_\pi(z)\sigma^2(z))$ obtained from (3) is exactly the integrand of (7.4), and (4) and (5) are its limits, up to orientation, as $x \to \infty$ and $x \to 0$. At $\pi = 1$ the computation reduces to the proof of the JKP criterion in [60], Corollary 4.3, as it must, since $V^1 = v_0 S/s_0$.

Corollary 2 (bubbles and default). Under the assumptions of Theorem 1, $I_0 < \infty$ if and only if S reaches zero with positive probability. Hence, on every finite horizon,

$$V^\pi \text{ is strict} \iff \Big(\pi > \tfrac{1}{2} \text{ and } S \text{ is strict}\Big) \text{ or } \Big(0 < \pi < \tfrac{1}{2} \text{ and } S \text{ can reach zero}\Big). \tag{6}$$

In particular $V^{1/2}$ is never strict, and the statement "V^π is strict if and only if S is strict and $\pi > 1/2$" holds for every $\pi > 0$ exactly when S cannot reach zero. If S can reach zero, it fails for every $0 < \pi < 1/2$, whether or not S is strict.

Status. Conditional on Theorem 1, Feller's test and the JKP criterion. The equivalence of $I_0 < \infty$ with reaching zero is not formalized; the scope statement in its last sentence, read with I_0 in place of reaching zero, is proved in Lean.

Proof. Apply Feller's test to S itself: $\mu = 0$, $s(x) = x - c$, $s(0) = -c$ is finite, and $(s(x) - s(0))/(\rho(x)\sigma^2(x)) = x/\sigma^2(x)$. So S exits at zero with positive probability if and only if $I_0 < \infty$. (The same equivalence is [45], Proposition 7.1(ii).) Combine with (2) and the JKP criterion. □

Read through Corollary 2, a constant-proportion investor's wealth has a bubble from one of two sources. Above one-half, wealth inherits the bubble of the asset. Below one-half, it inherits the asset's default: a positive probability that S is absorbed at zero, which a fixed-proportion investor rides down to zero wealth, is enough to make V^π lose expected value, even when S itself is a true martingale. This is an effect of rebalancing continuously into default: the continuously rebalanced investor keeps selling cash for the asset as its price falls to zero. Propositions 11 and 12 show that it does not carry over to funds rebalanced at discrete dates, where the threshold $1/2$ does not either. In particular, an exposure at or below one-half need not give bubble-free wealth, and the threshold is not universal across volatility functions: if $\sigma(x) = 1 + x^2$, both I_0 and I_∞ are finite, and V^π is strict for every $\pi \neq 1/2$.

Example 3 (counterexamples to the conjecture).

(a) *Brownian motion absorbed at zero.* Take $\sigma \equiv 1$, $s_0 = v_0 = c = 1$ and $\pi = 1/4$. Then $I_0 = \int_0^1 x\,dx = 1/2$ and $I_\infty = \infty$. By the JKP criterion S is a martingale; by Theorem 1, $V^{1/4}$ is a strict local martingale on every finite horizon. Both the integrals and this conclusion, conditional on the two cited criteria, are proved in Lean.

(b) *The driftless square-root diffusion.* Take $\sigma(x) = \sqrt{x}$, the constant-elasticity case $dS = S^{1/2}\,dW$. Then $x/\sigma^2(x) = 1$, so $I_0 < \infty$ and $I_\infty = \infty$, and by Corollary 2 S is a martingale that reaches zero with positive probability, and V^π is strict for every $0 < \pi < 1/2$.

As numerical evidence about the cases computed, not as proof: for $dS = S^{1/2}\,dW$ with $S_0 = 1$, two independent methods (a finite-difference solution of the Feynman–Kac equation for $E_Q[V_T^\pi]$ on truncated domains, and a closed form in incomplete gamma functions from the hitting law of a Bessel-type transform of the auxiliary diffusion) give the martingale defect $1 - E_Q[V_T^\pi]/V_0^\pi$ as 0.0455 at $\pi = 0.25$ and 0.0130 at $\pi = 0.4$ for $T = 1$, and 0.371 and 0.148 for $T = 5$, with agreement to better than 10^{-6}. On a grid of elasticities $\{0.5, 1, 1.5, 2\}$ and exposures from 0.25 to 2, the defect is positive exactly in the cells predicted by (2). At $\pi = 1/2$ the finite-difference method gives only an upper bound of order 10^{-4} and the closed form gives zero.

Over an infinite horizon the classification collapses. Allow now $\pi \geq 0$, including $\pi = 0$, and take $V^\pi = v_0 Z^\pi$ at all times, with the exit convention of Theorem 1.

Proposition 4 (uniform integrability only at zero exposure). Under the standing assumptions and (W), V^π is a martingale and $\{V_t^\pi : t \geq 0\}$ is uniformly integrable if and only if $\pi = 0$. With Theorem 1, the classification reads:

Exposure	**Martingale on every finite horizon exactly when**	**Uniformly integrable martingale on** $[0, \infty)$
$\pi = 0$	always	yes
$0 < \pi < 1/2$	$I_0 = \infty$	no
$\pi = 1/2$	always	no
$\pi > 1/2$	$I_\infty = \infty$	no

Status. Conditional on the uniform-integrability criterion of Mijatović and Urusov ([60], Theorem 2.3), stated as a hypothesis with a vacuous instance, and on the inputs of Theorem 1 for the finite-horizon column; the endpoint computation is proved unconditionally. Machine-checked (Appendix A).

Proof. The criterion gives a uniformly integrable martingale exactly when the integrand vanishes almost everywhere, or a suitable endpoint is good. For $\pi > 0$ the integrand $\pi\sigma(x)/x$ vanishes nowhere, and both endpoints are bad, as in the proof of Theorem 1: infinity because the original scale is unbounded, and zero because the goodness integrand π^2/x is not integrable there, whether or not S reaches zero. For $\pi = 0$ the integrand vanishes. The constant $v_0 > 0$ preserves both properties. □

So every positive exposure fails over an infinite horizon, whatever the two endpoint integrals. The statement concerns the conjunction of the martingale property and uniform integrability over all $t \geq 0$; it does not say that a strict local martingale's values cannot be uniformly integrable, since some are, and it does not bear on uniform integrability over a fixed finite interval. On geometric Brownian motion, a martingale for every π, $E[V_t^\pi 1\{V_t^\pi > 10\}]$ with $v_0 = 1$ tends to one as t grows for every $\pi > 0$, for example 0.006, 0.629 and 1.000 at $t = 10$, 100 and 1,000 for $\pi = 1/4$, so its values are not uniformly integrable although every finite-horizon mean is one. On Brownian motion absorbed at zero, $E[V_t^\pi] < 1$ for $0 < \pi < 1/2$, for example 0.8465, 0.5166 and 0.2931 at $t = 1$, 10 and 100 for $\pi = 1/4$, and $E[V_t^\pi] = 1$ for $\pi \geq 1/2$, as Theorem 1 says.

Letting the exposure decay over time restores uniform integrability in the geometric model. Let the price be a weak solution of $dS = \sigma(S)\,dW$ on the canonical path space with cemetery time ζ, with σ as in the standing assumptions, let π_t be a predictable exposure with $|\pi_t| \leq M$ on every path, and let wealth be $V = v_0 Z$, with Z the stochastic exponential of $\int \pi_t\, dS_t/S_t$, extended at the end of its lifetime as in Ruf's criterion ([73], Proposition 3.1 and Theorem 3.3). That criterion says that V is a martingale on every finite horizon, or a uniformly integrable martingale on $[0,\infty)$, exactly when the energy $\int \pi_s^2\sigma(S_s)^2/S_s^2\, ds$ is finite almost surely under an auxiliary law Q_{aux}, up to every finite time or over the whole lifetime respectively.

Proposition 5 (time-varying exposure). (a) If $\pi_s^2\sigma(x_s)^2/x_s^2 \leq h(s)$ on every path and for almost every s before the end of its lifetime, for a deterministic locally integrable $h \geq 0$, then V is a martingale on every finite horizon, and a uniformly integrable martingale on $[0,\infty)$ if $\int_0^\infty h < \infty$. (b) For geometric Brownian motion, $\sigma(x) = \kappa x$, the lifetime is infinite under the pricing law and under every auxiliary law, V is a martingale on every finite horizon, and it is a uniformly integrable martingale exactly when $Q_{\text{aux}}(\int_0^\infty \pi_s^2\, ds < \infty) = 1$. For a deterministic bounded schedule $\pi(t)$ this is $\int_0^\infty \pi(t)^2\, dt < \infty$: e^{-t} gives a uniformly integrable martingale, $\pi \equiv 1$ a martingale that is not uniformly integrable, and $(1+t)^{-\gamma}$ a uniformly integrable one exactly when $\gamma > 1/2$. (c) Bounded exposure alone is not enough: with $\sigma(x) = x^2$ and $\pi \equiv 1$, V is strict on every finite horizon.

Status. Conditional on Ruf's criterion, stated as a hypothesis with a degenerate instance, and, for (c), on Theorem 1 and its inputs. Machine-checked (Appendix A).

Proof. (a) The pathwise bound gives a deterministic bound on the energy, finite at every finite time, and finite over the whole lifetime when h is integrable, under any auxiliary law. (b) Each path's range before ζ is compact, so the lifetime of the criterion is ζ. Since the energy is at most $\kappa^2 M^2 t$, the localizing times exceed every finite T once their index is large, so the event $\{\zeta \leq T\}$ is in the stopped σ-field, and the criterion's density identity carries its Q-nullity to every auxiliary law. Then the energy is $\kappa^2 \int_0^\infty \pi_s^2\, ds$. (c) $\int_c^\infty x/\sigma(x)^2\, dx = 1/(2c^2) < \infty$ and $\pi = 1 > 1/2$, so Theorem 1 applies. □

On geometric Brownian motion with $\kappa = 1$ and $v_0 = 1$, $E[V_T 1\{V_T > 10\}]$ at $T = 10^{12}$ is 0.0018 for e^{-t} and 0.1785 for $(1+t)^{-3/4}$, but 0.9857 and still rising for $(1+t)^{-1/2}$, whose energy grows logarithmically, and 1.0000 for $(1+t)^{-0.4}$ and for $\pi \equiv 1$. The auxiliary law matters: the bounded adaptive exposure $\pi_t = 1\{S_t > s_0\}$

on geometric Brownian motion has finite energy almost surely under the pricing law, because the price eventually stays below s_0, but under the auxiliary law the path escapes upwards with probability one half, and wealth is not uniformly integrable, with $E[V_\infty] = 1/2$; in a simulation the mean of V_T truncated at 100 fell from 0.888 at $T = 5$ to 0.503 ± 0.002 at $T = 100$ and 200. Finite energy under the pricing law does not suffice, so the classification by $\int \pi(t)^2\,dt$ holds for deterministic schedules only.

Two independent assets each keep their own endpoint test. Let S_1 and S_2 satisfy the standing assumptions with their own σ_i, driven by Brownian motions in a common filtration, and let Z_i be the exponential of Theorem 1 for exposure $\pi_i \geq 0$, with its own exit convention, and $Z_i \equiv 1$ when $\pi_i = 0$. Assume the full systems (S_1, Z_1) and (S_2, Z_2) are independent under Q, and take wealth $V = v_0 Z_1 Z_2$ at all times, supplied as a continuous local martingale. Write $m_i(T) = E_Q[Z_{i,T}]$ and $\delta_i(T) = 1 - m_i(T)$.

Proposition 6 (two independent assets). (a) For every $T \geq 0$, $m_i(T) \in [0, 1]$ and $1 - E_Q[V_T]/v_0 = \delta_1(T) + \delta_2(T) - \delta_1(T)\delta_2(T)$. For $T > 0$, V is a martingale on $[0, T]$ if and only if each coordinate passes the test of Theorem 1 with its own endpoint integrals; otherwise V is strict on every finite horizon. (b) V is a martingale and $\{V_t : t \geq 0\}$ is uniformly integrable if and only if $\pi_1 = \pi_2 = 0$. (c) With $\sigma_1 \equiv 1$, $\sigma_2(x) = x$ and $\pi_1 = \pi_2 = 1/4$, both prices are martingales, but V is strict on every finite horizon, with $m_2 \equiv 1$ and defect $\delta_1(T) > 0$.

Status. Conditional on the supermartingale property of nonnegative local martingales, stated as a hypothesis with a degenerate instance, and on the inputs of Theorem 1 and Proposition 4 for each factor, supplied as premises. Machine-checked (Appendix A).

Proof. (a) Independence factorizes the mean. Each factor and the product are supermartingales from a constant, so each is a martingale on $[0, T]$ exactly when it keeps its mean at T, and $m_1 m_2 = 1$ with $m_i \leq 1$ forces both means to one. (b) For independent nonnegative X, Y with $E[Y] = 1$, $E[X1\{X \geq 2C\}] \leq 2E[XY1\{XY \geq C\}]$, so uniform integrability of the product, with both means one at every time, passes to each factor, and Proposition 4 forces both exposures to zero. (c) For $\sigma_1 \equiv 1$, $I_0 < \infty$, so $\pi_1 = 1/4$ fails its test; for $\sigma_2(x) = x$ both endpoint integrals diverge, so the second factor is a martingale. □

An independent true-martingale factor does not cure the zero-endpoint obstruction. With $\sigma_1 \equiv 1$, $\sigma_2(x) = x$ and $\pi_1 = \pi_2 = 1/4$, both prices are true martingales, but the product's defect equals that of the absorbed Brownian factor: 0.0005, 0.1535, 0.4834 and 0.7069 at $T = 0.1$, 1, 10 and 100, in agreement with a Monte Carlo check.

Theorem 1 therefore extends in three directions, and stops at a fourth. Over an infinite horizon, constant exposure gives a uniformly integrable martingale only at zero exposure (Proposition 4). On geometric Brownian motion, bounded time-varying exposure gives a martingale on every finite horizon, and a deterministic schedule gives uniform integrability exactly when its square is integrable, under the auxiliary law rather than the pricing law (Proposition 5); bounded exposure alone is not enough in general. For independent assets, wealth is a martingale exactly when each factor passes its own test, and uniformly integrable exactly when both exposures are zero (Proposition 6); an absorbed Brownian factor at exposure $1/4$ keeps its defect even when paired with a true-martingale geometric Brownian motion. Correlated assets and liquidation through default are not covered.

3.2 What the volatility on a bounded range identifies

Fix $0 < c < R < \infty$ and σ satisfying standing assumption 1. Let $\mathscr{C}_R(\sigma)$ be the class of all coefficients g satisfying standing assumption 1 with $g(x) = \sigma(x)$ for every $0 < x \leq R$. For $g \in \mathscr{C}_R(\sigma)$ write $F_c(g) = \int_c^\infty x/g^2(x)\,dx \in [0, \infty]$, let $\theta(g) = 1$ if $F_c(g) < \infty$ and $\theta(g) = 0$ otherwise, and let $J_{cR} = \int_c^R x/\sigma^2(x)\,dx$. The

identified set of a quantity is the set of its values over all coefficients compatible with the observation, here σ on $(0, R]$.

Proposition 7 (the volatility tail is not identified). Under these assumptions,

$$\{F_c(g) \,:\, g \in \mathscr{C}_R(\sigma)\} = (J_{cR}, \infty], \qquad \{\theta(g) \,:\, g \in \mathscr{C}_R(\sigma)\} = \{0, 1\}. \tag{7}$$

The lower endpoint J_{cR} is not attained. If σ is positive and continuous, the completions exhibited in the proof are positive and continuous as well.

Status. The identities (7) are unconditional given standing assumption 1 and are machine-checked, together with the completions (9) and the tail value (10). Their reading as a statement about bubbles is conditional on the JKP criterion: for each g, and for an absorbed driftless diffusion with coefficient g as in standing assumption 2, $\theta(g) = 1$ exactly when that diffusion is a strict local martingale. We do not construct those diffusions (Appendix A).

Proof. J_{cR} is finite, because $x \le R$ on $[c, R]$ and $1/\sigma^2$ is integrable there. For $g \in \mathscr{C}_R(\sigma)$,

$$F_c(g) = J_{cR} + \int_R^\infty \frac{x}{g^2(x)}\, dx. \tag{8}$$

The last integral is strictly positive, possibly infinite, because its integrand is measurable and strictly positive on $(R, R+1)$. Hence $F_c(g) \in (J_{cR}, \infty]$. Conversely, put $k = \sigma(R) \neq 0$ and, for $p \ge 1$,

$$g_p(x) = \sigma(x) \quad (0 < x \le R), \qquad g_p(x) = k\,(x/R)^p \quad (x > R). \tag{9}$$

Each g_p is Borel and nowhere zero, and $1/g_p^2 \le 1/k^2$ above R, so $g_p \in \mathscr{C}_R(\sigma)$; it is positive and continuous when σ is. Above R the integrand is $x/g_p^2(x) = (R^{2p}/k^2)\, x^{1-2p}$. For $p = 1$ its integral over $[R, H]$ is $(R^2/k^2)\log(H/R)$, which tends to infinity, so $F_c(g_1) = \infty$. For $p > 1$,

$$\int_R^\infty \frac{x}{g_p^2(x)}\, dx = \frac{R^2}{2(p-1)k^2}, \tag{10}$$

and the choice $p = 1 + R^2/(2k^2 a)$ gives the value a for any $0 < a < \infty$. By (8) every value in $(J_{cR}, \infty]$ is attained, and θ takes both values. □

Proposition 7 is the exact, set-valued form of the extrapolation problem of [41] (§1 and §5). There it is stated informally: the volatility function can be estimated only at levels the price has visited, so its tails cannot be known, nor whether the JKP integral is finite. The two remedies proposed there, a parametric fit checked against a nonparametric estimate and an extension in a reproducing kernel Hilbert space, each choose one completion g. Proposition 7 says that without such a structural choice the observation leaves every value in $(J_{cR}, \infty]$ open.

What Proposition 7 does not say matters as much. Its observation is the coefficient σ known exactly on $(0, R]$. That is not a finite sample of prices, nor even a continuously observed price path. It does not compare the laws of price observations under two completions, and it does not control the probability that the price reaches R. So it is not a theorem that tests of "no bubble" against "bubble" have no power, and $\{0, 1\}$ in (7) is not a confidence statement; Theorem 18 below is a statement of that kind for price data, over a broader null class. Two features of the observation should be noted. The result is monotone in information: an observation that reveals no more than σ on $(0, R]$, for example σ on a subrange, has a compatible class containing $\mathscr{C}_R(\sigma)$, so it does not identify θ either. On the other hand, the observation is

generous at the lower end, because σ on $(0, c]$ determines I_0. Under this observation, then, the condition of Theorem 1 for $0 < \pi < 1/2$ is identified and the condition for $\pi > 1/2$ is not. Proposition 17 shows that with σ known only on $[r, R]$, $r > 0$, neither endpoint integral is identified.

3.3 Discretely rebalanced and inverse funds

Exchange-traded leveraged and inverse funds rebalance at discrete dates, pay fees and financing costs, and track their target imperfectly. Between two rebalancing dates such a fund holds a fixed number of shares and a cash position. Fix a horizon $T > 0$, a pricing probability Q with a filtration satisfying the usual conditions, and a continuous nonnegative Q-local martingale X with $X_0 = 1$: the underlying price discounted by a positive reference cash account and normalized by its initial value. For a real exposure π, the fixed-share-plus-cash contract over one holding interval, before costs, is, in the same normalized units,

$$G_t^\pi = (1 - \pi) + \pi X_t, \qquad 0 \le t \le T. \tag{11}$$

The net asset value is modelled as

$$Y_t = G_t^\pi - c_t + e_t, \tag{12}$$

where the costs c and tracking deviations e are specified adapted processes, integrable at each time, with $c_0 = e_0 = 0$ and $E_Q|e_T| \le \varepsilon$. Write $\delta_T = 1 - E_Q[X_T]$ and $d_T = E_Q[c_T]$. A concrete daily convention takes $T = \Delta$ one day, the reference gross return $B = 1 + r\Delta$, an expense rate f, a stock-borrowing spread λ and a spread ℓ on borrowed cash, and sets

$$C = \big[f + (-\pi)^+\lambda + (\pi - 1)^+\ell\big]\Delta, \qquad c_T = C/B. \tag{13}$$

With $\ell = 0$ this is the daily return relation of [6] (its equation 1 for $\pi > 1$ and equation 2 for $\pi \le -1$); for other exposures, and for $\ell > 0$, it is our stated convention.

Proposition 8 (one holding interval: a signed gap). G^π is a local martingale. For $\pi \ne 0$ it is a martingale on $[0, T]$ exactly when X is, and

$$1 - E_Q[G_T^\pi] = \pi\,\delta_T, \quad 0 \le \delta_T \le 1, \qquad \big|(1 - E_Q[Y_T]) - \pi\,\delta_T - d_T\big| \le \varepsilon, \tag{14}$$

with $\delta_T > 0$ exactly when X is strict on $[0, T]$. If the underlying has a bubble, the gap is positive for $\pi > 0$ and negative for $\pi < 0$.

Status. Conditional on bounded stopping and the supermartingale property [38]. Machine-checked in full, from those two inputs stated as hypotheses, including the cost convention (13), its two specializations to [6], and the deterministic-expense example below (Appendix A).

Proof. The stopped processes $X_{\cdot\wedge\tau_n}$, $\tau_n = \inf\{t : X_t > n\}$, are bounded martingales, so their affine images localize G^π. For $\pi \ne 0$, $X = (G^\pi - 1 + \pi)/\pi$, so either is a martingale when the other is. The supermartingale property gives $0 \le E_Q[X_T] \le 1$; if $E_Q[X_T] = 1$, the nonnegative gaps $X_u - E_Q[X_t \mid \mathcal{F}_u]$ have mean zero and vanish, so X is a martingale. Taking expectations in (11) and (12) gives (14). □

So the defect of a leveraged or inverse fund over one interval is $\pi\delta_T$ plus its costs, up to tracking error. Two cautions follow. A fall in the mean value of a fund is not by itself a bubble: with $X \equiv 1$, $e = 0$ and $c_t = t/(2T)$, the value $Y_t = 1 - t/(2T)$ is positive and deterministic and not a local martingale, and its shortfall is an expense. And for $\pi < 0$ the gap $\pi\delta_T$ is negative: G^π is a signed strict local martingale, not a positive price with a bubble.

Proposition 9 (inverse funds need a liquidation rule). The value $1 - \pi + \pi x$ is nonnegative for every $x \geq 0$ exactly when $0 \leq \pi \leq 1$. If X is strict on $[0, T]$ and $\pi < 0$, then

$$Q\big(G_t^\pi < 0 \text{ for some } t \leq T\big) > 0. \tag{15}$$

For $\pi < 0$ and $1 < K < 1 - 1/\pi$, let $\tau_K = \inf\{t \geq 0 : X_t > K\}$ and liquidate into cash at τ_K: $H_t = 1 - \pi + \pi X_{t \wedge \tau_K}$. Then H is a uniformly integrable martingale with

$$0 < 1 + \pi(K - 1) \leq H_t \leq 1 - \pi. \tag{16}$$

This holds even when X is strict on every finite horizon, for example $dX = X^2\, dW$, $X_0 = 1$, with $\pi = -1$ and $K = 3/2$, where $1/2 \leq H \leq 2$.

Status. Conditional on bounded stopping and the supermartingale property; the example also on the JKP criterion. Machine-checked, with the example as a conditional corollary (Appendix A).

Proof. The positivity range is read off the two coefficients. If (15) failed, G^π would be a nonnegative local martingale started at one, so $E_Q[G_T^\pi] \leq 1$, contradicting $E_Q[G_T^\pi] = 1 - \pi\delta_T > 1$. The stopped $X_{\cdot \wedge \tau_K}$ is a martingale with values in $[0, K]$, and H is its affine image with $\pi < 0$, which gives (16). In the example $\int_1^\infty x/x^4\, dx = 1/2 < \infty$, so X is strict by the JKP criterion. □

A nonnegative inverse fund is therefore a different contract from (11): it needs a stated rule for what happens when the underlying rises far enough, and under the rule above it has no bubble even though its underlying does. Whether an inverse fund "inherits" its underlying's bubble depends on that rule, not on π alone.

The same rule works on a finite grid.

Proposition 10 (finite-grid inverse funds with a liquidation floor). Let $\pi < 0$, let W be the signed fixed-share recursion that rebalances to exposure π at the dates $0 = t_0 < \cdots < t_N = T$ of a finite grid (the recursion (17) below, now with $\pi < 0$), and for a floor $0 \leq a < 1$ let τ_a be the first time W enters $(-\infty, a)$, capped at T, monitored continuously between rebalancing dates. The liquidated wealth $H^a = W_{\cdot \wedge \tau_a}$ is a continuous local martingale with $H_0^a = 1$, equal to a from τ_a on when $\tau_a < T$, and

$$a \leq H_t^a \leq (1 - \pi)^N \quad \text{for every } t.$$

Hence H^a is a uniformly integrable martingale with $E_Q[H_T^a] = 1$, strictly positive if $a > 0$, even when the underlying is strict. With one interval, τ_a is the first time X exceeds $1 + (1 - a)/(-\pi)$, the rule of Proposition 9, and if X is strict on $[0, T]$ then $Q(\tau_a < T) > 0$.

Status. An application of bounded stopping [38], with the finite-trading fact, the stopping-time property of exit times and the stopping property of stochastic integrals; not a new closure theorem. Machine-checked from these inputs stated as hypotheses (Appendix A).

Proof. W is a continuous local martingale by the finite-trading fact, and τ_a, an entry time of an open set, is a stopping time, so H^a is a continuous local martingale. Before τ_a the wealth is at least a, and at a finite entry it equals a. Each interval factor $1 - \pi + \pi X_t / X_{t_i}$ is at most $1 - \pi$, so induction over the intervals gives $H^a \leq (1 - \pi)^N$. A bounded local martingale is a uniformly integrable martingale. With one interval the floor is reached exactly when X exceeds the stated level; if τ_a never occurred before T, H^a would agree with the signed contract, which would make X a martingale by Proposition 8. □

Three limits should be stated plainly. The floor holds only under continuous monitoring with continuous prices and exact execution: a fund that checks its floor at discrete times, or whose price gaps, can

overshoot the floor and go negative. The upper bound is for each fixed grid and grows with N, so nothing uniform in the number of rebalances is claimed. And the rule is a model of liquidation, not a description of any issuer's actual practice.

For $0 \le \pi \le 1$ and a finite grid $0 = t_0 < \cdots < t_N = T$, rebalancing at each t_i to exposure π gives, with $W_0 = 1$ and for $t_i \le t \le t_{i+1}$,

$$W_t = W_{t_i}\big[(1-\pi) + \pi X_t / X_{t_i}\big] \text{ if } X_{t_i} > 0, \qquad W_t = W_{t_i} \text{ if } X_{t_i} = 0. \tag{17}$$

The second case keeps the fund in cash after the underlying has been absorbed at zero.

Proposition 11 (finite grids remove the zero-endpoint effect). If X is a martingale on $[0,T]$, then for every finite grid W is a nonnegative martingale on $[0,T]$. In the setting of Example 3(a), with $\pi = 1/4$, every finite-grid W is a martingale while the continuous wealth $V^{1/4}$ of Theorem 1 is strict, and

$$E_Q|W_T - V_T^{1/4}| \ge 1 - E_Q[V_T^{1/4}] > 0 \quad \text{for every grid.} \tag{18}$$

Status. Conditional on bounded stopping and the supermartingale property, and for the example on Theorem 1 and the JKP criterion. Machine-checked, with the example as a conditional corollary (Appendix A).

Proof. On $\{X_{t_i} > 0\}$ the holding $\pi W_{t_i}/X_{t_i}$ is nonnegative and $\mathscr{F}_{t_i}$-measurable; pulling it out of the conditional expectation (through bounded truncations and monotone convergence) gives $E_Q[W_t \mid \mathscr{F}_{t_i}] = W_{t_i}$, and on $\{X_{t_i} = 0\}$ the wealth is constant. The tower property carries this across rebalancing dates. In Example 3(a) the underlying is a martingale by the JKP criterion and $V^{1/4}$ is strict by Theorem 1, so $E_Q[W_T] = 1 > E_Q[V_T^{1/4}]$, and (18) is the triangle inequality. □

At a default the discretely rebalanced fund keeps its cash fraction, while the continuously rebalanced wealth is absorbed at zero with the price (Section 2). The lower-endpoint case of Theorem 1 is thus a property of rebalancing continuously into default, absent on every finite grid. By (18), finite-grid wealth cannot converge to $V^{1/4}$ in L^1 at T; we do not assert or exclude convergence in a weaker sense.

Over a strict underlying, the bubble survives every finite grid.

Proposition 12 (finite-grid funds inherit the underlying's bubble). Let X be any continuous nonnegative Q-local martingale with $X_0 = 1$, let $0 < \pi \le 1$, and let W be defined by (17) on a finite grid. Then W is a nonnegative continuous local martingale, and it is a martingale on $[0,T]$ exactly when X is, and strict there exactly when X is. In the driftless setting of Section 2, every such W is strict exactly when $I_\infty < \infty$, with no condition at the lower endpoint. For $\sigma(x) = x^2$ and $\pi = 1/4$, every finite-grid W is strict, whereas $V^{1/4}$ of Theorem 1 is a martingale.

Status. Conditional on the supermartingale property [38] and on the fact that a finite sum of predictable multiples of increments of a continuous local martingale is again one ([26], Theorem 7.11 and Proposition 7.14); the diffusion statements also on the JKP criterion and, for the comparison with $V^{1/4}$, on Theorem 1. Machine-checked from these inputs stated as hypotheses, with the diffusion statements as conditional corollaries (Appendix A).

Proof. On each interval $W_t = W_{t_i} + h_i(X_t - X_{t_i})$ with the $\mathscr{F}_{t_i}$-measurable holding $h_i = \pi W_{t_i}/X_{t_i}$, zero when $X_{t_i} = 0$, so W is a continuous local martingale by the cited fact, and nonnegative because $0 < \pi \le 1$. With the conditional losses $A_i = X_{t_i} - E_Q[X_{t_{i+1}} \mid \mathscr{F}_{t_i}] \ge 0$, pulling out h_i gives the exact decomposition $1 - E_Q[W_T] = \sum_{i=0}^{N-1} E_Q[h_i A_i]$. If X is a martingale, every A_i vanishes and W is a martingale. Conversely, $W_{t_{i+1}} \ge \pi(W_{t_i}/X_{t_i})X_{t_{i+1}}$ gives by induction $W_{t_i} \ge \pi^i X_{t_i}$, hence $h_i \ge \pi^{i+1}$, and the decomposition is at least

$\pi^N(1-E_Q[X_T])$, which is positive when X is strict. The diffusion statements follow from the JKP criterion; for $\sigma(x)=x^2$, $\int_1^\infty x/x^4\,dx = 1/2$ while $\int_0^1 x/x^4\,dx=\infty$, so by Theorem 1 $V^{1/4}$ is a martingale. □

The result is the classification, not the bound $\pi^N(1-E_Q[X_T])$ used in its proof, which is far too small to be informative at realistic N (daily rebalancing over a year has N about 250). Together with Proposition 11, it shows that the continuous-rebalancing criterion of Theorem 1 differs from every finite grid in both directions: $\sigma\equiv 1$, $\pi=1/4$ gives martingale funds and strict continuous wealth, and $\sigma(x)=x^2$, $\pi=1/4$ the reverse. Costs and tracking enter as in Proposition 8: if the terminal value is $Y_T = W_T - c_T + e_T$, with c_T, e_T the terminal difference from W and $E_Q|e_T|\le\varepsilon$, then $|(1-E_Q[Y_T])-(1-E_Q[W_T])-E_Q[c_T]|\le\varepsilon$.

With Proposition 11, Proposition 12 means that the threshold $1/2$ and the zero-endpoint case of Theorem 1 can hold at most as limits as the rebalancing becomes continuous.

Such a limit holds away from zero.

Proposition 13 (grid refinement before zero). Let X be a continuous nonnegative Q-local martingale with $X_0=1$, absorbed at zero, let π be any real number, and let $W^{(n)}$ be the fixed-share recursion (17) on finite grids of $[0,T]$ whose mesh tends to zero, not necessarily nested. For $m>1$ let ρ_m be the first time X leaves $(1/m,m)$, capped at T, let $L^m=\int dX^{\rho_m}/X^{\rho_m}$ and $V^{\pi,m}=\exp(\pi L^m-\pi^2\langle L^m\rangle/2)$. Then $W^{(n)}_{\cdot\wedge\rho_m}\to V^{\pi,m}$ uniformly on $[0,T]$ in probability. If $Q(X_t>0 \text{ for all } t\le T)=1$, then $W^{(n)}\to V^\pi=\exp(\pi L-\pi^2\langle L\rangle/2)$, $L=\int dX/X$, uniformly on $[0,T]$ in probability, and $Q(\inf_{t\le T} W^{(n)}_t\le 0)\to 0$. In either case, expectations of bounded functionals of the path that are Lipschitz in the uniform norm converge, and if the terminal values are uniformly integrable, $E_Q|W^{(n)}_T-V_T|\to 0$ and the bubble sizes $1-E_Q[W^{(n)}_T]$ converge to $1-E_Q[V_T]$. In the driftless setting of Theorem 1 with σ locally bounded and $\pi>0$, the exponent of V^π is that of the stochastic exponential of $\int b_\pi(X)\,dW$, the wealth of Theorem 1.

Status. An application of Émery's stability theorem for stochastic differential equations ([84], Theorem 2), with the stochastic integral and exponential as in [26] and the stopping-time property of exit times; it is not a new approximation theorem. Machine-checked from these inputs and the supermartingale property stated as hypotheses (Appendix A).

Proof. Stopped at ρ_m, the recursion is the solution of $U=1+\int F_n(U)_{s-}\,dX^{\rho_m}_s$ with $F_n(U)_t=\pi U_{\kappa_n(t)}/X^{\rho_m}_{\kappa_n(t)}$, $\kappa_n(t)$ the last grid point at or before t; these operators are non-anticipating and Lipschitz with the common constant $1+|\pi|m$, because $X^{\rho_m}\ge 1/m$, and converge at the limiting solution to $F(U)_t=\pi U_t/X^{\rho_m}_t$, whose equation is solved by $V^{\pi,m}$. Émery's theorem gives the first convergence. Under strict positivity $Q(\rho_m<T)\to 0$ and $V^{\pi,m}=V^\pi_{\cdot\wedge\rho_m}$, which removes the stop; the breach limit uses the positive minimum of V^π. The expectation statements are the bounded-Lipschitz comparison and Vitali's theorem. □

Three limits of its scope should be stated plainly. First, this convergence does not preserve strictness: for $0<\pi\le 1$ every finite-grid fund over a strict price is strict (Proposition 12), while for a strictly positive price the limit V^π is strict only when $\pi>1/2$. Second, bubble sizes converge only if the terminal values are uniformly integrable, which Proposition 13 assumes rather than proves (Proposition 14 proves it when the limit is a martingale); expectations of bounded Lipschitz functionals of the path, such as a put payoff for $0\le\pi\le 1$, converge in any case. Third, prices that can reach zero are not covered, consistent with (18).

When the continuous limit is a martingale, uniform integrability is not an extra hypothesis.

Proposition 14 (bubble sizes converge when the limit is a martingale). In the setting of Proposition 13 with $Q(X_t>0 \text{ for all } t\le T)=1$ and $0\le\pi\le 1$, suppose $E_Q[V^\pi_T]=1$. Then $E_Q|W^{(n)}_T-V^\pi_T|\to 0$, $E_Q[W^{(n)}_T]\to 1$, the family $\{W^{(n)}_T\}$ is uniformly integrable, and $E_Q[g(W^{(n)}_T)]\to E_Q[g(V^\pi_T)]$ for every Lipschitz g, calls and puts included. At $\pi=0$ and $\pi=1$ the grid wealth equals the limit (one, and the price itself), so these conclusions hold even over a strict price, with common bubble size $1-E_Q[X_T]$ at $\pi=1$.

Without the hypothesis $E_Q[V_T^\pi] = 1$, only $\limsup_n(1 - E_Q[W_T^{(n)}]) \le 1 - E_Q[V_T^\pi]$ holds. In the driftless setting of Theorem 1 with zero inaccessible ($I_0 = \infty$), the hypothesis holds for $0 < \pi \le 1/2$ for every σ, and for $0 < \pi \le 1$ when $I_\infty = \infty$.

Status. An application of Fatou's lemma, dominated convergence and the converse of Vitali's theorem ([26], Appendix B, for the probabilistic forms), conditional on the inputs of Propositions 12 and 13; the diffusion regimes are conditional in addition on the martingale criterion and (W). Machine-checked; the formal proof takes the convergence theorems from the Lean library rather than as cited inputs (Appendix A).

Proof. The grid wealths are nonnegative with mean at most one (Proposition 12 and the supermartingale property) and converge in probability to V_T^π (Proposition 13). If $E_Q[V_T^\pi] = 1$, Fatou's lemma forces their means to one, dominated convergence applied to the shortfall $(V_T^\pi - W_T^{(n)})^+ \le V_T^\pi$ gives L^1 convergence, and the converse of Vitali's theorem gives uniform integrability; the Lipschitz bound $|Eg(W) - Eg(V)| \le C\,E|W - V|$ gives the payoff prices. At $\pi = 1$ the grid factors telescope to X. The diffusion regimes are those where Theorem 1 makes V^π a martingale. □

So the loss of strictness in the limit is compatible with uniform integrability: for $\sigma(x) = x^2$ and $\pi = 1/4$, every grid fund is strict (Proposition 12), yet its bubble size tends to zero. When the limit is strict, here $1/2 < \pi < 1$ with $I_\infty < \infty$, only the one-sided bound holds, and whether the bubble sizes converge to that of the limit is open. As evidence only, a registered numerical experiment for $\sigma(x) = x^2$, $\pi = 3/4$, $T = 1$ (Appendix A) gives $E_Q[W_1] \approx 0.786$, 0.819 and 0.833 for $N = 4$, 64 and 1008 grid intervals, approaching $E_Q[V_1^{3/4}] \approx 0.8465$ from below, with small tail expectations; it cannot exclude mass escaping in rare paths.

Leveraged funds can be brought in with the liquidation rule of Proposition 10. For $\pi > 1$, let W be the fixed-share recursion on a finite grid with N intervals, H^a its liquidation the first time it enters $(-\infty, a)$, $0 \le a < 1$, monitored continuously, $M = \pi^N$, and $d_H = 1 - E_Q[H_T^a]$.

Proposition 15 (liquidated leveraged funds). H^a is a continuous local martingale with $H_0^a = 1$, $a \le H_t^a \le a + MX_t$ at every time, $0 \le d_H \le 1 - a$, and $d_H \le M\delta_T$; so $d_H > 0$ implies $\delta_T > 0$. If X is a martingale on $[0, T]$, so is H^a. With one interval, τ_a is the first time X falls below $1 - (1 - a)/\pi$ and $d_H = \pi(1 - E_Q[X_{\tau_a}])$.

Status. Conditional on the inputs of Propositions 10 and 12. Machine-checked (Appendix A).

Proof. On each interval before liquidation the wealth is at most π times its value at the last rebalancing date times the price ratio, so by induction $W \le MX$ up to τ_a, and $H^a = a$ afterwards. Then $R = MX - H^a + a$ is a nonnegative continuous local martingale started at $M - 1 + a$, and the supermartingale property for X, H^a and R gives the bounds on d_H. □

A positive bubble in a liquidated leveraged fund thus implies one in the underlying; no converse is asserted. The bound $M = \pi^N$ is enormous at realistic N, so d_H/M is a negligible lower bound on δ_T. The floor holds only under continuous monitoring with continuous prices and exact execution; with discrete monitoring the fund can overshoot it. And the rule is a model of liquidation, not a description of any issuer's practice.

Continuous monitoring cannot be simulated exactly, but it can be bounded from one side. For a liquidated fund of Proposition 10 or Proposition 15, and a finite set $F \subseteq [0, T]$ of monitoring times containing every rebalancing date, let $Z_F = a$ if $W_r < a$ at some $r \in F$, and $Z_F = W_T$ otherwise: follow the signed recursion until the first sampled breach and replace the sampled value by a. This is a numerical device, not the value of a fund, since the replacement is an artificial cash adjustment.

Proposition 16 (clipped monitoring bounds the liquidated defect). (a) On every continuous nonnegative price path, $a \le H_T^a \le Z_F$, with $Z_F - H_T^a = (W_T - a)1_{\{\tau_a<T,\ W_r\ge a \text{ for all } r\in F\}}$; a sampled breach gives $Z_F = H_T^a = a$, and $F \subseteq F'$ gives $Z_{F'} \le Z_F$. (b) $d_F = 1 - E_Q[Z_F]$ satisfies $d_F \le d_H \le 1 - a$; for $\pi < 0$, $E_Q[Z_F] \ge 1$; for $\pi > 1$, $d_F > 0$ implies $\delta_T > 0$, and $E_Q[Z_F] \ge 1$ when X is a martingale. (c) If the monitoring sets F_m become dense in $[0, T]$, for a fixed rebalancing grid, then $Z_{F_m} = H_T^a$ eventually on every path, $E_Q|Z_{F_m} - H_T^a| \to 0$ and $d_{F_m} \to d_H$. (d) No bound that tends to zero with the mesh holds uniformly over continuous paths: for every finite F, with $\pi = 2$, $a = 1/2$, one interval and $T = 1$, a path equal to one on F that dips to $1/2$ in a gap of F has $Z_F = 1$ and $H_T^a = 1/2$.

Status. Parts (a) and (d) are deterministic and unconditional; parts (b) and (c) are conditional on the inputs of Propositions 10 and 15. Machine-checked (Appendix A).

Proof. Before the open entry the wealth is at least a, so a sampled $W_r < a$ lies strictly after it and forces $H_T^a = a$; if no breach is sampled, $T \in F$ gives $W_T \ge a$, which is the identity in (a). Taking expectations gives (b), with Propositions 10 and 15. Under dense monitoring an open excursion below the floor is eventually sampled, so $Z_{F_m} = H_T^a$ eventually, and dominated convergence gives (c). The tent path gives (d). □

A positive d_F therefore shows a positive bubble in the continuously liquidated fund, and for $\pi > 1$, by Proposition 15, in the underlying; d_F itself can be negative. The convergence in (c) has no rate, and (d) shows that none is possible from the samples alone. Freezing the sampled wealth at a breach, without clipping it to a, gives no such one-sided bound, so simulated defects computed that way are not bounds on d_H. A positive simulated estimate of d_F is, moreover, evidence, not a proof of a positive expectation.

3.4 What price data cannot identify

We begin with coefficients and then turn to price data. Proposition 7 assumed σ known down to zero; a realistic observation covers neither endpoint.

Proposition 17 (neither endpoint integral is identified). Let $0 < r < c < R < \infty$, and let σ be Borel and nowhere zero on $[r, R]$ with $1/\sigma^2$ integrable there. Among Borel nowhere-zero extensions g of σ to $(0, \infty)$ with $1/g^2$ locally integrable, the lower tail $\int_0^r x/g^2(x)\,dx$ takes every value in $(0, \infty]$, and, independently, the upper tail $\int_R^\infty x/g^2(x)\,dx$ takes every value in $(0, \infty]$. Hence I_0 ranges over $\int_r^c x/\sigma^2 + (0, \infty]$, I_∞ over $\int_c^R x/\sigma^2 + (0, \infty]$, and both conditions of Theorem 1 are unidentified.

Status. Unconditional, deterministic, and machine-checked (Appendix A).

Proof. Below r put $g_p(x) = \sigma(r)(x/r)^p$ for real p. Then

$$\int_0^r \frac{x}{g_p^2(x)}\,dx = \frac{r^{2p}}{\sigma^2(r)}\int_0^r x^{1-2p}\,dx = \begin{cases} \dfrac{r^2}{2(1-p)\sigma^2(r)}, & p < 1, \\ \infty, & p \ge 1, \end{cases} \tag{19}$$

and $p = 1 - r^2/(2a\sigma^2(r))$ gives any $a \in (0, \infty)$. The value 0 is excluded because the integrand is positive. With the lower part fixed, the coefficient is known on $(0, R]$, and the completions (9) of Proposition 7 supply any upper tail in $(0, \infty]$ without changing it. □

This is a statement about coefficients. It does not say that a realized price path reveals σ exactly on its visited range, and it compares no laws of observations; the rest of this section does that.

Fix $s_0 > 0$, a finite horizon $T > 0$ and a number k of observed paths. Prices are discounted and pay no dividends. Let E_T be the space of continuous nonnegative paths on $[0, T]$ starting at s_0, with the uniform topology. An admissible model consists of a filtered probability space satisfying the usual conditions, a physical probability P, a pricing probability Q equivalent to P on $\mathcal{F}_T$, and an adapted nonnegative contin-

uous Q-local martingale S with $S_0 = s_0$. No coefficient form, tail bound or uniqueness of Q is imposed, and stopped or frozen prices are allowed. The data are k independent copies of S on $[0,T]$ under P. A test is a Borel map $\phi : E_T^k \to [0,1]$, the probability of rejecting; a finite sampling grid, or any other measurable function of the paths, is included by composition. The null hypothesis H_0 is the set of admissible models in which S is a Q-martingale on $[0,T]$, and the alternative H_1 the set in which it is strict there. The size of ϕ is $\mathrm{size}(\phi) = \sup_{H_0} E_{P^{\otimes k}}[\phi]$.

Theorem 18 (power is at most uniform size). For every test ϕ and every model in H_1,

$$E_{P^{\otimes k}}[\phi(S^1, \dots, S^k)] \le \mathrm{size}(\phi). \tag{20}$$

The bound is attained by $\phi \equiv \varepsilon$. Consequently, for fixed T, if the uniform size of a sequence of tests has limit superior at most ε as the number of paths grows, then so does its power at each fixed alternative. Nothing is claimed for tests whose size is controlled only pointwise over the null.

Status. Conditional on bounded stopping [38]. Machine-checked, from that input stated as a hypothesis, including sharpness and the asymptotic statement (Appendix A). The null class contains frozen prices (below), and Theorem 22 gives the bound with local-volatility nulls.

Proof. Fix a model in H_1 and $n > s_0$. Stop S at $\tau_n = \inf\{t : S_t > n\}$. By bounded stopping S^n is a bounded Q-martingale, so the same P, Q and the price S^n form a model in H_0. Couple each observed path with its stopped version on the same sample point, and let A_n be the event that some original path reaches n on $[0,T]$. Off A_n the original and stopped paths coincide, so $|\phi(S^1, \dots, S^k) - \phi((S^1)^n, \dots, (S^k)^n)| \le 1_{A_n}$, and

$$E_{P^{\otimes k}}[\phi(S^1, \dots, S^k)] \le \mathrm{size}(\phi) + P^{\otimes k}(A_n). \tag{21}$$

Continuous paths are bounded on $[0,T]$, so $P^{\otimes k}(A_n) \to 0$ for finite k, which gives (20). The asymptotic statement follows by taking limits superior of (20). □

The mechanism is the unobserved tail of Section 3.2 in statistical form: the null model differs from the alternative only above a level that the data reach with small probability. The abstract form, that a null approximation agreeing with an alternative with probability close to one caps the power of every bounded test, is standard, and [8] may anticipate it; we have not read that paper against this problem. Theorem 18 is the application of bounded stopping to this observation scheme. The stopped null model is frozen above n. It is not a diffusion with nowhere-zero volatility, so Theorem 18 does not say that tests confined to the local-volatility family of Section 2, whose null models are true-martingale diffusions, have no power. Theorem 22 supplies that restricted statement.

For the size of the bubble, take the unconditional defect $b_T = s_0 - E_Q[S_T]$, the bubble at time zero for maturity T when initial information is trivial. Fix a law P of the price path on E_T, with the canonical price and its natural filtration and no other traded assets, and set

$$\mathscr{M}(P) = \{Q \sim P : S \text{ is a } Q\text{-local martingale on } [0,T]\}, \qquad B_T(P) = \{s_0 - E_Q[S_T] : Q \in \mathscr{M}(P)\}. \tag{22}$$

Proposition 19 (the identified set of the bubble depends on the law of prices). Let $\mathscr{M}(P) \neq \emptyset$. Then $B_T(P)$ is the exact identified set of b_T given the law P: every admissible model with physical price law P, including models with hidden factors, has its defect in $B_T(P)$, and every point of $B_T(P)$ is attained. It is an interval in $[0, s_0]$ with infimum $s_0 - \sup_{\mathscr{M}(P)} E_Q[S_T]$ and supremum $s_0 - \inf_{\mathscr{M}(P)} E_Q[S_T]$, containing each endpoint exactly when the corresponding extremum is attained, and it excludes s_0 if $S_T > 0$ P-almost surely. If only the law R of a Borel observation O is known, for example prices on a finite grid, the identified

set is the union of $B_T(P)$ over all P with $O_\# P = R$ and $\mathscr{M}(P) \neq \emptyset$. For all maturities jointly, the set of defect curves $(s_0 - E_Q[S_t])_{0 \le t \le T}$, $Q \in \mathscr{M}(P)$, is convex, and each curve starts at zero and is nondecreasing; one common Q is required for all maturities, so this set is not the product of its marginals. There is no universal answer: a constant price path gives $B_T(P) = \{0\}$, and a known pricing law gives a singleton.

Status. Conditional on bounded stopping, the supermartingale property [38] and the fact that exit times of open sets by continuous adapted processes are stopping times ([26], its equation 14). Machine-checked, from those inputs stated as hypotheses, including the exactness of (22) for every path law (Appendix A).

Proof. A model with physical law P pushes its pricing measure forward to a probability equivalent to P on the path space. The bounded stopped prices S^n are martingales in the model's filtration, hence in the natural filtration of the price, and they localize the canonical price; so the pushforward lies in $\mathscr{M}(P)$ with the same terminal mean. Conversely each $Q \in \mathscr{M}(P)$ is a model. For $Q_1, Q_2 \in \mathscr{M}(P)$ and $\lambda \in [0,1]$, the same pathwise stopping times localize S under both, and the martingale identities for S^n are linear in the measure, so $\lambda Q_1 + (1-\lambda)Q_2 \in \mathscr{M}(P)$. Terminal means are affine in λ, so $B_T(P)$ is an interval, and the same argument makes the set of curves convex. The supermartingale property gives $0 \le E_Q[S_t] \le s_0$, nonincreasing in t, and $E_Q[S_T] > 0$ when $S_T > 0$ almost surely. Equivalence preserves the constant-path event, which forces $\{0\}$; a known pricing law fixes the mean. □

A known pricing measure identifies the bubble, and Theorem 18 is not contradicted: it concerns physical observations with Q unspecified. Under stochastic volatility even the full physical law can leave the bubble undetermined.

Proposition 20 (one physical law, a bubble and no bubble). There are a finite horizon T_0 and a single physical law P of a continuous, strictly positive price on $[0, T_0]$ such that $\mathscr{M}(P)$ contains a measure under which S is a martingale on $[0, T_0]$ and a measure under which it is strict there. Every test has the same rejection probability under both models, for every k. The identified set of the binary label is $\{0, 1\}$, and $[0, b] \subseteq B_{T_0}(P)$ for some $b > 0$; $s_0 \notin B_{T_0}(P)$.

Status. An application of Sin's model [75], conditional on its Theorems 3.2 and 3.6 as stated in Section 2, and on bounded stopping and the supermartingale property. Machine-checked from a configuration with one physical law and a strict and an equivalent martingale pricing measure; that configuration is derived from a hypothesis structure for Sin's model whose instance is vacuous, so this result is exactly as conditional as the prose (Appendix A). We do not determine $\sup B_{T_0}(P)$.

Proof. Take $\alpha = \kappa = L = 1$, $\gamma = (1, 0)$ and $a = (1, 1)$. Then $a \cdot \gamma = 1 > 0$ and $1 < \sqrt{2} = |a||\gamma|$, so the vectors are not parallel. Sin's theorems give a finite T_0 on which S is strict under Q^s, and $Q^m_{T_0} \sim Q^s$ under which it is a martingale on $[0, T_0]$. Let the physical measure be $P = Q^s$ in both models; the null prices with $Q^m_{T_0}$, the alternative with Q^s. The price process and its physical law are the same, so rejection probabilities agree exactly; the volatility drifts under the two pricing measures are $1 - \nu^2$ and $1 - \nu$. Let $b = s_0 - E_{Q^s}[S_{T_0}]$. If $b = 0$, the supermartingale property would give $E_{Q^s}[S_t] = s_0$ for all $t \le T_0$, and a supermartingale with constant mean is a martingale, a contradiction; so $b > 0$. The two measures give 0 and b in $B_{T_0}(P)$, and Proposition 19 gives $[0, b]$. Strict positivity of S_{T_0} excludes s_0. □

That bubbles depend on the choice among pricing measures in incomplete markets is not new in substance: Jarrow, Protter and Shimbo note that bubbles can be born in incomplete markets ([44], Section 1) and treat them in a companion paper [43], which we have not read. Proposition 20 is an exact instance in which the physical law of prices is held fixed. It also bears on the open stochastic-volatility question for Theorem 1, as far as identification goes: under stochastic volatility, the physical price law need not determine whether the asset itself has a bubble.

Corollary 21 (confidence sets). If C is a measurable random set with $P^{\otimes k}(b_T \in C) \geq 1-\varepsilon$ for every admissible model, then the test $1_{\{0 \notin C\}}$ has power at most ε at every alternative.

Status. As Theorem 18; machine-checked. This is a coverage implication, not a confidence procedure.

Proof. Under the null $b_T = 0$, so the test has size at most ε; apply (20). □

The frozen nulls of Theorem 18 can be replaced by genuine diffusions. Fix the setting of Section 2 and let the pricing law of a coefficient σ be the law ν_σ on E_T of the absorbed solution of $dS = \sigma(S)\,dW$, $S_0 = s_0$. A local-volatility model is a pair (σ, P) with σ satisfying standing assumption 1 and a physical law P on E_T equivalent to ν_σ; nothing is assumed about the physical drift or the likelihood ratio $dP/d\nu_\sigma$, and no tail bound or parametric form is imposed on σ. The null hypothesis is $I_\infty(\sigma) = \infty$, the alternative $I_\infty(\sigma) < \infty$, and size is the supremum over the null local-volatility models.

Theorem 22 (local volatility alone gives no power above size). For every test ϕ of k paths and every local-volatility model in the alternative, $E_{P^{\otimes k}}[\phi] \leq \mathrm{size}(\phi)$. The same holds when the coefficients are restricted throughout to positive continuous functions, when the physical law is restricted throughout to $P = \nu_\sigma$, or both, with size taken over the corresponding nulls. Explicitly, for $n > s_0$ let $\sigma_n = \sigma$ on $(0, n]$ and $\sigma_n(x) = \sigma(n)x/n$ above n, the completion g_1 of Proposition 7 with $R = n$, and let $p_n = P(\max_{[0,T]} S \geq n)$. Then there is a null model (σ_n, P_n) with $|E_{P^{\otimes k}}[\phi] - E_{P_n^{\otimes k}}[\phi]| \leq 1-(1-p_n)^k \leq k p_n$, and $p_n \to 0$.

Status. Conditional on uniqueness in law of the stopped equation ([45], Proposition 9.1) and on the existence of the absorbed pricing laws ([60], Section 2, with Feller's test excluding explosion), as stated in Section 2. Machine-checked from these inputs stated as premises (Appendix A).

Proof. Each σ_n satisfies standing assumption 1, is positive and continuous when σ is, and has $\int_n^H x/\sigma_n^2(x)\,dx = (n^2/\sigma(n)^2)\log(H/n) \to \infty$, so its absorbed diffusion exists, cannot explode because the scale function $s(x) = x - c$ is unbounded above, and is a martingale by the JKP criterion. Let A_n be the event that the path stays below n on $[0, T]$. Stopping at the first hit of n is the identity on A_n and maps paths that reach n to paths that still reach n, and the two coefficients agree on $(0, n)$, so uniqueness in law of the stopped equations gives $\nu_\sigma(B \cap A_n) = \nu_{\sigma_n}(B \cap A_n)$ for every Borel B. Put $r_n = \nu_{\sigma_n}(A_n^c)$, which is positive exactly when p_n is, and $P_n(B) = P(B \cap A_n) + (p_n/r_n)\,\nu_{\sigma_n}(B \cap A_n^c)$, or $P_n = P$ if $r_n = 0$. Then P_n is a probability equivalent to ν_{σ_n} that agrees with P on A_n, and it equals ν_{σ_n} when $P = \nu_\sigma$. The k-fold products agree on A_n^k, whose complement has mass $1-(1-p_n)^k$ under both, which gives the bound; $p_n \to 0$ because continuous paths are bounded on $[0, T]$. Letting $n \to \infty$ proves the theorem. □

Theorem 22 is the local-volatility specialization of Theorem 18 and closes its scope gap: every null is a genuine driftless local-volatility diffusion, positive and continuous whenever the alternative is, under $P = Q$ or any equivalent physical law. So local volatility, positivity and continuity alone do not make the presence of a bubble testable. It is the statistical form of the extrapolation problem of [41] (§1 and §5), with Proposition 7 supplying the completions; we claim no novelty for the general mechanism, which [8], unread, may anticipate. Two limits of scope should be stated plainly. First, the physical null P_n replaces the physical law above n by a rescaled pricing law, so it need not preserve the physical dynamics of the alternative. That does not mean a restriction on the physical dynamics escapes Theorem 22: the theorem already holds within $P = Q$, where the physical drift is zero and the likelihood ratio is one. A test escapes only if its model class excludes the linear-tail completions σ_n of the proof; a restriction such as unspecified eventual regularity of the tail need not do so. Second, Theorem 22 concerns tests from finitely many paths with size controlled uniformly over the null; it is not a statement about population identification. Under $P = Q$, with the full path law known, the bubble is identified (Proposition 19).

3.5 A usable price-based test

Sections 3.2 and 3.4 say what price data cannot identify. This section collects what a test from prices alone can use: the visited part of the JKP integral, an estimator of it, identification under a declared tail, pooling across names, calibrated bands, and a false-discovery list. None of it yet gives a test with useful power on realistic samples (Sections 5, 6 and 8).

What a single price path does reveal is the visited part of the JKP integral, through its local time. Let L_t^x be the semimartingale local time of S at level x up to a fixed time t, normalized so that the occupation formula reads $\int_0^t f(S_u)\,\sigma^2(S_u)\,du = \int f(x)L_t^x\,dx$ before absorption, and let m_t and M_t be the minimum and maximum of the path on $[0,t]$.

Proposition 23 (the visited part of the JKP integral from local time). Under standing assumptions 1 and 2, with S a supplied weak solution absorbed at zero, fix $t \geq 0$ and set the integrand below to zero when $S_u = 0$ or $L_t^{S_u} = 0$. Then, outside one Q-null set and simultaneously for every Borel $B \subseteq (0,\infty)$,

$$\int_0^t 1_B(S_u)\,\frac{S_u}{L_t^{S_u}}\,du = \int_{B\cap(m_t,M_t)} \frac{x}{\sigma^2(x)}\,dx < \infty.$$

The same holds almost surely under any probability P absolutely continuous with respect to Q on $\mathscr{F}_t$.

Status. An application of the occupation formula ([10], Lemmas 1–3), the quadratic variation of a stochastic integral ([26], Theorem 7.11), the positivity of diffusion local time after a level is hit ([60], Propositions A.1–A.2), and Feller's test for finiteness when the price can reach zero; not a new stochastic result. Machine-checked from these inputs stated as hypotheses (Appendix A).

Proof. On one event of probability one, the occupation formula on rational intervals and uniqueness of finite measures on the line give equality of the occupation measure of the path with $L_t^x\,dx$, before any path-dependent weight is chosen. Local time is positive at almost every level strictly between m_t and M_t: each such level is hit before t and before absorption, and positivity at a rational time after the hit carries to t by monotonicity, then Tonelli's theorem passes to almost every level. Local time vanishes outside $[m_t, M_t]$. Dividing the density L_t^x out on the set where it is positive gives the identity. □

With $B = (0,\infty)$ the right side is the integral of x/σ^2 over the whole visited range; with $B = (a,b)$ the indicator on the left cannot in general be dropped. The identity holds through absorption at zero, where time spent at zero contributes nothing, and needs no recurrence or martingale assumption. The local time in the denominator is the semimartingale local time; the chronological local time $L_t^x/\sigma^2(x)$ of [10] is a different object. Two limits should be stated plainly. The identity determines only the visited part of the JKP integral, $\int_{m_t}^{M_t} x/\sigma^2(x)\,dx$, and says nothing about the tail beyond M_t, which by Propositions 7 and 17 and Theorem 18 is what decides whether there is a bubble. And it is an identity for the continuous path and its exact local time: it provides no estimator from sampled prices and no rate.

Over a fixed compact interval, part of the JKP integral can be estimated from sampled prices. Suppose $\log S$ has a representation $d\log S_t = b_t\,dt + a(\log S_t)\,dW_t$ with diffusion coefficient $a(y) = \sigma(e^y)/e^y$, continuously differentiable and positive, whose law on each finite horizon is equivalent to that of the driftless equation $dY = a(Y)\,dW$, and observe S at $n+1$ equally spaced times in $[0,1]$. Fix $u < v$ and a lower bound $c > 0$ on $a^2 = \sigma(x)^2/x^2$ over $[e^u, e^v]$, supplied in advance. Let Γ_n be the kernel estimator of a^2 of [39], built from the sampled log prices on a spatial grid of mesh $2/\sqrt{n}$ with its occupation threshold, and let D be its event that the path spends enough time near every level of $[u,v]$, and D^n the sample version of that event. Set $B = \int_{e^u}^{e^v} x/\sigma^2(x)\,dx$ and $\hat{B}_n = \int_u^v 1/\max(\Gamma_n(y), c)\,dy$.

Proposition 24 (a compact part of the JKP integral from sampled prices). In this setting: (a) $0 < B \le (v-u)/c$ and $0 < \hat{B}_n \le (v-u)/c$, and $\hat{B}_n$ is a finite sum over the cells of the grid that meet $[u,v]$, of each cell's length in $[u,v]$ divided by $\max(\Gamma_n, c)$ at the cell, so it is a measurable function of the sampled prices. (b) For every sample, $|\hat{B}_n - B| \le c^{-2} \int_u^v |\Gamma_n(y) - a^2(y)|\, dy$. (c) With $D_n = D$ for every n, or $D_n = D^n$, $n^{1/3}(\hat{B}_n - B)1_{D_n}$ is bounded in probability, and $P(D_n \cap \{|\hat{B}_n - B| > \varepsilon\}) \to 0$ for every $\varepsilon > 0$. Also $P(D^n) \to P(D)$, and if $P(D) > 0$ the same rate and consistency hold conditionally on D^n.

Status. Conditional on Jacod's kernel estimation theorem ([39], Theorem 2), stated as a hypothesis with a vacuous instance; the change of variables, the bounds, the finite sum and the clipping bound (b) are proved without probability. Machine-checked (Appendix A).

Proof. (a) The substitution $x = e^y$ gives $B = \int_u^v a(y)^{-2}\, dy$, and $a^2 \ge c$ gives the bounds; Γ_n is constant on each grid cell, which gives the finite sum. (b) The map $x \mapsto 1/\max(x,c)$ is c^{-2}-Lipschitz, and $\max(a^2, c) = a^2$ on $[u,v]$. (c) The cited theorem gives the uniform rate $n^{-1/3}$ for Γ_n on the occupation event and the convergence of the sample events, and (b) transfers it to $\hat{B}_n$. □

This is an application of Jacod's theorem together with the bound (b), not new estimation theory. Its premises matter. The floor c must be known in advance and cannot be fitted from the same data: if it exceeds the true minimum of a^2, the estimator is biased for every n (with $a^2 \equiv 1$ and $c = 2$, $\hat{B}_n = 1/2$ while $B = 1$). The interval is fixed, not the random visited range of the identity above; estimators over data-chosen intervals, such as the reciprocal-local-time estimator of a registered numerical experiment (Appendix A), are different and inherit nothing from this rate. Off the occupation event no consistency is asserted. And the estimator gives no confidence interval, says nothing about the tail beyond e^v, and is not a test: Theorem 22 still applies.

A declared tail is the kind of restriction Theorem 22 calls for, and with it identification becomes exact. Fix, before seeing any data, an onset $R > 0$ and a ratio $r > 1$, and write $B_1 = \int_R^{Rr} x/\sigma^2\, dx$, $B_2 = \int_{Rr}^{Rr^2} x/\sigma^2\, dx$ and $I_R = \int_R^\infty x/\sigma^2\, dx$.

Proposition 25 (identification under a declared power tail). Let σ be positive, continuous and admissible, with $\sigma(x) = \kappa x^p$ for every $x \ge R$, for some $\kappa > 0$ and p.

(a) $B_1, B_2 > 0$, $q = B_2/B_1 = r^{2-2p}$, $p = 1 - \log q/(2\log r)$, $I_R = B_1/(1-q)$ if $q < 1$ and $I_R = \infty$ otherwise, and $I_R < \infty$ exactly when $B_2 < B_1$. Every pair of positive values of (B_1, B_2) is attained, so these identifications are sharp.

(b) If only $B_i \in [L_i, U_i]$ with $0 < L_i \le U_i$ is known, the identified set of the label $1_{\{I_R < \infty\}}$ contains 0 exactly when $L_1 \le U_2$ and contains 1 exactly when $L_2 < U_1$. If the bands are measurable and cover both integrals jointly with probability at least $1 - \alpha$ at every null model of a specified family, the rule that rejects when $U_2 < L_1$ has size at most α over that family.

(c) With sampled prices, if the estimators of Proposition 24 for the two blocks satisfy their $n^{-1/3}$ rate on the occupation events, the test rejecting when the two estimates differ by more than h_n, with $h_n \to 0$ and $n^{1/3}h_n \to \infty$, has rejection probability tending to zero at each fixed null, and to the probability of the occupation event of $[R, Rr^2]$ at each fixed alternative.

(d) If the onset is not declared, the identified set of the label is $\{0, 1\}$ over positive continuous coefficients that are eventually power laws, even with σ known on $(0, R_0]$ for any finite R_0, and the bound of Theorem 22 holds within that class.

Status. Parts (a), (b) and the coefficient statement of (d) are unconditional and machine-checked, except that the size statement in (b) assumes the supplied joint coverage. Part (c) is machine-checked with the rate of the estimator of Proposition 24 and the convergence of its occupation events as premises; Proposition 24

proves them in that form, conditionally on Jacod's kernel estimation theorem. The test statement of (d) is conditional on the inputs of Theorem 22. The reading of the label as the presence of a bubble is conditional on the JKP criterion (Appendix A).

Proof. (a) The substitution $x = ry$ gives $B_2 = r^{2-2p}B_1$ on the power tail, and the tail integral is a power integral that converges exactly when $p > 1$, that is, when $q < 1$; the attaining coefficients are the completions of Proposition 7 of a line below R. (b) Given positive values in the bands, (a) realizes every ratio in $[L_2/U_1, U_2/L_1]$, and the label is zero exactly when the ratio can be at least one; under joint coverage a rejection at a null forces an uncovered integral. (c) On the occupation event both estimates are within $o(h_n)$ of their targets, so the threshold separates $B_1 - B_2 > 0$ from $B_1 - B_2 \le 0$. (d) The completions with exponents 1 and 2 above any level are eventually power laws with opposite labels, and the null completions in the proof of Theorem 22 are exact power laws beyond n. □

This is the parametric extrapolation of [41] (§5.1) made exact; no gain over that method is claimed. The band rule's size rests on joint coverage that must be supplied, and the estimator of Proposition 24 does not supply it. The threshold test is pointwise, not uniform, and its power is capped by how often the path visits the whole interval. The identification comes from the declared onset and the exact form, and the data cannot test them: the tail above Rr^2 is never observed.

A first cross-sectional step declares a common exponent: for names $i = 1, \dots, m$ with declared onsets R_i and a common r, $\sigma_i(x) = \kappa_i x^p$ above R_i, with one unknown p and separate scales κ_i. Write $B_{i,1}, B_{i,2}$ for name i's blocks, and for rectangles $B_{i,j} \in [L_{i,j}, U_{i,j}]$ put $\ell_i = L_{i,2}/U_{i,1}$, $u_i = U_{i,2}/L_{i,1}$, $\ell_* = \max_i \ell_i$ and $u_* = \min_i u_i$.

Proposition 26 (a common power tail across names). (a) Every name has $B_{i,2}/B_{i,1} = q = r^{2-2p}$ and the same label $1_{\{q<1\}}$. If only the first blocks are observed, every p is compatible and the label set is $\{0, 1\}$ for every panel; any common $q > 0$ is attained with any positive first blocks. (b) With rectangles, the sharp set for q is $[\ell_*, u_*]$, empty if $\ell_* > u_*$. When it is nonempty, the exponent set is $[1 - \log u_*/(2\log r), 1 - \log \ell_*/(2\log r)]$, the label set contains 0 exactly when $u_* \ge 1$ and 1 exactly when $\ell_* < 1$, and $u_* < 1$ exactly when some single name has $U_{i,2} < L_{i,1}$. (c) If measurable rectangles cover all $2m$ integrals simultaneously with probability at least $1-\alpha$ at every null of the common-exponent class, the rule rejecting when $\ell_* \le u_* < 1$ has size at most α; per-name coverage failures α_i with $\sum_i \alpha_i \le \alpha$ give that coverage without independence, and there is a two-name example in which each name's coverage is exactly $1 - \alpha$ and the pooled rejection probability is exactly 2α. (d) With sampled prices and the premises of Proposition 25(c) for every name under one joint law, the rule rejecting when any name's threshold test rejects has rejection probability tending to zero at each fixed common null and to $P(\bigcup_i D_i)$ at each fixed common alternative, where D_i is name i's occupation event; $P(\bigcup_i D_i) \ge \max_i P(D_i)$.

Status. Parts (a)–(c) are unconditional, except that (c) assumes the supplied coverage; part (d) is conditional on the premises of Proposition 25(c) for each name. Machine-checked (Appendix A). The reading of the label as the presence of a bubble is conditional on the JKP criterion.

Proof. (a) and (b) follow from Proposition 25 name by name, with the separate scales realizing any first blocks; the intersection of intervals is $[\ell_*, u_*]$, and $\min_i u_i < 1$ exactly when some $u_i < 1$. (c) A rejection at a null forces an uncovered integral; the union bound gives the per-name version, and an explicit three-point space gives the example. (d) Apply Proposition 25(c) to each name and the union bound, using that the symmetric difference of the unions lies in the union of the symmetric differences. □

So the intersection can narrow the exponent range and signal incompatibility when it is empty, but it cannot produce a positive verdict that no single name produces, and calibrating each name at α does not control the pooled size. The only gain of the pooled threshold test is occupation: its power is the probability that some name's interval is fully visited, at least, and possibly equal to, the best single name's,

with no improvement in rate and no uniformity. The common exponent is itself a strong assumption that the data cannot test and that membership of an index does not imply; a nonempty intersection does not confirm it. Methods that average weak evidence across names are a different route and are not analysed.

Propositions 25 and 26 take jointly covering bands as given. They can be constructed by split likelihood ([82]) when the observations have a known likelihood. Suppose each name i has a nonempty finite catalogue Θ_i of parameters, each specifying a coefficient of the declared power-tail form, with exponent $p_i(\vartheta)$ and block integrals $b_{i,1}(\vartheta), b_{i,2}(\vartheta)$, and an exact, strictly positive conditional likelihood $g_{i,\vartheta}(v \mid u)$ of an evaluation block V_i given a training block U_i. Fit $\hat{\vartheta}_i$ on U_i alone, let $T_i(\vartheta) = g_{i,\hat{\vartheta}_i}(V_i \mid U_i)/g_{i,\vartheta}(V_i \mid U_i)$, keep $C_i = \{\vartheta : T_i(\vartheta) \le 1/\alpha_i\}$ with $\sum_i \alpha_i \le \alpha$, and take the extreme block integrals and the exponents $A_i = p_i(C_i)$ over C_i.

Proposition 27 (calibrated bands in a finite catalogue). Suppose each name's observations have the likelihood of its true parameter, with arbitrary dependence across names. (a) Every C_i is nonempty, and the bands $[\min_{C_i} b_{i,j}, \max_{C_i} b_{i,j}]$ have positive endpoints and cover all $2m$ block integrals simultaneously with probability at least $1 - \alpha$; in particular each name's band rule of Proposition 25(b) has size at most α_i whatever the other names are. (b) If all true exponents are equal, $P(\bigcap_i A_i = \emptyset) \le \alpha$. (c) If the catalogue has null parameters $p_i \le 1$, then $E_i = \min_{p_i(\vartheta)\le 1} T_i(\vartheta)$ is a positive e-value for name i's null, $E_P[E_i] \le 1$, and $\min(1, 1/E_i)$ is a valid p-value; C_i excludes the null parameters exactly when $E_i > 1/\alpha_i$.

Status. An application of the split-likelihood bound [82], stated as a hypothesis; there is not yet a hypothesis structure with an instance for it. Machine-checked from that input (Appendix A). The link between each likelihood and the pricing coefficient it lists is a modelling premise and appears in no formal statement.

Proof. At the true parameter, $E_P[T_i(\vartheta_i^*)] \le 1$ by the split-likelihood bound, so Markov's inequality gives $P(\vartheta_i^* \notin C_i) \le \alpha_i$, and the union bound gives simultaneous coverage without independence. The bands, the exponent intersection and the e-value are images of C_i, and $E_i \le T_i(\vartheta_i^*)$ at a null. □

So the size statements of Propositions 25(b) and 26(c) hold with derived rather than supplied coverage, rejecting a common exponent when the retained exponent sets do not intersect tests that restriction within the catalogue without verifying the power form outside the observed range, and not rejecting does not establish it. Its limit is severe and should be stated plainly: everything rests on a correctly specified finite catalogue of exact likelihoods for the observations, each tied to the pricing coefficient it lists, and for price data such a catalogue is available only under strong assumptions (Proposition 28), since an equivalent physical law alone does not specify a transition likelihood. Checks on finite examples confirm the arithmetic, not validity for price data, and the per-name e-values are inputs for a later multiple-testing procedure, not such a procedure.

A first instance with actual price laws exists under strong assumptions. Suppose the physical law equals the pricing law ($P = Q$), and each name's price follows exactly one of a finite, declared catalogue of global power laws $\sigma(x) = \kappa x^p$ with $p \in \{1, 2\}$ and $\kappa > 0$, containing at least one candidate with $p = 1$: geometric Brownian motion, a martingale, or the reciprocal of a three-dimensional Bessel process, a strict local martingale. Observe each name at the dates of a declared grid, train on its first prices by maximum likelihood, and evaluate on the rest, starting the evaluation chain at the last training price.

Proposition 28 (a price-law instance of the catalogue calibration). In this setting, with arbitrary dependence across names: (a) the products of the explicit lognormal and Bessel transition densities over the evaluation dates are exact, positive, normalized conditional densities of the evaluation prices given the training prices, the maximum-likelihood selector is measurable, and every candidate coefficient has the declared power tail with explicit block integrals; so all premises of Proposition 27 hold, and $p = 1$ is a

martingale while $p = 2$ is strict. (b) All conclusions of Proposition 27 hold: calibrated bands, the common-exponent test, and per-name e-values and p-values for "no bubble", unconditionally, with no occupation event. (c) On a fixed grid any two candidates have mutually equivalent laws, so no test with size below one has power one at any alternative, and a test of size zero has no power.

Status. An application of Proposition 27 with the cited lognormal and Bessel transition laws, which enter as hypothesis structures whose instances are vacuous; the laws of the prices on the grid are supplied as premises (Appendix A). Machine-checked.

Proof. The finite products of positive normalized one-step kernels are positive and normalized by iterated integration, and the true candidate's product is the conditional density of the observed evaluation prices. The block integrals are those of Proposition 25 for exact power laws. Then Proposition 27 applies. All candidate grid densities are positive, so the grid laws are mutually equivalent, which gives (c). □

Its limits are severe. It assumes no physical drift, exact global power laws rather than a tail above a level, and a correct finite catalogue; physical drift, other powers or scales, stochastic volatility or jumps void the calibration. By (c), power at realistic sampling, such as two years of monthly prices, may be low, and it is not quantified here.

The same construction works for any finite catalogue of constant-elasticity powers, with absorption at zero retained. Let each candidate be $dS = \kappa S^p\, dW$ for some $p > 0$, $\kappa > 0$, absorbed at zero, with the null catalogue the candidates with $p \le 1$. Let the reference measure be Lebesgue measure on $(0, \infty)$ plus a unit atom at zero, and the one-step kernel the cited constant-elasticity density on $(0, \infty)$, with an atom at zero of the cited mass for $p < 1$ and none for $p \ge 1$, and trapped at zero from zero. Split, fit and evaluate as in Proposition 28, with $\hat{G}_i$ the fitted evaluation likelihood and $G_{i,0}$ the null maximum.

Proposition 29 (a constant-elasticity catalogue with absorption). In this setting, with $P = Q$ and arbitrary dependence across names: (a) every C_i is nonempty and covers the true pair with probability at least $1 - \alpha_i$, jointly at least $1 - \alpha$; $\hat{G}_i > 0$ implies $G_{i,0} > 0$, so the score $E_i = \hat{G}_i / G_{i,0}$, or 0 if $\hat{G}_i = 0$, is finite on every nonnegative array, with $E_P[E_i] \le 1$ at every true null and a valid p-value. (b) The bands from the closed-form block integrals cover jointly with probability at least $1 - \alpha$, the band rule has size at most α_i, and the common-exponent test has size at most α. (c) The e-BH rule of Proposition 40 applied to these E_i has FDR $\le q\, m_0/m$, also after any zero-masked screen, and the labels are bubble labels: $p \le 1$ is a martingale and $p > 1$ a strict local martingale. (d) A name absorbed by the end of training has $E_i = 1$ and C_i the whole catalogue; a name absorbed later has $E_i \le 1$; so a name observed at zero is never flagged.

Status. An application of the cited constant-elasticity kernels, the support-aware split-likelihood bound (proved), the JKP criterion and the e-BH theorem, the stochastic inputs stated as hypothesis structures with vacuous instances and the grid laws supplied as premises. Machine-checked (Appendix A).

Proof. The chain densities are normalized from every starting value by iterated integration against the reference measure, and the support-aware bound gives coverage and the null moment. If the fitted likelihood is positive, either the fitted candidate is null, or a positive $p > 1$ chain has an all-positive or all-zero pattern, which some null candidate also gives positive likelihood. The bands use the block integrals of Proposition 25; the labels are the JKP criterion, $\int^{\infty} x/x^{2p}\, dx < \infty$ exactly when $p > 1$. □

A default thus counts as evidence for an absorbing null, by the structure of the procedure rather than by discarding defaulted names. The premises are those of Proposition 28: $P = Q$ and a correct finite catalogue. The misspecification bounds of Proposition 43 show that an off-grid volatility makes the guarantee nearly vacuous at high sampling frequency, and power at monthly sampling may be low; it is not quantified here.

Off-grid volatility and physical drift, for the geometric-Brownian part of the null, can be absorbed into the null itself. Add to each name's null the continuous family of Gaussian conditional laws for the evalua-

tion prices, with log returns independent and normal with mean $(b-\kappa^2/2)\Delta_j$ and variance $\kappa^2\Delta_j$ over steps Δ_j, for any drift b and volatility $\kappa > 0$. With evaluation prices $y_1, \dots, y_n$ after the last training price y_0, log returns R_j, $\tau = \sum_j \Delta_j$, $a = \sum_j R_j/\tau$ and $Q = \sum_j (R_j - a\Delta_j)^2/\Delta_j$, let D_G be the supremum of these densities over $b \in \mathbb{R}$ and $\kappa > 0$, and set $E_i^{\text{drift}} = \hat{G}_i / \max(G_{i,0}, D_G)$, or 0 if $\hat{G}_i = 0$ or $D_G = \infty$.

Proposition 30 (a continuous Gaussian null). In the setting of Proposition 29, with the actual law P arbitrary across names and in training, suppose each name without a bubble has, given its training prices, either a catalogue null law or one of these Gaussian laws for its evaluation prices. (a) On positive paths with $Q > 0$, the supremum is attained at $\kappa^2 = Q/n$, $b = a + Q/(2n)$, and

$$\log D_G = -\frac{n}{2}\Big[\log \frac{2\pi Q}{n} + 1\Big] - \frac{1}{2}\sum_j \log \Delta_j - \sum_j \log y_j;$$

if $Q = 0$, $D_G = \infty$. The score is finite, nonnegative and measurable on every nonnegative array. (b) At every name without a bubble, $E_P[E_i^{\text{drift}} \mid \text{training}] \le 1$, uniformly over all drifts and volatilities, so $E_i^{\text{drift}} > 1/\alpha_i$ has size at most α_i, $\min(1, 1/E_i^{\text{drift}})$ is a valid p-value, and the probability of some false rejection is at most $\sum \alpha_i$, without independence. (c) The e-BH rule of Proposition 40 applied to these scores has FDR $\le q\, m_0/m$, also after any zero-masked screen with the original m. (d) $E_i^{\text{drift}} \le E_i$ on every path; a name absorbed by the end of training has score one, one absorbed later has score at most one, a positive path with $Q = 0$ has score zero, and with a single evaluation step the score is at most one everywhere, so none of these is ever flagged.

Status. An application of the composite-null split-likelihood bound ([82], Theorem 3), which is proved here, and, for (c), the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the normalization of the Gaussian family is proved. Machine-checked (Appendix A).

Proof. (a) For fixed $\kappa^2 = w$, the log density is maximized at $b = a + w/2$, leaving $-(n/2)\log w - Q/(2w)$ up to constants, maximized at $w = Q/n$; if $Q = 0$ it is unbounded as $w \to 0$. (b) The denominator dominates every null density, catalogue or Gaussian, so the score's conditional mean under any null is at most the selected density's mass, which is at most one; Markov's inequality gives the tests. (c) is the e-BH theorem. (d) A larger denominator gives a smaller score; the absorbed cases are Proposition 29(d), and with one step $Q = 0$. □

This answers the off-grid failure of Proposition 43 for the geometric-Brownian part of the null, with no grid and no total-variation budget. The cost is power: rejections under E_i^{drift} are a subset of those under E_i, and the continuous Gaussian null absorbs much of the evidence a mild constant-elasticity alternative leaves in returns; a simulation below illustrates how much. The link to bubbles is a premise: reading such a name as bubble-free needs a supplied pricing law under which its price is driftless geometric Brownian motion, whose existence is not proved here, and neither a good Gaussian fit nor the sign of the drift supplies it. Drift at other powers, stochastic volatility, jumps and rounding are not covered.

The drift need not be left free. Declare in advance, for each name, a closed interval $B_i = [b_{\text{lo}}, b_{\text{hi}}]$ of physical drifts, let $D_{G,B}$ be the supremum of the Gaussian densities of Proposition 30 over drifts in B_i and all $\kappa > 0$, and set $E_i^B = \hat{G}_i / \max(G_{i,0}, D_{G,B})$, or 0 if $\hat{G}_i = 0$ or $D_{G,B} = \infty$.

Proposition 31 (a declared drift interval). In the setting of Proposition 30, suppose each name without a bubble has, given its training prices, either a catalogue null law or a Gaussian law with drift in B_i and any volatility for its evaluation prices. (a) On positive paths $D_{G,B}$ is infinite exactly when $Q = 0$ and $a \in B_i$; otherwise it is the largest of the Gaussian likelihoods at the two drift endpoints, each with its profiled variance $w_c = 2A(c)/(n + \sqrt{n^2 + \tau A(c)})$, where $A(c) = Q + \tau(a-c)^2$, and at the unconstrained maximizer

of Proposition 30(a) when its drift lies in B_i. The score is finite and measurable on every nonnegative array. (b) At every name without a bubble the score has conditional mean at most one, so the per-name tests, the p-value and the union bound hold as in Proposition 30, and the e-BH rule of Proposition 40 has FDR $\le q\,m_0/m$, also after a zero-masked screen with the original m. (c) On every path $E_i^{\mathrm{drift}} \le E_i^{B} \le E_i$; a narrower interval gives a larger score, and as the intervals grow to the whole line the scores decrease to E_i^{drift}. Absorbed names are never flagged.

Status. An application of the composite-null split-likelihood bound ([82]), which is proved here, and, for the false-discovery bound, the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) For a fixed variance the best drift in B_i is the clipped unconstrained one, so the supremum lies on an edge of the interval or on the ridge $b = a + w/2$; along each edge the profile in the variance is unimodal with peak w_c, and along the ridge it peaks at $w = Q/n$. If $Q = 0$ and $a \in B_i$, the likelihood is unbounded as $w \to 0$. (b) The denominator dominates every null density, as in Proposition 30(b). (c) Larger parameter sets give larger denominators. □

This ordering is not a power guarantee, and the interval must be fixed before the data are seen; choosing it after inspecting the scores voids the guarantee. In one simulation, of daily prices over two years with the first year for training, the interval $[-0.10, 0.20]$, a catalogue of powers $p \in \{1/2, 1, 3/2\}$ and scales $\kappa \in \{0.2, 0.3, 0.45\}$, $\alpha = 0.1$ and 10,000 paths per law, the score of Proposition 29 had null mean 28 and a 6.9% false-alarm rate under geometric Brownian motion with off-grid volatility 0.35, while the free-drift and bounded-drift scores had means about 0.13 and 0.14 and false-alarm rates below 0.1% under every Gaussian law tried, including one with drift outside the interval. Against $p = 3/2$ the price is power: with $\kappa = 0.3$ it fell from 41% for the score of Proposition 29 to 5% with free drift and 7% with bounded drift, and with $\kappa = 0.45$ from 60% to 21% and 24%. Bounding the drift recovers little; most of the loss comes from leaving the volatility scale free. This is one design and one interval, not a systematic study. Outside the interval the guarantee is lost. For a Gaussian law whose drift lies at distance δ from the interval, the in-bound law with the nearest drift and the same volatility σ is, by Proposition 43(c), within total variation $\sqrt{1 - \exp(-\delta^2\tau/(4\sigma^2))}$ of it, whatever the number of observations over the evaluation horizon τ: with the interval $[-0.2, 0.2]$, $\sigma = 0.3$ and one year, a drift of 0.21 gives about 0.017 and a drift of 0.25 about 0.083. This is a sufficient budget only: comparing bounded functions as in Proposition 43(a) turns it into a per-name size of at most α_i plus the budget, not α_i, it does not restore the false-discovery bound for the unclipped scores, and it tends to one as $\sigma \to 0$, so the interval alone gives no small uniform budget. Stochastic volatility, jumps, time-varying drift and rounding are not covered.

Restricting the volatility scale recovers much of the lost power, at the price of fragility. Declare in advance, for each name, a closed drift interval $B_i = [b_{\mathrm{lo}}, b_{\mathrm{hi}}]$ and a closed scale interval $[s_{\mathrm{lo}}, s_{\mathrm{hi}}]$ with $s_{\mathrm{lo}} > 0$, let $D_{G,B,C}$ be the maximum of the Gaussian densities of Proposition 30 over this rectangle, and set $E_i^{B,C} = \hat{G}_i / \max(G_{i,0}, D_{G,B,C})$, or 0 if $\hat{G}_i = 0$, where $G_{i,0}$ keeps every catalogue null, including those with scales outside the interval.

Proposition 32 (a declared drift and scale rectangle). In the setting of Proposition 30, suppose each name without a bubble has, given its training prices, either a catalogue null law or a Gaussian law with drift in B_i and scale in $[s_{\mathrm{lo}}, s_{\mathrm{hi}}]$ for its evaluation prices. (a) On positive paths $D_{G,B,C}$ is the largest of the Gaussian likelihoods at at most five explicit points of the rectangle: each drift endpoint with its profiled variance, the positive root of a quadratic, clipped to the scale interval; each variance endpoint with its profiled drift clipped to the drift interval; and the unconstrained maximizer of Proposition 30(a) when it lies in the rectangle. It is positive and finite there, and the score is finite and measurable on every nonnegative array. (b) At every name without a bubble the score has conditional mean at most one,

so the per-name tests, the p-value and the union bound hold as in Proposition 30, and the e-BH rule of Proposition 40 has FDR $\le q\,m_0/m$, also after a zero-masked screen with the original m. (c) On every path $E_i^{\mathrm{drift}} \le E_i^{B} \le E_i^{B,C} \le E_i$, where E_i^{B} is the bounded-drift score of Proposition 31, so the rejections are nested; enlarging the rectangle decreases the score, and as the scale interval grows to $(0,\infty)$ the score decreases to E_i^{B}. Absorbed names are never flagged.

Status. An application of the composite-null split-likelihood bound ([82]), which is proved here, and, for the false-discovery bound, the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) Along each drift edge the profile in the variance is unimodal, with its peak at the displayed root; along the ridge $b = a + w/2$ it peaks at $w = Q/n$; every point of the rectangle is dominated by one of the five candidates, and all five are feasible. (b) The denominator dominates every null density, so the argument of Proposition 30(b) applies. (c) Larger parameter sets give larger denominators. □

In the simulation design used above for the bounded-drift score, with drift 0.05 for the Gaussian laws and scale intervals $[0.25, 0.35]$ and $[0.28, 0.32]$, power against $p = 3/2$ rose from 20% with free drift to 44% and 52% at $\kappa = 0.45$, against 60% for the score of Proposition 29, and from 6% to 12% and 24% at $\kappa = 0.3$, against 41%; Gaussian nulls inside the rectangle kept means below one and false-alarm rates below 0.3% at $\alpha = 0.1$. A true volatility outside the declared interval loses the guarantee: at 0.40, the null mean was 1.2 and 5.5 for the two intervals, with false-alarm rates of 2% and 4%, and at 0.25, outside the narrower interval, 2.2 and 2%. The narrower the interval, the more power and the more fragility. The interval must be declared before the data are seen; choosing it from the training data voids the guarantee. One honest way to declare it is described next. A continuum of elasticity exponents or scales outside the finite catalogue, stochastic volatility, jumps and rounding are not covered.

An honest declaration costs most of that power. Declare the rectangle of Proposition 32 from a disjoint earlier record W, by measurable rules fixed in advance, with a fallback to the free-drift score of Proposition 30, and call the resulting score E_i^{dec}. For a name without a bubble, the coverage event is that its true null law is retained by the declared rectangle or by the fallback.

Proposition 33 (declarations from an earlier window). Suppose that, given W and its training prices, each name without a bubble has a fixed catalogue null or Gaussian law for its evaluation prices. (a) On the coverage event, E_i^{dec} has conditional mean at most one given W and the training prices, so $E_i^{\mathrm{dec}} > 1/\alpha_i$ has conditional size at most α_i; if coverage holds almost surely, the e-BH rule of Proposition 40 has FDR $\le q\,m_0/m$, also conditionally on W. (b) If coverage fails with probability at most δ, the rule $E_i^{\mathrm{dec}} > 1/\alpha_i$ has size at most $\alpha_i + (1-\alpha_i)\delta$, the threshold $(1-\delta)/(\alpha_i - \delta)$ has size at most α_i when $\delta < \alpha_i$, and the safe score $\min(E_i^{\mathrm{dec}}, c_i)/(1 + (c_i - 1)\delta)$, for $c_i \ge 1$, has mean at most one, so e-BH applied to it keeps FDR $\le q\,m_0/m$ under any dependence; it can reject only if $\delta < \alpha_i$ and $c_i > (1-\delta)/(\alpha_i - \delta)$. (c) If the earlier window has $n \ge 2$ Gaussian log returns with the same parameters as the later law, the parameters whose likelihood is at least δ times that of a reference Gaussian law fixed in advance lie in an explicit rectangle, with drift and scale endpoints computed from the two roots of a one-variable equation in the variance and with a positive lower scale; this rectangle fails to retain the true parameters with probability at most δ, and the fallback has probability zero. (d) Any measurable upper scale that covers every scale almost surely is infinite almost surely; a finite upper scale misses a later scale with probability tending to one as that scale grows, so a change of scale between the windows breaks (c); and for every $\gamma \in (0, 1]$ there is a catalogue and a declaration failing with probability γ under which the raw score of a true driftless geometric Brownian motion has infinite null mean.

Status. An application of the composite-null split-likelihood bound and likelihood confidence inversion ([82]), both proved here, the mixture bound of Proposition 46 and, for the false-discovery bounds, the e-

BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the counterexample in (d) needs no cited input. Machine-checked (Appendix A).

Proof. (a) On the coverage event the denominator dominates the true null density, so the argument of Proposition 30(b) applies conditionally on W. (b) Integrate: the conditional mean is at most one on the coverage event and the capped score is at most c_i off it. (c) The likelihood set is a confidence set by the split-likelihood bound with a single-point null; for each variance, the drift section is an interval around $a + w/2$, and the variance range is between the two roots. (d) A positive density gives every set of positive measure positive probability; the counterexample pairs a rare narrow declaration with the strict power-two catalogue law, whose density bound makes the raw mean diverge. □

The first premise is substantive: given the earlier record as well as the training prices, the later evaluation law must be the same law with the same parameters. Disjoint timestamps do not ensure it, and a bound on parameter change, or an envelope as in Proposition 43, would have to be supplied. Because coverage is only probabilistic, the raw score must not be used. In one simulation, with a one-year daily earlier window, $\delta = 0.05$, the reference law with drift 0 and volatility 0.3, the test design above on the following two years, $\alpha = 0.1$, cap 50 and 2,000 paths per law, the rectangle covered the true parameters in at least 99.6% of earlier windows, so the budget δ is conservative. Under Gaussian nulls the safe score never rejected. Against $p = 3/2$ its power was 8% at $\kappa = 0.3$ and 15% at $\kappa = 0.45$, against 24% and 52% for the narrower interval after Proposition 32, chosen knowing the true scale; to reject, the uncapped score had to exceed 34.5 rather than 10. Much of the power recovered by a scale interval thus depends on knowing the scale, and the free-drift score of Proposition 30 remains the simplest valid choice.

Variance need not accrue in calendar time. Overnight and weekend gaps, and the open and close, carry different amounts of variance per unit of calendar time, while the physical drift accrues in calendar time. Declare in advance a finite catalogue $\mathscr{V}_i$ of variance clocks $v = (v_1, \dots, v_n)$, with $v_j > 0$, that contains the calendar clock $v = \Delta$, and consider the laws whose log returns are independent normal with mean $b\Delta_j - wv_j/2$ and variance wv_j, for any drift b and scale $w > 0$. On a positive path put $A_v = \sum_j \Delta_j^2/v_j$, $B_v = \sum_j \Delta_j R_j/v_j$, $Q_v = \sum_j R_j^2/v_j - B_v^2/A_v$ and $H_v = \sum_j v_j - (\sum_j \Delta_j)^2/A_v$, let D_v be the supremum of these densities over b and w, and set $E_i^{\mathrm{clock}} = \hat{G}_i / \max(G_{i,0}, \max_{v \in \mathscr{V}_i} D_v)$, or 0 if $\hat{G}_i = 0$ or the denominator is infinite.

Proposition 34 (declared variance clocks). In the setting of Proposition 30: (a) if $Q_v > 0$, the supremum is attained uniquely at $w^* = 2Q_v/(n + \sqrt{n^2 + H_v Q_v})$ and $b^* = (B_v + w^* \sum_j \Delta_j/2)/A_v$, with a closed-form value; if $Q_v = 0$, $D_v = \infty$. $Q_v = 0$ exactly when all return rates R_j/Δ_j coincide, whatever the clock; this set has measure zero when $n \geq 2$, and contains every positive path when $n = 1$. (b) Suppose that, given its training prices, each name without a bubble has an evaluation law that is a mixture of catalogue null laws and such clock laws. Then the score is finite and measurable, has conditional mean at most one, $E_i^{\mathrm{clock}} > 1/\alpha_i$ has size at most α_i, and the e-BH rule of Proposition 40 keeps FDR $\leq q\, m_0/m$. (c) $D_{cv} = D_v$ for every $c > 0$, and D_Δ is the Gaussian supremum of Proposition 30; so $E_i^{\mathrm{clock}} \leq E_i^{\mathrm{drift}}$, adding clocks lowers the score, and if every clock is proportional to Δ the score is the free-drift score.

Status. An application of the split-likelihood bound, which is proved here, with an explicit Gaussian profile, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) Completing the square in b leaves Q_v and H_v; in w, the gap from the root w^* of $H_v w^2 + 4nw - 4Q_v = 0$ is $-(n/2)(\log t + 1/t - 1) - (H_v/8)(w - w^*)^2/w$ with $t = w/w^*$, which is negative for $w \neq w^*$. Q_v is the weighted residual of regressing the returns on Δ, so it vanishes exactly when the returns are

proportional to Δ. (b) The denominator dominates every null component; integrate over the latent law. (c) Scaling v by c and w by $1/c$ leaves each density unchanged. □

Covering a clock costs power, as other enlargements of the null do. In a simulation with two years of weekday closes, weekend gaps of three calendar days, training on the first year, $\alpha = 0.1$ and 2,000 paths per law, the null was a geometric Brownian motion whose variance accrues only in trading time. The free-drift score, which does not cover it, had null mean 0.23 and a false-alarm rate of 0.05%. With the trading clock added, the score covered the null, but its power against $p = 3/2$, $\kappa = 0.45$ fell from 6.2% to 0.55%. The constant-elasticity forecast in that design ignores the clock, which weakens both scores; a forecast that follows a clock is given in Proposition 64. The guarantee is for exact prices and declared clocks: a clock fitted to the data has no coverage guarantee, and observation noise, rounding and jumps are not covered. At intraday frequency, with a forecast that follows the clock, the declared clock pays for itself. A simulation used sessions of 6.5 hours with overnight gaps and a declared U-shaped clock, 2,000 paths per null and 1,000 per alternative, and constant-elasticity alternatives moving in clock time. There, the score of Proposition 34 had no rejections at $\alpha = 0.05$ on any of five Gaussian nulls, including two whose clocks lie outside the declared set, which carry no guarantee. Its power against the four alternatives with $p \in \{3/2, 2\}$ and $\kappa \in \{0.3, 0.45\}$ was 85 to 100% over 250 sessions of five-minute bars, 3 to 65% over 60 sessions, and 11 to 77% at hourly sampling over 250 sessions. The free-drift score with a calendar-time forecast and null also had essentially no false alarms on these paths, but found at most 33% and mostly under 7%. A clock-aware forecast over a calendar-time Gaussian null, in contrast, rejected nulls with clocks outside the declared set in 26 to 40% of five-minute paths. The 250 sessions include the training half, and real intraday prices may not follow any declared clock.

Implied volatility can serve as the clock, known one step ahead. Let prices S_t and implied volatilities $IV_t > 0$ be adapted to a filtration $(\mathscr{F}_t)$, with the training record in $\mathscr{F}_0$, deterministic durations $\Delta_t > 0$, returns $R_t = \log(S_t/S_{t-1})$ and $v_t = IV_{t-1}^2 \Delta_t$. Declare, before the evaluation record, a band $0 < w_L \le w_H$ for the ratio of realized to implied variance. The null is that, for one drift b and one ratio $w \in [w_L, w_H]$ over the whole record,

$$R_t \mid \mathscr{F}_{t-1} \sim N\big(b\Delta_t - wv_t/2,\ wv_t\big), \qquad 1 \le t \le N;$$

how implied volatility itself evolves is unrestricted. Choose normalized forecasts q_t of R_t measurable in $\mathscr{F}_{t-1}$, let D_t^{IV} be the supremum over b and $w \in [w_L, w_H]$ of the product of these conditional normal densities over the returns up to t, and set $E_t^{IV} = \prod_{j \le t} q_j(R_j)/D_t^{IV}$, with $E_0^{IV} = 1$.

Proposition 35 (an implied-volatility-scaled null). (a) For $t \ge 1$, D_t^{IV} is positive, finite and measurable, and is attained at the scale of Proposition 34(a) computed on the realized v_j and clipped to $[w_L, w_H]$, or at w_L if the returns are proportional to Δ, with the corresponding drift; in particular it is finite with one step. (b) Under any law satisfying the null, E^{IV} is nonnegative and adapted, and for every stopping time $\tau \le N$ and $0 < \alpha \le 1$, $E[E_\tau^{IV} \mid \mathscr{F}_0] \le 1$ and $P(E_t^{IV} \ge 1/\alpha \text{ for some } t \le N \mid \mathscr{F}_0) \le \alpha$; the same holds for any score with a smaller numerator and a larger denominator. (c) A local-lognormal law whose returns are $N(b\Delta_t - a_t/2, a_t)$ given the past, with $a_t = \kappa^2 (S_{t-1}/s)^{2p-2} \Delta_t$, and with $IV_{t-1} = \kappa (S_{t-1}/s)^{p-1}/\sqrt{w_0}$ for some $w_0 \in [w_L, w_H]$, satisfies the null for every p, so (b) holds for it.

Status. An application of the product rule for conditional e-values, proved from Mathlib, with the profile of Proposition 34. Machine-checked (Appendix A).

Proof. (a) Completing the square in b as for Proposition 34 leaves a function of w that increases up to the free root and decreases after it, so its maximum over the band is at the clipped root; the lower end keeps it finite when the weighted residual vanishes. (b) Under the null with the true (b, w), each

factor $q_t(R_t)/p_{b,w,t}(R_t)$ has conditional mean $\int q_t = 1$ given $\mathcal{F}_{t-1}$, so their product is a test martingale; D_t^{IV} dominates the true product, so E^{IV} is at most that martingale, and optional stopping and Ville's inequality apply, conditionally on each event of $\mathcal{F}_0$. (c) Substituting gives $w_0 v_t = a_t$. □

The conditioning must be on the past only. Treating the whole implied-volatility path as given fails: a later implied volatility can reveal the sign of an earlier return, and a forecast that bets on that sign then rejects with probability one at threshold 2. By (c), level dependence that implied volatility tracks belongs to the null, so a rejection shows a departure from the null, such as non-Gaussian returns, a changing variance ratio or a changing drift, and not necessarily a bubble; E^{IV} need not itself be a supermartingale. In a check with 250 daily returns, the band $[0.5, 1]$, a forecast with $p = 2$ level dependence that ignores implied volatility, and 2,000 paths per law, the score never reached 20 under the local-lognormal law with $p = 2$, $\kappa = 0.45$ and implied volatility tracking the level, where it stayed at or below one at every time on every path, nor under a random predictable implied volatility. With the same returns but implied volatility frozen at its starting value, it reached 20 on 77% of paths: a departure from the null, not a bubble. A variance ratio drifting inside the band, which the null does not cover, was not flagged by this forecast. The band must cover the actual ratio of realized to implied variance, and its width decides how much level dependence is absorbed. For monthly S&P 500 data from 1990 to 2007, the mean realized variance was 14.93 against a mean squared implied volatility of 33.23 ([15], Table 1), a ratio of about 0.45, below a band such as $[0.5, 1]$. Returns with jumps, regime switches, stochastic volatility beyond the implied volatility, or a time-varying variance premium are outside the null.

Other names can narrow the clock catalogue. For a tested name, take r reference names observed over the same evaluation steps, and let A be all the training information. Form a normalized forecast q^{ref} of the references' joint evaluation prices from A alone, for example a mixture over clocks of products of Gaussian densities. For each declared clock v, let D_v^{ref} be the product of the references' own Gaussian suprema on that clock, as in Proposition 34, and keep the calendar clock and every clock with $q^{\mathrm{ref}}/D_v^{\mathrm{ref}} \le 1/\delta$; call the kept set C. Score the tested name with the denominator of Proposition 34 over the clocks in C, giving E^{sel}, with any normalized forecast that does not use its own evaluation prices, and let $E^{\mathrm{safe}} = \min(E^{\mathrm{sel}}, c)/(1+(c-1)\delta)$ for a fixed $c \ge 1$.

Proposition 36 (a clock confidence set from other names). Suppose that, for a name without a bubble, given A there is a latent component that is either a catalogue null for the tested name, or a clock v together with drifts and scales for which the tested evaluation law is the Gaussian law of Proposition 34 on v and, given A and the component, the references are independent Gaussian laws on the same clock v, independent of the tested name. (a) The calendar clock is always kept, C grows as δ decreases, and on every Gaussian component the true clock is excluded with probability at most δ, with no union cost over the catalogue. (b) Given the references, the raw score has conditional mean at most one when the true clock is kept. In all cases E^{safe} has mean at most one given A, $E^{\mathrm{sel}} > 1/\alpha$ has probability at most $\alpha+(1-\alpha)\delta$, and the e-BH rule of Proposition 40 applied to the safe scores with the original m keeps FDR $\le q\, m_0/m$. (c) E^{sel} is at least the score over the full catalogue; the safe score is at most $c/(1 + (c-1)\delta)$, and can reject at level α only if $\delta < \alpha$ and $c > (1-\delta)/(\alpha-\delta)$.

Status. An application of the split-likelihood bound, which is proved here, to the references and to the tested name, of the correction of Proposition 33 and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) Each reference density is at most its Gaussian supremum, so $q^{\mathrm{ref}}/D_v^{\mathrm{ref}}$ has mean at most one under the true clock, and Markov's inequality bounds the exclusion probability; only the true clock matters. (b) Given the references, the tested law is still the component's law, since the tested name is independent

of the references; on the coverage event the denominator dominates it, and off it the capped score is at most c. Integrating over the references and the component gives the bounds, as in Proposition 33. (c) A smaller set of clocks gives a smaller denominator. □

The premise is strong: the references must share the tested name's clock, with no common return shock after conditioning on it, which a market factor violates. Name-specific changes of the clock's shape, and clocks outside the catalogue, are not covered. In the weekend design after Proposition 34, with 20 reference names on the trading clock, $\delta = 0.01$, cap 100 and 2,000 panels per law, the references always reduced the catalogue to the calendar and trading clocks, and no null panel raised a false alarm. But against $p = 3/2$, $\kappa = 0.45$ at $\alpha = 0.05$, the safe score had power 12.3%, against 14.9% for the uncorrected score over the full catalogue. The excluded clocks barely constrained the tested path, so the narrowing gained little and the safety correction cost more. A gain would need wrong clocks that would otherwise absorb the evidence of a bubble.

Drift and volatility can also be allowed to change from day to day. Split the N evaluation steps into consecutive days, day d with n_d steps and a declared finite set of clock shapes, each day starting from the previous day's last price. Let P_d be the largest of the Gaussian suprema of Proposition 34 over day d's shapes, $P_{\text{day}} = \prod_d P_d$, and $E_{\text{day}} = q/P_{\text{day}}$, or 0 if $q = 0$ or $P_{\text{day}} = \infty$, for a normalized forecast q of the evaluation prices chosen on the training record.

Proposition 37 (daily drift and scale profiles). Suppose that, given the training record and the earlier days, each day's evaluation prices follow a finite mixture of the Gaussian clock laws of Proposition 34 for that day, with drift, scale and weights that may depend on the history but are fixed before the day's returns. (a) P_{day} is the supremum of the product of daily Gaussian densities over independent daily choices of shape, drift and scale; it is infinite exactly when some day's returns are proportional to its time steps, which always happens for a one-step day, and this set has measure zero when every day has at least two steps; rescaling a day's shapes leaves it unchanged. (b) E_{day} has conditional mean at most one, $P(E_{\text{day}} > 1/\alpha) \le \alpha$, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$ at a fixed horizon. (c) Grouping days to share parameters, or a single profile over the whole record with clocks whose daily restrictions are declared, gives a denominator at most P_{day}, so the same forecast scores at least as high. (d) With the calendar clock, let Q_d be day d's weighted residual and Q the residual of one profile over the whole record, which also counts the spread of the daily return rates. Then

$$\log P_{\text{day}} - \log P_{\text{common}} = \frac{N}{2} \log \frac{Q}{N} - \sum_d \frac{n_d}{2} \log \frac{Q_d}{n_d} \ge 0,$$

with equality exactly when all days have the same return rate and the same residual variance per step.

Status. An application of the split-likelihood bound with a supremum that separates over days, using the profile of Proposition 34, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter only through an optional forecast. Machine-checked (Appendix A).

Proof. (a) The product of daily densities has independent parameters for each day, so its supremum is the product of the daily suprema; a singular day can be driven to infinity with the others fixed. (b) Each mixture component of each day's law is at most that day's supremum, so the null density is at most P_{day} at every history. (c) Fewer free parameters give a smaller supremum. (d) The pooled residual decomposes as $\sum_d Q_d + \sum_d T_d(\hat{a}_d - \hat{a})^2$, with T_d day d's duration and $\hat{a}_d$, $\hat{a}$ the daily and pooled return rates; insert both profiles of Proposition 34. Equality needs every day's factor to attain its own maximum at the common optimum. □

So the daily score can only lose to the single profile with the same forecast, by exactly that amount, while switches inside a day are excluded by the premise. In a simulation with days of seven equal steps, 60 training and 60 evaluation days, a training-selected constant-elasticity forecast and 2,000 records per law, a null whose scale switched between 0.15 and 0.45 with probability $1/10$ a day and whose drift was drawn afresh each day gave the single-profile score a mean of about 5×10^7 and a 3.2% rate of scores above 20, while the daily score stayed calibrated. But the resets cost about 1.1 in log terms per day even under a constant geometric Brownian motion (a median J of 68 over 60 days), so against $p = 2$, $\kappa = 0.45$ the rate of scores above 20 fell from 26.8% for the single profile to zero. A forecast that adapts to each day would be needed for power, and none is supplied.

The scale need not reset every day. Declare one clock shape for each day and a budget K on the number of days at which the variance scale changes, keeping a free drift for every day; the budget and the shapes are declared before the data and may not be fitted. For a run I of consecutive days, let P_I be the supremum of the product of the daily Gaussian densities over one common scale and a free drift for each day, and let $P_{\mathrm{sw}}(K)$ be the supremum over paths with at most K scale changes, with $E_{\mathrm{sw}} = q/P_{\mathrm{sw}}(K)$, or 0, for a normalized forecast q chosen on the training record.

Proposition 38 (a budget on daily scale switches). (a) With the daily statistics of Proposition 34 summed over the run, P_I has a closed form, attained at a scale that is the positive root of a quadratic and at each day's own profiled drift; it is infinite exactly when every day of the run has returns proportional to its time steps. (b) $P_{\mathrm{sw}}(K)$ is the maximum, over partitions of the days into at most $K+1$ consecutive runs, of the product of the P_I, computed exactly by a recursion on the last run; it is infinite exactly when an allowed partition has a singular run, and otherwise attained. (c) If, given the training record, the evaluation prices follow a mixture of the daily Gaussian laws over admissible paths, then E_{sw} has conditional mean at most one, $P(E_{\mathrm{sw}} > 1/\alpha) \le \alpha$, and the e-BH rule of Proposition 40 keeps FDR $\le q\,m_0/m$; the same holds for a finite catalogue of full-record shapes. (d) $P_{\mathrm{common}} \le P_{\mathrm{sw}}(0) \le \ldots \le P_{\mathrm{sw}}(D-1) = P_{\mathrm{day}}$, so the scores decrease with the budget.

Status. An application of the split-likelihood bound to a finite maximum of profiles, using the daily statistics of Proposition 34 and the daily profile of Proposition 37, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) Completing the square in each day's drift leaves the summed residual, and the remaining scalar profile in the scale is that of Proposition 34. (b) A path with at most K scale changes is constant in scale on at most $K+1$ runs; the supremum exchanges with the product, and grouping by the last run gives the recursion. (c) Each admissible path's density is at most $P_{\mathrm{sw}}(K)$. (d) Larger parameter sets give larger suprema. □

The recursion takes of order $(K+1)D^2$ operations for D days. In the simulation after Proposition 37, a null with free daily drifts and exactly three scale switches gave the single-profile score a mean of about 1.6×10^7 and a 2.8% rate of scores above 20, while the score with $K = 3$ stayed calibrated. But the free daily drifts alone, at $K = 0$, raised the log denominator by about 31.5 over 60 days, and the rate of scores above 20 against $p = 2$, $\kappa = 0.45$ fell from 27.2% for the single profile to 0.35%, before any scale switch was allowed. The cost is in the free drifts, not in the switch budget; a common or budgeted drift with a switching scale would be the natural variant. The mixture is over whole parameter paths, not a switching rule driven by the returns, and switches inside a day are not covered.

Budgeting changes of drift and scale together cuts that cost, but not enough. Count a day boundary when the pair of drift and scale changes, a change of either or both counting once, and allow at most K such boundaries, so that drift no longer resets freely each day. For a run I of consecutive days, let P_I^{joint} be the

Gaussian supremum of Proposition 34 on the concatenated run, with one drift and one scale; let $P_{\text{joint}}(K)$ be the supremum of the daily product of Gaussian densities over pair paths with at most K changes, and $E_{\text{joint}} = q/P_{\text{joint}}(K)$, or 0, for a normalized forecast q chosen on the training record.

Proposition 39 (a budget on joint drift and scale changes). (a) The concatenated run's residual is the sum of the within-day residuals and the spread of the daily return rates about the run's pooled rate; P_I^{joint} is infinite exactly when all return rates in the run agree. (b) $P_{\text{joint}}(K)$ is the maximum, over partitions of the days into at most $K+1$ consecutive runs, of the product of the run profiles, computed exactly by a recursion on the last run; it is infinite exactly when an allowed partition has a singular run, and otherwise attained. (c) If, given the training record, the evaluation prices follow a mixture over pair paths with at most K changes of the corresponding Gaussian laws, then E_{joint} has conditional mean at most one, $P(E_{\text{joint}} > 1/\alpha) \le \alpha$, and the e-BH rule of Proposition 40 keeps FDR $\le q\,m_0/m$. (d) With $P_{\text{sw}}(K)$ the free-daily-drift denominator of Proposition 38, $P_{\text{common}} = P_{\text{joint}}(0) \le P_{\text{joint}}(K) \le P_{\text{sw}}(K) \le P_{\text{day}}$, with $P_{\text{joint}}(D-1) = P_{\text{day}}$, and the scores reverse this order. (e) With the calendar clock, on a fixed run of N_I steps with concatenated residual Q_I and summed within-day residuals $Q_{\text{within},I} > 0$, the cost of free daily drifts against a common drift is $(N_I/2)\log(Q_I/Q_{\text{within},I}) \ge 0$, and these costs add over a fixed partition.

Status. An application of the split-likelihood bound to a finite maximum of profiles, using the profile of Proposition 34 on concatenated runs and the daily profile of Proposition 37, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) Pool the run's weighted statistics and complete the square in the drift. (b) A pair path with at most K changes is constant on at most $K+1$ runs, so its density is a product of run densities, and the supremum exchanges with the product; grouping by the last run gives the recursion. (c) Each admissible path's density is at most $P_{\text{joint}}(K)$. (d) Larger parameter sets give larger suprema. (e) Insert (a) into both profiles of Proposition 34. □

In the simulation after Proposition 37, a null with a common drift and three scale switches gave the single-profile score a mean of about 1.5×10^3 and a 2.6% rate of scores above 20, while the score with $K=3$ stayed calibrated. On the alternative, the median cost at $K=3$ was 12.4 in log terms, against about 31 for free daily drifts, but the rate of scores above 20 against $p=2, \kappa=0.45$ was still 27.8%, 1.2% and 0 at $K=0$, 1 and 3. Daily drift redraws are not covered at a budget below the number of days.

Allowing each day its own volatility, and drift, thus keeps the scores calibrated under day-to-day regime switches where a single profile breaks, under declared block laws and budgets. But each free daily nuisance absorbs the evidence of a bubble: on these simulated hourly alternatives, power fell from about 28% to between 0 and 1%. With a free daily scale, only changes of level within a day remain informative: in a constant-elasticity model the variance ratio between two prices in a day is their price ratio to the power $2(p-1)$, so a 5% range gives a ratio of about 1.05 for $p=3/2$ and 1.10 for $p=2$. Switches within a day, jumps and noise remain uncovered, and five-minute sampling and a predictive forecast are untested.

The per-name e-values give a list of bubbly names with a false-discovery guarantee. For a level $0 < q < 1$, let k^* be the largest $k \le m$ such that at least k names have $E_i \ge m/(qk)$, or 0 if there is none, and report $\mathscr{R} = \{i : E_i \ge m/(qk^*)\}$ when $k^* > 0$, and nothing otherwise; equivalently, sort the E_i in decreasing order and report the first k^*. Let N_0 be the set of names without a bubble, $m_0 = |N_0|$, and FDR $= E_P[|\mathscr{R} \cap N_0|/\max(|\mathscr{R}|, 1)]$.

Proposition 40 (a list of bubbly names with false-discovery control). In the setting of Proposition 27, with arbitrary dependence across names and bubbly and non-bubbly names mixed: (a) each E_i is positive and measurable, and $E_P[E_i] \le 1$ for $i \in N_0$; (b) FDR $\le q\,m_0/m \le q$, and if every name is non-bubbly,

$P(\mathscr{R} \neq \emptyset) \leq q$; (c) for any measurable random screen S, setting $E_i = 0$ for $i \notin S$ and rerunning the rule with the original m gives $\mathscr{R} \subseteq S$, $\mathscr{R} = \emptyset$ when $S = \emptyset$, and still FDR $\leq q\, m_0/m$.

Status. An application of the e-BH theorem of Wang and Ramdas ([81]), stated as a hypothesis whose structure has a degenerate instance, together with the split-likelihood bound of Proposition 27, which is proved; not a new multiple-testing result. Machine-checked (Appendix A).

Proof. At a null name the true parameter lies in the null catalogue, so $E_i \leq T_i(\vartheta_i^*)$, whose mean is at most one; this is (a). The e-BH theorem turns null e-values into (b), under any dependence. Masking multiplies each null e-value by an indicator, which keeps its mean at most one, and (b) applies again with the original m. □

The guarantee concerns the expected proportion, not each realized list, and it is not a guarantee conditional on the screen; replacing m by the number of screened names at the same q does not preserve it. Inclusion uses the weak inequality $E_i \geq m/(qk^*)$, whereas Proposition 27's confidence set excludes the null only when $E_i > 1/a$, so a name can be reported while its closed confidence set at the random level $a = qk^*/m$ still contains a null. For price data, the list inherits all the premises of Proposition 28 and its low power. It is also conservative, because each name's e-value is a minimum over its null catalogue and so lies below the likelihood ratio at the true parameter; how conservative, and how many bubbly names it finds at realistic sampling, is not quantified here.

Pooling the names into one global test does not help, at least in its simplest valid form. Model the m names' evaluation log-return vectors over n steps as independent Gaussian vectors with means $a\Delta_j$ and covariances $\Sigma\Delta_j$, for any drift vector a and any positive-definite Σ; this allows any correlation across names, including an unknown factor structure. Let $\hat{a}$ be the pooled return rate and $S = n^{-1}\sum_j (R_j - \hat{a}\Delta_j)(R_j - \hat{a}\Delta_j)^{\mathsf{T}}/\Delta_j$ the weighted residual covariance, with residual correlation matrix C. For the numerator q, take the forecast that fits, on the training prices, one common power $p > 1$ for all names, with name-specific scales and independent names, by the first maximizer over a declared finite catalogue. Let D_{pool} be the supremum of the Gaussian price densities over a and Σ, and set $E_{\text{pool}} = q/D_{\text{pool}}$, or 0 if $D_{\text{pool}} = \infty$ or a price is not positive.

Proposition 41 (a pooled Gaussian score). Suppose that, given the training prices, the evaluation law of the panel is a mixture of these Gaussian laws, with latent parameters. (a) If S is positive definite, $D_{\text{pool}} = \prod_{i,j} y_{ij}^{-1}\,(2\pi e)^{-nm/2}(\det S)^{-n/2}\prod_j \Delta_j^{-m/2}$, attained at $(\hat{a}, S)$; otherwise $D_{\text{pool}} = \infty$. The rank of S is at most $n-1$, so $D_{\text{pool}} = \infty$ on every path when $m \geq n$; when $n \geq m+1$, it is finite almost surely. (b) E_{pool} is finite and measurable, has conditional mean at most one given the training prices, and $P(E_{\text{pool}} \geq 1/\alpha) \leq \alpha$; it is identically zero when $m \geq n$, and positive on positive full-rank records when $n \geq m+1$. (c) With $D_{G,i}$ the Gaussian supremum of Proposition 30 for name i and q_i its forecast factor, $D_{\text{pool}} = (\prod_i D_{G,i})(\det C)^{-n/2}$ and $E_{\text{pool}} = (\prod_i q_i/D_{G,i})(\det C)^{n/2}$, with $\det C \leq 1$; for equal correlations ρ, the log score loses $-(n/2)[(m-1)\log(1-\rho) + \log(1+(m-1)\rho)]$.

Status. An application of the split-likelihood bound, with the Gaussian maximum ([78], Section 3) proved here; the constant-elasticity kernels of the forecast enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) After the drift is profiled out, $\log\det\Sigma + \operatorname{tr}(\Sigma^{-1}S) \geq \log\det S + m$, by $\log t \leq t-1$ on the eigenvalues of $S^{1/2}\Sigma^{-1}S^{1/2}$; if S is singular, $\Sigma = S + \varepsilon I$ makes the density unbounded as $\varepsilon \to 0$. The residuals sum to zero, so S has rank at most $n-1$; when $n \geq m+1$, singular S forces one price to be determined by the others, a null set. (b) The denominator dominates every null component, and the forecast is normalized. (c) $\det S = (\prod_i S_{ii})\det C$, and the diagonal subfamily attains $\prod_i D_{G,i}$. □

The factors $q_i/D_{G,i}$ in (c) are algebraic comparisons, not the per-name scores of Proposition 30, and their product is not asserted to be valid. In practice naive pooling loses to per-name testing. In a simulation with five names, one year of daily training and one year of evaluation, $\alpha = 0.1$ and 1,000 panels per law, the forecast chose one power from $\{1.25, 1.5, 2\}$ and per-name scales from $\{0.2, 0.3, 0.45\}$. Under independent and one-factor geometric Brownian motions, neither the pooled test nor a per-name Bonferroni rule raised a false alarm. Against independent constant-elasticity names with $p = 3/2$ and $\kappa = 0.45$, the pooled test rejected in 32% of panels against 43% for the per-name rule. With a common factor of correlation 0.5, it rejected in 0.1% against 37%. Sampling noise in the correlations alone cost a median of 4.6, about $m(m-1)/4$, and the common factor about 175, which a forecast with independent names cannot recover. A useful pooled score would need a forecast that models the dependence across names; one is tried next.

A forecast that models the dependence recovers some of the loss, but not enough. Keep the null and its denominator, and replace the forecast with independent names by a Gaussian-copula forecast with the same constant-elasticity marginals ([56], its equations 3, 4 and 8). At each evaluation step, map each name's next price through its fitted conditional distribution function and then the normal quantile function, giving normal scores z_j, and multiply the product of the marginal densities by the Gaussian copula density $c_R(z) = (\det R)^{-1/2} \exp(-z^{\mathsf{T}}(R^{-1} - I)z/2)$. The correlation matrix R is the correlation of the training normal scores, shrunk towards the identity by a fixed weight $\gamma \in (0, 1]$, or the identity if a training score is degenerate. Let q_{cop} be the resulting whole-path forecast, $q_{\mathrm{mix}} = w_0 q + (1 - w_0) q_{\mathrm{cop}}$ for a fixed $w_0 \in (0, 1)$, and E_{cop}, E_{mix} the corresponding pooled scores.

Proposition 42 (a copula forecast for the pooled score). In the setting of Proposition 41: (a) R depends on the training prices only, has unit diagonal and $R \succeq \gamma I$; each step of the copula forecast is a probability density with the fitted constant-elasticity marginals, so q_{cop} and q_{mix} are normalized. (b) E_{cop} and E_{mix} have every conclusion of Proposition 41(b). (c) $E_{\mathrm{mix}} = w_0 E_{\mathrm{pool}} + (1 - w_0) E_{\mathrm{cop}} \geq w_0 E_{\mathrm{pool}}$, and on positive full-rank records $\log(E_{\mathrm{cop}}/E_{\mathrm{pool}}) = -(n/2) \log\det R - \frac{1}{2} \operatorname{tr}[(R^{-1} - I) \sum_j z_j z_j^{\mathsf{T}}]$, which can have either sign; if $R = I$, all three scores coincide.

Status. An application of the Gaussian copula construction, cited as a hypothesis on each step's normalization and marginals, and of Proposition 41; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) The distribution function of each marginal is continuous and strictly increasing, so the normal scores are finite; the copula construction gives each step mass one with the stated marginals, and a chain of normalized steps is normalized. (b) Proposition 41(b) holds for any normalized forecast. (c) The forecasts differ by the factor $\prod_j c_R(z_j)$, whose logarithm is the displayed gain. □

With $w_0 = 1/2$, the mixture loses at most $\log 2$ against the forecast with independent names. In the simulation after Proposition 41, with shrinkage weight 0.1, none of the scores raised a false alarm under the Gaussian nulls. With a common factor of correlation 0.5 and $p = 3/2$, $\kappa = 0.45$, the copula score rejected in 19% of panels and the mixture in 18%, where the forecast with independent names had none, against 39% for per-name Bonferroni. With independent names, the copula score rejected in 17%, because its correlation is fitted to noise, the mixture in 29% and the independent forecast in 32%, against 44%.

Pooling across names is thus valid but does not beat per-name testing. The per-name scores with the e-BH rule of Proposition 40 remain the method, and they also name the bubbly names.

What happens when the true law is not in the catalogue? The practical answer is discouraging: at high sampling frequency the available robustness bounds become nearly vacuous. Misspecification is measured by a total-variation envelope: for a name without a bubble, $\varepsilon_i(u)$ bounds the distance between the actual

conditional law of its evaluation prices given training prices u and that of some null catalogue member chosen measurably from u.

Proposition 43 (robustness of the catalogue scores). In the setting of Proposition 27, but with the actual law P arbitrary, suppose a valid envelope ε_i exists for each name without a bubble. (a) The rule $E_i > 1/\alpha_i$, and the band rule, reject such a name with probability at most $\min(1, \alpha_i + E_P[\varepsilon_i])$; with a deterministic $e_i < \alpha_i$ bounding ε_i, the threshold $1/(\alpha_i - e_i)$ has size at most α_i; and the probability that some such name is rejected is at most the sum of these bounds, without independence. (b) For caps c_i fixed in advance, $E_i^{\mathrm{rob}} = \min(E_i, c_i)/(1+c_i\varepsilon_i)$ has mean at most one at every name without a bubble, so the e-BH rule of Proposition 40 applied to these scores keeps FDR $\le q\, m_0/m$ under any dependence; a strict rejection at level α_i requires $\varepsilon_i < \alpha_i$ and $c_i > 1/(\alpha_i - \varepsilon_i)$. (c) For Gaussian log returns with drift b and volatility σ_* over n increments of total length τ, against a null with volatility κ_0 and no drift, the Bhattacharyya affinity is

$$A_G = \Big(\frac{2\sigma_*\kappa_0}{\sigma_*^2 + \kappa_0^2}\Big)^{n/2} \exp\Big(-\frac{(b + (\kappa_0^2 - \sigma_*^2)/2)^2\, \tau}{4(\sigma_*^2 + \kappa_0^2)}\Big),$$

and $1 - A_G \le \mathrm{TV} \le \sqrt{1 - A_G^2}$, so $\sqrt{1 - A_G^2}$ is a valid envelope. With $\sigma_* = \kappa_0$ it is at most $|b|\sqrt{\tau}/(2\kappa_0)$ for every n; with $\sigma_* \ne \kappa_0$ and τ fixed, $A_G \to 0$ as $n \to \infty$, against every member of a finite volatility grid that omits σ_*. (d) The raw scores are not e-values off the catalogue: in a two-point example at distance e, the raw score's mean is $1 + e[1/(2p) - 1/(2(1-p))]$, unbounded as $p \to 0$, and an arbitrarily small distance can push a raw level-α test above α.

Status. Applications of Markov's inequality, the comparison of bounded functions and, for (b), the e-BH theorem stated as a hypothesis with a degenerate instance; the split-likelihood bound and the Gaussian computations are proved. Machine-checked (Appendix A).

Proof. (a) Disintegrate P over the training prices; under the reference law the score has conditional mean at most one, Markov's inequality gives α_i, and comparing the bounded indicator under the two conditional laws adds ε_i. (b) For $0 \le Y \le c$, $\int Y\, dH \le \int Y\, dG + c\varepsilon$, which gives the clipped mean, and e-BH applies. (c) The affinity is the product of one-step Gaussian Bhattacharyya coefficients; the bounds are $\min \le \sqrt{hg}$ and Cauchy–Schwarz. (d) is a direct computation. □

Numerically, from (c): with drift 0.05, $\kappa_0 = 0.3$ and $\tau = 1$, the envelope is about 0.083, so a nominal 5% test has size at most about 13%; with $\sigma_* = 0.31$ against $\kappa_0 = 0.30$ over one year it is about 0.080 with 12 observations, 0.356 with 252 and 0.999987 with 19,656. These evaluate the formulas; they are bounds, not the test's actual false-alarm rates. Robustness at realistic sampling therefore needs a richer null family, or measured false-alarm rates outside the catalogue, not these bounds.

The same bounds hold for the constant-elasticity catalogues with absorption of Proposition 29, whose densities have an atom at zero and so fall outside Proposition 43's positive-density setting, and they can be stated so that data failures are kept rather than dropped. Let each observation be a nonnegative real or a symbol $\perp$ for a missing or unparseable record, call a training record valid when every entry is a nonnegative real, and set $\bar{E}_i = E_i$ when the training record is valid and the evaluation record is a nonnegative price vector, and $\bar{E}_i = 0$ otherwise. For a name without a bubble, let the envelope $\varepsilon_i(u)$ bound, at almost every valid training record u under the actual law, the total-variation distance between the actual conditional law of the evaluation record and a catalogue null reference, both on the full observation space, so that actual mass on failed records counts against the envelope.

Proposition 44 (robustness with absorption and retained failures). In the setting of Proposition 29, with the actual law P arbitrary and J_i the event of a valid training record: (a) the rule $\bar{E}_i > 1/\alpha_i$, and the band rule, which abstains on invalid records, reject a name without a bubble with probability at most

$\min(1, \alpha_i P(J_i) + E_P[1_{J_i}\varepsilon_i])$; with $\varepsilon_i \le e_i < \alpha_i$ on valid records, the threshold $1/(\alpha_i - e_i)$ has size at most α_i; and the probability of some false rejection is at most the sum of these bounds, without independence. (b) For caps c_i fixed in advance and any measurable screen M_i with values in $[0, 1]$, which may use the whole panel, $M_i \min(\bar{E}_i, c_i)/(1 + c_i\varepsilon_i)$ has mean at most $P(J_i) \le 1$ at every name without a bubble, so the e-BH rule of Proposition 40 applied to these scores, with the original m, keeps FDR $\le q\, m_0/m$ under any dependence. (c) Both conclusions hold with $\bar{E}_i$ replaced by any measurable computed score that is almost surely at most $\bar{E}_i$. (d) A valid record with a zero price at the end of training or in evaluation has $\bar{E}_i \le 1$, so such a name is never flagged.

Status. An application of the support-aware split-likelihood step (proved), Markov's inequality, the comparison of bounded functions and, for the false-discovery bound, the e-BH theorem, with the constant-elasticity and lognormal kernels as for Proposition 29. Machine-checked (Appendix A).

Proof. At every training record the selected density, restricted to the support of a null reference, has mass at most one, so $\bar{E}_i$ has reference mean at most one; then argue as in Proposition 43, with the comparison made on the full observation space, so that events on failed records are included. On invalid training records $\bar{E}_i = 0$, which gives the factor $P(J_i)$. (d) is Proposition 29(d). □

A default or a failed record is thus zero evidence: the name stays in the panel, m is unchanged, and there is no guarantee conditional on the records that succeeded. Rounding computed scores without a one-sided bound need not preserve the guarantee. How much the raw scores inflate off the catalogue may depend on the direction of the misspecification, but this has not been measured in a registered experiment.

The same total-variation comparison governs screens with empirically calibrated thresholds. Fix $n \ge 1$ and $\alpha \in (0, 1)$, and fix the scoring protocol, including window lengths, selection and any fitted score or simulated reference, before the calibration and evaluation scores are drawn. Larger scores are more extreme. Given n calibration scores c, let $k = \lceil (n+1)(1-\alpha) \rceil$ and let $q(c)$ be the k-th smallest of them, with repetitions kept, or $+\infty$ if $k > n$; an evaluation score S is flagged when $S > q(c)$. Let U be the whole recorded calibration data, including any selection information, on a standard Borel space, with actual law μ, calibration scores $C(U)$, and $K(u, \cdot)$ the actual conditional law of S given $U = u$. The realized threshold's risk is $R(u) = K(u, \{s : s > q(C(u))\})$. All laws are those of the windows actually selected and evaluated. Total variation is the supremum over events of the difference in probabilities.

Proposition 45 (calibrated screens under post-selection score laws). (a) If the joint law of $(C(U), S)$ is within total variation ε of an exchangeable law on $n+1$ real scores, the flag probability is at most $\min(1, \alpha + \varepsilon)$. In particular $\varepsilon = 0$ when the selected windows are exchangeable and every score is the same measurable function of its window, whatever their dependence. (b) If instead, for a fixed score law ν, the law of $C(U)$ is within total variation η of ν^n, and $K(u, \cdot)$ is within ρ of ν for μ-almost every u, the flag probability is at most $\min(1, \alpha + \eta + \rho)$. (c) Under the premises of (b), let $r = \lfloor \alpha(n+1) - 1 \rfloor = n - k$ and, for $E, \delta \in (0, 1)$, $B = \sum_{j=0}^{r} \binom{n}{j} E^j (1-E)^{n-j}$, with $B = 0$ if $r = -1$. If $B \le \delta$, then $\mu\{R > E + \rho\} \le \min(1, \delta + \eta)$, and with $b = \min(1, E + \rho)$ and $d = \min(1, \delta + \eta)$ the flag probability is at most $b + (1-b)d$. (d) Selection on past information alone does not supply these premises. Take $n = 19$ and $\alpha = 1/20$, and independent blocks, each with a fair bit observed before its window and a binary score equal to that bit with probability 0.99; retain calibration blocks whose bit is zero. An independent unselected evaluation block is flagged with probability $(1/2)(0.99)^{19} > 2/5$, and one retained by the same rule with probability $(1/100)(0.99)^{19} < 1/20$. The premises of (b) hold in the first design with ν Bernoulli$(1/100)$, $\eta = 0$ and $\rho = 49/100$, and in the second with both budgets zero.

Status. An application of the exchangeable quantile lemma of conformal prediction ([77], Lemma 1) and the training-conditional guarantee of [79] (Theorem 1(a)), each at the reference law used, with the total-

variation comparison of Proposition 43. Machine-checked, conditional on those two cited results, which enter as hypothesis structures with only degenerate instances (Appendix A).

Proof. The k-th order statistic is a minimum of maxima of coordinates, so the flag event is Borel, and R is measurable with mean equal to the flag probability. (a) Under the exchangeable reference law the cited lemma, whose rank, tie and infinity conventions are those above, bounds the flag probability by α; the comparison of Proposition 43, applied to this one event, transports the bound. A common measurable map sends an exchangeable law of windows to an exchangeable law of scores. (b) The kernel premise moves the flag probability by at most ρ when $K(u,\cdot)$ is replaced by ν. What remains is the integral of the bounded function $c \mapsto \nu\{s > q(c)\}$ against the law of $C(U)$, which moves by at most η when that law is replaced by ν^n; the result is the flag probability under the independent, hence exchangeable, law ν^{n+1}, at most α by (a). (c) Let G be the set of c with $\nu\{s > q(c)\} > E$. The cited guarantee gives $\nu^n(G) \le \delta$ when $B \le \delta$, so $\mu\{C(U) \in G\} \le \min(1, \delta+\eta)$; and on the full-measure set of the kernel premise, $R(u) \le \nu\{s > q(C(u))\}+\rho$, so $\{R > E+\rho\}$ lies in $\{C(U) \in G\}$ up to a null set. Splitting the mean of R on $\{R \le b\}$ gives the last bound. (d) A retained block scores one with probability 0.01, independently. With $n = 19$ and $\alpha = 1/20$, $k = 19$ and the threshold is the maximum, so a binary evaluation score is flagged exactly when all calibration scores are zero and it is one; the stated probabilities and total variations follow by direct computation. □

The budgets are not estimated here; neither (b) nor (c) holds automatically for real windows. Equal lengths, no overlap, selection on the past and similar marginal laws do not supply them, as (d) shows: the violated premise is a comparable law of the evaluation score after selection, not the rank rule. The result gives no guarantee conditional on every calibration sample, none uniform over levels chosen afterwards, and no power, false-discovery, repeated-launch or anytime guarantee. Its null is a law of observations; reading it as the absence of a bubble needs a pricing interpretation. Sections 5.9 to 6.13 use this rank rule but establish none of its premises for real windows, and the interpolated, tie-inclusive thresholds of Section 6.5 are not instances of it.

Jumps are a misspecification that the Gaussian null of the free-drift score of Proposition 30 does not absorb. Contamination is a known danger for likelihood-ratio inference ([64], Section 3.1, Example 3); here it can be budgeted. For a name without a bubble, suppose that at almost every training record u under the actual law, its evaluation law is $(1-p)G + pJ$ for some $p \le \rho_i(u)$, with G one of the Gaussian null laws of Proposition 30 and J an arbitrary law.

Proposition 46 (jump-contamination budgets). In the setting of Proposition 30, with this mixture premise at each name without a bubble: (a) the total-variation distance from G is at most ρ_i, and $E_i^{\mathrm{drift}} > 1/\alpha_i$ rejects with conditional probability at most $\alpha_i+(1-\alpha_i)\rho_i$, and unconditionally at most $\alpha_i+(1-\alpha_i)E[\rho_i]$; with $\rho_i \le e_i < \alpha_i$, the threshold $(1-e_i)/(\alpha_i-e_i)$ has size at most α_i. (b) For a cap $c_i \ge 1$, $\min(E_i^{\mathrm{drift}}, c_i)/(1+(c_i-1)\rho_i)$ has conditional mean at most one, so the e-BH rule of Proposition 40 applied to these scores, or to any measurable scores below them, including zero-masked ones with the original m, keeps FDR $\le q\, m_0/m$ under any dependence; such a score can reject at level α_i only if $\rho_i < \alpha_i$ and $c_i > (1-\rho_i)/(\alpha_i-\rho_i)$. (c) The premise holds when the log returns are Gaussian with drift and scale as in Proposition 30 plus jump terms independent of them, with p the probability of any jump; then a supplied no-jump probability $e^{-\lambda\tau}$ gives $\rho_i = 1-e^{-\lambda\tau}$. (d) If the jumps may depend on the Gaussian part, the total-variation distance is still at most the probability of a jump, so the bounds of Proposition 43 hold, with normalization $1+c_i\rho_i$.

Status. An application of linearity of integration, Markov's inequality, the comparison of bounded functions and, for the false-discovery bound, the e-BH theorem, stated as a hypothesis with a degenerate instance; the reference moment of the free-drift score is proved at every training record, and the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) Under G the score has mean at most one, so the rejection event has G-probability at most α_i and J-probability at most one. (b) The capped score has G-mean at most one and is at most c_i under J. (c) On the event of no jump the prices are those of the Gaussian law, by induction over the returns. (d) The two price maps agree when there is no jump. □

The normalization $1 + (c_i - 1)\rho_i$ needs the mixture form, which can fail for dependent jumps: for a reference law with mass $1 - 1/c$ on a point where the score is zero and $1/c$ on a point where it is c, moving mass ρ to the second point gives a clipped mean of $1 + c\rho$. However finely the window is sampled, ρ_i is the probability of any jump in it. In practice the budget is uninformative. In one simulation, of daily prices over two years with the first year for training, the catalogue of Proposition 29 with powers $\{1/2, 1, 3/2\}$ and scales $\{0.2, 0.3, 0.45\}$, and 20,000 paths per law, the truth was bubble-free geometric Brownian motion with drift 0.05 and volatility 0.3 plus independent compound-Poisson log jumps. Without jumps the free-drift score had mean 0.38; with jumps at rates from one every two years to ten a year, its mean ranged from about 7 to 10^{12}, so the raw scores must not enter the discovery list when jumps are possible. Their actual false-alarm rates at $\alpha = 0.1$ stayed at or below 1.6%, so the failure is of the e-value property, not evidence of large inflation of the per-name size. But even one jump every two years gives $\rho \approx 0.39$ over a one-year window, above any usual α, so the clipped test can never reject, and its mean was about 0.01 to 0.02. The guarantee is valid but powerless over realistic windows; it becomes useful only when $\lambda\tau$ is much smaller than α. A null family that models jumps is needed; that of Proposition 52 is valid but powerless.

Volatility that is random but constant within the evaluation window costs nothing, and volatility that changes within it can be budgeted. Let E_i be the free-drift score of Proposition 30 or the rectangle score of Proposition 32, chosen in advance. Suppose that, at almost every training record u under the actual law, a bubble-free name's evaluation law is a mixture, over a latent variable z with law $\nu_i(u, dz)$, of laws with independent Gaussian log returns whose drift $b_j(u, z)$ and volatility $\sigma_j(u, z) > 0$ may vary across the steps j, and compare each component with a constant reference law $(b_{\mathrm{ref}}, \kappa)(u, z)$ in the score's null.

Proposition 47 (volatility-path budgets). In this setting: (a) each component is within total variation $\varepsilon = \sqrt{1 - A^2}$ of its reference, with

$$A = \prod_j \sqrt{\frac{2\sigma_j\kappa}{\sigma_j^2 + \kappa^2}}\ \exp\Big(- \sum_j \frac{\Delta_j\,[\,b_j - b_{\mathrm{ref}} + (\kappa^2 - \sigma_j^2)/2\,]^2}{4(\sigma_j^2 + \kappa^2)}\Big),$$

on prices as well as log returns, and $\varepsilon = 0$ when the component equals its reference. (b) With $\bar{\varepsilon}_i(u) = \int \varepsilon\, \nu_i(u, dz)$, the rule $E_i > 1/\alpha_i$ has size at most $\alpha_i + \bar{\varepsilon}_i$, conditionally and unconditionally; with $\bar{\varepsilon}_i \le e_i < \alpha_i$, the threshold $1/(\alpha_i - e_i)$ has size at most α_i; $\min(E_i, c_i)/(1 + c_i r_i)$, for any measurable budget $r_i \ge \bar{\varepsilon}_i$, has mean at most one, so the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$ for these scores under any dependence. If every component equals its reference, as when the volatility is random but constant in the window, the raw E_i has null mean at most one. (c) If all volatilities lie in $[s_{\min}, s_{\max}]$, $|\log(\sigma_j/\kappa)| \le h$ and $|b_j - b_{\mathrm{ref}}| \le B$, then $\bar{\varepsilon}_i \le \sqrt{1 - e^{-L}}$ with $L = n \log\cosh h + \tau[B/(2s_{\min}) + s_{\max}\sinh(h)/2]^2$.

Status. An application of the Gaussian affinity of Proposition 43(c), the comparison of bounded functions and the split-likelihood moment, integrated over the latent law, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) The affinity is the product of one-step Gaussian coefficients, and on prices each step's coefficient does not depend on the current price. (b) The score has mean at most one under each reference, since its denominator dominates the reference density; compare bounded functions under each component and integrate over z. (c) Write $2\sigma\kappa/(\sigma^2 + \kappa^2) = 1/\cosh\log(\sigma/\kappa)$ and $(\sigma^2 - \kappa^2)/2 = \sigma\kappa\sinh\log(\sigma/\kappa)$. □

The budget of the realized component cannot be used when that component is unobserved. The bound in (c) tends to one as sampling gets finer at any fixed $h > 0$; it is sufficient, not evidence of failure. In the jump simulation's design, with 10,000 paths per law and $\alpha = 0.1$, a latent even mixture of geometric Brownian motions with volatilities 0.25 and 0.40 gave the free-drift score a null mean of 0.14 and a false-alarm rate of 0.03%, as the zero-budget case requires. But a switch from 0.25 to 0.40 halfway through the evaluation window gave a null mean of 13 and a false-alarm rate of 2.8%, with budget one, and even a smooth seasonal volatility $0.30 \pm 0.05 \sin(2\pi t)$ gave a budget of 0.91 at daily sampling. So there is still no informative guarantee for genuine within-window variation. Volatility correlated with returns, as in Heston-type models with leverage, is outside this representation: in one such model, with correlation -0.7, the score had null mean 0.28 and no false alarms, but this is a measurement, not a guarantee.

A single volatility switch can be put into the null, and it restores validity but leaves almost no power on daily data. In a second simulation of the same design, a switch from 0.25 to 0.40 in the middle of the evaluation window gave the free-drift score a null mean of 17, 263 for a switch a few days later, and 83 when the switch date was random; such heavy-tailed means vary between runs. Declare in advance a set K_i of cut points c, each leaving at least two evaluation steps on either side, and add to the null every law whose log returns are independent Gaussian with drift b_L and volatility κ_L up to step c and b_R, κ_R after it. Let D_{sw} be the supremum of these densities over the cuts and parameters, and set $E_i^{\mathrm{sw}} = \hat{G}_i / \max(G_{i,0}, D_{\mathrm{sw}})$, or 0 if $\hat{G}_i = 0$ or the denominator is infinite.

Proposition 48 (a single-switch null). In the setting of Proposition 30, suppose that, given its training prices, each name without a bubble has an evaluation law that is a mixture of catalogue null laws and such switch laws, with the cut and parameters latent and possibly dependent on the training prices. (a) On positive paths, D_{sw} is the maximum over $c \in K_i$ of the product of the closed-form Gaussian maxima of Proposition 30(a) on the two segments; it is finite exactly when every permitted segment has positive residual variation, and the set where it is infinite has probability zero under every such law. A cut leaving a single step makes it infinite on every positive path. (b) The score is finite and measurable on every nonnegative array and has conditional mean at most one at every name without a bubble, so the per-name tests, p-values and the union bound hold, and the e-BH rule of Proposition 40 has FDR $\le q\, m_0/m$, also after a zero-masked screen with the original m. (c) On every path $E_i^{\mathrm{sw}} \le E_i^{\mathrm{drift}} \le E_i$; enlarging K_i decreases the score; it is not identically zero, and absorbed names are never flagged.

Status. An application of the composite-null split-likelihood bound ([82]), which is proved here, the Gaussian profile of Proposition 30 on each segment and the mixture integration of Proposition 47, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) Given the cut, the density is a product over the two segments with separate parameters, so its supremum is the product of the segment suprema. A segment with zero residual variation has its returns on a line through the origin, a set of measure zero when it has at least two steps; a one-step segment always has zero residual. (b) The denominator dominates every null component, and the argument of Proposition 47 with zero budget integrates over the latent law. (c) Equal segment parameters recover the Gaussian laws of Proposition 30. □

The guarantee repairs the within-window failure after Proposition 47, but on every path the score is at most the free-drift score, so it cannot gain power. In the same simulation, with cuts every month or at every permitted step, the null means fell below 0.001, with no false alarms. But power against $p = 3/2$ fell from 6% to 0% at $\kappa = 0.3$, and from 20% to at most 0.3% at $\kappa = 0.45$. A Gaussian law with one switch fits the level-dependent volatility of a constant-elasticity path about as well as the alternative does, so on daily data a change of volatility over time and a dependence of volatility on the price level are nearly confounded.

This echoes, at the level of finite samples and of a measured design, the mechanism of Proposition 20, where stochastic volatility makes one physical law compatible with both labels; it is not a proof that the two are unidentified. A usable test would need a restriction that separates the two, or more informative data. Repeated switches, switches driven by the returns themselves, general stochastic volatility, jumps and rounding are not covered.

The confounding comes from discrete sampling, not from the coefficient. Fix $T > 0$, $s_0 > 0$ and $\beta \in \mathbb{R}$, and suppose the log price X, with $X_0 = 0$, has on $[0, T]$ the continuous representation $X_t = M_t + B_t$, $M_t = \int_0^t \sqrt{v_u}\, dW_u$, $v_t = \exp(a(t) + \beta X_t)$, with B continuous, adapted and of finite variation, and a an unknown deterministic continuous function of total variation at most V; existence of such a solution is a premise. The coefficient $\sigma(t, s) = s \exp((a(t) + \beta \log(s/s_0))/2)$ then combines a time effect with a level dependence of elasticity $1 + \beta/2$. Write $Q(t) = [X]_t$, $L = \log v$ and $[L, X]_T = ([L + X]_T - [L - X]_T)/4$. On a grid, given arrays x_j, a_j with $\sum_j |a_j - a_{j-1}| \le V$, errors $|e_j| \le \varepsilon$ and $\hat{l}_j = a_j + \beta x_j + e_j$, write $d_j = x_j - x_{j-1}$, $w = \max_j |d_j|$, $H = |d_1| + |d_n| + \sum_{j<n} |d_j - d_{j+1}|$, and, when $\sum_j d_j^2 > 0$, $\hat{\beta} = \sum_j (\hat{l}_j - \hat{l}_{j-1}) d_j / \sum_j d_j^2$ and $r = (Vw + \varepsilon H) / \sum_j d_j^2$.

Proposition 49 (level dependence is identified from a continuous record). (a) Almost surely, simultaneously for $t \le T$: $Q(t) = \int_0^t v_u\, du$, $Q(T) > 0$, Q has (one-sided at the ends) derivative v, so $L = \log Q'$, and $[L, X]_T = \beta Q(T)$; hence $\beta = [\log Q', X]_T / Q(T)$, whatever the time effect. A Lipschitz a with constant ℓ has total variation at most ℓT. (b) $|\hat{\beta} - \beta| \le r$, and for each nonconstant array x and each β the bound is attained by some admissible a and e. (c) On the accuracy event, $\beta \le 0$ implies $\hat{\beta} \le r$ and $\beta > 2r$ implies $\hat{\beta} > r$; for measurable arrays whose accuracy event has probability at least $1 - \delta$ under every null with $\beta \le 0$, rejecting when $\hat{\beta} > r$ has size at most δ, and its power at an alternative is at least $1 - \delta - P(\sum_j d_j^2 = 0 \text{ or } \beta \le 2r)$.

Status. Conditional on the quadratic variation of a Brownian stochastic integral plus a continuous finite-variation term ([26], Proposition 6.25), stated as a hypothesis with a vacuous instance; that a continuous finite-variation process alone has zero quadratic variation is proved. Part (a) is a statement about the supplied representation, not a test. Parts (b) and (c) are deterministic or need only the supplied accuracy event. Machine-checked (Appendix A).

Proof. (a) Finite variation adds nothing to quadratic variation, so $[X]_t = \int_0^t v$, whose derivative is $v > 0$; and $L \pm X = a + (\beta \pm 1)(M + B)$, whose brackets are $(\beta \pm 1)^2 Q(T)$ because a is deterministic, continuous and of finite variation; polarizing gives $\beta Q(T)$. (b) Summation by parts gives $\sum_j (\hat{l}_j - \hat{l}_{j-1}) d_j - \beta \sum_j d_j^2 = \sum_j (a_j - a_{j-1}) d_j + \sum_j e_j c_j$ with $\sum_j |c_j| = H$; choosing a and e with the signs of d and c attains the bound. (c) follows from (b). □

It is an identification statement, not a test: no estimator of the log variance from discrete prices, no event on which its error is controlled, no minimax rate and no table of required sampling frequencies is supplied. In a simulation with $\beta = 1.5$, a time effect $a(t) = \log 0.09 + 0.5 \sin 2\pi t$ of total variation 2 on $[0, 1]$, and 200 paths, the estimate with exact log variance recovered β at every grid, and the median guaranteed radius r shrank from 1.36 to 0.24 as the number of increments grew from 256 to 16,384. With a log-variance error of 0.05, the radius grew from 4.3 to 23.4, because the error term grows with the number of increments while $\sum_j d_j^2$ does not. The bound gives a usable test only with a log-variance estimator whose error falls faster than $n^{-1/2}$.

A time effect alone can also be made the null, with a budget on its variation. Let $\Delta_j > 0$ be deterministic durations, R_j the n evaluation log returns, and $V \ge 0$ a budget declared in advance, and let Z_V be the set of $z \in \mathbb{R}^n$ with $\sum_{j<n} |z_{j+1} - z_j| \le V$. The null is that, for some $z \in Z_V$, the returns are independent with $R_j \sim N(-\Delta_j e^{z_j}/2,\ \Delta_j e^{z_j})$, or a mixture of such laws: zero physical drift and a log variance that changes over time only. With $A_j = R_j^2/(2\Delta_j)$, $B_j = \Delta_j/8$ and $f_j(s) = -s/2 - A_j e^{-s} - B_j e^{s}$, the log likelihood is

$\ell_R(z) = c_R + \sum_j f_j(z_j)$ for a constant c_R; let $P_V = \sup_{z \in Z_V} e^{\ell_R(z)}$. For $u \in \mathbb{R}^{n-1}$ put $c_1 = -u_1$, $c_j = u_{j-1} - u_j$ for $1 < j < n$, $c_n = u_{n-1}$ (and $c = 0$ if $n = 1$), $h_j(c_j) = \sup_s (f_j(s) - c_j s)$ and $U_V(u) = c_R + V \max_j |u_j| + \sum_j h_j(c_j)$. For a normalized density q of the returns fixed before they are observed, set $E_V = q/P_V$ and $E_{\mathrm{cert}} = q\, e^{-U_V(u)}$, each zero where its denominator is infinite.

Proposition 50 (a variation budget on a time-only variance). (a) The null is the grid law of a driftless geometric Brownian motion run on a deterministic clock $A(t) = \int_0^t e^{a(s)}\, ds$, with z_j the log of the clock's average rate over step j: every continuous a of total variation at most V gives a $z \in Z_V$, and every $z \in Z_V$ arises from a clock rate constant between observations. (b) P_V is Borel and positive; if some return is nonzero it is finite with a unique maximizer, and if all returns are zero it is infinite; it is nondecreasing in V, and at $V = 0$ it is the common-scale profile, attained at $e^z = 2\sum_j A_j / (n/2 + \sqrt{n^2/4 + 4\sum_j A_j \sum_j B_j})$. (c) Each h_j is explicit, finite or $+\infty$, and for every feasible z and every u, $\ell_R(z) \le \log P_V \le U_V(u)$, with the gap $U_V(u) - \ell_R(z)$ a sum of nonnegative terms; a finite certificate exists whenever some return is nonzero, and at $V = 0$ one with zero gap. (d) Under every null, $E[E_V] \le 1$ and $P(E_V > 1/\alpha) \le \alpha$, and the same holds for $E_{\mathrm{cert}} \le E_V$, for any choice of u, and for any score that is zero on records without a certificate; a feasible pair with gap at most η gives $e^{-\eta} E_V \le E_{\mathrm{cert}}$.

Status. An application of the composite-null split-likelihood bound ([82]), which is proved here; the grid law of the time-changed price is derived from the independent Gaussian increments of Brownian motion in Mathlib. Machine-checked (Appendix A).

Proof. (a) The clock's increment over step j is Δ_j times the mean of e^a there, a value of e^a at an intermediate point, so consecutive z_j differ by at most the variation of a between those points; conversely a step rate realizes any z. (b) Each f_j is strictly concave and tends to $-\infty$ at both ends when $A_j > 0$, and Z_V bounds every $z_j - z_1$, so an upper level set is compact. (c) Fenchel–Young: $\sum_j f_j(z_j) = \sum_j (f_j(z_j) - c_j z_j) + u \cdot Dz$, with Dz the vector of differences, and $u \cdot Dz \le V \max_j |u_j|$ on Z_V; each difference to the bound is a nonnegative term. (d) The denominator dominates every null density; Markov's inequality. □

A numerical maximizer gives only a lower bound on P_V, and so a score that may be too large; the certificate gives the upper bound the test needs. The scalar supremum is attained at $e^s = 2A_j / (\sqrt{d^2 + 4A_jB_j} + d)$ with $d = 1/2 + c_j$; the equivalent textbook form $(\sqrt{d^2 + 4A_jB_j} - d)/(2B_j)$ cancels catastrophically when d is large, and a naive evaluator then fails rather than erring conservatively. In a check with 40 daily returns, a constant-elasticity forecast with $p = 2$ and $\kappa = 0.45$, and 400 paths per law, the certificate gaps were at most about 3×10^{-7} in median. The certified score's mean stayed well below one under geometric Brownian motion (0.051, 0.017 and 0.002 at $V = 0$, 0.5 and 2) and under a sinusoidal schedule of variation about 0.5, but it never exceeded 20 against the constant-elasticity alternative, even at $V = 0$ (mean score 1.18); the budget shrinks the evidence further. Rates, power, validity at every time and certified floating-point evaluation are not claimed, and a rejection does not identify a bubble: the null is a deterministic, time-only variance schedule with zero drift.

Without a floor on the volatility scale, no finite grid separates elasticities uniformly. Take, with $P = Q$ and start $S_0 = 1$, a fixed grid of durations $\Delta_1, \dots, \Delta_n$ and two of the price laws of Proposition 28 at scale $\kappa > 0$: $P_{1,\kappa}$, geometric Brownian motion $dS = \kappa S\, dW$, a martingale, and $P_{2,\kappa}$, $dS = \kappa S^2\, dW$, the reciprocal of a scaled three-dimensional Bessel process and a strict local martingale. Their elasticities are 1 and 2 and neither has a time effect. Write the prices in normalized coordinates, $\log(S_j/S_{j-1}) = \kappa\sqrt{\Delta_j}\, z_j - \kappa^2 \Delta_j/2$ with z standard normal under $P_{1,\kappa}$, let $L_\kappa(z)$ be the likelihood ratio of $P_{2,\kappa}$ to $P_{1,\kappa}$ in these coordinates, an explicit product over the steps, and let $d(\kappa) = \int |L_\kappa - 1|\, d\Gamma_n$, with Γ_n the standard normal law on $\mathbb{R}^n$.

Proposition 51 (no uniform separation without a scale floor). (a) L_κ is positive with mean one, $L_\kappa(z) \to 1$ for every z and $d(\kappa) \to 0$ as $\kappa \to 0$; for every measurable test ψ with values in $[0, 1]$, even one

that depends on κ, $|E_{2,\kappa}\psi - E_{1,\kappa}\psi| \le d(\kappa)$. (b) If $E_{1,\kappa}\psi \le \alpha$ for every $\kappa > 0$, then for every $\varepsilon > 0$ some κ has $E_{2,\kappa}\psi \le \alpha + \varepsilon$; the best worst-case power over scales is α, attained by rejecting at random with probability α. (c) For any sequence of finite grids and tests of size at most α at every scale, there are scales $\kappa_n < 1/n$ at which the power on the n-th grid is at most $\alpha + 1/n$, so no sequence has worst-case power tending to one.

Status. Conditional on the reciprocal-Bessel transition density and its normalization, as in Proposition 28, and on the varying-dominator convergence theorem ([26]), each stated as a hypothesis with a vacuous instance; the coordinate identity for the geometric Brownian law is proved from Mathlib's Gaussian law and change of variables. Machine-checked (Appendix A).

Proof. (a) Each factor of L_κ is the ratio of the two one-step kernels at the coordinates, and tends to one because the scaled difference quotient of $1/S$ tends to $-z_j$; L_κ is nonnegative with mean one, so pointwise convergence gives convergence in L^1 (Scheffé), and $|E_2\psi - E_1\psi| = |\int \psi(L_\kappa - 1)\, d\Gamma_n|$. (b) and (c) follow, the latter grid by grid. □

The sharp bound in (a) is the total variation $d(\kappa)/2$, since $\int (L_\kappa - 1)\, d\Gamma_n = 0$. The same obstruction holds for any null and alternative classes containing these families, so for the time-and-level models of Propositions 49 and 50 with an unrestricted scale and any time budget. This is the pattern of the nonexistence results of Bahadur and Savage ([8], which we have not read against this problem), alternatives that approach the null in total variation, made explicit for these kernels. Both laws depend on the grid only through $\kappa^2\Delta_j$, so a small scale is the same as a short record: from a fixed start, a record of small integrated variance $\kappa^2 T$ cannot separate the elasticities, and a scale floor is a floor on $\kappa^2 T$. Numerically, with one step of unit length the total variation is 0.381 at $\kappa = 1$ and 0.075 at $\kappa = 0.2$, nearly linear in κ; at $\kappa = 0.5$ over a unit horizon, refining from 1 to 5, 25 and 125 steps raises the L^1 distance from 0.37 to 0.68, 1.20 and 1.35, toward the continuous-record identification of Proposition 49. The statement gives no rate, does not say that the two families overlap, and says nothing with a positive scale floor, a continuous record or $P \ne Q$; the bubble's size also vanishes as $\kappa \to 0$, so it does not separate bubble values either.

Jumps can be put into the null in the same way as the switch of Proposition 48, with the same verdict: valid, and without power. In a further run of the daily design after Proposition 46, with volatility 0.3 and normal log jumps of mean -0.05 and standard deviation 0.1 at one or five a year, the free-drift score had null means 341 and 561. Declare in advance a diffusion-scale floor $\sigma_{\min} > 0$ and an intensity cap $\Lambda \in [0, \infty]$, and add to the null every law whose log returns over each step are, independently, a Poisson mixture of normal laws, as in Merton's jump model ([59]): log drift a, diffusion scale $\sigma \ge \sigma_{\min}$, intensity $\lambda \le \Lambda$, and normal jump sizes. On positive paths put $M = \prod_j 1/(y_j \sigma_{\min}\sqrt{2\pi\Delta_j})$, $\rho = 1 - e^{-\Lambda\tau}$ (one if $\Lambda = \infty$) and $U = (1-\rho)\min(D_G, M) + \rho M$, and let E_i^J divide the fitted likelihood by the largest of $G_{i,0}$, D_G and U.

Proposition 52 (a Poisson-jump envelope). In the setting of Proposition 30, with $n \ge 2$ evaluation steps: (a) every such jump law has density at most U on every array; U is finite and measurable, increases with the cap and decreases with the floor, is at most D_G when $\Lambda = 0$, and equals M on positive paths when $\Lambda = \infty$. (b) Suppose that, given its training prices, each name without a bubble has an evaluation law that is a mixture of catalogue null laws, Gaussian laws and such jump laws. Then the score is finite almost surely, has conditional mean at most one, $E_i^J > 1/\alpha_i$ has size at most α_i, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$ under any dependence. (c) $E_i^J \le E_i^{\text{drift}} \le E_i$; raising the cap or lowering the floor lowers the score; with $\Lambda = 0$ it is the free-drift score; and absorbed names are never flagged. (d) Truncating the Poisson series at K terms changes one step's density by at most $P(\text{Poisson}(\lambda\Delta) > K)/(\sigma_{\min}\sqrt{2\pi\Delta})$.

Status. An application of the composite-null split-likelihood bound ([82]) with an upper bound on the null maximum, which is proved here, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. The

domination holds for the series formula itself, without its interpretation as Merton's law. Machine-checked (Appendix A).

Proof. (a) In each step's series the term without jumps is a Gaussian density, and every term with jumps is a normal density with variance at least $\sigma_{\min}^2\Delta_j$, hence at most $1/(\sigma_{\min}\sqrt{2\pi\Delta_j})$. Expanding the product over steps, the all-zero-count term is at most $e^{-\lambda\tau}\min(D_G, M)$, and the others sum to at most $(1-e^{-\lambda\tau})M$; and $1-e^{-\lambda\tau} \le \rho$. (b) The denominator dominates every null component; integrate over the latent law as in Proposition 47. □

The ceiling is the problem: for a year of daily data, the logarithm of M exceeds that of the Gaussian supremum by about 230, so the envelope is dominated by ρM unless ρ is below about e^{-230}. In the same simulation, with $\sigma_{\min} = 0.2$, the score stayed valid under jumps at every cap, but power against $p = 3/2$ was essentially zero for any cap of 10^{-60} a year or more. Only caps of about 10^{-120} a year, which exclude every real jump process, kept part of the free-drift score's power, at 3 to 4%. The quantity ρ must be computed as $-\mathrm{expm1}(-\Lambda\tau)$: computed directly, it underflows to zero for caps below about 10^{-16} and silently removes the jump envelope. Like the single-switch null, this enlargement is valid but powerless. A usable jump null would need the exact supremum over the constrained jump family, or a much tighter bound than a per-step variance floor. Stochastic volatility, regime changes, jumps to zero and rounding are not covered.

Sharing the drift and scale across the steps tightens the envelope and recovers a little power. For a finite cap, put $q_j = e^{-\Lambda\Delta_j}$, $p_j = 1-q_j$ and

$$F(a,s) = \prod_j \frac{1}{\sqrt{2\pi\Delta_j}}\, s^{-n} \prod_j \Big[p_j + q_j \exp\Big(-\frac{(r_j - a\Delta_j)^2}{2s^2\Delta_j}\Big)\Big],$$

let $V = \prod_j y_j^{-1} \sup_{a\in\mathbb{R},\, s\ge\sigma_{\min}} F(a,s)$ (and $V = M$ for an infinite cap), and let E_i^{prof} divide the fitted likelihood by the largest of $G_{i,0}$, D_G and V.

Proposition 53 (a shared-parameter jump envelope). In the setting of Proposition 52: (a) every jump law of Proposition 52 has density at most V, and $V \le U$; V is finite and measurable, so E_i^{prof} has every conclusion of Proposition 52(b), and $E_i^J \le E_i^{\mathrm{prof}} \le E_i^{\mathrm{drift}} \le E_i$, with equality to the free-drift score at $\Lambda = 0$ and to E_i^J at $\Lambda = \infty$. (b) The supremum of F is attained in an explicit rectangle $[a_-, a_+] \times [\sigma_{\min}, s_+]$ computed from the evaluation returns: $a_\pm$ are the extreme return rates and s_+ is determined by the value of F at the sample rate. (c) On any closed box, an explicit upper bound for F is obtained by putting the smallest scale in the power and the largest in the exponent, at the distance from each return to the box's drift range. For any finite cover of the rectangle by boxes, chosen measurably from the data, the largest box bound B_P gives a valid envelope $\min(U, \prod_j y_j^{-1} B_P)$ and a valid score between E_i^J and E_i^{prof}; refining the cover does not lower that score, and the box bounds converge to the supremum as the boxes shrink.

Status. An application of the split-likelihood bound with an upper bound on the null maximum, together with the extreme-value theorem and uniform continuity on compact sets, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Machine-checked (Appendix A).

Proof. (a) In each step's Poisson mixture, the term without jumps is the Gaussian density at the law's own drift and scale, and every term with jumps is at most the ceiling at that scale; replacing the jump probability by its cap gives F at the law's parameters, and the count expansion of Proposition 52 gives $V \le U$. (b) Moving the drift into the range of the return rates brings every return closer, and above s_+ the factor s^{-n} alone keeps F below its value at the sample rate. (c) Each factor is monotone in the scale and in the distance from the drift range, and F is uniformly continuous on the compact rectangle. □

Only returns that look like jumps now pay the ceiling. Any finite cover gives a valid score, so the computation may stop early, but values of F at chosen points are lower bounds on the maximum and must not be used as denominators. In the same daily design, with $\sigma_{\min} = 0.2$ and 1,000 paths per case, jump nulls within the cap gave the refined score null means of at most 0.02, with no false alarms. Against $p = 3/2$ with $\kappa = 0.45$, its power was 5.6% at a cap of one jump a year and 0.2% at five, against zero for the envelope of Proposition 52 and 21% for the free-drift score, which does not cover jumps; with $\kappa = 0.3$ it was 0.6% and 0, against 5%. Adaptive refinement in the scale is needed in practice: with it, the remaining gap between the upper bound and the best point value was 0.3 to 0.7 in log terms on these one-year daily blocks, while a fixed grid of boxes leaves a much larger gap, because errors in the scale compound over the steps.

Jump and switch nulls are thus feasible for validity but leave little power on daily data. They repair the failures after Propositions 46 and 47, bringing raw null means of 17 to 561 back below one. But against $p = 3/2$ on two years of daily data, power falls from 5 to 21% for the free-drift score to 0 to 0.3% with a switch, and to 0.6 to 5.6% with jumps capped at one a year, about zero at five. A switch or a jump can imitate level-dependent volatility on daily data, as in the mechanism of Proposition 20. Further gains would need a tighter bound on the jump family's supremum, an honestly small cap, or richer, intraday data.

Removing returns that look like jumps does not escape this. Fix a cutoff $a > 0$ and let $H_a(r) = r$ for $|r| \le a$ and $H_a(r) = 0$ otherwise. Let p be a strictly positive probability density of a return R, q any probability density, and $\rho_p = \int_{|r|>a} p$, $\rho_q = \int_{|r|>a} q$. On the filtered value $Y = H_a(R)$ take the reference measure ν_a, Lebesgue measure on $\{0 < |y| \le a\}$ plus a unit mass at zero, and the densities p_a and q_a equal to ρ_p and ρ_q at zero, to p and q on $\{0 < |y| \le a\}$, and to zero elsewhere.

Proposition 54 (filtered returns lose Gaussian calibration). (a) Y has density p_a with respect to ν_a, and the filtered forecast has density q_a; both have total mass one. $P(Y = 0) = \rho_p \in (0, 1)$, so Y has no Lebesgue density and is not a nondegenerate Gaussian, and changing p at zero does not change ρ_p. (b) Take one return in Proposition 35 with $\Delta = \nu = 1$, $w_L = w_H = 1$ and a free drift, at the null with R standard normal, so that the profiled denominator is the constant $1/\sqrt{2\pi}$, and the forecast $N(0, \varepsilon^2)$. The unfiltered score has mean $1/\sqrt{1+\varepsilon^2} \le 1$, and the same score applied to Y has mean at least ρ_p/ε. For every $0 < \alpha < 1$, the cutoff $a = (1-\alpha)/2$ and any $0 < \varepsilon < \alpha$ make its rejection probability at threshold $1/\alpha$ exceed α. (c) If a family of strictly positive densities contains p, and a measurable $D_a > 0$ on the support of ν_a dominates every filtered density of the family, then $q_a(Y)/D_a(Y)$ has mean at most one; with $D_a = p_a$ the mean is exactly one, and the ratio at zero is ρ_q/ρ_p, not $q(0)/p(0)$.

Status. Elementary measure theory and the split-likelihood bound, proved here with the supplied densities; nothing is assumed. Machine-checked (Appendix A).

Proof. (a) For a Borel set C, $P(Y \in C) = \int_{C \cap \{0<|y|\le a\}} p + \rho_p 1_{0 \in C}$, which is the integral of p_a against ν_a. (b) The score equals $\varepsilon^{-1} e^{-y^2/(2\varepsilon^2)}$, which is $1/\varepsilon$ at every filtered zero; a Gaussian integral gives the unfiltered mean, and $\rho_p \ge (1+\alpha)/2 > \alpha$ at the stated cutoff. (c) The denominator dominates p_a, and q_a integrates to one against ν_a. □

A score that treats filtered returns as unfiltered ones can thus lose calibration at any prescribed level, even when the unfiltered null is exactly Gaussian and the cutoff is known; at zero, the denominator must dominate the null family's tail masses, not its densities at zero. In a check at $\alpha = 0.05$ with cutoff 1.5, so that $P(|R| > a) = 0.134$, and $\varepsilon = 0.04$, the filtered score rejected 15.5% of the time against 2.1% for the unfiltered score. The statement concerns one fixed observation and constructs no sequential filter. It refutes only that a filtered monitor inherits the unfiltered guarantee automatically, and says nothing about the false-alarm rates of particular filtered forecasts or adaptive cutoffs.

Rounding of quoted prices is a misspecification of the observation law, not of the price model, and it defeats both the total-variation envelope and the naive use of the catalogue densities. To model it, round log prices to a grid of step $h > 0$, $Q_h(x) = \exp(h\lfloor \log x/h + 1/2 \rfloor)$, so that each observation reports the cell containing the price.

Proposition 55 (rounded prices). (a) For every law of a real observation and every atomless reference law, the law of the rounded observation is at total variation one from the reference, at every $h > 0$; the same holds for vectors against a reference absolutely continuous with respect to Lebesgue measure. (b) Let the true price be geometric Brownian motion with volatility one from 1 over one unit of time, let the catalogue consist of this law and a reciprocal-Bessel alternative of scale κ, and let the selector always choose the alternative. The score that evaluates the continuous densities at the rounded terminal price Y equals $e^{1/8}(1-e^{-2/\kappa^2})/\kappa$ on $\{Y = 1\}$, which has null probability $p_h = \Phi(1/2+h/2)-\Phi(1/2-h/2) > 0$, so its null mean is at least $p_h e^{1/8}(1 - e^{-2/\kappa^2})/\kappa$, which tends to infinity as $\kappa \to 0$. At $h = 1/2$ and $\kappa = 1/100$ the nominal 5% rule rejects with probability more than $1/20$, while the ratio of the two laws' cell probabilities has null mean one and is below 20 on that cell. (c) In the setting of Proposition 28 with every observed price rounded and the start known, let $g^h_\vartheta(v \mid u)$ be the ratio of the probabilities, under candidate ϑ, of the observed training and evaluation cells jointly and of the training cells alone, computed from the joint price density. These are strictly positive conditional probabilities summing to one, and with them, for any selector based on the training cells, the conclusions of Propositions 27 and 28 hold for the rounded data, and so does the false-discovery bound of Proposition 40. (d) This conditional law is not that of a chain started at the rounded training price: for geometric Brownian motion with volatility one, start 1, grid 0, 1, 2 and training cell 0, at every $h > 0$ some evaluation cell has a different probability under the two.

Status. An application of the price kernels, stated as hypothesis structures with vacuous instances and the grid laws supplied as premises, of the split-likelihood bound (proved) and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; (a) and the constants in (b) are proved outright. Machine-checked (Appendix A).

Proof. (a) A rounded law is concentrated on countably many points, which an atomless or absolutely continuous law gives measure zero. (b) On $\{Y = 1\}$ the rounded price is 1, where the density ratio is the displayed value, and p_h is the normal probability of the cell. (c) The cell probabilities are the actual probabilities of the observed labels, so they are conditional densities against counting measure, and the split-likelihood argument applies. (d) Conditioning on the training cell leaves positive mass on latent prices above the representative, where the transition-density ratio against starting at the representative grows with the evaluation cell. □

Numerically, $p_h \approx 0.175$ at $h = 1/2$, so the nominal 5% rule in (b) rejects with probability about 0.175; that grid is a counterexample, not a realistic tick. The correction in (c) is exact but not practical: its cell probabilities are integrals over the cells of the whole training prefix, since the conditional law of the evaluation prices averages over the unobserved exact training prices, and no practical scorer is given here. As a heuristic, not a bound, the size of the problem is set by the tick relative to the one-step spread of the narrowest catalogue law: a one-cent tick on a fifty-dollar stock is about 0.02% in log price, small against daily volatility, so the plug-in error is probably small for daily data, but it need not be for intraday data or for catalogues with tight alternatives.

For Gaussian nulls, the unobserved exact price inside the last training cell can be budgeted instead of filtered. Fix a finite catalogue Γ_i of Gaussian null laws with smallest volatility $\kappa_{\min}$, the log tick h and the first evaluation step Δ_1. Let $g^{\mathrm{restart}}_\theta(v \mid u)$ be the probability that the whole rounded evaluation block is v when prices follow $\theta \in \Gamma_i$ from the representative of the last training cell, and set $E_i^{\mathrm{restart}} = q_i(v \mid$

$u)/\max_\theta g_\theta^{\mathrm{restart}}(v \mid u)$, for any normalized forecast q_i of the cell labels based on the rounded training record u. Put $\varepsilon^{\mathrm{round}} = \sqrt{1-\exp(-h^2/(16\kappa_{\min}^2\Delta_1))}$.

Proposition 56 (a rounding budget without filtering). Suppose that a bubble-free name's future log returns, given its rounded training record and its exact last training price, are Gaussian under one of the catalogue laws. (a) The restarted probabilities are positive and sum to one, and the score has mean at most one under each restarted law. (b) At almost every rounded training record, the actual conditional law of the rounded evaluation block is within total variation $\sqrt{1-\exp(-h^2/(16\kappa^2\Delta_1))} \le \varepsilon^{\mathrm{round}}$ of the restarted law of its catalogue law, whatever the conditional law of the exact price within its cell and however many observations follow. (c) Hence $E_i^{\mathrm{restart}} > 1/\alpha_i$ has size at most $\alpha_i + \varepsilon^{\mathrm{round}}$, the threshold $1/(\alpha_i - \varepsilon^{\mathrm{round}})$ has size at most α_i when $\varepsilon^{\mathrm{round}} < \alpha_i$, and $\min(E_i^{\mathrm{restart}}, c_i)/(1 + c_i\varepsilon^{\mathrm{round}})$ has mean at most one, so the e-BH rule of Proposition 40 keeps FDR $\le q\,m_0/m$ for it under any dependence. (d) $\varepsilon^{\mathrm{round}} \le e$ exactly when $h \le 4\kappa_{\min}\sqrt{\Delta_1}\sqrt{-\log(1-e^2)}$, and $\varepsilon^{\mathrm{round}}$ tends to one as Δ_1 or $\kappa_{\min}$ tends to zero.

Status. An application of the Gaussian affinity of Proposition 43(c), through the rounding map and integrated over the exact price, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. Starting from the exact price instead of the representative moves only the first log return, by at most $h/2$; the two Gaussian laws of the returns then have affinity at least $\exp(-h^2/(32\kappa^2\Delta_1))$, which bounds total variation after the rounding map. The actual law is a mixture of such laws over the exact price, so the bound survives averaging. (c) follows as in Proposition 43. □

Reading a one-cent tick as a log tick of one cent divided by the price, as the logarithmic grid requires, and with $\kappa_{\min} = 0.2$, the budget is about 0.002 at daily sampling and 0.018 at five minutes on a hundred-dollar stock, 0.020 and 0.17 on a ten-dollar stock, and 0.10 and 0.73 on a two-dollar stock: essentially free for daily data on normally priced stocks. It applies only to this cell likelihood, not to the scores of Propositions 29–39, 41, 42, 48, 50, 52 and 53, which evaluate continuous densities at rounded prices and can fail by Proposition 55(b); computing it needs joint integrals over the cells of the evaluation block, and no scalable implementation is given. Constant-elasticity nulls, absorption, jumps and stochastic volatility are not covered.

A cheaper route avoids both the latent price and the joint cell integrals, at a cost that grows with every observation. For a Gaussian null law θ and evaluation step j, let $K_{\theta,j}$ be the largest probability of the observed cell z_j over all exact starting prices in the previous cell z_{j-1}, let $B_\theta = \prod_j K_{\theta,j}$, and set $E_i^{\mathrm{cell}} = q_i(v \mid u)/\max_{\theta\in\Gamma_i} B_\theta$, for a finite catalogue Γ_i of Gaussian null laws and any normalized forecast q_i of the cell labels based on the rounded training record.

Proposition 57 (a Gaussian cell envelope). Under the premise of Proposition 56: (a) $K_{\theta,j}$ is a difference of two normal distribution functions evaluated at the start that puts the mean step nearest the middle of the target cell, clipped to the previous cell, and it is the supremum over starts in that cell; it lies strictly between zero and one. (b) At almost every rounded training record, the actual probability of the observed evaluation cells is at most B_θ for the true law θ, whatever the conditional law of the exact price in the last training cell; so E_i^{cell} is finite, has conditional mean at most one, and $E_i^{\mathrm{cell}} > 1/\alpha_i$ has size at most α_i, and the e-BH rule of Proposition 40 has FDR $\le q\,m_0/m$ for these scores under any dependence, with no total-variation budget and no clipping. (c) For a law with zero log drift and equal steps Δ, each row of transition maxima sums to $S = 2\Phi(h/(2\kappa\sqrt{\Delta})) > 1$, so the path envelope has total mass S^n; and dividing the envelope by this excess is invalid: with one step from the representative of the last cell, a forecast concentrated on the central cell then has null mean $S > 1$.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) A normal cell probability, as a function of the mean, increases up to the cell's midpoint and decreases after it. (b) By induction along the path, from every exact start in the last training cell the probability of the remaining observed cells is at most the product of the step maxima; averaging over the exact start keeps the bound; then $\sum_v (q_i/B_\theta) P(v \mid u) \le \sum_v q_i = 1$. (c) The row sum is a direct summation of normal distribution functions over the cells. □

Computing the envelope needs one normal-distribution difference per step and null law. In a simulation with a geometric-Brownian null of volatility 0.3 and, as numerator, the true law's one-step forecast from each rounded price, the score kept its null mean below one. But its log evidence fell by about the logarithm of the total excess mass: about 0.5 over a year of daily data at a log tick of 10^{-4} (about one cent on a hundred-dollar stock, on the logarithmic grid), 5 at 10^{-3}, and 30 and about 280 over a month of five-minute data, against a rejection threshold of $\log 20 \approx 3$ at the 5% level. The two routes are complementary: this one is cheap and nearly free for daily data with fine ticks, but ruinous at high frequency or with coarse ticks, while the restarted cell likelihood of Proposition 56 has a small budget at high frequency but needs joint cell integrals. Neither covers five-minute data cheaply as it stands; the refinement of Proposition 58 reduces the envelope's cost. Constant-elasticity nulls, absorption, jumps and stochastic volatility are not covered, and rounding to a fixed number of cents differs from this logarithmic grid.

Subdividing the cells makes this valid rounded-price likelihood affordable; it does not create power. At level ℓ, split each cell into $K = 2^\ell$ equal subcells. For each subcell a of the cell observed at step j and each null law θ, define backward along the observed path $U^\ell_{\theta,n,a} = 1$ and

$$U^\ell_{\theta,j,a} = \min\Big(1,\ \sum_d W^\ell_{\theta,j,a,d}\, U^\ell_{\theta,j+1,d},\ U^{\ell-1}_{\theta,j,\lfloor a/2 \rfloor}\Big),$$

where $W^\ell_{\theta,j,a,d}$ is the largest probability of reaching subcell d at step $j+1$ from an exact price in subcell a, and the last term is omitted at level zero. Let $B^\ell_\theta = \max_a U^\ell_{\theta,0,a}$ and $E^\ell_i = q_i / \max_{\theta \in \Gamma_i} B^\ell_\theta$, with the numerator and catalogue of Proposition 57.

Proposition 58 (subdividing the cell envelope). Under the premise of Proposition 56: (a) $W^\ell_{\theta,j,a,d}$ is the clipped-midpoint difference of normal distribution functions of Proposition 57(a), for subcells of any width, and lies in $(0, 1]$. (b) Every $U^\ell_{\theta,j,a}$ lies in $(0, 1]$ and bounds the probability of the remaining observed cells from every exact price in its subcell; hence, at almost every rounded training record, the actual probability of the observed evaluation cells is at most B^ℓ_θ for the true law, at every level simultaneously. (c) B^0_θ is the envelope of Proposition 57, and $B^{\ell+1}_\theta \le B^\ell_\theta$, so $E^{\ell+1}_i \ge E^\ell_i$. (d) For any level $L(u, v)$, even one chosen after seeing the evaluation cells, E^L_i has conditional mean at most one and $E^L_i > 1/\alpha_i$ has size at most α_i, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$ under any dependence.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (b) For any one-step law and any finite partition of the next cell, the probability of the remaining path is at most $\sum_d W_d U_d$, by induction backward along the path; the parent bound is valid by the previous level, and the minimum of valid bounds is valid. (c) The parent term gives monotonicity. (d) Since every level dominates, so does any choice among them, and $\sum_v (q_i/B^L_\theta) P(v \mid u) \le 1$. □

The cost is of order nK^2 normal-distribution differences per path and null law. In the simulation after Proposition 57, each level roughly halved the loss of log evidence. At level five it fell from 5.3 to 0.18 over a year of daily data at a log tick of 10^{-3}, and from 30 to about 1 over a month of five-minute data at 10^{-4}, at about 1.7×10^6 one-step probabilities per path and null law. A loss below 0.1 at five minutes would need about level eight, roughly 10^8 evaluations per path. The limits stated after Proposition 57 still apply.

Propositions 56–58 assume a logarithmic grid of fixed width; real quotes are rounded to a fixed number of cents, and can be zero. For a tick δ, let the quote of a price $x \geq 0$ be $\delta\lfloor x/\delta + 1/2 \rfloor$, so that the cells are $[0, \delta/2)$ and $[\delta(z - 1/2), \delta(z + 1/2))$ for $z \geq 1$. Suppose that, given the rounded training record, the exact last price lies in the last observed cell with an arbitrary law, possibly with an atom at zero; that from a positive price the future log returns are independent Gaussian under one catalogue law, with μ and s^2 the step's mean and variance; and that from zero the price stays at zero. For a source interval $[a, b)$ and a target $[c, d)$, let W be one if $a = c = 0$; $\Phi((\log d - \log a - \mu)/s)$ if $c = 0 < a$; and otherwise the difference of normal distribution functions at the log midpoint of the target clipped to $[\log a + \mu, \log b + \mu]$, unbounded below when $a = 0$. Run the backward recursion of Proposition 58 with W on subcells of equal price width, the first subcell of the zero cell containing zero.

Proposition 59 (cent ticks and zero quotes). (a) W lies in $(0, 1]$ and is the supremum of the one-step probability of reaching the target over every exact source price, including zero; the product of these maxima along the path, and each level of the recursion, bounds the probability of the observed cells from every exact price in the last cell, and the bounds decrease with the level. (b) So the score with any normalized forecast of the labels, at a level chosen measurably from the data, has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\leq q\, m_0/m$; the product of one-step cell probabilities from the representatives δz, or $\delta/4$ at a zero quote, is such a forecast. (c) A positive cell's log width $\log((y + \delta/2)/(y - \delta/2))$ at quote y is asymptotic to δ/y but not equal to it, and the zero cell has no finite log width. (d) From a positive price $\delta/4$ inside the zero cell, the next quote is positive with probability $1 - \Phi((\log 2 - \mu)/s) > 0$, and from an exact zero with probability zero; on an all-zero path from a zero quote, the bound is one at every level.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the Gaussian cell results of Propositions 57 and 58 are reused. Machine-checked (Appendix A).

Proof. (a) For a positive target, the normal cell probability is unimodal in the log start, so its supremum over the source is at the clipped midpoint; a zero target is reached most easily from the lowest source price, and from an exact zero with certainty. The recursion and its monotonicity are as for Proposition 58, the trapped zero entering through the first subcell. (b) As for Proposition 57, by averaging over the exact last price. (c) and (d) are direct computations. □

So the earlier results do not apply to cents without this change. A zero quote does not reveal an exact zero, but at daily volatility the probability in (d) is about 10^{-295}, so zero quotes are effectively absorbing there. In a simulation with a geometric-Brownian null of volatility 0.3, one year of daily data and a cent tick, the loss of log evidence fell from 10.3 at level zero to 0.72 at level four for a stock starting at five dollars, and from 77.5 to 8.6 for one starting at fifty cents. So penny stocks need deeper refinement, at a cost of order nK^2. An implementation of this score in outward-rounded ball arithmetic (‘pythonflint080balls‘) returns a value no larger than the exact score, conditional on that library and on the code; it is not machine-checked. Gaussian nulls only are covered: not constant-elasticity or default nulls, jumps, stochastic volatility, or real quote rules such as bid and ask prices and changes of tick size.

The model-mismatch budgets of Propositions 43 and 47 carry over to the cent-tick score without any extra cost for rounding. Let E_i be the cent-tick score of Proposition 59, at any level, with a finite Gaussian

catalogue Γ_i. Suppose that, given its rounded training record, a bubble-free name's evaluation law is a mixture, over a latent exact last price x in the last observed cell and a latent variable z, of laws that start at x, stay at zero if $x = 0$, and otherwise have independent Gaussian log returns with drift b_j and volatility $\sigma_j > 0$ at step j, all rounded to cents. Let $\varepsilon(u, x, z)$ be the smallest over $\theta \in \Gamma_i$ of the affinity budget of Proposition 47(a) for that component, zero at $x = 0$, and let $\bar{\varepsilon}_i(u)$ be its average over the latent law.

Proposition 60 (model mismatch for the cent-tick score). In this setting, with a supplied measurable $e_i \geq \bar{\varepsilon}_i$: (a) there is a reference mixture G_i, with the same latent law and each positive component replaced by its closest catalogue law, such that the rounded actual law is within total variation $\bar{\varepsilon}_i$ of G_i and E_i has mean at most one under G_i. (b) Hence $E_i > 1/\alpha_i$ has size at most $\alpha_i + e_i$, conditionally and unconditionally; the threshold $1/(\alpha_i - e_*)$ has size at most α_i when $e_i \leq e_* < \alpha_i$; and $\min(E_i, c_i)/(1 + c_i e_i)$ has mean at most one, so the e-BH rule of Proposition 40 keeps FDR $\leq q\, m_0/m$ for it, and it can reject only if $e_i < \alpha_i$ and $c_i > 1/(\alpha_i - e_i)$. (c) If every positive component is a constant-parameter catalogue law, possibly a different one for each component, then $\bar{\varepsilon}_i = 0$ and the raw score has mean at most one; and the uniform envelope of Proposition 47(c) bounds $\bar{\varepsilon}_i$ without reference to the tick or the level.

Status. An application of Propositions 43 and 47 and the domination of the cent-tick envelope from every exact starting price, and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) From each exact start, the score has mean at most one under every catalogue law, by the domination from every exact start and the normalization of the forecast. The actual and reference components start from the same price and pass through the same rounding map, so their distance is at most that of the Gaussian return laws, which is the budget of Proposition 47(a); averaging over the latent law gives (a). (b) follows as in Proposition 43, and (c) as in Proposition 47. □

The budget is as weak as that of Proposition 47. In a check with cent ticks from one dollar, fifteen daily steps, a single null law with volatility 0.30, subdivision level two and 300 paths, a true volatility of 0.30 gave budget zero and a false-alarm rate of 1.3% at $\alpha = 0.1$. A true volatility of 0.40 gave a budget of 0.68, above α, so the clipped test could not reject at all; its mean was 0.001. Rounding may shrink the true distance, but this sufficient budget does not exploit that. It gives no informative guarantee for volatility that changes within the window; stochastic volatility outside this representation, jumps and constant-elasticity or default nulls are not covered.

Misspecification bounds thus exist, but they are useful only for small, slow departures from the null. An error in the drift costs a budget that does not grow with the sampling frequency. Errors in the volatility, volatility that changes within the window, and jumps cost budgets that grow with the frequency and quickly leave the clipped test without power, while the raw scores lose validity. Rounding, by contrast, is handled for Gaussian nulls: cent ticks and zero quotes are covered, and the computation is affordable for intraday data. Nulls enlarged to absorb jumps and switches are valid but leave little power on daily data.

Quotes can also carry errors before they are rounded. Fix an error bound $a \geq 0$. Under the *support contract*, each pre-rounding price $Y_j \geq 0$ differs from the latent price X_j by at most a, with errors that may be dependent across dates and on the path. The premise is now stated given the noisy, rounded training record: the catalogue law and the exact last price have an arbitrary joint law given that record, and from a positive last price the latent future has that law's independent Gaussian log returns; from zero it stays at zero. An unconditional Gaussian specification alone does not supply this. Under the stronger *uniform contract*, also $Y_j = \max(0, X_j + \varepsilon_j)$, with $\varepsilon_1, \ldots, \varepsilon_n$ independent and uniform on $[-a, a]$ given the record and the latent path. Widen each observed cell by a on each side, and run the backward recursion of the cent-tick envelope on subcells of the widened cells. Under the support contract each step's factor is the

cent-tick maximum; under the uniform contract it is multiplied by the supremum, over the subcell, of the explicit probability that a uniform error moves the latent price into the observed cell.

Proposition 61 (bounded and uniform observation noise). (a) Under each contract, with its own recursion, the actual probability of the observed evaluation cells is at most the resulting bound D, with $0 < D \le 1$, at almost every training record and at every level; so the score with any normalized forecast of the labels, at any level chosen measurably from the data, has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$. (b) The bounds decrease with the level, and the uniform bound is at most the support bound, so the uniform score is at least the support score for the same forecast; at $a = 0$ both recursions are the cent-tick recursion. (c) On a path of zero quotes from a zero quote, the support bound is one at every level, and the uniform bound is g^n with $g = \min(1, 1/2 + \delta/(4a))$ for tick δ. In particular, with $a = \delta$, $n = 12$ and an exact price of zero, which satisfies the support contract, the uniform score of the forecast concentrated on the all-zero path is $(4/3)^{12} > 20$ with probability one: used under the support contract alone, the uniform score has size one.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) A bounded error puts the latent price in the widened cell. The probability of the observed labels is at most (support) or equal to (uniform) the expected product of emission factors along the latent path, and that expectation is bounded on every subcell by backward induction, as for Proposition 58. (b) Each emission factor is at most one, and the parent bound gives monotonicity. (c) From zero the latent path stays at zero, where the uniform emission of the zero cell is g. □

A quote of zero is thus not evidence of default under noise. Noise is expensive. In a check with a five-dollar geometric-Brownian null of volatility 0.3, a cent tick, errors of up to half a tick, 60 daily steps and the cent-tick forecast, the median log evidence at subdivision level three was -0.4 without noise, -42.3 under the support bound and -14.1 under the uniform law. The support bound costs about $\log 2$ a day at this error size, close to $n \log 2$ over n steps. The uniform law recovers most of that, and was still improving with the level. Neither mode supplies a real bound on the noise, and jumps and stochastic volatility are not covered.

A much stronger declaration removes almost all of that cost. Suppose that each error is exactly $+a$ or $-a$, like a bid–ask bounce: $Y_j = \max(0, X_j + aH_j)$ with a sign $H_j \in \{-1, +1\}$. Given the previous sign h and the latent move from x to y, the next sign is positive with probability $\rho\, 1\{h = +1\} + (1-\rho)\,\mathrm{logistic}(\beta \log(y/x))$, for declared $\rho \in [0, 1)$ and β, with the logistic replaced by $1/2$ at a zero price. As before, the law is stated given the noisy training record, with an arbitrary joint posterior for the catalogue law, the exact last price and the current sign. Given the sign, each label pins the latent price to the cell shifted by the sign times a, one tick wide, rather than to a cell widened by $2a$. Run the backward recursion over feasible sign and subcell pairs, multiplying the cent-tick transition bound by an explicit upper bound on each sign transition over the pair of subcells and keeping the parent of the same sign, and let the bound be the smaller of this one and the support bound of Proposition 61.

Proposition 62 (persistent and return-dependent noise signs). (a) Under the declared law, the actual probability of the observed evaluation cells is at most the bound at almost every training record and every level, and the bound lies in $(0, 1]$; so the score with any normalized forecast of the labels, at a level chosen measurably from the data, has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$. (b) The bound decreases with the level, and the score is at least the support score of Proposition 61. With $\beta = 0$ the sign stays the same with probability $(1 + \rho)/2$; with also $\rho = 0$ the signs are independent fair signs; and with $a = 0$ the bound is the cent-tick bound, for every β.

(c) With a equal to one tick, $\beta = 0$ and an all-zero label path of length n, the bound is $((1+\rho)/2)^n$, which is the actual probability from an exact zero with a negative sign. In particular, with $\rho = 0.9$ and $n = 5$, a forecast concentrated on the all-zero path scores $32 > 20$ against the bound for independent fair signs, $(1/2)^5$, on an event of probability $0.95^5 \approx 0.77$: treating persistent signs as independent gives a nominal 5% test of size about 0.77.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the cent-tick transition bound and the support bound of Proposition 61 are reused. Machine-checked (Appendix A).

Proof. (a) A label from sign t puts the latent price in the cell shifted by t. The probability of the label path is an integral over the returns of a sum over sign paths, bounded on every subcell by backward induction, since each sign transition is at most its upper bound and the cent-tick bound covers the price step. The support bound holds because the returns' marginal law is still Gaussian, the sign weights summing to one. (b) Each row of sign bounds sums to at least one, which gives the reduction at $a = 0$. (c) From zero the latent path stays at zero, where every step's sign stays negative with probability $(1+\rho)/2$. □

In the noise check after Proposition 61, with signs from the declared rule with $\rho = 0.5$ and $\beta = 0$, the median log evidence at level three was -0.4, essentially the noiseless level, against -42.3 under the support bound. The gain comes entirely from the strong declaration: errors of exactly $\pm a$, with a known radius and a known sign rule. That is an empirical premise that free price data cannot check, and estimating ρ or β from training gives no conditional coverage. A bounded but continuous error still costs about $\log 2$ a step.

The declared variance clocks extend to rounded prices, over a compact parameter box. Declare in advance a finite set $\mathscr{V}$ of clocks, as in Proposition 34, and a closed box $\mathscr{B} = [b_-, b_+] \times [w_-, w_+]$ of drifts and variances, with $w_- > 0$. For a subbox $B = [b_L, b_H] \times [w_L, w_H]$, a clock increment v and price intervals $S = [a, d_S)$ and T, let $W_{\mathrm{clock}}(B; S, T)$ be the largest normal probability of reaching T over log means in $[\log a + b_L\Delta - w_H v/2,\ \log d_S + b_H\Delta - w_L v/2]$ and scales in $[\sqrt{w_L v}, \sqrt{w_H v}]$. It is a difference of normal distribution functions at a clipped mean with an explicit scale, with the trapped-zero conventions of the cent-tick bound. Run the backward recursion of the cent-tick envelope with W_{clock} for each subbox of the h-fold bisection of $\mathscr{B}$ and each clock, at cell level ℓ, and let $D^{h,\ell}$ be the maximum over subboxes and clocks.

Proposition 63 (rounded variance clocks over a compact box). Suppose that, given the whole rounded training record, the exact last price, the clock and (b, w) have an arbitrary joint law supported on the last observed cell, $\mathscr{V}$ and $\mathscr{B}$, and that, given these, the future log returns are independent normal with mean $b\Delta_j - wv_j/2$ and variance wv_j, rounded to cents. (a) W_{clock} lies in $(0, 1]$ and bounds the one-step probability of reaching T from every price in S, including zero, and every $(b, w) \in B$; for a single parameter it is the cent-tick bound. (b) $D^{h,\ell}$ lies in $(0, 1]$, bounds the probability of the observed cells at almost every training record, and decreases in h and in ℓ. (c) With any normalized forecast of the labels and any levels (h, ℓ) chosen measurably from the data, the score has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$. On an all-zero path from a zero quote the bound is one; with the calendar clock alone and singleton boxes, the recursion is the cent-tick recursion.

Status. An application of the split-likelihood bound with an upper bound on the null probability and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance. Machine-checked (Appendix A).

Proof. (a) A normal cell probability is unimodal in the mean, and in the log scale it increases up to $\sqrt{(r^2 - u^2)/(2\log(r/u))}$ and decreases after it, where $u < r$ are the distances to the cell's ends; so the

maximum over the rectangle is at the clipped mean and clipped scale. (b) By induction backward along the path, as for Proposition 58; a child box lies in its parent's rectangle, which gives monotonicity. (c) As for Proposition 58. □

Only the computation is refined; the declared family is not narrowed. Refining the parameter box removes most of its cost. In a check with a five-dollar geometric-Brownian null of volatility 0.3, a cent tick, 60 daily steps and the cent-tick forecast at the true parameters, the median log evidence was -0.67 for the true parameter alone. For the box $b \in [-0.1, 0.2]$, $\sqrt{w} \in [0.15, 0.45]$ at cell level two, it was -22.8, -19.2, -11.3 and -5.4 after zero to three bisections of the box, and still falling, at a cost growing as 4^h per step. Coverage by the compact box, including its lower variance, is a premise. Constant-elasticity nulls are not retained, and observation noise is treated in Proposition 61. In the intraday simulation after Proposition 34, at hourly sampling with two bisections of a wide box and no cell subdivision, this score had power of at most 3.6%.

The forecast can follow a clock too. Declare an ordered finite catalogue of constant-elasticity forecasts $\xi = (p, \kappa, \nu)$, with $p > 1$, $\kappa > 0$ and their own positive clock increments ν for the training and evaluation steps, which need not be null clocks. For exact prices, choose the first maximizer $\hat{\xi}$ of the training likelihood $\prod_j K_{p,\kappa}(\nu_j, y_{j-1}, y_j)$, with the constant-elasticity kernel K of Proposition 29, and forecast the evaluation prices with the same product over the evaluation steps. For rounded prices, use instead the probabilities of the cells from the representative price of the previous cell, both to choose $\hat{\xi}$ on the training labels and to forecast the evaluation labels. For a training-only weight $\lambda \in (0, 1)$, mix either forecast with an earlier normalized one.

Proposition 64 (clock-aware constant-elasticity forecasts). (a) Both forecasts, and their mixtures, are normalized for every training record; the cell rows have positive masses summing to one. (b) Divided by the unchanged denominator of Proposition 34, the exact-price forecast gives a score with every conclusion of Proposition 34(b); divided by the unchanged bound of Proposition 63, the label forecast gives a score with every conclusion of Proposition 63(c). (c) The mixture's score is at least λ times the earlier score, so it loses at most $-\log \lambda$. An absorbed name is never flagged, and with one exact evaluation step neither score exceeds one. (d) The forecast must be chosen on the training data: with 21 outcomes, a uniform null and the 21 forecasts that each put all mass on one outcome, choosing the forecast after seeing the outcome gives score 21 on every outcome, so a 5% test always rejects, while their fixed uniform mixture gives score one.

Status. An application of the split-likelihood bound with a training-selected forecast, through Propositions 34 and 63, whose validity holds for any normalized forecast; the constant-elasticity kernels enter as for Proposition 29, and the false-discovery bound is conditional on the e-BH theorem. Machine-checked (Appendix A).

Proof. (a) Each kernel and each row of cell probabilities has mass one, and a product of normalized steps is normalized. (b) Propositions 34 and 63 hold for any normalized forecast chosen on the training data. (c) The mixture's score is the mixture of the two scores. (d) is a direct computation. □

The forecast does not change the null coverage, and the label forecast is not a likelihood of the latent price given its rounded history; there is no general ordering between the clock-aware and earlier scores. In the weekend design after Proposition 34, with the null clocks unchanged and a forecast catalogue of powers $\{1.25, 1.5, 2\}$, scales $\{0.2, 0.3, 0.45\}$ and the calendar and trading clocks, no null path scored above 10. At $\alpha = 0.05$, power against a trading-time alternative with $p = 3/2$, $\kappa = 0.45$ rose from 0.6% with the calendar forecast to 14.5% with the clock-aware forecast, and 10.8% with the half-and-half mixture; with $p = 2$, $\kappa = 0.3$, from 2.7% to 29.4% and 24.6%. This is optimistic for the forecast, since the alternative's clock is exactly one of the forecast clocks; whether real clocks match a declared one is an empirical question.

The pieces compose. Declare clocks and a compact box as in Proposition 63, and a two-point error with a persistent, return-dependent sign as in Proposition 62. For each subbox and clock, run the sign recursion of Proposition 62 with the box-and-clock transition bound of Proposition 63 in place of the cent-tick bound, and run the support recursion of Proposition 61 on the widened cells with the same transition bound. Let D be the smaller of the two maxima over subboxes and clocks. For the numerator, declare a finite catalogue of constant-elasticity forecasts with their own clocks and their own sign persistence. Score the training labels by the sign chain started uniformly on the feasible signs, choose the first maximizer, and forecast the evaluation labels from the final sign weights of the training record; optionally mix it with any other normalized forecast by a training-only weight λ.

Proposition 65 (clocks, box and noise signs together). Suppose that, given the whole noisy rounded training record, the clock, (b, w), the exact last price and the current sign have an arbitrary joint law supported on the declared clocks, the box and the compatible price and sign pairs, and that, given these, the future returns and signs follow the law of Proposition 62 with means $b\Delta_j - wv_j/2$ and variances wv_j. (a) D lies in $(0, 1]$, decreases with the box depth and the cell level, and bounds the probability of the observed cells at almost every training record, for every refinement, including any cover of the box and any levels chosen measurably from the data that keep all declared clocks. (b) The sign-chain forecast and its mixtures are positive and normalized for every training record. (c) With any normalized forecast, the score has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\leq q\, m_0/m$; the mixture's score is at least λ times the other. (d) With no noise, D is the bound of Proposition 63; with a single parameter and the calendar clock, it is the bound of Proposition 62; and on an all-zero path from a zero quote with radius one tick and $\beta = 0$, it is $((1 + \rho)/2)^n$.

Status. A composition of Propositions 61–64 with the split-likelihood bound; the forecast's constant-elasticity kernels enter as for Proposition 29, and the false-discovery bound is conditional on the e-BH theorem. Machine-checked (Appendix A).

Proof. (a) The transition bound of Proposition 63 covers every parameter in a subbox and every source price, so it can replace the cent-tick bound in both recursions of Propositions 61 and 62, whose suffix bounds hold for any such step bound; the monotonicity of Proposition 63 carries through the sign-weighted sums. (b) Each forecast row has mass one, and the final sign weights form a probability vector. (c) As for Proposition 64. (d) Each reduction substitutes the special case into the recursion. □

This adds no coverage beyond its parts: the clocks, the box, the radius and the sign law must all hold given the noisy record. In a check with half-tick two-point noise on a five-dollar stock over 40 days, the median log evidence was -0.43 for the true parameters alone, so the noise stays nearly free. For a wide box it was -16.3, -13.4 and -8.9 after zero to two bisections: the box, not the noise, is the binding cost, and refining it is expensive.

Valid scores thus exist under declared observation noise, and their cost depends on the declaration. A bound on the error size alone costs about $\log 2$ a step at half a tick, which leaves the score nearly uninformative. A declared two-point bid–ask error with a persistent, return-dependent sign is nearly free. Combined with a declared intraday clock and cent ticks, the binding cost is the parameter box, not the noise. Every such declaration is an empirical premise that price data without quotes cannot check.

Under a stronger declaration, the exact cell likelihood of Proposition 55(c) can replace the envelopes. Suppose the whole record, training and evaluation, follows a declared Gaussian law from a known exact starting price s: independent log returns with mean $b\Delta_j - wv_j/2$ and variance wv_j, for a clock v in a finite declared set and (b, w) in a compact box, or any mixture of such whole-record laws, observed through cent ticks. For each parameter θ, let $\alpha_{\theta,0}$ be the point mass at s and let $\alpha_{\theta,j}$ be the price distribution after step j, carried forward by the transition and restricted to the observed cell, without restarting at a representative

price; let $L_{\theta,j}$ be its total mass. Set $g_\theta = L_{\theta,N}/L_{\theta,k}$ for k training and N total steps, $D_{\text{filter}} = \max_\theta g_\theta$ over the clocks and the box, and $E = q/D_{\text{filter}}$ for a normalized training-selected forecast q of the evaluation cells.

Proposition 66 (a filtered cell likelihood). (a) Every finite string of cells has positive mass, g_θ is the conditional probability of the evaluation cells under θ, it sums to one over them and is continuous in (b, w), and the maximum is attained, with $0 < D_{\text{filter}} \le 1$. (b) Under the declared whole-record law, E has conditional mean at most one and size at most α, and the e-BH rule of Proposition 40 keeps FDR $\le q\, m_0/m$; the same holds with any denominator between D_{filter} and one. (c) On positive cells, the midpoint quadrature $Q_{\theta,J}$ on M subintervals of each log cell satisfies $e^{-\varepsilon_J} Q_{\theta,J} \le L_{\theta,J} \le e^{\varepsilon_J} Q_{\theta,J}$, with an explicit ε_J of order $1/M$. (d) For a finite cover of the box by subboxes and explicit parameter-sensitivity allowances η, the maximum over subboxes and clocks of $\min(1, e^{\varepsilon_N+\varepsilon_k+\eta_N+\eta_k} Q_N/Q_k)$ at the subbox centres, set to one on records with a zero quote, is a bound D_{cert} with $D_{\text{filter}} \le D_{\text{cert}} \le 1$, and the corresponding lower expression bounds D_{filter} from below; both converge to D_{filter} on a fixed positive record as the cover and the quadrature are refined. (e) For matched parameters and clocks, D_{filter} is at most the supremum of the exact evaluation likelihood over all endpoints in the last training cell, which is at most the bound of Proposition 63.

Status. An application of the split-likelihood bound to the exact cell likelihood and, for the false-discovery bound, of the e-BH theorem, stated as a hypothesis with a degenerate instance; the Gaussian laws are Mathlib's, and the cent-tick cells and the box bound of Proposition 63 are reused. Machine-checked (Appendix A).

Proof. (a) The forward recursion is the label mass, and integrating the evaluation suffix against the training endpoint law gives the joint mass. (b) The conditional law of the evaluation cells is a mixture of the g_θ, each at most the denominator, and the forecast is normalized. (c) By induction along the string, the density ratio between a point and its subinterval midpoint is at most $e^{C_j(r_{j-1}+r_j)}$ at each step. (d) The one-step log density changes by at most the stated allowance over each subbox, by algebra in the drift and $\log x \le x - 1$ in the scale. (e) The filtered conditional law is an average over endpoints in the last cell. □

This is stronger than the contract of Proposition 63, which allows an arbitrary joint posterior for the parameters and the exact last price given the training record. Records with a zero quote fall back to the bound one, and the minimum of D_{cert} and the bound of Proposition 63 is valid under the present contract; D_{cert} can be computed in outward-rounded arithmetic, which is not machine-checked. In a check with a five-dollar geometric-Brownian null of volatility 0.3, a cent tick, 30 training and 30 evaluation days and the cent-tick forecast at the true law, the median log evidence was -6.05 with the box bound of Proposition 63. What is proved is only that the ideal filtered score is no worse than that bound. Evaluating the filtered maximum on a grid of 41×41 parameters gave a median of -0.23. A grid underestimates the maximum over the box, so this is only an upper bound on the filtered score's log evidence; it suggests a substantial improvement but does not certify it. The certified cover alone gives a median of about -8 on a comparable design, worse than the box bound, because it bounds the training and full masses separately; its cost is of order nM^2 per box and clock. Noise, jumps and power are not covered.

A sharper certified bound uses the conditional ratio directly, since a parameter-dependent factor common to every training path cancels in it. For a latent log path y through the observed cells and a set I of steps, let $B_I(y) = \sum_{j\in I} \Delta_j (y_j - y_{j-1})/v_j$ and $C_I(y) = \sum_{j\in I} (y_j - y_{j-1})^2/v_j$; the Gaussian path log density depends on the parameters only through these and a parameter constant. For a subbox with centre (b_g, w_g), let c_1 and c_2 bound $|b/w - b_g/w_g|$ and $|1/w - 1/w_g|/2$ over the subbox, let $\Omega_{B,T}$ and $\Omega_{C,T}$ be explicit bounds on the oscillation of B and C over the training cells, and $d_{B,E}$, $d_{C,E}$ bounds on their deviation over the evaluation cells from the path through the cell midpoints, and let T_E be an explicit second-order bound on the

evaluation log density ratio at that midpoint path. Set $\eta_{\mathrm{cond}} = c_1(\Omega_{B,T} + d_{B,E}) + c_2(\Omega_{C,T} + d_{C,E}) + T_E$, and let D_{cond} be the certified bound of Proposition 66(d) with the allowance $\eta_N + \eta_k$ replaced by $\min(\eta_N + \eta_k, \eta_{\mathrm{cond}})$.

Proposition 67 (a conditional parameter bound). Under the contract of Proposition 66, on records with positive cells: (a) for every parameter in a subbox, $\log(g_\theta / g_{\theta_g}) \le \eta_{\mathrm{cond}}$, where θ_g is its centre with the same clock. (b) $D_{\mathrm{filter}} \le D_{\mathrm{cond}} \le D_{\mathrm{cert}} \le 1$, so q/D_{cond} is at least q/D_{cert} and has every conclusion of Proposition 66(b); the same holds for outward-rounded evaluations, for the minimum with the bound of Proposition 66(d), and for covers and resolutions chosen from the data. Records with a zero quote keep the fallback one. (c) On a fixed record, η_{cond} is at most a constant times the subbox's largest radius, so D_{cond} converges to D_{filter} as the cover and the quadrature are refined.

Status. An application of Proposition 66, whose validity holds for any denominator between the filtered maximum and one; the false-discovery bound is conditional on the e-BH theorem. Machine-checked (Appendix A).

Proof. (a) The log ratio of path densities is a parameter constant plus $(b/w - b_g/w_g)B + (1/(2w_g) - 1/(2w))C$. Over the paths compatible with the training cells it varies by at most $c_1\Omega_{B,T} + c_2\Omega_{C,T}$ about its infimum m_T, and the evaluation part adds at most $T_E + c_1 d_{B,E} + c_2 d_{C,E}$. So the joint mass grows by at most $e^{m_T + \eta_{\mathrm{cond}}}$ and the training mass by at least e^{m_T}, and m_T cancels in the ratio. (b) Each subbox bound is at most that of Proposition 66(d). (c) Each term is linear or quadratic in the radii. □

Computing η_{cond} costs of order n per subbox and clock. In the design after Proposition 66, with a 41×41 cover and $M = 16$, the median allowance per subbox fell from 1.79 to 0.20, about nine times smaller. The certified median log evidence was -1.31 at 30 + 30 days, against -6.81 for the box bound of Proposition 63 on the same paths and -7.96 for the previous certificate. At 125 + 125 days it was -4.63, against -41.9. These are floating-point evaluations, not outward-rounded certificates. They support a substantial gain over Proposition 63 from both sides at 30 days, since the ideal filtered score is at least the certified one, but the remaining cost still grows with the length of the record. The contract is unchanged, and no coverage is added.

The integration term can be sharpened too. On a cell, the Gaussian exponent is linear in the displacement from the midpoint minus a nonnegative quadratic form, and averaging over a reflection about the midpoint cancels the linear term. For the midpoint quadrature of Proposition 66(c), with subinterval radii r_j (and $r_0 = 0$), a subbox with lower variance w_-, and the step bounds C_j of that quadrature (set $C_{J+1} = 0$ for a prefix of length J), let

$$\varepsilon_J^- = \sum_{j \le J} \frac{(r_j + r_{j-1})^2}{2 w_- v_j}, \qquad \varepsilon_J^+ = \sum_{i \le J} \frac{r_i^2 (C_i + C_{i+1})^2}{2},$$

and let D_{center} and $D_{\mathrm{center}}^{\mathrm{low}}$ be the bounds of Proposition 67(b) with these allowances, each intersected with that of Proposition 66(c), in place of the integration allowance.

Proposition 68 (centred integration). Under the contract of Proposition 66, on records with positive cells: (a) for every parameter in a subbox, $e^{-\varepsilon_J^-} Q_{\theta,J} \le L_{\theta,J} \le e^{\varepsilon_J^+} Q_{\theta,J}$, and both allowances are their values at $M = 1$ divided by M^2. (b) $0 < D_{\mathrm{center}}^{\mathrm{low}} \le D_{\mathrm{filter}} \le D_{\mathrm{center}} \le 1$, and at a common M, $D_{\mathrm{center}} \le D_{\mathrm{cond}} \le D_{\mathrm{cert}}$ and $D_{\mathrm{center}}^{\mathrm{low}}$ is at least the lower bound of Proposition 66(d); so q/D_{center} has every conclusion of Proposition 66(b), and so does its minimum with any earlier certified bound. (c) On a fixed record, as the cover is refined and every subbox's M grows, both bounds converge to D_{filter}.

Status. An application of Propositions 66 and 67 with the bound $\cosh x \le e^{x^2/2}$ from Mathlib; the false-discovery bound is conditional on the e-BH theorem. Machine-checked (Appendix A).

Proof. (a) A unitriangular change of variables writes the label mass as an integral of the Gaussian path density over the product of the observed log cells, which splits into boxes around midpoint paths. Reflecting each box about its midpoint preserves it and cancels the linear term, which gives the lower bound since $\cosh \geq 1$. For the upper bound, drop the quadratic form, factor the exponential over coordinates, and bound each factor by the cosh inequality; summation by parts gives the coefficients $C_i + C_{i+1}$. (b) and (c) follow from (a) as in Propositions 66 and 67. □

For a fixed record the integration error is of order M^{-2}, at a cost of order nM^2 per centre. In the cent-tick design after Proposition 66, at the true parameters the integration allowance fell from 0.53 to 0.0018 at 30 + 30 days, about three hundred times smaller, because cent cells are narrow against the daily spread. The certified median log evidence was -0.71, against an upper bound of -0.33 on the ideal filtered score, and -6.81 for the box bound of Proposition 63. At 125 + 125 days it was -1.50, against an upper bound of -0.75. These are floating-point evaluations, not outward-rounded certificates, and the remaining cost at 125 days comes mostly from the parameter allowance, which still grows with the record. On one simulated record of 1,999 hourly steps, about a year of hourly bars, the computed bounds enclosed the filtered maximum within 0.82 in log terms in about 12 seconds, before exhausting their work budget. That is one record, not a runtime guarantee.

The exact filtered likelihood of the observed tick cells therefore gives a valid split-likelihood score, under a declared whole-record Gaussian law with a known start and declared clocks. It removes the per-step penalty that made the envelopes powerless: on the records tested, the certified score came within about one unit of log evidence of the ideal. The cost is of order nM^2 per parameter centre, with an integration error of order M^{-2}, and certifying individual real records looks feasible. But no uniform runtime, no power comparison with the intraday simulations above, and no verified production evaluator is established, and the parameter allowance still grows with the length of the record.

The filtered likelihood also gives a score that can be monitored as the evaluation cells arrive. With the notation of Proposition 66, k training cells and n evaluation cells, let $G_{\theta,t} = L_{\theta,k+t}/L_{\theta,k}$ be the filtered conditional probability of the first t evaluation cells, $D_t^{\mathrm{seq}} = \max_\theta G_{\theta,t}$, and $d_t = \max_\theta G_{\theta,t}/G_{\theta,t-1}$. At each evaluation step choose a probability q_t on the next cell label that depends only on the cells seen so far, let Q_t be the product of these one-step forecasts at the observed labels, and set $E_t^{\mathrm{seq}} = Q_t/D_t^{\mathrm{seq}}$ and $E_t^{\mathrm{step}} = \prod_{i\leq t} q_i/d_i$, both equal to one at $t = 0$.

Proposition 69 (an anytime-valid filtered score). Suppose that, given the training cells, the law of the whole evaluation suffix is a single mixture of the whole-record laws of Proposition 66, with a mixing law that may depend on the training cells. (a) $G_{\theta,t} > 0$, the maxima are attained, $0 < D_t^{\mathrm{seq}} \leq 1$, and the scores are finite, nonnegative and adapted. (b) For every stopping time $\tau \leq n$, $E[E_\tau^{\mathrm{seq}} \mid \text{training}] \leq 1$, and the probability that $E_t^{\mathrm{seq}} \geq 1/\alpha$ for some $t \leq n$ is at most α. The same holds for any adapted score between zero and E^{seq}, in particular for a lower forecast divided by an adapted upper bound on D_t^{seq}, such as the certified bound of Proposition 66(d) at training length k and evaluation length t. (c) E^{step} is a test supermartingale, and $E_t^{\mathrm{step}} \leq E_t^{\mathrm{seq}}$ on every prefix. (d) Smoothing the observed label frequencies of the whole history towards a fixed positive probability, such as 2^{-z-1} on label z, gives an admissible forecast with positive mass on every label.

Status. An application of the product rule for conditional e-values ([33], Section 1.3), which is proved here from Mathlib, to the anytime form of the split-likelihood bound ([82], Section 8), with the filtered likelihood of Proposition 66; the false-discovery bound is conditional on the e-BH theorem. Machine-checked (Appendix A).

Proof. (b) Under the mixture, the probability of the observed first t evaluation cells is a mixture of $G_{\theta,t}$, hence at most D_t^{seq}; so E_t^{seq} is at most the ratio of the forecast to the actual conditional probability, whose

one-step factors have conditional mean at most one, and the product rule applies. (c) Each step factor q_i/d_i has conditional mean at most one, and $\prod_i d_i \geq D_t^{\mathrm{seq}}$. (d) is a direct computation. □

Parameter changes after training are not covered. The ratio of the maximized probability after t steps to the maximized probability after training is not such a process and must not be used, and certificates may not be revised with later cells. In a deliberately coarse check, with start 1, tick 0.05, ten weekly training and ten evaluation steps, a null mixed uniformly over a box of drifts and variances, a predictable plug-in forecast and 4,000 records, the monitored score reached 2 at some step in 0.7% of records, against 2.4% for the unlicensed ratio of maxima. The smaller test supermartingale reached 2 in 0.5% and lay below the monitored score on every prefix. The forecasts used there only show that admissible forecasts exist; one that smooths the observed label frequencies is always defined. No power, detection-delay or runtime result is claimed. Daily parameter changes, jumps and noise are not covered.

Monitoring can also be restarted at declared dates. Before the data, declare training lengths $k_1 < \cdots < k_J$ and weights $\lambda_r > 0$ with $\sum_r \lambda_r \leq 1$. At each anchor k_r, start a score of the form of Proposition 69: a predictable forecast from k_r on, which may forget the earlier record, divided by $\max_\theta L_{\theta,t}/L_{\theta,k_r}$, the filtered conditional probability given the whole observed prefix up to k_r; set this score to one before its anchor. Let the scan be $E_t = \max_r \lambda_r E_{r,t}$.

Proposition 70 (declared restarts). Under the law of Proposition 69 with training length k_1: (a) each restart denominator is an attained maximum in $(0, 1]$, equal to one at its anchor, and the anchor scores and the scan are nonnegative, finite and adapted, with $E_t \geq \lambda_r E_{r,t}$ for every r. (b) For every stopping time, the stopped scan has conditional mean at most $\sum_r \lambda_r$, and the probability that it ever reaches $1/\alpha$ is at most $\alpha \sum_r \lambda_r$. (c) The same holds with adapted lower forecasts and adapted upper bounds on the restart denominators.

Status. An application of Proposition 69, the union bound and finite-sum domination. Machine-checked (Appendix A).

Proof. Under the null, the probability of the observed cells up to t is at most the restart denominator times the probability of the cells up to the anchor. So each anchor score is at most the oracle product of Proposition 69 for the forecast that follows the actual conditional law before the anchor and the restart forecast after it. The weighted sum of these has mean at most $\sum_r \lambda_r$, and the crossing event of the maximum is the union of the anchors' crossings at thresholds $1/(\alpha\lambda_r)$. □

So each anchor pays an explicit $\log(1/\lambda_r)$, or $\log J$ with equal weights. A restart lets only the forecast forget, not the null: the law is the single mixture fixed after the first training length, and no parameter change is licensed. The weights are needed: for two independent fair bits X and Y, the scores $2 \cdot 1\{X = 1\}$ and $2 \cdot 1\{Y = 1\}$ have mean one, but their unweighted maximum reaches 2 with probability 3/4. Dates and weights must be declared, not chosen afterwards, and the running maximum is not itself an e-value for later use. In the coarse check after Proposition 69, extended to 24 labels with anchors at 10 and 16 and weights one half, the rate of reaching 2 was 0.97% for the first anchor alone, 0.27% for the weighted scan and 1.25% for the unweighted maximum, which is not licensed. No power against a late-starting bubble is claimed, nor any useful runtime or panel guarantee.

Restarts can also be launched adaptively, from one lifetime budget. A policy has finitely many slots; slot r launches at a stopping time σ_r with an allocation $\lambda_r \geq 0$ chosen from the cells seen up to and including σ_r, and $\sum_r \lambda_r \leq 1$ on every record. After its launch, slot r scores only later cells: a predictable forecast divided by $\max_\theta L_{\theta,t}/L_{\theta,\sigma_r}$, the filtered conditional probability given the whole prefix up to σ_r, giving a score $E_{r,t}$ equal to one at launch. The total evidence is $C_t = 1 - \sum_{\sigma_r \leq t} \lambda_r + \sum_{\sigma_r \leq t} \lambda_r E_{r,t}$, the unused capital plus the launched slots' allocations times their scores.

Proposition 71 (funded adaptive restarts). Under the law of Proposition 69: (a) C is finite, nonnegative and adapted, starts at one, and for every stopping time its stopped value has conditional mean at most one, and the probability that $C_t \geq 1/\alpha$ for some t is at most α. (b) Once slot r is launched, $C_t \geq \lambda_r E_{r,t}$, so it pays the realized $\log(1/\lambda_r)$; with fixed dates and weights, C_t is the mixture of the restart scores of Proposition 70 with the unused capital, and dominates their weighted maximum. (c) The same holds with adapted lower forecasts and adapted upper bounds on the denominators.

Status. An application of the product rule for conditional e-values, proved from Mathlib, with the filtered likelihood of Proposition 66 and the restarts of Proposition 70. Machine-checked (Appendix A).

Proof. Under the null, the probability of the observed cells up to t is at most each launched slot's denominator times the probability up to its launch. So C_t is at most a weighted evidence divided by the actual probability of the cells, and summing that weighted evidence over the next cell leaves it unchanged: launched slots multiply by a normalized forecast, and new launches take exactly the capital they leave, at score one. The product rule then gives (a); (b) and (c) are direct. □

Refilling capital is not allowed: restarting a valid test with fresh unit capital after each failure brings back the two-fair-bits failure of Proposition 70, with probability 3/4 of reaching 2. In the coarse check after Proposition 69, with a first slot at the start and a second launched when the first score fell below one half, with an allocation depending on the triggering cell, the funded total reached 2 in 0.75% of records, against 1.43% for free refills with fresh unit capital. The lifetime budget is binding, a launch never resets the null, and no power or detection delay is claimed.

Calibrated monitoring on rounded prices is thus possible. A predictable plug-in score over the current filtered null maximum is anytime-valid under one whole-record Gaussian law with declared clocks and box, and restarts at declared dates or at adaptive stopping times funded from one budget let the forecast forget early data at an explicit $\log(1/\lambda)$ threshold cost. Power is not established (Section 5.6). In an hourly feasibility pilot with 100 paths per law, 30 sessions and certified denominators at 20 checkpoints, the score never crossed $1/\alpha$ under the in-contract null (a 95% upper bound of 3.7% on the crossing rate at $\alpha = 0.05$). It detected no late-starting constant-elasticity bubble within 15 sessions of its onset, whether the forecast was refitted on the growing history or frozen. Daily switches, jumps and noise remain outside the null.

4. Conditional results under option-pricing conventions

The results of this section use option prices or quantities that only option prices would supply, and each rests on a convention about how option quotes relate to the bubble. Two conventions appear. Under the first, option quotes are European expectation prices under Q, and the bubble appears as the residual of put-call parity (Section 4.1). Under the second, call prices carry the bubble: a call at any strike is worth at least the bubble, $\beta \leq C(K)$, in the form used here through the American exercise value (Section 4.3). Neither convention is derived from how options are priced. The second is chosen to restore put-call parity when the underlying is a strict local martingale ([18]), and under it the parity residual of the first vanishes. It also has a consequence that makes it circular as evidence: far out-of-the-money calls are quoted at the one-cent tick routinely, in recent index option chains and, by practitioners' accounts that we have not verified, during 2000, so under this convention the episodes the literature calls bubbles had essentially no bubble. Moreover, dealers price far strikes with models that have no bubble, so quotes there do not measure one under either convention. The results below are therefore conditional theory. Each states the convention it uses, and none is offered as a usable test.

4.1 Options on funds and on the underlying

Options on funds give a test of the underlying's bubble, under stated conditions. Under the convention that option quotes are European expectation prices, a call and a put on a fund at strike K are $E_Q[(W_T-K)^+]$ and $E_Q[(K-W_T)^+]$; for strict local martingales this convention matters (Section 4 introduction and Section 7). Put-call parity then gives the fund's bubble size $d = 1 - E_Q[W_T] = 1 - K - C + P$, so intervals $[C_{\mathrm{lo}}, C_{\mathrm{hi}}]$ and $[P_{\mathrm{lo}}, P_{\mathrm{hi}}]$ for the two prices give $L = 1 - K - C_{\mathrm{hi}} + P_{\mathrm{lo}} \le d \le U = 1 - K - C_{\mathrm{lo}} + P_{\mathrm{hi}}$. For funds j over one underlying, with exposures $0 < \pi_j \le 1$ on grids with N_j intervals, let $a_j = \pi_j^{N_j}$, and let B_{out} be the set of $\delta \in [0,1]$ for which there are $z_j \in [L_j, U_j] \cap [0,1]$ with $a_j\delta \le z_j$ and $z_j = 0$ exactly when $\delta = 0$.

Proposition 72 (an option-interval test from funds). Under the convention that option quotes are European expectation prices: (a) if every price interval contains its expectation price, $\delta_T = 1 - E_Q[X_T] \in B_{\mathrm{out}}$. With $b_j = \min(1, U_j)$, the set contains 0 exactly when $L_j \le 0 \le U_j$ for all j; its positive part is empty if some $b_j \le 0$ or $L_j > b_j$, and otherwise equals $(0, \min(1, \min_j b_j/a_j)]$. Zero is excluded exactly when some $L_j > 0$ or some $U_j < 0$. (b) If the interval endpoints are measurable on an observation space and cover all their prices simultaneously with probability at least $1-\alpha$, then $\delta_T \in B_{\mathrm{out}}$ with that probability, and the tests rejecting when $0 \notin B_{\mathrm{out}}$, or when some $L_j > 0$, have probability at most α of rejecting when $\delta_T = 0$. (c) If the endpoints converge in probability to their prices, both tests reject with probability tending to one at every $\delta_T > 0$. (d) With costs and tracking, $Y^j = W_T^j - c^j + e^j$, $E_Q c^j = d_{T,j}$ and $E_Q|e^j| \le \varepsilon_j$, parts (a) and (b) hold with $[L_j - d_{T,j} - \varepsilon_j, U_j - d_{T,j} + \varepsilon_j]$ in place of $[L_j, U_j]$, and (c) holds if some fund has $d_j > 2\varepsilon_j$.

Status. Conditional on the inputs of Proposition 12 and, for (b) and (c), on the stated coverage and convergence of the observations; the call-put identity is proved algebra. Machine-checked; the set computation in (a) is a separate unconditional statement (Appendix A).

Proof. Proposition 12 gives $0 \le d_j \le 1$, $a_j\delta_T \le d_j$ and $d_j = 0 \iff \delta_T = 0$, so $z_j = d_j$ witnesses $\delta_T \in B_{\mathrm{out}}$, and eliminating the z_j gives the explicit set. On the coverage event every $L_j \le 0$ when $\delta_T = 0$. Convergent endpoints make $L_j \to d_j > 0$. The cost correction shifts each residual by $d_{T,j}$ and at most ε_j. □

Two kinds of rejection must be kept apart: some $L_j > 0$ is evidence of a positive bubble, while some $U_j < 0$, or an empty set, means the observations are incompatible with the model and is not evidence of a bubble. Joint coverage of exact expectation prices is an assumption, not something market quotes provide: a bid-ask spread is not such a coverage interval. Proposition 72's practical reach is limited. The bound $a_j = \pi_j^{N_j}$ is useless at realistic N ($0.75^{52} \approx 3 \times 10^{-7}$ and $0.75^{252} \approx 3 \times 10^{-32}$), so in practice only zero exclusion is informative. For $\pi \le 1/2$ with zero inaccessible the fund's bubble size tends to zero as rebalancing becomes frequent (Propositions 13 and 14), so the test is weak for daily funds of low exposure. It covers only $0 < \pi \le 1$, not the leveraged and inverse funds traded on exchanges. And if exact options on the underlying itself are observed, they already identify δ_T by the same parity, and options on funds cannot improve on them.

The same parity residual brings in options on the liquidated leveraged funds of Proposition 15.

Proposition 73 (an option test from liquidated leveraged funds). Under the convention that option quotes are European expectation prices, take funds j as in Proposition 15 with $\pi_j > 1$, $M_j = \pi_j^{N_j}$, floors a_j and option intervals $[L_j, U_j]$ obtained by parity as in Proposition 72, and let $\ell_j = \max(0, L_j)$ and $u_j = \min(U_j, 1 - a_j)$. (a) If price containment holds, δ_T lies in B_{lev}, which is empty if some $\ell_j > u_j$ and otherwise equals $[\max_j \ell_j/M_j, 1]$. (b) Under simultaneous coverage at least $1-\alpha$, both zero-exclusion tests have size at most α; with converging endpoints they reject with probability tending to one at models where some $d_j > 0$. With costs and tracking, the intervals are widened as in Proposition 72(d).

Status. Conditional on the inputs of Proposition 15, with the call-put identity as proved algebra; the set computation in (a) is a separate unconditional statement. Machine-checked (Appendix A).

Proof. By Proposition 15, $0 \le d_j \le 1 - a_j$ and $d_j \le M_j \delta_T$; eliminating the auxiliary values $z_j \in [\ell_j, u_j]$ with $z_j \le M_j \delta$ gives B_{lev}, and coverage and convergence argue as in Proposition 72. □

The outer set mirrors the upper bound of Proposition 72 for $0 < \pi \le 1$, and power requires a positive fund defect, which a positive δ_T alone does not guarantee. Because M_j is enormous at realistic N_j, the lower bound is negligible and in practice only zero exclusion is informative.

What do options on the underlying alone give, and what do fund options add? Under the convention that option quotes are European expectation prices, suppose only calls on the underlying are observed, at strikes $0 < K_1 < \cdots < K_r$, with intervals $[C_i^{\text{lo}}, C_i^{\text{hi}}]$ for their prices, and no put. Let I_{call} be the set of values $1 - \int x\, \mu(dx)$ over terminal laws μ on $[0, \infty)$ with mean at most one and calls in the intervals, and B_{call} the set of $1 - c_0$ over vectors $(c_0, c_1, \dots, c_r)$ with $0 \le c_0 \le 1$, c_i in its interval and nonnegative, and secant slopes $(c_i - c_{i-1})/(K_i - K_{i-1})$, with $K_0 = 0$, increasing within $[-1, 0]$; c_0 stands for the unobserved $E_Q[X_T]$.

Proposition 74 (calls on the underlying, and what fund options add). (a) $I_{\text{call}} \subseteq B_{\text{call}} \subseteq [0, 1]$, and B_{call} is empty or a closed interval, computable from two linear programs or in closed form for exact prices. It need not be sharp: for $K = (11/10, 7/5)$ and both calls equal to $1/10$, $1/2 \in B_{\text{call}}$, yet no terminal law has those calls. (b) If $K_1 \ge 1$, every nonempty I_{call} and every nonempty B_{call} contains 0; an in-the-money call can exclude it ($K = 1/4$, $c = 1/4$ gives $B_{\text{call}} = [1/2, 3/4]$). (c) With the liquidated leveraged funds of Proposition 73, the combined set is $B_{\text{call}} \cap [\max_j \ell_j/M_j, 1]$, empty if some $\ell_j > u_j$; when $K_1 \ge 1$, a nonempty combined set excludes zero exactly when some $L_j > 0$. (d) A quote at a tick floor h, under the rule that the quote is $\max(h, c)$, means $c \in [0, h]$; widening intervals only enlarges the sets; under joint coverage of the calls and the fund residuals with probability at least $1 - \alpha$, both sets cover δ_T with that probability and the rule rejecting when a set is nonempty and excludes zero has size at most α.

Status. Parts (a)–(d) are finite inequalities, integrals of explicit payoffs over terminal laws and one event inclusion, and are unconditional; their application to δ_T is conditional on the supermartingale property, and the fund part on the inputs of Proposition 73. Machine-checked (Appendix A).

Proof. (a) The secant slopes of $K \mapsto E(x - K)^+$ lie in $[-1, 0]$ and increase, and these facts integrate over μ; the feasible set is a compact convex set, so its image is a closed interval. (b) For a feasible law of mean $m < 1$, mixing it with the law of $\max(x, K_1)$, with weight $(1 - m)/(K_1 + c_1 - m)$, keeps every observed call and has mean one. (c) is the elimination of Proposition 73. (d) follows by inverting the quote rule and by event inclusion. □

A panel of out-of-the-money calls therefore only bounds δ_T from above; a design must specify its lowest strike, not only its highest. Adding liquidated leveraged funds can exclude zero only through a positive fund residual, its upper endpoint is unchanged, and the enormous M_j make its lower bound negligible. An exact underlying put already identifies δ_T by parity, so a gain over calls alone is not a gain specific to leverage. Bid-ask intervals do not supply joint coverage. A sharp set would need a characterization of which call prices are consistent, which is not used here. No gain from fund options over underlying options is shown for any attainable model, and nothing here justifies acquiring fund-option data.

4.2 Baskets: additivity, survival-weighted hazards and killed weights

The results of this subsection use no options themselves; they are here because applying them to single names needs an option-based input, the index's pricing law in Proposition 77 or a budget from Proposition 80. Take finitely many names with fixed holdings $n_i > 0$, strictly positive continuous Q-local martingale prices S_i started at s_i, the basket $I_t = \sum_i n_i S_{i,t}$, initial weights $\omega_i = n_i s_i / I_0$, and bubbles $\beta_i(t) = s_i - E_Q[S_{i,t}]$

and $\beta_I(t) = I_0 - E_Q[I_t]$ at zero rates. At a fixed t let $q_{i,t}(A) = E_Q[(S_{i,t}/s_i)1_A]$ and $q_{I,t}(A) = E_Q[(I_t/I_0)1_A]$ on $\mathscr{F}_t$, with masses $p_i(t)$ and $p_I(t)$.

Proposition 75 (basket additivity and survival-weighted mixtures). For every t, $\beta_I(t) = \sum_i n_i\beta_i(t)$ and $0 \le n_i\beta_i(t) \le \beta_I(t) < I_0$; $q_{I,t} = \sum_i \omega_i q_{i,t}$ as finite measures, with $p_i(t), p_I(t) \in (0,1]$, $p_I = \sum_i \omega_i p_i$ and $1 - p_I(t) = \beta_I(t)/I_0$. If the p_i are absolutely continuous on $[0,T]$, with hazards $h_i = -p_i'/p_i$, then so is p_I, and $h_I = \sum_i(\omega_i p_i/p_I)\,h_i$ almost everywhere.

Status. Conditional on the supermartingale property (for the bounds); machine-checked (Appendix A). Additivity is a corollary of the definition of the fundamental value as an expectation under one pricing measure ([44], Definition 3), not a new pricing principle.

Proof. Linearity of expectation and $I_t/I_0 = \sum_i \omega_i S_{i,t}/s_i$ give the first two statements, and the supermartingale property gives the bounds; differentiating $p_I = \sum_i \omega_i p_i$ and dividing by $p_I > 0$ gives the hazards. □

A name with a bubble at t has mass below one, so the $q_{i,t}$ are not probabilities, and a global mixture of the names' share probabilities does not follow. Using the share probability of a positive local martingale, defined only up to its explosion time τ and satisfying $\tilde{Q}(A \cap \{t < \tau\}) = E_Q[(X_t/X_0)1_A]$ ([47], Theorem 2.12, on an uncompleted filtration), the identity says exactly that the index share probability is the ω-mixture of the names' share probabilities on survival events. The hazards mix with the weights $\omega_i p_i(t)/p_I(t)$, not with the initial weights ω_i; the difference grows with t when one name's bubble grows.

Proposition 76 (killed weights carry the bubble; ordinary weights cannot). Let $\tilde{Q}_I$ be a share probability of the normalized index, with lifetime τ^I, and $\mu_i^\dagger(t) = \mu_i(t)1_{\{t<\tau^I\}}$ the cap weight of name i killed at the index's explosion. Under $\tilde{Q}_I$, each $\mu_i^\dagger$ is a bounded nonnegative supermartingale started at ω_i, $\sum_i \mu_i^\dagger(t) = 1_{\{t<\tau^I\}}$ almost surely, $\tilde{Q}_I(\tau^I \le t) = \beta_I(t)/I_0$, and

$$\omega_i - E_{\tilde{Q}_I}[\mu_i^\dagger(t)] = \frac{n_i\beta_i(t)}{I_0},$$

so $\mu_i^\dagger$ is a martingale on $[0,T]$ exactly when $\beta_i(T) = 0$. In contrast, weights that are supermartingales under one probability and sum to one at each time are all martingales; and two names with proportional strict prices have ordinary weights $1/2$, with no expected loss under any probability, while their killed weights lose $(1 - E_Q[S_T]/s_0)/2 > 0$.

Status. Conditional on the supermartingale property and the existence of the share probability ([47], Theorem 2.12); machine-checked (Appendix A). The example's strict name is taken from the setting of Theorem 22.

Proof. The defining identity of the share probability is an equality of finite measures on $\mathscr{F}_t$; integrating the bounded weight against it gives $E_{\tilde{Q}_I}[\mu_i(t)1_A 1_{\{t<\tau^I\}}] = E_Q[(n_i S_{i,t}/I_0)1_A]$, whence the expectation and loss, and the supermartingale inequality transfers from that of S_i under Q. The null event of a nonpositive price is also null on survival, which gives the sum. For the contrast, nonnegative conditional losses summing to zero all vanish. □

So the bubble of name i, in index units, is the expected loss of its killed weight under the index share probability, and the hazard weights of Proposition 75 are the expected killed weights given survival. The natural statement that each name's bubble is the expected loss of its ordinary cap weight is false. A decomposition of bubble shares through ordinary weights, for example $\mu_{i,0} - E_{\tilde{Q}_I}[E_P(\mu_{i,T} \mid I_T)]$, therefore needs restating with killed weights, or with explicit conditioning on survival, before it can hold.

Recovering single-name bubbles from the index. The restated form holds under a strong identifying restriction. Fix the maturity T and a physical measure P equivalent to Q on $\mathscr{F}_T$. Let ν_T be the pricing

law of I_T, m_i Borel functions with $m_i(I_T) = E_P[\mu_i(T) \mid I_T]$, the physical conditional mean weights, $p_I(T) = E_Q[I_T]/I_0$, $\eta_T(dx) = x\,\nu_T(dx)/I_0$ the size-biased law of the index, of total mass $p_I(T)$, and R_T the index share probability restricted to $\{T < \tau^I\}$ and normalized by $p_I(T)$.

Proposition 77 (single-name bubbles from the index law). With ν_T the pricing law of I_T, which under the convention that option quotes are European expectation prices index options at all strikes would determine, suppose $dQ/dP = k_T(I_T)$ on $\mathscr{F}_T$ for a Borel k_T. (a) For every bounded Borel function h of the weight vector, $E_Q[h(\mu(T)) \mid I_T] = E_{R_T}[h(\mu(T)) \mid I_T] = E_P[h(\mu(T)) \mid I_T]$. (b) For every name,

$$\frac{n_i\beta_i(T)}{I_0} = \omega_i - \int m_i\,d\eta_T = \omega_i - \frac{1}{I_0}\int x\,m_i(x)\,\nu_T(dx) = \omega_i - p_I(T)\,E_{R_T}[m_i(I_T)],$$

and the right side depends only on the holdings, the starting prices, m_i and ν_T. (c) The weaker restriction $E_Q[\mu_i(T) \mid I_T] = m_i(I_T)$ already gives (b) and $0 \le \int x\,m_i\,d\nu_T \le n_i s_i$; and $\sum_i \int x\,m_i\,d\nu_T = \int x\,d\nu_T$ holds from the definition of m_i alone. (d) In the example of two proportional names with $P = Q$, the expression without the survival factor, $\omega_i - E_{\tilde{Q}_I}[m_i(I_T)]$, is zero, while each name's bubble in index units, and the right side of (b), is $(1 - p_I(T))/2 > 0$.

Status. Conditional on the supermartingale property and the existence of the share probability, with the setting and narrowings of Propositions 75 and 76. Machine-checked (Appendix A).

Proof. Under the restriction, R_T has a density with respect to P that is a function of I_T, $k_T(I_T)I_T/(I_0 p_I(T))$, so conditioning on I_T is unchanged by the change of measure, which is (a). Proposition 76 gives $\omega_i - E_{\tilde{Q}_I}[\mu_i^\dagger(T)] = n_i\beta_i(T)/I_0$, and the survival identity of the share probability turns $E_{\tilde{Q}_I}[\mu_i^\dagger(T)]$ into $p_I(T)E_{R_T}[\mu_i(T)] = \int m_i\,d\eta_T$, which is (b). Only the conditional mean is used in that step, which gives (c), and the weights sum to one. The example is that of Proposition 76 with $k_T \equiv 1$ and $m_i \equiv 1/2$. □

So among models satisfying the restriction, the physical joint law of the index and the weights, together with ν_T, determines each name's bubble, and the survival factor cannot be dropped. Its limits should be stated plainly. The restriction is a terminal density that depends on the index level only, much stronger than saying the pricing kernel depends on the index, and it is not shown to be testable. The recovered contributions always sum to the index bubble, because the weights sum to one, so that sum is not a check of the restriction; the bounds in (c) are a necessary check. And ν_T and the physical joint law of the index and the weights are idealized population inputs: ν_T is what expectation-priced index options would determine at all strikes, and it is not estimated here; Proposition 78 estimates the contributions from samples of the physical law only when ν_T is supplied.

The recovered contributions of Proposition 77 can be estimated, and their necessary inequalities tested, in a synthetic sampling experiment. In the setting of Proposition 77, let $m_i(x) = E[\mu_i(T) \mid I_T = x]$ under the physical law, chosen simplex-valued, let ν_T be equivalent to the physical index law with $p_I \le 1$, and put $\theta_i = \int m_i\,d\eta_T$; under the restriction of Proposition 77, $n_i\beta_i(T)/I_0 = \omega_i - \theta_i$ and $0 \le \theta_i \le \omega_i$. Observe N independent draws of $(I_T, \mu(T))$ from the physical law and, independently, M draws X_l from ν_T, which is supplied: an experiment with repeated terminal panels, not overlapping windows of one historical path, and with no estimate of ν_T from option quotes assumed valid. For a regression $\hat{m}_i$ fitted on the physical draws, measurable and simplex-valued, set $\hat{\theta}_i = (MI_0)^{-1}\sum_l X_l\hat{m}_i(X_l)$. For the histogram regression on a fixed range $B = [a, R]$, with J cells of diameter at most h, counts N_j, the value ω on empty cells and outside B, and a supplied Lipschitz bound L on the m_i over B, let $e_N(\delta) = \min(1, Lh + \max_j \sqrt{\log(2dJ/\delta)/(2N_j)})$, or 1 if a cell is empty, and $\text{lower}_i = \hat{\theta}_{i,B} - e_N(\alpha_{\text{reg}}) - (R/I_0)\sqrt{\log(2d/\alpha_{\text{price}})/(2M)}$, where $\hat{\theta}_{i,B}$ is the plug-in restricted to $X_l \in B$.

Proposition 78 (estimating the recovered contributions). (a) The estimated contributions $\omega_i - \hat{\theta}_i$ sum to $1-(MI_0)^{-1}\sum_l X_l$ on every sample. If the fitted regression converges to m_i in L^1 of the physical index law, then $\hat{\theta}_i \to \theta_i$ in probability as $N, M \to \infty$, with no second moment of ν_T needed. (b) With probability at least $1-\delta$, the histogram is within $e_N(\delta)$ of every m_i on B, simultaneously, with no independence across names; with ranges, cells and Lipschitz bounds chosen so that $Lh \to 0$ and the smallest cell count grows faster than $\log(2dJ/\delta)$, the histogram is L^1-consistent. (c) At every law with $\theta_i \le \omega_i$ for all i, the rule that rejects when some $\text{lower}_i > \omega_i$ has size at most $\alpha_{\text{reg}} + \alpha_{\text{price}}$, and so does any rule using a value below lower_i. (d) Against a fixed law with some $\theta_i > \omega_i$, some compact range captures the excess, and on it the rule rejects with probability tending to one as the cells shrink and the samples grow. Such laws exist with $\sum_i \theta_i = 1$: tilting a compatible model whose constituents are true martingales by $1 + \varepsilon\mu_1(T)$ raises θ_1 above ω_1 whenever the conditional variance of $\mu_1(T)$ given the index is not zero under η_T.

Status. Proved from Mathlib's sub-Gaussian concentration for bounded independent variables, the strong law of large numbers, conditional expectation, Fubini and Radon–Nikodym; the reading of θ as a bubble contribution is Proposition 77's and carries its conditions. Machine-checked (Appendix A).

Proof. (a) The sum identity is finite summation of a simplex-valued fit. The error splits into the regression error integrated against η_T, small because η_T is absolutely continuous with respect to the physical index law, and the pricing-sample average, which converges by the strong law. (b) Given which draws fall in a cell, they are independent with the conditional law there, so Hoeffding's inequality bounds each cell mean, and a union bound over d names and J cells gives the radius; the Lipschitz bound handles the cell's width. (c) On the joint event, $\text{lower}_i \le \int_B m_i \, d\eta_T \le \theta_i$, since the omitted tail is nonnegative. (d) Monotone convergence captures the excess, and both radii tend to zero. The tilt is a direct computation of the conditional mean. □

Non-rejection is not acceptance of the restriction of Proposition 77, nor evidence that a pricing model exists, and the test targets only violations of single coordinates. In a simulation with two names, a log-normal pricing law and a compatible null exactly at the boundary $\theta_1 = \omega_1 = 0.5$, the test with 20 cells on $[0.6, 1.7]$ made no null rejection in 450 tests, and rejected a tilted alternative with $\theta_1 = 0.65$ in 0 of 200, 134 of 200 and 50 of 50 runs at $N = M = 10^4$, 10^5 and 10^6. The radius uses the sparsest cell everywhere, so a wider range with nearly empty edge cells gave no power even at 10^6; per-cell radii integrated against the pricing sample are also valid and reached full power, 200 of 200, at 10^5.

4.3 Upper bounds on bubbles from American calls

Option prices can bound a bubble from above. Let d_t be a deterministic, continuous, positive discount factor with $d_0 = 1$, $\bar{S}_t = d_t S_t$ the discounted ex-dividend price, and D_t the cumulative discounted distributions, adapted, nonnegative and nondecreasing with $D_0 = 0$. Suppose the discounted gain $G = \bar{S} + D$ is a continuous nonnegative Q-local martingale started at $g_0 > 0$, with bubble $\beta_G(T) = g_0 - E_Q[G_T]$. Let $C_E(K,T) = E_Q[(\bar{S}_T - d_T K)^+]$ be the European expectation call and $A_Q(K,T)$ the supremum of $E_Q[(\bar{S}_\tau - d_\tau K)^+]$ over all stopping times $\tau \le T$, the American call with continuous exercise.

Proposition 79 (American calls bound the bubble). (a) $0 \le \beta_G(T) \le \beta_G(T) + C_E(K,T) \le A_Q(K,T)$, dividends included; both call values decrease in K, so any finite $U \ge A_Q(K,T)$, at any strike, bounds $\beta_G(T)$. (b) Neither the European call nor exercise on a fixed finite set of dates suffices: $C_E(K,T) \to 0$ as $K \to \infty$, so $C_E < \beta_G$ at large strikes when $\beta_G > 0$, and the supremum over stopping times with values in a finite set also tends to zero.

Status. Conditional on bounded stopping, the supermartingale property and the stopping-time property of exit times; machine-checked (Appendix A).

Proof. (a) Since $(\bar{S} - dK)^+ = G - \min(G, D + dK)$, exercising at κ_n, the first time G exceeds n capped at T, is worth at least $g_0 - E_Q[\min(G_{\kappa_n}, D_T + d_{\kappa_n}K)]$, because G_{κ_n} has mean g_0 by bounded stopping and D is nondecreasing. As $n \to \infty$, $\kappa_n = T$ eventually on each path, and dominated convergence, with dominator $D_T + K \max_{[0,T]} d$, gives the limit $\beta_G + C_E$. (b) The payoffs are bounded by G_T, or by a finite sum of values of G, and tend to zero pointwise. □

For a fixed basket with holdings n_i, initial prices s_i, gains $G_i = \bar{S}_i + D_i$ of the same kind, $I_0 = \sum_i n_i s_i$, $\omega_i = n_i s_i / I_0$ and bubbles $\beta_i = s_i - E_Q[G_{i,T}]$, let the option be on a fund Y that tracks the basket, with scale a ($I_0 = aY_0$) and a bound $E_Q|G_{B,T} - aG_{Y,T}| \le \eta$ on the gap between the basket's gain and a times the fund's.

Proposition 80 (a tolerance certificate for every name). Under the convention that call prices carry the bubble, in the form that a quoted call price U bounds the American exercise value, $A_Q \le U$ for the fund's American call, then with $B_{\text{cap}} = aU + \eta$, simultaneously for every name,

$$0 \le n_i\beta_i \le \beta_B \le B_{\text{cap}}, \qquad \frac{\beta_i}{s_i} \le \frac{B_{\text{cap}}}{\omega_i I_0},$$

and the same bounds hold at every horizon $T' \le T$. If $B_{\text{cap}} = 0$, every β_i is zero; if $B_{\text{cap}} > 0$, a positive and arbitrarily small bubble is compatible with the bound.

Status. Conditional as Proposition 79; the bounds U and η are premises. Machine-checked (Appendix A).

Proof. Additivity as in Proposition 75 gives $\beta_B = \sum_i n_i\beta_i$ with nonnegative terms, and $\beta_B = a\beta_{G_Y} + E_Q[aG_{Y,T} - G_{B,T}] \le aU + \eta$ by Proposition 79. Defects are nondecreasing in the horizon by the supermartingale property. For the last statement, a one-interval fund with small positive exposure over any strict underlying has a positive defect below any given tolerance (Proposition 8). □

The relation between American calls and bubbles is studied by Bayraktar, Kardaras and Xing ([14]); Cox and Hobson ([21]), which we have not read, may contain the inequality of Proposition 79(a), so we do not present it as new.

Four limits should be stated plainly. First, Proposition 80 is a certificate up to a monetary tolerance, not exact ground truth: with $U > 0$ it bounds β_i above, so it contradicts a declaration $\beta_i > \varepsilon_i s_i$ when the bound is below $\varepsilon_i s_i$, but it cannot contradict a bare $\beta_i > 0$, and counting such a flag as a false alarm of an exact test is not justified. Second, the bound needs the American value with continuous exercise (Proposition 79(b)). Third, U must be an upper bound on that value; a bid, a midpoint or a quote at the tick does not establish it. Fourth, η must bound the gap under Q, including fees, rebalancing and premiums or discounts, with the holdings frozen over the horizon; physical tracking data do not supply it. In practice the tracking term dominates. For illustration, a far-strike call at the one-cent tick on a fund trading near 450 gives $U/Y_0 \approx 2 \times 10^{-5}$, while an expense ratio alone makes η/I_0 of order 10^{-3} a year; the resulting relative tolerance is about 0.013 for a name of weight 7%, but of order 10, and so meaningless, for a name of weight 10^{-4}. The certificate is informative only for the largest names, and only once both bounds are actually supplied.

Which names a basket budget can single out. Combine intervals $\beta_i \in [\ell_i, u_i]$ for the names with a budget $\beta_B = \sum_i n_i\beta_i \in [B_-, B_+]$, put $l_i = \max(0, \ell_i)$ and $v_i = \min(s_i, u_i)$, and let D be the set of vectors b with $l_i \le b_i \le v_i$ for all i and $B_- \le \sum_i n_i b_i \le B_+$.

Proposition 81 (dispersion sets from a basket budget). For any supplied intervals, and in particular for an upper budget from Proposition 80 under the convention that call prices carry the bubble: (a) D is nonempty exactly when $l_i \le v_i$ for all i, $B_- \le B_+$, $\sum_i n_i l_i \le B_+$ and $B_- \le \sum_i n_i v_i$, and then its projection on name i is exactly

$$\Big[\max\Big(l_i, \frac{B_- - \sum_{j\neq i} n_j v_j}{n_i}\Big),\ \min\Big(v_i, \frac{B_+ - \sum_{j\neq i} n_j l_j}{n_i}\Big)\Big],$$

with both endpoints attained; the projections need not form a product set. (b) With $\ell_i = 0$, $u_i = s_i$, $B_- = 0$ and $B_+ \geq 0$, name i's projection is $[0, \min(s_i, B_+/n_i)]$, so every name has a compatible allocation with zero bubble. In general zero is excluded for name i exactly when $l_i > 0$ or $B_- > \sum_{j\neq i} n_j v_j$. (c) If the intervals and the budget contain their targets jointly with probability at least $1 - \alpha$, then D and all its projections cover the true bubbles simultaneously with that probability, and the set of names whose projection excludes zero contains a name without a bubble with probability at most α, without independence.

Status. Parts (a)–(c) are finite algebra and one event inclusion, unconditional except that (c) assumes the stated coverage; their application to bubbles, $0 \leq \beta_i \leq s_i$ and the additivity of Proposition 75, is conditional on the supermartingale property. Machine-checked (Appendix A).

Proof. D is a box cut by a slab in the weighted sum, which is nonempty exactly when the slab meets the range of the sum over the box; fixing b_i and filling the other coordinates at their bounds gives the extreme values of b_i. On the coverage event the true vector lies in D, and a name without a bubble then has lower endpoint at most zero. □

So an upper budget on the basket's bubble never shows that any particular name has a bubble, whatever the budget, including that of Proposition 80. Excluding zero for a name needs a positive monetary lower bound, and the identification results of Sections 3.4 and 3.5 supply none, since they concern tail integrals and labels, not amounts. In the two-name example with $n = s = (1, 1)$ and $B_+ = 1$, each projection is $[0, 1]$ but $(1, 1)$ is not compatible. Marginal coverage of each interval does not give (c), and a budget taken from Proposition 80 carries all of that proposition's premises.

5. Simulation evidence

The results of Section 3.5 are guarantees under stated null laws. They say nothing about power, and little about what happens when the truth lies outside the null. This section reports simulations in which the answer is known. Every design was fixed before it was run, and each experiment's figures were reproduced, within Monte Carlo error, by an independent implementation. Power is pointwise, at the stated alternatives: by Theorems 18 and 22, power taken uniformly over a broad null class is at most size, so no figure here is a uniform statement. Rates are proportions of simulated paths or panels. The Monte Carlo standard error of a proportion over n paths is at most $1/(2\sqrt{n})$, and the bound is given with each table. Unless stated otherwise, prices start at 1, a year has 252 trading days, a score rejects when it exceeds $1/\alpha = 20$, and the forecast is the constant-elasticity likelihood fitted on the first half of the record, from the 18 candidates $dS = \kappa S^p\, dW$ with $p \in \{0.5, 0.75, 1, 1.25, 1.5, 2\}$ and $\kappa \in \{0.2, 0.3, 0.45\}$. We write E for the catalogue score of Proposition 29 and E^{drift} for the score of Proposition 30 with its continuous Gaussian null, and call E^{drift} with the e-BH rule of Proposition 40 the drift-robust test.

5.1 The four existing tests

We compare with four published procedures. The SADF and GSADF tests of Phillips, Shi and Yu [66] test for an explosive autoregressive root. The volatility-tail test of Jarrow, Kchia and Protter (JKP) [41] estimates σ on the visited range and extrapolates its tail, by a parametric fit (Method 1) or a reproducing-kernel fit (Method 2). The panel t and J tests of Liu, Phillips and Yu [53] test for explosive roots group by group after clustering. Obayashi, Protter and Yang [63] date bubbles by smoothing a daily JKP signal from

five-minute prices with a two-state hidden Markov model. An explosive root is a different notion of bubble from a strict local martingale, so the explosive-root tests' rates are set beside the others for comparison only.

Each implementation was first checked against its authors' published design.

- **SADF and GSADF.** With 10,000 replications per cell, all 48 published critical values and all 10 published sizes are reproduced within Monte Carlo tolerance. The published power under the Evans collapsing-bubble process is reproduced in 2 of 10 cells, with ours 1 to 8 points higher. The text does not fix the process completely: plausible readings of its innovation scale, burn-in and initial values move power by up to 60 points. GSADF has more power than SADF in every cell under every reading.
- **JKP.** The paper's tick data are licensed, so the method was run on 200 simulated paths per case with the paper's stated tails. Method 2 returns the paper's verdict in four of five cases. It recovers the stated exponents in median (3.58 for 3.58, and 4.64 for 4.63), with interquartile ranges as wide as 2.3 to 4.9. It calls a bubble on 42% of geometric Brownian paths and on 26.5% of bounded-volatility paths, both true martingales (standard error at most 0.035). Its interpolation step lowers the fitted exponent m by about 0.5 on a price range spanning a factor of 2, in the direction of "no bubble", and by more on wider ranges. Method 1, with a correctly specified power law, is never conclusive.
- **Panel tests.** With the t statistic as printed, 54 of 72 targets are reproduced. The printed statistic is not scale invariant, and over-rejects at 0.19 to 0.23. With the scale-invariant reading consistent with its long-run variance, all 72 are reproduced at 5,000 replications per cell. In the published design the clustering step separates the groups trivially, so it is not tested here.
- **The dating procedure.** The data are licensed, so the hidden-Markov emission probabilities and the lifetime distribution were re-estimated from data simulated at the paper's parameters (200 estimates). All five targets are recovered within tolerance, for example emission probabilities 0.89995 and 0.01002 against the published 0.90 and 0.01. The daily signal itself is JKP's.

Scope. These are reproductions in their authors' designs, or on data simulated at their stated parameters where the data are licensed. None reproduces a real-data figure.

5.2 Known-answer models

The first comparison uses driftless constant-elasticity prices $dS = \sigma S^{\beta}\, dW$, true martingales for $\beta \leq 1$ and strict local martingales for $\beta > 1$, and the stochastic-volatility model of Sin [75], a martingale or a strict local martingale according to the sign of its correlation parameter. The local volatility at the start is 0.3. The existing tests see daily closes over one or five years, or five-minute bars over the last month.

Truth (label)	**GSADF, daily, 1 year**	**GSADF, daily, 5 years**	**JKP Method 2, five-minute month**	**Dating procedure, 84 days**
constant elasticity, $\beta = 1$ (martingale)	0.073	0.187	0.600	0.487
constant elasticity, $\beta = 2$ (strict)	0.123	0.345	0.709	0.398
Sin's model, martingale	0.158	0.475	0.268	0.195
Sin's model, strict	0.295	0.708	0.772	0.292

Scope. Rejection rates at nominal 5% for GSADF; rates of a "bubble" verdict for JKP and of at least one dated bubble for the dating procedure, which have no nominal level. 1,000 paths per cell for the single-

series tests (standard error at most 0.016) and 400 for the dating procedure (at most 0.025). Exact paths; pointwise alternatives.

On the constant-elasticity models, no existing test rejects strict prices much more often than true martingales. GSADF holds its size over one year of daily data, but over five years it rejects 0.08 to 0.19 of constant-elasticity martingales and 0.48 of Sin's martingale: volatility that rises with the level makes a long random walk look explosive to a homoskedastic critical value, and its higher rates on strict models are mostly the same effect. On daily data JKP's Method 2 has one or two grid points and says "bubble" on 90 to 100% of paths in every model; on the five-minute month it says so on 55 to 60% of martingales. The panel tests, applied to price levels, reject 58 to 98% of null panels; applied to $S - S_0$, the t test has no power (at most 0.05 on strict panels) and the J test rejects 0.06 to 0.21 of null panels against 0.11 to 0.29 of strict ones. Under Sin's model JKP's Method 2 and the dating procedure do separate the two labels, though neither holds its size elsewhere.

The threshold test of Proposition 25(c), with the compact-interval estimator of Proposition 24 on two declared blocks of log price $[-0.15, 0.15]$, acts only on paths that spend enough time at every level of the blocks. On the same paths no constant-elasticity path did so over a year of daily data or a month of five-minute bars, and about 30% did over five years of daily data. There the rejection rate rises with β, from 0.081 ± 0.009 at $\beta = 1$ to 0.205 ± 0.013 at $\beta = 2$, but 27 to 29% of the null paths that pass the occupation check are rejected. The threshold test is not calibrated, and it rarely acts.

The catalogue scores are calibrated when the truth is in the catalogue, and their power is set by sampling frequency far more than by horizon. For the two-exponent catalogue of Proposition 28 ($p \in \{1, 2\}$, three scales), the largest null rejection rate over all horizons and frequencies is 0.001 at $\alpha = 0.05$, and the true parameter is missed at most 0.3% of the time. Against $p = 2$ its power is 0.00 to 0.03 with a year of monthly data and 0.14 to 0.21 with five years, 0.24 to 0.46 with a year of daily data and 0.96 to 0.98 with five, and 0.99 to 1.00 with five-minute data. The next table compares the three catalogue scores in and out of their catalogues.

Truth (label)	**Relation to the catalogue**	**Daily, 5 years**	**Five-minute, 1 year**
geometric Brownian, $\kappa = 0.3$ (martingale)	in	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
$p = 1.25$, $\kappa = 0.3$ (strict)	in	0.49 / 0.07 / 0.00	0.83 / 0.39 / 0.00
$p = 1.5$, $\kappa = 0.3$ (strict)	in	0.76 / 0.43 / 0.12	0.98 / 0.86 / 0.22
$p = 2$, $\kappa = 0.3$ (strict)	in	0.95 / 0.82 / 0.97	1.00 / 0.99 / 1.00
geometric Brownian, $\kappa = 0.35$ (martingale)	off the scale grid	0.09 / 0.00 / 0.01	0.24 / 0.00 / 0.27
Merton jumps (martingale)	outside	0.02 / 0.00 / 0.00	0.08 / 0.02 / 0.03
Sin's model (strict)	outside	0.30 / 0.20 / 0.28	0.42 / 0.19 / 0.42

Scope. Each cell gives the rejection rates of E / E^{drift} / Proposition 28's score at $\alpha = 0.05$, over 1,000 paths (standard error at most 0.016); exact observations, $P = Q$ except where labelled. Proposition 28's catalogue is $p \in \{1, 2\}$; "off the scale grid" is a volatility between the catalogue's scales; "outside" means outside every score's null.

Off the catalogue, the raw score E loses its calibration at high frequency, as Proposition 43 predicts: an off-grid geometric Brownian price is flagged 24% of the time with a year of five-minute data, and e-BH applied to panels of 20 such names makes a false discovery in essentially every panel (mean false-discovery proportion 0.98 to 1.00). The drift-robust score E^{drift} removes every one of these false alarms, and keeps

the false-discovery proportion at zero. Its cost is power on daily data: against $p = 1.25$ over five years it rejects 0.07 where E rejects 0.49. A physical drift of 0.05 a year produced no rejection under any score, Sin's martingale at most 0.01, and Merton jumps at most 0.03 under E^{drift}. The clipped scores of Proposition 43 never rejected: the computed envelopes are 0.34 to 0.68 on monthly data and about 1 from daily data on, so they are nearly vacuous, as that proposition warns.

Declaring the drift does not buy the power back; declaring the scale does, but fragilely. With the drift confined to $[-0.2, 0.2]$ (Proposition 31), the paired gain in daily power over E^{drift} is at most 0.015 ± 0.004 on any alternative, and even a driftless Gaussian null gains at most 0.048. The power is lost to the free scale. With the scale also confined, to $[0.20, 0.45]$ (Proposition 32), the paired gain reaches 0.40, and with the scale fixed at its true value 0.3 the score nearly matches E (0.761 against 0.772 for $p = 1.5$ over five years). But Gaussian nulls with a scale just outside the declared interval are then flagged at nearly the raw score's rates: 0.21 to 0.67 at $\kappa = 0.15$ for most drifts, and up to 0.41 at $\kappa = 0.60$, on daily data. A drift just outside its interval costs at most 0.001. The scale interval therefore buys power only if the volatility truly lies in it, which is what Proposition 33's declaration from an earlier window cannot promise when volatility changes between windows.

Scope. Exact observations of constant-elasticity and Gaussian laws, 1,000 paths per cell (standard error at most 0.016); the intervals are illustrative.

5.3 Misspecification and panels

The drift-robust test was then run on correlated panels with realistic nulls. Each panel has $m = 20$ or 100 names observed daily for 5 or 10 years. Null names load on one market factor (pairwise correlation 0.25) and are true martingales from four families: geometric Brownian motion with drift and volatility drawn from $[0.15, 0.60]$, inside the null of Proposition 30; Merton jumps, about one every one to two years; a two-regime volatility switching between 0.15 and 0.45 about once a year; and Heston stochastic volatility with leverage -0.7. Bubbly names, a fraction 0, 0.1 or 0.3 of the panel, are constant-elasticity prices with $p = 1.5$ or 2 and scales drawn from $[0.20, 0.45]$, so off the catalogue's scale grid.

Null family	Drift-robust test	E with e-BH	GSADF with BH	Panel t
geometric Brownian with drift	0.00 / 0.00	0.76 / 1.00	0.80 / 1.00	0.04 / 0.25
Merton jumps	0.02 / 0.17	0.43 / 1.00	0.81 / 1.00	0.08 / 0.38
two-regime volatility	0.50 / 0.88	0.71 / 0.95	0.99 / 1.00	0.15 / 0.15
Heston with leverage	0.01 / 0.03	0.02 / 0.20	0.67 / 0.97	0.20 / 0.75

Scope. Share of all-null panels with at least one discovery at $q = 0.05$ (for the panel t, rejection at 5%), for $m = 20$ over 5 years / $m = 100$ over 10 years. 100 panels per cell for $m = 20$ and 40 for $m = 100$ (standard errors at most 0.05 and 0.08). Daily exact prices; only the first family is inside the drift-robust test's null.

On null names inside its class the drift-robust test made no false discovery, in any panel, under the factor correlation. Outside its class the outcome depends on the misspecification. Heston volatility barely moves it: the all-null rate is at most 0.03, and the false-discovery rate on panels with bubbles at most 0.01. Poisson jumps give a modest excess that grows with panel size and horizon, to 0.17 at $m = 100$ over 10 years, with mean false-discovery rate 0.04 and maximum 0.12 on panels with bubbles. Repeated volatility regimes break it: 45 to 98% of all-null panels contain a false discovery, and the false-discovery rate on panels with bubbles averages 0.22 and reaches 0.54. This is the failure Proposition 47 predicts for volatility that changes within the evaluation window; the single-switch null of Proposition 48 does not cover it,

since the evaluation window here contains at least two switches with probability about 0.71 over 5 years and 0.96 over 10. The test finds a share of the bubbly names that grows with the exponent and the horizon: for $m = 100$ at $q = 0.10$, 0.12 for $p = 1.5$ and 0.49 for $p = 2$ over 5 years, and 0.32 to 0.42 and 0.70 over 10. The raw score E finds more (0.38 to 0.91) at a false-discovery rate of 0.16 to 0.56.

No existing test controls false discoveries on these nulls. With the Benjamini–Hochberg rule, SADF and GSADF flag 45 to 100% of all-null panels, geometric Brownian ones included, with mean false-discovery rates of 0.33 to 0.86; their per-name rates on null names (0.19 to 0.58) are about as high as on bubbly names (0.21 to 0.37). JKP's Method 2 flags about 97% of null names and 93% of bubbly ones, Method 1 2 to 10% of null names against 17% of bubbly ones, and the panel test rejects 4 to 75% of all-null panels. In this design the drift-robust test is the only procedure tested whose false discoveries stay controlled on realistic nulls; its weakness is repeated changes of volatility.

A single switch is mild, and the null that covers it costs most of the power. With one volatility switch in the evaluation window, the drift-robust score's false-alarm rate stayed at most 0.015, and the single-switch score of Proposition 48 held every null, including switches at dates it does not admit. Its power falls as the set of admitted switch dates grows: against $p = 1.5$, $\kappa = 0.3$ over five years of daily data it is 0.447 for E^{drift}, 0.173 with only the midpoint admitted, and 0.021 with the middle 80% of the window admitted; against $p = 2$, $\kappa = 0.45$ it falls from 0.911 to 0.451. A Gaussian law with one switch tracks the level-dependent volatility of a constant-elasticity path closely, so on daily data variation of volatility over time and its dependence on the price level are nearly confounded.

Scope. Exact prices; bubbly names are constant-elasticity and independent of the factor; parameter ranges are illustrative; 1,000 paths per single-switch cell (standard error at most 0.016).

5.4 Intraday clocks and rounding

Intraday volatility follows a daily pattern, high at the open and close, with an overnight gap. With 6.5-hour sessions, a U-shaped intraday clock and an overnight variance share, the declared-clock score of Proposition 34 was compared with the drift-robust score on calendar time, and with scores whose forecast follows the clock but whose Gaussian null does not. The clock-aware forecasts choose among the 18 constant-elasticity candidates on either clock shape.

Truth	Clock score, five-minute	Clock score, hourly	Drift-robust, calendar time, five-minute	Clock-aware forecast without the clock null, five-minute
Gaussian on the declared clock	0.000	0.000	0.000	0.000
Gaussian, stronger U-shape (undeclared)	0.000	0.000	0.001	0.325
Gaussian, overnight share 0.5 (undeclared)	0.000	0.000	0.000	0.263
$p = 1.5, \kappa = 0.3$	0.845	0.106	0.070	0.985
$p = 1.5, \kappa = 0.45$	0.963	0.296	0.002	0.995
$p = 2, \kappa = 0.3$	0.997	0.537	0.326	1.000
$p = 2, \kappa = 0.45$	0.998	0.771	0.003	0.999

Scope. Rejection rates at $\alpha = 0.05$ over 250 sessions (about a year), exact prices, constant-elasticity alternatives in clock time, no weekends or noise. 2,000 paths per null cell and 1,000 per alternative (standard errors at most 0.011 and 0.016). The undeclared clocks are outside every score's null.

A declared clock leaves useful power: a year of five-minute bars detects these alternatives 85 to 100% of the time, with no false alarm on any tested null; over 60 sessions the power is 0.03 to 0.65, and with hourly bars over a year 0.11 to 0.77. Ignoring the clock cost power rather than false alarms on these paths: the calendar-time score found at most a third of the alternatives. A forecast that follows the clock, without a null that does, false-alarms on 26 to 40% of undeclared-clock paths at five-minute sampling.

Real intraday prices are rounded to the cent. Applied to rounded prices as if they were exact, which no result here licenses (Proposition 55), the clock score raised no false alarm above 0.002 on any null and kept its power on a \$50 price; on a \$5 price its five-minute power over a year fell from 1.000 to 0.492 (paired difference -0.508 ± 0.016). The valid rounded-price score of Proposition 63 is far weaker. Its best hourly power over a year, with a declared volatility box $\kappa \in [0.20, 0.45]$ and the deeper refinement, is 0.297 against the exact-price clock score's 0.772 (paired difference -0.475 ± 0.027); with a wide box it stays at or below 0.10. A narrow box gives false alarms when the volatility lies outside it: a Gaussian null with $\kappa = 0.60$ was flagged on 18 to 24% of paths (600 paths; 95% Wilson interval $[0.21, 0.28]$ at one setting). The most powerful rounded score costs about 10 seconds per path for a year of hourly bars, against 0.06 seconds for the exact-price score.

A common clock can be learned from other names, but here that bought nothing. On daily panels of 20 names sharing one of 15 four-block clocks, a reference panel identified the true clock in every one of 200 panels per cell (95% upper bound 0.018 on a wrong exclusion), but excluding the other clocks never changed a null name's denominator, and the selected score of Proposition 36 found no more bubbles than the full clock catalogue (paired differences in true-discovery proportion between -0.005 ± 0.004 and 0.000 ± 0.004). The comparison does show the cost of a wrong clock: under a common non-calendar clock the drift-robust test on calendar time had a false-discovery rate of 0.345 on all-null panels, while the full clock catalogue stayed at or below 0.001 with true-discovery proportions of 0.53 to 0.83. With name-specific switches instead of a common clock, selection from the reference panel was harmful: false-discovery rate 0.255 ± 0.031 against 0.065 ± 0.017 for the full catalogue.

Resetting the nuisance parameters each day buys robustness at the cost of all the power at these horizons. On hourly bars over 20, 60 and 250 sessions, the daily-reset score of Proposition 37 rejected in no null or alternative cell. Its denominator exceeds the whole-record clock profile by about 1.1 to 1.2 nats per day, combining each day's drift, scale and clock shape. The whole-record clock profile has power over 250 sessions (0.32 to 0.77 against $\kappa = 0.45$ or $p = 2$) but false-alarms under volatility switches between days (0.061 ± 0.005) and within days (0.073 ± 0.006). Budgets on switches sit between the two, and in the following comparison they keep the robustness and lose the power.

Truth	**Common profile (Proposition 34)**	**Joint budget** $K = 1$	**Joint budget** $K = 3$	**Free daily drift,** $K = 3$ **(Proposition 38)**
constant Gaussian	0	0	0	0
three scale switches, common drift	0.0235	0	0	0
three joint drift and scale switches	0.0190	0	0	0
three scale switches, drift redrawn daily	0.0225	0	0	0
$p = 2, \kappa = 0.45$	0.2735	0.0045	0	0

Scope. Rejection rates at threshold 20 (a zero is an observed count); 60 training and 60 evaluation days of seven exact increments each; 2,000 records per row (standard error at most 0.011). The joint budgets are those of Proposition 39; the switching nulls are covered at $K \geq 3$ and not at $K = 1$, and the daily-redrawn drift is outside the joint budgets.

The joint budget cuts the alternative's profile penalty from 42 nats (free daily drift) to 12.5, but restores no power: 9 of 2,000 rejections at $K = 1$, none at $K = 3$, against 547 for the common profile, which has no guarantee under any of the switching nulls.

5.5 An implied-volatility clock

The implied-volatility null of Proposition 35 was run on daily simulated records: 126 training returns, then 252 or 504 scored returns monitored at every step. The score's forecast is a mixture of 28 local-lognormal forecasts whose variance scales the previous day's implied volatility by the price level to a power $p - 1$, $p \in \{1, 1.25, 1.5, 2\}$, with Bayesian weights and a drift fitted on past returns only. The implied volatility is a synthetic, predictable proxy, not an option-implied volatility. The primary band for the realized-to-implied variance ratio is $w \in [0.3, 1.0]$, which covers the published ratio near 0.45, with $[0.25, 1.25]$ and $[0.5, 1.0]$ reported separately. Controls inside the null have constant drifts $b \in \{-0.1, 0, 0.1\}$, ratios w from 0.25 to 1.25, and implied volatility that is constant, switches between 0.15 and 0.40 about every 20 days (revealed a day ahead), or rises with the price level.

p	$\kappa = 0.2$	$\kappa = 0.4$	$\kappa = 0.8$	**Switching background,** $\kappa = 0.2$	**Switching background,** $\kappa = 0.4$
1.25	0.0 / 1.0	1.3 / 8.8	11.8 / 44.4		
1.5	1.2 / 9.9	14.0 / 45.8	46.3 / 84.9	7.9 / 26.9	31.1 / 74.1
2	12.8 / 44.7	47.8 / 86.9	88.0 / 99.5	32.5 / 73.3	75.0 / 97.5

Scope. Percentage of records crossing $1/\alpha = 20$ at any time over one / two years of scored daily returns, primary band. The first three columns are exact constant-elasticity prices with implied volatility tracking only the initial scale; the last two are a local-lognormal model whose background variance switches between 1 and 4 times the base about every 20 days, with implied volatility tracking the background but not the level. 2,000 records per cell (standard error at most 0.011); synthetic implied volatility; exact prices.

Inside its contract the score is calibrated: the 72 control cells whose ratio lies inside the primary band cross at most 1.45% at $\alpha = 0.05$ (3.4% at $\alpha = 0.10$), including a revealed switching scale and an implied volatility that rises with the level; the other two bands give at most 1.15% and 0.95%. A ratio below the band is the costly error: controls with $w = 0.25$ crossed the primary band 12 to 61% of the time, and the band $[0.5, 1]$ flagged 94 to 100% of controls with $w \leq 0.3$. Outside the null, Merton jumps (diffusion share 0.83) crossed 7.7 to 17.4% and switching within the day 3.8 to 8.3%, growing with the horizon, while Heston volatility with implied volatility built from its state crossed at most 0.1%. When implied volatility tracks the level dependence fully, the score absorbs it: on the same exact constant-elasticity price records, power falls to at most 0.2% in every cell, and with partial tracking to 0 to 66%. Proposition 35(c) proves the absorption exactly for the discrete local-lognormal law; for exact constant-elasticity transitions these rates are numerical evidence. Level dependence that implied volatility does not track is detected even with a switching background that it does track: 27 to 98% over two years and 8 to 75% over one, at nominal α and in-band false alarms of at most 1.3%. A threshold calibrated empirically on switching controls raises that power but is not a guarantee: held out, it gave pooled false alarms of 3 to 6%, but 12.2% in the worst in-band cell. On the same paths GSADF crossed 4 to 42% of every kind of control, and the drift-robust test, which ignores implied volatility, flagged 9 to 53% of the controls whose implied volatility rises with the level.

Scope. A synthetic implied volatility that reveals the background state exactly and predictably, with a constant ratio inside the band; real implied volatility is noisy and its ratio varies. Exact prices; floating-point scores, not certified. Non-bubble departures also fire, so a rejection is a departure from the null, not necessarily a bubble.

5.6 Sequential monitoring

The monitored score of Proposition 69, with a predictable local-lognormal forecast, was piloted on hourly bars rounded to the cent: 10 training sessions, then 20 monitored sessions with a check at the end of each, certified denominators over the box of drifts in $[-0.5, 0.5]$ and variances in $[0.01, 0.64]$, and 100 paths per law. Under a null inside the box the score never crossed $1/\alpha$ at $\alpha = 0.05$ (95% upper bound 0.037). Under volatility switching between sessions, outside the null, it crossed once in 100 paths. Against late-starting constant-elasticity bubbles ($p = 1.5$ or 2, $\kappa = 0.45$, from the sixth monitored session) it detected none within 15 sessions of onset, whether the forecast was refitted on the growing history or frozen; refitting raised the median final log score from about -21 to -11. A weighted scan over restarts at three declared dates with weights $1/3$ (Proposition 70), on the same paths, raised the median final log score further (-9.0 against -12.7 for $p = 1.5$), but no restart's score, weighted or not, reached $1/\alpha$ on any path. For the selected forecasts these misses are not numerical: an enclosure of every score at every check certified that none of the ideal procedures crosses. Many denominators reached their computation budget and kept valid but looser bounds; a 30-session record with 20 certified checks costs about 15 seconds per path.

Scope. A feasibility pilot: one horizon, one box, hourly sampling, 100 paths per law (standard error at most 0.05), size evidence from the in-box null only. It shows no detection for this forecast and design, not a limit on what any forecast could detect.

5.7 Numerical checks of Sections 3.1 to 3.3

The wealth and fund results were also checked numerically, at the cases computed.

- **Theorem 1.** For constant-elasticity prices absorbed at zero, the defect $1 - E_Q[V_T^\pi]$, computed by two numerical methods that agree to 2.5×10^{-6}, is positive exactly in the cells the theorem predicts, including $\beta = 0.5$ with $\pi \in \{0.25, 0.4\}$, where the conjecture fails. It falls continuously to zero as π approaches $1/2$. For Brownian motion absorbed at zero and $\pi = 1/4$ it is 0.1535 at $T = 1$ and 0.3917 at $T = 5$; for $\sigma(x) = 1 + x^2$, which is strict and reaches zero, it is positive on both sides of $\pi = 1/2$.
- **Finite grids.** A fund rebalanced daily, weekly or monthly over a martingale price that reaches zero has zero defect for every exposure, while the continuous wealth's defect is positive (Proposition 11). Over a strict price that does not reach zero, the grid defect lies above the continuous one and approaches it as the grid refines: 0.0624, 0.0535 and 0.0491 for monthly, weekly and daily rebalancing at $\beta = 1.5$, $\pi = 0.75$, against 0.0455. With the price stopped on leaving $(1/5, 5)$, the distance between grid fund and continuous wealth shrinks at about the rate $N^{-1/2}$ in N rebalancing dates, consistent with the convergence of Proposition 13; on paths absorbed at zero it is unresolved. The bubble sizes behave as uniform integrability requires in every case computed (Proposition 14).
- **Liquidated leveraged funds.** With a liquidation floor monitored continuously, leveraged long funds on two strict prices keep a defect well above zero at daily rebalancing: 0.40 to 0.96 for exposures 1.5 to 3 with the floor at zero, and 0.36 to 0.50 with the floor at half the starting value (standard error at most 0.009), while a geometric Brownian control is within 1.23 standard errors of zero and inverse funds are zero within error. The clipped bounds of Proposition 16 confirm these as lower bounds, within 0.011 of

the continuous values, and remain positive (0.16 to 0.96) when the floor is monitored only at every 64th fine point. The bound $\pi^N \delta_T$ of Proposition 15 is vacuous beyond one interval.

- **The visited integral.** The left side of Proposition 23, computed from n sampled prices with a kernel local time, has relative error falling from 16 to 43% with a year of daily prices to 2 to 4% with a year of one-minute prices, with a residual bias of about -2% and a delta-method interval that undercovers as n grows (coverage 0.57 at the finest sampling for Brownian motion).

Scope. Each item is evidence for the cases computed, at the stated horizons and grids, not a proof.

5.8 Recovering basket contributions in synthetic panels

The estimator and compatibility test of Proposition 78 were run in its synthetic sampling experiment, with $N = M \in \{10^4, 10^5, 10^6\}$ physical panels and pricing draws, 1,000 replicates per cell, $\alpha_{\text{reg}} = \alpha_{\text{price}} = 0.025$, and both the common radius and the per-cell radius. Two models were used. The first is a two-name terminal-law model with a lognormal pricing law, conditional weights of Beta form and an analytic Lipschitz bound, a compatible null at the boundary $\theta_1 = \omega_1$, and a tilted alternative with $\theta_1 = 0.646$ against $\omega_1 = 0.5$; it satisfies the proposition's premises. The second is a three-name basket priced by exact laws, one constant-elasticity name with $p = 2$ and $\kappa = 0.40$, whose recovered contribution is $b_1 = 0.0041$, and two geometric Brownian names, under the restriction and under weight tilts that exceed ω_i by at most 0.019. Its Lipschitz bound is a Monte Carlo finite difference rather than a supplied bound, so its rows are diagnostic.

Model, cell	$N = M = 10^4$	10^5	10^6
terminal-law null, both radii	0 / 0	0 / 0	0 / 0
terminal-law alternative, common / per-cell radius	0 / 0	0 / 100	100 / 100
price basket, restriction and all tilts, both radii	0	0	0

Scope. Rejection rates in percent over 1,000 replicates (95% Wilson interval for a zero rate: 0 to 0.4%). Synthetic terminal panels with a supplied pricing law; floating-point evaluation; the terminal-law model is not a process model.

Under the proposition's premises the test made no null rejection in 3,000 replicates per rule, the lower bound covered the target in every replicate, and power appeared from $N = 10^5$ with the per-cell radius and from 10^6 with the common radius; these fixed-partition runs illustrate the size and power statements, they do not prove them. On the price basket neither rule rejects anything, and the separations, at most 0.019, are far below the radii (median common radius 0.16 to 1.0 across cells and sample sizes). Recovery has a persistent bias at fixed bins: about $+0.0035$ on the constant-elasticity name and -0.0016 to -0.0019 on the others, at every sample size and with 20 or 40 cells, the same size as the contribution $b_1 = 0.0041$ being recovered. The estimator with the true conditional means on the same pricing sample has bias at most 10^{-4}, so the error comes from the fixed-range histogram regression rather than the pricing sample. In the terminal-law model, most of that bias comes from assigning ω outside the fixed range (0.0015 of a first-coordinate bias of 0.0016 at the null), which shrinking the cells cannot remove; consistency needs the range to grow as well as the cells to shrink, as Proposition 78(b) requires. The only rigorous Lipschitz bound known for this basket is of order 10^{16}, which would saturate every radius.

Scope. Non-rejection is not acceptance of the restriction of Proposition 77; the test is a necessary-compatibility test only, and no two-sided interval for the contributions is claimed.

5.9 A repaired score

Section 6.8 traces the drift-robust test's silence to its construction: a forecast that is often itself a null law, and a forecast scale fixed on training data against a null refitted on evaluation data. A repaired score gives both sides one model for the nuisance parameters, so that they differ only in the level exponent. Returns follow an asymmetric conditional-variance recursion with standardized Student-t innovations, with variance multiplied by $(S_{t-1}/S_0)^{\beta}$; the null fits $\beta \in [-10, 0]$ and the alternative $\beta \in [0, 10]$, each with the same fitter for drift, the recursion's parameters and the degrees of freedom $\nu \in \{6, 10\}$ on the 252 scored daily returns after 126 training returns. The score is a monotone transform of the fitted log-likelihood gap. It is a numerical likelihood ratio from a finite fitter, not an e-value, so it was calibrated empirically by the conservative rank rule of Proposition 45, on 499 independent calibration paths per reference law, and evaluated on 500 paths per cell. The same rule calibrated the drift-robust score, GSADF, and a likelihood ratio at known nuisance parameters, all on the same paths.

Cell (calibrated on)	Repaired score	Drift-robust	GSADF	Known nuisance
null, two-regime volatility (itself)	7.4	4.6	3.4	4.0
null, Heston with leverage (itself)	6.4	0.2	3.2	4.6
$\beta = 0.25$, $\nu = 6$ ($\beta = 0$ reference)	10.0	4.8	4.6	19.2
$\beta = 0.5$, $\nu = 6$	19.4	4.6	8.0	43.6
$\beta = 0.25$, $\nu = 10$	10.8	4.6	5.0	30.8
$\beta = 0.5$, $\nu = 10$	22.0	4.4	6.4	53.6
CEV $p = 1.5$ with Heston volatility (Heston)	8.6	0.4	3.8	6.4

Scope. Flag rates in percent at $\alpha = 1/20$ over 500 evaluation paths (standard error about 1 point at 5%; Wilson intervals in the record); each cell reuses one realized calibration sample, so its interval is conditional on that cutoff, and intervals are not simultaneous across cells or scores. Simulated daily returns from a statistical conditional-return model; the known-nuisance ratio fixes the reference law's parameters and is not an oracle bound.

Under calibration on the matching law, the repaired score flags 3.4 to 7.4% of paths in all ten null laws, including laws outside its family (an extreme negative exponent, Student-t with $\nu = 4$) and Merton jumps, two-regime volatility and Heston volatility. The edge case is two-regime volatility at 7.4% (5.4 to 10.0%), conditional on its one cutoff; the training-conditional bound of Proposition 45(c) puts the chance that a realized cutoff's risk exceeds 7% at about 0.03, if the reference windows are independent draws from the evaluated law. The drift-robust score's matched rates are often far below α because many calibration windows score exactly zero, the null-member forecasts of Section 6.8, so its cutoff lands on a tie. Against in-model level alternatives the repaired score flags 10 to 11% at $\beta = 0.25$ and 19 to 22% at $\beta = 0.5$, against 4 to 8% for the drift-robust score and calibrated GSADF; on the same paths, flags by the repaired score alone outnumber flags by GSADF alone 45 to 18, 73 to 16, 50 to 21 and 93 to 15, a finite-case advantage, not uniform superiority. These rates are modest, two to four times α, and against a constant-elasticity diffusion with $p = 1.5$ and Heston volatility the score flags 8.6% against 6.4% on Heston itself: the level effect is weak and easily absorbed by a flexible volatility state, as Section 6.8 found on real data.

Transporting the calibration to a mismatched law is mostly conservative: cutoffs from the $\beta = 0$ reference give 1.0 to 3.2% on negative exponents, lower persistence, jumps and Heston. But on two-regime volatility they give 9.0% (6.8 to 11.8%), so validity on real data rests on the premise of that remark that the reference windows share the evaluated window's law after selection; matching the volatility regime alone does not establish it. A sequential version, with forecasts refitted from past returns only, had a negative endpoint log ratio on all but one of 744 diagnostic paths (medians −5.0 to −11.4), dominated by the cost

of fitting the nuisance from the past; the one exception, at 7.52 on a $\beta = 0.5$ path, is uncertified, crossings before the endpoint were not computed, and no sequential test is validated.

Scope. A pilot in simulation only; no certified profile or sequential e-value; the fitter's boundary selections are numerical near-ties, not features of the data. Whether calibration on real reference windows meets the premise above is the decisive open question.

5.10 A daily statistic with a market scale

Section 6.11 found that controlling a stock's own slowly moving volatility understates a level effect. A daily statistic was therefore built whose volatility scale comes from the market instead. The variance of a daily log return is $w\, m_t^g \exp(\beta x)$, where m_t is the mean squared SPY return over the previous 20 sessions, normalized once on training data; x is the log price relative to the start of scoring; and β is the full exponent of variance in the price, so that a CEV diffusion with exponent p has $\beta = 2(p-1)$. Innovations are Student-t. The nuisance (μ, w, g, ν) is fitted at $\beta = 0$ on the 252 prior returns and then frozen, so that only β differs between the two sides. The batch score compares the best $\beta \geq 0$ with the best $\beta \leq 0$ over 252 scored days. A daily monitor accumulates an equally weighted mixture over eight values of β from 1/8 to 16 against the profiled $\beta \leq 0$ maximum and alarms at its first crossing. Each is calibrated by the conservative rank rule of Proposition 45 on 499 paths of a null law, and evaluated on 500. GSADF and the repaired score of Section 5.9 were calibrated the same way on the same paths; the repaired score was run on only 100 calibration and 100 evaluation paths per cell, for runtime.

The laws are of three kinds. In the statistic's own family, the market's log variance is autoregressive and the stock's variance is 2.25 times the market's times $\exp(\beta x)$, with $\beta \in \{-1, 0, 1, 2\}$. CEV diffusions with $p \in \{1, 1.5, 2\}$ were simulated with constant volatility, with a stochastic-volatility factor that also drives the market, or with the stock's own stochastic volatility, each at a correlation $\rho \in \{-0.7, 0, 0.7\}$ between price and volatility shocks. And the ten null laws of Section 5.9 were rerun with a market attached. Each alternative is calibrated on the null with the same volatility, correlation and market ($p = 1$ for the CEV cells).

Law (calibrated on its null)	Daily statistic	No market scale	GSADF	Repaired score
own family, $\beta = 1$, $\nu = 6$	39	19	7	27
own family, $\beta = 2$, $\nu = 6$	66	41	11	52
own family, $\beta = 2$, $\nu = 10$	71	32	14	21
CEV $p = 1.5$, constant volatility	41	42	6	11
CEV $p = 2$, constant volatility	72	74	10	31
CEV $p = 2$, market volatility, $\rho = 0$	23	12	11	44
CEV $p = 2$, own volatility, $\rho = 0$	10	11	10	16
CEV $p = 2$, own volatility, $\rho = 0.7$	20	20	8	28

Scope. Percentages of 500 evaluation paths flagged at $\alpha = 1/20$ by the batch score (standard error at most 2.2 points); the repaired score's column uses 100 paths and a 100-path calibration (standard error up to 5 points), and its matched null rates scatter from 1 to 18% (that tier omitted a few early simulation failures in three null laws). Each cell reuses one realized calibration sample. Discretized diffusions (10 steps a day; 50 steps change the fitted β by at most 0.07 in the median) are not certified strict-local-martingale samples, and a Student-t return model is not a pricing law.

In its own family the statistic recovers a known exponent. Every estimate is finite, and the median error is at most 6% of the truth. The positive share is at least 0.88 in every cell but one. In that cell $\beta = 1$ is already active in the training year, and only 79% of estimates are positive, against 90% with the true nuisance; the

fit at $\beta = 0$ plausibly absorbs part of the effect, though that cause is not isolated. On constant-volatility CEV it reports 1.02 at $p = 1.5$ and 1.89 at $p = 2$. Under calibration on the matching law, its flag rate averages 4.85% (batch) and 4.61% (monitor) over 22 null laws, with four single-cell rates whose Wilson intervals lie above 5% (7.0 to 8.8%) and one stress law, an exponent of -40, whose 15% simulation failures leave it uninformative. An average near nominal is not a guarantee for every law. Its power is 37 to 72% in its own family and on constant-volatility CEV, far above GSADF over these 252-day windows (2 to 14% in every alternative). Without the market scale the power roughly halves in its own family and under market-driven volatility. Under the stock's own stochastic volatility, power at $p = 2$ is only 10 to 20%. The monitor's rates are a little lower than the batch score's. Among alarmed paths its median first alarm comes on day 130 to 170 of 252, against 24 to 135 for GSADF, which alarms far less often.

The fitted exponent is not specific to a level effect. With no level effect ($p = 1$) and the stock's own volatility correlated +0.7 with its price, the median estimate is 2.54 (76% positive), and -2.58 at $\rho = -0.7$. The market-driven cells give 1.00 and -1.12. An independent implementation on fresh paths gives 2.58. So a positive daily estimate is equally consistent with correlated stochastic volatility, the limit found within the day in Section 6.11. The calibration does not transport across volatility laws either. A cutoff set on the in-family $\beta = 0$ null flags 8 to 70% of paths when applied to nulls with the stock's own volatility clustering, positive correlation or regime switching (for example 33% on the two-regime law of Section 5.9, and 63% under own volatility at $\rho = 0.7$), though only 3.8% under Merton jumps. A real-data reading therefore needs references with the target's own kind of volatility history, and even then the transport error is unknown.

Scope. Simulation only. The gates on recovery and calibration were declared in advance but are exploratory and not simultaneous. Matched calibration is the exchangeable case of the rank rule for complete simulated paths, and the transported rates measure misspecification.

6. Real-data results

This section applies the tests to real prices. Three caveats govern every number in it. The labels are consensus, not ground truth: a "positive" episode is one that the cited literature or the press calls a bubble, not one shown to be a strict local martingale under any pricing law, and a "negative" control is calm by a rule fixed before its window, not certified free of bubbles. A flag rate on positives is therefore not power, and one on negatives is not size. And every application lies outside the guarantees of Section 3.5: real prices are rounded, are not known to follow a law in any null of that section, and are not known to have $P = Q$. Each series is normalized by its first price in the window, and the tests are those of Section 5 with the same catalogue and split; all designs, episode lists and control rules were fixed before any test was run on them.

6.1 The data and their limits

Free daily closes cover most listed names, but not those that failed. Reconstructing the S&P 500 membership month by month, 57 of 506 members in January 2020 have no free daily series, falling to 40 of 505 by December 2021, 27 of 503 in January 2023 and none today. About one member in ten is missing in 2020–21, against one in 33 over 2023–26. The missing names are not random: about half of the 2020–21 losses were acquisitions, and the losses include failed banks such as Silicon Valley Bank, First Republic and Signature. For scale, Greenwood and Sammon ([31]) list 750 S&P 500 deletions between 1980 and 2020 and match 749 of them to licensed price data; free daily prices exist for few of those names.

Daily data see the range of prices visited but not the time spent near its top. In a two-week sample of one-minute bars, SPY spanned 2.1% of its level and QQQ 3.6%; ten daily closes show only 1.4% and 1.8%. Within 1% of the one-minute maximum, QQQ spent 429 regular-hours minutes (11% of the time), which daily closes record as 2 days of 10. A volatility tail can be estimated only on the visited range, and the local-time weights of Proposition 23 are thinnest at its top.

The option and fund inputs of Sections 3.3 and 4 are measurable only coarsely. On end-of-day SPY chains, the largest-strike call at one to three months is quoted at the one-cent floor on most dates, at median strikes of 1.07 to 1.41 times the price, so these data bound the far wing only to the tick and cannot show how call prices decay in the strike. For eight leveraged and inverse exchange-traded funds on QQQ and SPY, the cost beyond the expense ratio, measured against the total-return underlying, is 0.6 to 1.7% a year for the leveraged long funds and 0.1 to 1.1% for the inverse ones (standard errors 0.1 to 0.4% a year), and tracking error compounds to about 0.5 to 1.2% a year; against the price index instead, dividends would be misread as cost.

6.2 Consensus episodes on daily data

The positive controls are nine episodes: the Nasdaq indices in 1995–2001; surviving dot-com leaders; US homebuilders and US financials in 2003–08; the 2021 meme stocks; the 2026 run in SK Hynix; the Nikkei in 1986–90; the Shanghai Composite in 2014–15; and the 2020–21 growth names. Each has 1 to 8 names, 33 in all. Negative controls follow a mechanical rule using only data before each window: for each episode's window length, the three calmest non-overlapping start dates by trailing S&P 500 volatility and drawdown (one for the longest windows), each with two indices and five Dow members, giving 103 names in 15 windows. Each window is tested in full, and also cut at the first name's maximum; that end date is chosen by the outcome, so the cut windows are outside every guarantee.

Set	Window	Names	Drift-robust, 5%	E, 5%	SADF, 5%	GSADF, 5%	JKP Method 2	JKP Method 1
positive	full	33	0.06	0.27	0.76	0.64	1.00	0.06
positive	to the peak	32	0.09	0.31	0.72	0.62	0.94	0.09
negative	full	103	0.00	0.01	0.16	0.19	0.85	0.04
negative	to the peak	103	0.00	0.02	0.18	0.19	0.91	0.09

Scope. Shares of names flagged, on adjusted daily closes. The binomial standard errors of the differences between positives and negatives are 0.03 to 0.09; names within a row, and rows sharing windows, are dependent. The surviving-leader, financial and meme rows contain only survivors, and the 1990–94 negative windows lost 11 to 13 Dow members to missing data.

The drift-robust test flags no negative control, and almost no positive one. Its largest log score on a negative is 1.07, against the threshold $\log 20 = 3.00$. It flags two positives in full windows (BlackBerry and SK Hynix) and three to the peak (GameStop, Koss and BlackBerry), all from the meme and SK Hynix rows, and none of the dot-com, housing, financial, Japanese, Chinese or growth names. The flagged meme windows contain the January 2021 squeeze, a jump no catalogue law has, and in Section 5.2 Merton jumps produced false alarms; these data cannot tell a strict local martingale from a jump. The raw score E flags 27 to 31% of positives and 1 to 2% of negatives. SADF and GSADF separate the groups most (differences 0.43 to 0.66, standard error about 0.09), but flag 13 to 19% of the calm controls at nominal 5%, mostly in the long bull markets after 2009 and in 2014–18: they detect persistent upward trends, which calm rising

markets also have. JKP's Method 2 flags almost everything and Method 1 almost nothing. The panel t test flags no positive row in full windows; cut at the peak it flags six of nine positive rows and three or four of 15 negative windows, because cutting at a maximum manufactures an explosive end.

Adding the names that actually collapsed does not change this. Ten collapsed names with free daily data were added to the dot-com and financial rows: InfoSpace, CMGI, JDS Uniphase, VA Linux and theglobe.com; Lehman, Wachovia, Countrywide, Merrill Lynch and Freddie Mac. The drift-robust test flags none of the ten in either window, with log scores from −1.6 to −2,473. E flags three of the five dot-com names and none of the financials, as it did the survivors. SADF and GSADF flag three or four of the five dot-com names and GSADF three of the five financials, but they flag 14 to 43% of the matched calm controls, so they separate the groups only weakly; JKP's Method 2 says bubble on most names in every group. The early split-adjusted prices of some of these names are rounded to the cent, with many zero returns, and the larger raw and explosive-root statistics rest partly on those segments.

JKP's published verdicts could be checked at daily frequency for two of their four dot-com stocks. On daily closes, Method 2 calls InfoSpace a bubble, as JKP did on tick bid prices, but with fitted exponents of 1.27 and 1.14 against their 6.17 and 3.58; it also calls GeoCities a bubble, where JKP found none. With 945 and 202 daily prices the method has two and one grid points, so these verdicts are near-degenerate and are not evidence about JKP's tick-data results. The drift-robust test flags neither name, and SADF and GSADF flag both.

6.3 Hourly data

Free hourly bars cover about two years, which contains one positive control, SK Hynix, and one calm negative window of seven names. Hourly, the drift-robust test on calendar time flags SK Hynix in both windows (log scores 138.6 and 125.5), but it also flags Coca-Cola in the calm window (log score 12.8, kept by e-BH), a false alarm under the consensus labels that does not recur in the cut window. The first hour of the session carries 60 to 69% of the day's variance time in these data, a clock that the calendar-time null ignores; Section 5.4 found that ignoring the clock cost power in simulation, and here it also produced a false alarm. The declared-clock score of Proposition 34, which has no guarantee for rounded prices, flags nothing in either group, and on SK Hynix is strongly negative (−5,762): its forecast, fitted on the first half, chose a null-type exponent, and the price then rose about thirteenfold. SADF and GSADF flag SK Hynix and no calm name at 5%. The certified filtered score of Propositions 66–68 also flags nothing among the calm names; it certified each two-year denominator to within 0.7 nats in 24 to 52 seconds, or stopped at its computation budget with valid bounds.

Recent run-ups were then tested on the same hourly window: 21 names that the press called bubbles or manias (among them GameStop and AMC in 2024, Super Micro, Palantir, Nvidia, the quantum-computing and nuclear names, and the crypto-treasury companies), 8 further crypto-linked and mega-cap names, and 98 calm large caps over the same dates. The registered rule for calm windows before the run-ups left none, so the controls overlap the positives in time.

Group	Names	Drift-robust	Clock	Certified	SADF	GSADF	JKP 2
bubble-narrative names	21	0	0	0	12	18	20
crypto-linked and other	8	0	0	0	4	6	8
calm large caps	98	0	0	0	4	13	58

Scope. Counts of names flagged at 5% on hourly bars over about two years, by the drift-robust test, the declared-clock score, the certified filtered score, SADF, GSADF and JKP's Method 2; e-BH rejected nothing in any group. Press labels; rounded prices, for which only the certified score has a guarantee, and that only inside its declared box.

The likelihood tests flag nothing, but on the positives this silence is mostly a limitation of the forecast. Their scores on the positives are far below zero, down to −106,126, because the forecast's scales stop at 0.45 at the starting price while these names moved several times faster. In a follow-up registered before it was run, on these already inspected records, with the forecast's scales widened up to 4 and everything else unchanged (the larger of the old and new scores is never used), the scores rise by orders of magnitude, and most named run-ups then look like a Gaussian law with a free scale. The drift-robust test still flags nothing. The clock score flags only GameStop (log score 21.5), whose 2024 window is dominated by jumps: about half its hourly realized variance is jump-like (the bipower variation of Barndorff-Nielsen and Shephard [12], which separates the continuous part of the variation from jumps, is 0.49 of realized variance), one hourly return carries 27% of it and five carry 69 to 78%, and its volatility in the second half was 2.23 against 0.65 in the first. At hourly sampling a burst of volatility can be mistaken for a jump ([9]), so this split is descriptive. The certified score flags seven names, but every one has volatility above its declared box's 0.80 (from 0.91 to 3.64), so these are failures of the box, as for the out-of-box null of Section 5.4, not evidence of bubbles. The calm controls stay unflagged in every column. GSADF separates the groups best (18 of 21 against 13 of 98) and JKP's Method 2 not at all, neither with a guarantee here.

Daily evidence does not accumulate on high-range days. Splitting each hourly record into daily contributions under the daily-reset score of Proposition 37, a day's contribution is positive on at most 1% of days in any band of daily range, and it grows more negative as the range widens, from about −4 on days with a range under 2% to −760 on days above 20% among the positives. GameStop's and AMC's days with a range of 10% or more sum to −852 and −1,464. With this forecast, the most volatile days carry the most evidence against a bubble, not for one; whether a forecast that can follow larger scales changes that was not tested.

Scope. One positive and seven negatives for the first comparison, and press labels for the second, so no rate is estimable; rounded hourly bars, with 23 to 54% of the closes from sub-penny trades in the first comparison.

6.4 An unselected universe

The separations of Section 6.2 are selected: the episodes and their cut-off ends were chosen with hindsight. To remove that selection, the drift-robust score E^{drift}, the raw score E and the GSADF statistic were computed on every window of a rolling grid fixed in advance: the 503 current S&P 500 constituents, windows of 13, 24 and 72 months starting on the first trading day of every month from January 1995, each kept when the name's adjusted daily closes cover at least 99% of the window and of the 126 trading days after it. That gives 416,380 name-windows. The outcomes are the maximum drawdown over the next 63 and 126 trading days, with means 0.13–0.14 and 0.18–0.19, and the forward log return. Each score enters a regression as its percentile rank among the windows of its start date. Standard errors are Driscoll–Kraay with $1.5L$ monthly lags for window length L, longer than both the horizon and the overlap of adjacent windows; the out-of-sample R^2 uses coefficients from starts before 2011 on starts from 2011, with a block-bootstrap standard error.

Window length	Drift-robust	E	GSADF
13 months	−0.027 [−4.3]; −0.001 ± 0.002	−0.007 [−0.7]; −0.004 ± 0.001	+0.028 [6.4]; +0.007 ± 0.001
24 months	−0.023 [−2.3]; −0.003 ± 0.004	−0.004 [−0.3]; −0.001 ± 0.001	+0.034 [6.8]; +0.009 ± 0.001
72 months	−0.017 [−2.0]; −0.006 ± 0.001	−0.002 [−0.3]; −0.001 ± 0.000	+0.046 [8.4]; +0.013 ± 0.002

Scope. For the 126-day forward maximum drawdown: the change in drawdown from the lowest to the highest rank, its Driscoll–Kraay t in brackets, then the out-of-sample R^2 with its bootstrap standard error. Current constituents only; adjusted daily closes; no guarantee for any of the scores on real prices.

Window length	Period of window end	Windows	Drift-robust	E	GSADF
13 months	1995–99	14,780	0.17%	4.2%	21.2%
13 months	2003–07	22,740	0.11%	1.7%	12.9%
13 months	2009–19	58,664	0.08%	0.6%	12.9%
13 months	2023–26	19,291	0.15%	1.6%	11.5%
24 months	1995–99	11,183	0.38%	5.0%	27.7%
24 months	2003–07	22,371	0.23%	1.9%	16.1%
24 months	2009–19	57,883	0.10%	0.7%	14.6%
24 months	2023–26	19,201	0.18%	2.0%	13.9%
72 months	2003–07	20,277	0.25%	2.8%	33.9%
72 months	2009–19	54,052	0.13%	0.7%	23.9%
72 months	2023–26	18,566	0.06%	0.6%	41.7%

Scope. Share of windows flagged at the nominal level: a log score above $\log 20$, or GSADF above its simulated 95% critical value. Windows overlap heavily, so these shares are descriptive.

The drift-robust test almost never fires: it flags 0.11%, 0.17% and 0.24% of all windows at the three lengths, and at most 2.5% in any single year (72-month windows ending in 2000, 285 of them). Where it ranks windows, the ranking does not warn: a higher rank goes with smaller forward drawdowns, its out-of-sample R^2 is negative for every drawdown outcome, and for returns it is at most 0.0001. Its log score is negative on 90 to 92% of windows and exactly zero on 7 to 10%, so the rank mostly orders how strongly a window favours the null. The raw score E flags 1.7 to 2.1% of windows, up to 13.4% for 13-month windows ending in 2000, and has no association with the outcomes ($|t| \leq 1.4$, out-of-sample $R^2 \leq 0$). GSADF flags 15.3%, 19.0% and 33.2% of windows, 13 to 24% in the calm market of 2009–19, and 44 to 52% of the windows ending in 2020 and 2021. Its rank has a small, clear and out-of-sample association with forward drawdowns: 2.2 to 4.6 percentage points from the lowest to the highest rank, t between 6.4 and 8.6, and out-of-sample R^2 of 0.006 to 0.014. It also goes with slightly higher forward returns (t from 1.9 to 2.4), which fits momentum rather than a crash warning. With all three scores together, GSADF's coefficient is unchanged, the two likelihood scores split by collinearity (their ranks correlate 0.77 to 0.95), and the joint out-of-sample R^2 is below GSADF's alone.

On the records of Section 6.2, a comparison that is optimistic because those episodes were chosen with hindsight, the probability that a consensus positive outscores a calm control is 0.16 for the drift-robust score (95% bootstrap interval 0.05 to 0.34), 0.30 for E (0.05 to 0.64) and 0.84 for GSADF (0.70 to 0.995). At

the controls' 99th percentile, the three flag 6.1%, 27.3% and 45.5% of the positives. The drift-robust score ranks the consensus bubbles below the calm controls: its log score is more negative on them.

Scope. Current constituents only: names removed from the index or failed, which tend to have large drawdowns, are absent, which the design expected to work against finding drawdown predictability; its effect on any particular estimate is not established. The per-window scores and outcomes and the headline slopes, out-of-sample R^2 and flag rates were reproduced independently; the combined regression, the bootstrap standard errors and the comparison on the episode records were not recomputed. GSADF is an explosive-root statistic, not a strict-local-martingale test.

6.5 Empirical critical values for GSADF

GSADF's nominal critical values come from a homoskedastic random walk, and on the calm controls above they flag 12 to 30% of records. Its critical values can instead be set from the calm windows themselves. Each record's GSADF statistic was turned into a p-value against 2,000 simulated random walks of the record's own length, and the threshold at level α is the α-quantile of the calm records' p-values, separately for daily and hourly data. Size is measured on held-out windows: each calm window's names are flagged at the threshold computed from the other windows (14 daily or 13 hourly), and the interval comes from a bootstrap that resamples whole windows. The records are those of Sections 6.2 and 6.3, with no new data: 103 calm and 33 positive daily names, the 10 collapsed names, and 98 calm and 29 run-up hourly names.

Records	α	Held-out size	95% interval	Nominal, calm	Positives
daily, full	0.05	6.8%	1.9–12.5%	21.4%	63.6% of 33
daily, full	0.10	10.7%	4.9–17.5%	30.1%	63.6% of 33
daily, to the peak	0.05	6.8%	1.0–14.3%	19.4%	59.4% of 32
hourly	0.05	7.1%	2.0–13.3%	12.2%	76.2% of 21
hourly	0.10	10.2%	4.1–18.4%	24.5%	85.7% of 21

Scope. Held-out size by leaving one calm window out, with a window-cluster bootstrap interval; the nominal critical value's flag rate on the same calm records; the positives' flag rate at the threshold calibrated on all calm windows. Only 15 daily and 14 hourly calm windows, which are dependent; consensus labels chosen with hindsight; the to-peak ends are chosen by the outcome.

With empirical thresholds (p-values at or below 0.0135 daily and 0.0084 hourly at $\alpha = 0.05$), GSADF's held-out false-alarm rate falls to about its nominal level, and every interval covers α. These thresholds are empirical calibrations, not guaranteed sizes: the calm windows were selected by their own trailing volatility and drawdown, a past-only selection that does not by itself make a calibrated threshold valid (Proposition 45(d)), and the thresholds are interpolated quantiles rather than the conservative rank rule of that proposition. It still flags 59 to 64% of the daily consensus positives and 76% of the hourly run-ups, and at most a 5% false-alarm rate on calm records it reaches 64%, 59% and 76% of them. Most of that separation comes from records whose statistic exceeds every simulated null: 15 of 33 daily and 12 of 21 hourly positives, against at most 2 calm records. Among the collapsed names, it flags 4 of the 5 dot-com names and 1 of the 5 financials; among the hourly crypto-linked names, 4 of 8. The probability that a positive's p-value is below a calm one's, optimistic because the positives were chosen with hindsight, is 0.82 for daily full windows (95% interval 0.68 to 0.99), 0.82 to the peak (0.63 to 0.99) and 0.91 hourly; the hourly figure has no reliable interval, because all 21 hourly positives come from one list and the bootstrap

cannot vary them as a group. On the same labelled records, the drift-robust score ranks the positives below the calm controls (Section 6.4), so calibrated GSADF is the benchmark the likelihood scores do not meet.

Scope. GSADF tests an explosive autoregressive root, not a strict local martingale; the p-values rest on simulated nulls of 2,000 replications, which were spot-checked, not regenerated, in the reproduction.

The calibrated threshold was then applied, unchanged, to the 416,380 windows of the unselected universe of Section 6.4, whose GSADF p-values are computed the same way. These windows are unlabelled, so a flag rate there is not a false-alarm rate.

Window length	All windows	1995–99	2003–07	2009–19	2023–26	Highest year
13 months	8.2%	11.7%	6.0%	6.4%	5.5%	2020: 30.9%
24 months	10.9%	16.1%	8.5%	7.4%	6.8%	2021: 34.1%
72 months	23.0%		23.4%	14.8%	31.6%	2021: 38.2%

Scope. Share of windows flagged by period of the window's end at the calibrated threshold for $\alpha = 0.05$ (p-value at most 0.0135); the nominal rates are in the second table of Section 6.4. Overlapping windows; current constituents only.

Against the nominal rates of 15.3%, 19.0% and 33.2% over all windows, calibration roughly halves the flag rate at 13- and 24-month windows, to 6 to 7% in the calm years 2009–19, close to the held-out 5% target, but it still flags about 15% of 72-month windows in those years. The calm windows behind the threshold are shorter than 72 months, and a random-walk p-value is not free of window length on real prices, so the calibration does not transfer evenly across lengths. Used as a 0/1 flag, calibrated GSADF keeps the association of Section 6.4: windows it flags have forward maximum drawdowns about 3 percentage points larger than unflagged ones (for example 0.166 against 0.138 over 63 days for 13-month windows), with Driscoll–Kraay t between 3.1 and 5.5, an out-of-sample R^2 of 0.005 to 0.011 (bootstrap standard errors 0.003 to 0.012, so not distinguishable from zero at 24 months), and no reliable association with forward returns ($t \leq 1.7$). It is a modest risk screen, not a reliable bubble detector.

6.6 Jumps behind the flags

A flag can come from a jump rather than from a sustained change in the price's dynamics, and a monitor should say which. Every flag at 5% reported in Sections 6.2 and 6.3 by the drift-robust score, E, SADF, GSADF or the declared-clock score, 185 in all, was described by three statistics of its record's log returns: the jump share $1 - BV/RV$, with bipower variation $BV = (\pi/2)\sum_j |r_j||r_{j-1}|$ and realized variance RV; the share of RV in the five largest squared returns; and the ratio of the second half's volatility to the first half's. Each was compared with the same statistics on that data set's calm controls, and with 2,000 Gaussian paths of constant volatility as a no-jump reference. Then a jump filter stated in advance set to zero every return larger than four local bipower standard deviations, from 42 neighbouring returns, the price path was rebuilt, and every record, positives and controls alike, was scored again with the same code.

Records	Column	Flags	Survive the filter
daily consensus episodes, full windows	SADF / GSADF / E / drift-robust	25 / 21 / 9 / 2	23 / 20 / 7 / 1
daily consensus episodes, to the peak	SADF / GSADF / E / drift-robust	23 / 20 / 10 / 3	21 / 19 / 6 / 1
daily collapsed names	SADF / GSADF / E	9 / 11 / 5	6 / 8 / 3
hourly SK Hynix, full window	SADF / GSADF / drift-robust	1 / 1 / 1	1 / 1 / 0

Records	Column	Flags	Survive the filter
hourly SK Hynix, to the peak	SADF / GSADF / drift-robust	1 / 1 / 1	1 / 1 / 1
hourly run-ups	SADF / GSADF	16 / 24	14 / 22
hourly GameStop, widened forecast	clock score	1	0

Scope. A flag survives when the same column still flags the record after the detected jump returns are set to zero. The filter is a stated heuristic without a proven size, it uses returns on both sides of each date, and by Proposition 54 zeroing returns breaks the scores' calibration, so the filtered scores are descriptive only. In an independent reimplementation, which read the filter's window slightly differently, two survival counts differ by one and the reading is unchanged. The certified score's flags of Section 6.3, all outside its box, were described but not rescored. Consensus labels.

Most flags are not explained by jumps: 156 of the 185 survive the filter, including 137 of the 153 explosive-root flags and 16 of the 24 flags of E. The exceptions are where earlier sections pointed: GameStop's widened clock-score flag, whose record has a jump share of 0.51 and five returns carrying 69% of its variance; SK Hynix's full-window hourly flag by the drift-robust test; and several daily flags of E. What sets the flagged daily records apart from the calm controls is rising volatility, not jumps: their median rank among the controls is 0.55 in jump share and 0.63 in the top-five share, but 1.00 in the second-half to first-half volatility ratio, and 69% of daily flags lie above the controls' 95th percentile of that ratio. On hourly records jumps and volatility changes both contribute, and neither separates the flags cleanly. The controls themselves are far from the no-jump reference: daily calm records have a median jump share of 0.07, against about zero for Gaussian paths, and hourly ones 0.28 to 0.31, inflated because bipower variation pairs the regular-hour returns with the much larger overnight return. Filtering does not improve calibration: on the daily calm controls it raises GSADF's flag rate from 19.4% to 25.2% and SADF's from 15.5% to 21.4%, and on the hourly controls it lowers GSADF's from 13.3% to 7.1% but raises SADF's from 4.1% to 8.2%. The drift-robust test's hourly false alarm on Coca-Cola disappears after filtering, and SADF then flags another calm name.

6.7 Index data with implied volatility

The implied-volatility null of Proposition 35 was applied to daily index closes, with the published implied-volatility index as the scale: SPX with VIX from 1990, and NDX with VXN from 2001. The return from the close of day $t-1$ to the close of day t uses the volatility index's close on day $t-2$, because the index's own close is published after the return interval begins; the close of day $t-1$ was run as a sensitivity and gave the same counts. VIX values before September 2003 were reconstructed later from historical option quotes, so SPX windows before then are a retrospective analysis, not a record of what could have been computed at the time. The score, bands and forecast are those of Section 5.5. Rolling windows of 126 training and 252 scored returns start each quarter, and a window is called calm, before any score, when its scored year has a drawdown below 10% and does not overlap a recession.

Index	Windows	Calm windows	Crossings, primary band [0.3, 1.0]	Crossings, band [0.5, 1.0]	GSADF flags, all / calm
SPX	141	65	0	1 at 5%, 2 at 10% (not calm)	13.5% / 0%
NDX	97	14	0	0	0% / 0%

Scope. Anytime crossings of $1/\alpha = 20$ in any window. Overlapping quarterly windows, so the counts are descriptive; the calm label uses the scored year's own drawdown, a hindsight label that does not give a size guarantee (Proposition 45(d)).

The score never crosses at $\alpha = 0.05$ or 0.10 in the primary band or in $[0.25, 1.25]$, in calm or stressed windows, including 2008 and March 2020; its largest running maximum is 1.54 on SPX and 1.30 on NDX, against $\log 20 = 3.00$. The crossing in the narrow band at $\alpha = 0.05$ is an SPX year scored from April 2017 that contains the volatility spike of February 2018, after a year in which realized variance was low against implied; a ratio below that band or a jump would each be outside the null, but the crossing does not identify which, or a bubble. Through the episodes it never crosses either: on SPX from 1999 to 2001 its maximum is -0.06 and its final value -5.14, and on SPX and NDX in 2020–21 its finals are -6.98 and -2.83. GSADF flags the 1999–2001 SPX record and 19 SPX windows, scored in years ending 2001–03 and 2015–19, but no NDX window. The ratio of realized to implied variance, over each window's scored year, has median 0.62 on SPX and 0.74 on NDX; 21% of SPX windows fall below 0.5, 6 to 16% exceed 1, four SPX and two NDX windows exceed 1.25, outside every band (windows ending in late 2008 and in 2020), and none falls below 0.3; a sample ratio inside or outside a band neither certifies nor refutes a constant conditional ratio. So under these fixed forecasts, bands, thresholds and windows, no departure is detected from level dependence that the implied-volatility index already prices, the outcome the absorption of Section 5.5 leads one to expect. No calibration of the score on historical indices follows from the zero count, and the two indices and the overlapping windows are not independent trials. It is not evidence of no bubble, nor that implied volatility prices all level dependence, nor that individual stocks had no bubble.

Scope. Two indices; daily closes, with no rounding model; floating-point scores, not certified; retrospective VIX before 2003; VXN starts after the 2000 peak, so the NDX dot-com episode cannot be tested.

6.8 Why the drift-robust test is silent

The drift-robust test almost never fires on real prices (Section 6.4). A diagnosis on data already in hand separates two explanations: no signal, or a score built so that it cannot show one. It uses the 33 consensus positives and 103 calm controls of Section 6.2 and 3,000 sampled 24-month windows of the unselected universe of Section 6.4, of which 297 are run-ups (the top tenth of their start date by return). First, the log score is split exactly into four terms: the misfit of the training-chosen scale κ on the evaluation half; the level dependence, the fitted exponent against geometric Brownian motion with each at its best scale on the evaluation half; the drift, zero drift against a fitted one; and the enlargement of the null.

Records	Log score	Scale misfit	Level dependence	Drift	Null enlargement
consensus positives	−229.6	−228.2	−1.47	−0.32	0.00
calm controls	−18.1	−17.6	+0.33	−0.50	0.00
unselected windows	−12.1	−10.6	+0.27	−0.28	0.00
run-ups	−18.4	−16.7	0.00	−1.06	0.00

Scope. Medians in nats; the four terms sum to the log score on every record. Daily adjusted closes; consensus labels; survivorship as in Section 6.4.

Two features of the construction keep the score silent. First, the forecast chosen on the training half is often itself a null law, with $p \leq 1$: in 15 of 33 consensus positives, 88 of 103 calm controls, 2,287 of 3,000 unselected windows and 192 of 297 run-ups. The null contains it, so the score cannot exceed one on those records. Second, the scale misfit: the forecast's scale is chosen on the training half from $\{0.2, 0.3, 0.45\}$ with

zero drift, while the null refits its scale and drift on the evaluation half. This term is the largest negative one in 76 to 83% of records and, in the median, 88 to 99% of the score, which is why the score is most negative on the consensus bubbles. It is a retrospective penalty for the fit of the scale: it does not separate a change of volatility between the halves from the coarseness of the three-point grid, and refitting the scale on the evaluation half is a diagnostic, not a valid repaired forecast.

Removing that asymmetry does not reveal a strong level signal. Even in sample, with every parameter fitted on the evaluation half, a zero-drift constant-elasticity alternative with $p > 1$ and a free scale beats the drift-robust null, whose constant-elasticity scales are then also free, by more than $\log 20$ in only 4 to 9% of records, and on the consensus positives it loses by 38 nats in the median. A local-lognormal model with free drift, scale and variance exponent $\beta \in [-20, 20]$ (variance proportional to the price to the power β, twice the exponent of the volatility), on bounded domains, beats the same model restricted to $\beta \in [-20, 0]$ by more than $\log 20$ in 12.1% of consensus positives and 11.7% of calm controls, 7.5% of unselected windows and 14.1% of run-ups; it gains anything beyond numerical tolerance in only 26 to 37% of records, and the fitted β is negative in the median in every group. A regression of monthly realized variance on the price level, controlling for the market's, gives a pooled elasticity of -0.27 (record-clustered standard error 0.03) on the unselected windows, the leverage effect, about zero on run-ups ($+0.05 \pm 0.04$) and on the consensus positives ($+0.02 \pm 0.08$), and $+0.59 \pm 0.10$ on the calm controls. So a shared model for the nuisance parameters on both sides removes the asymmetric fit of the scale, but that alone does not guarantee that the score fires more often, and in these daily data the level-elasticity signal these models target is weak and not specific to the consensus bubbles. A finite family failing in sample does not show that every bubble signal is absent, and no size or calibration follows from an in-sample comparison with $\log 20$.

Scope. A diagnosis, not a test: the in-sample comparisons use the evaluation data and are not e-values, and their optimizers are not certified globally (four records hit the variance-collapse bound and are counted as zero). Realized variance is a noisy proxy. One finite family failing to separate does not show that no bubble signal exists.

6.9 The repaired score on real data

The repaired score of Section 5.9, with its fitter unchanged, was applied to daily real data together with the drift-robust score and GSADF, all three calibrated by the conservative rank rule on one fixed reference: 999 windows ending in 1995–2010, judged calm on their 126 training returns alone (the stock's annualized volatility at most 0.30 and drawdown at most 0.15, and the S&P 500's at most 0.20 and 0.10 on the same dates), at most one per date within a name. Each window has 379 consecutive prices, 126 training and 252 scored returns, ending on a month start. The evaluation sample is 10,000 windows drawn at random from the unselected universe of Section 6.4, without any filter; windows whose whole input starts in 2011 or later form the holdout, and earlier ones are retrospective. The consensus records of Section 6.2 were rebuilt to the same fixed length at their original end dates.

Windows	Number	Repaired score	Drift-robust	Calibrated GSADF
all	10,000	4.9	0.8	5.9
holdout, calm starts	1,899	7.5	1.1	5.2
holdout, other starts	3,134	3.5	0.5	6.2
retrospective	4,967	4.7	0.9	5.9
consensus positives (of 33)	30 scored	3	0	6

Windows	Number	Repaired score	Drift-robust	Calibrated GSADF
calm controls (of 103)	103	20	2	6

Scope. Percentages of windows flagged at the 5% reference cutoff, except the last two rows, which are counts. All windows share one calibration, so flags are dependent and no binomial interval is given; the rates are empirical, not certified sizes, because no real-data law for the reference windows is established. Current S&P 500 constituents; daily adjusted closes.

The repaired score fires on real data, unlike the drift-robust score, whose cutoff falls on its tie at zero so that it flags almost nothing. Over the four periods used earlier its 5% flag rate is 4.0 to 4.9%, but by year it ranges from 1.4% to 16.2% (55 of 339 windows ending in 2018). Calm windows after 2010, judged by the same predicate as the reference, flag at 7.5%, so the 1995–2010 reference does not transport cleanly; that rate at one shared cutoff neither estimates a total-variation budget nor refutes the marginal guarantee, but it shows that a zero budget cannot simply be assumed. The score shows no useful prediction of forward drawdowns: regressed as a rank within its end date, its coefficient on the 126-day forward maximum drawdown has Driscoll–Kraay $t = 0.96$, and its out-of-sample R^2 on the holdout is about zero; its return slope at 126 days is negative and marginal ($t = -1.81$). GSADF keeps the clear positive association of Sections 6.4 and 6.5 ($t = 4.9$ and 5.5, out-of-sample R^2 about 1%), and in a regression with all three scores the repaired score adds nothing; the drift-robust score's rank goes with smaller drawdowns (t about -3). On the consensus records, optimistically, the repaired score ranks the bubbles below the calm controls (AUC 0.34, bootstrap interval 0.19 to 0.57, which includes one half) and flags 3 of 33 positives against 20 of 103 controls, in line with the near-zero or positive variance elasticities of Section 6.8; GSADF at the 1% cutoff flags 5 positives and no control.

So the repaired score, calibrated empirically, fires at about the nominal rate on average but shows no useful prediction of drawdowns and does not separate the consensus episodes, for this fixed daily algorithm; nonsignificance does not show that no signal exists, and a sequential version remains uncertified.

Scope. Ranks within end date give tied scores distinct ordinal ranks in symbol order (the repaired score is exactly zero on 6,550 windows); an average-rank variant gives drawdown t of 0.8 to 0.95 and return t of -1.4, not a replacement result. The consensus inputs differ from the original episode windows in length, and three positives lack a complete history. The Student-t normalization of the model's density is still being checked against its source.

6.10 The level signature within the day

Daily data could not see a variance that rises with the price (Section 6.8). Five-minute consolidated-tape bars, regular session only, were used to look for it within the day, on the run-ups of Section 6.3 and on calm controls over the same dates. The positives are the 2020–21 meme stocks (GameStop, AMC, Koss, BlackBerry), the 2019–22 growth names (ARK Innovation, Tesla, Zoom, Peloton) and 20 of the 21 bubble-narrative names of 2023–26 (one had too short a history); the controls are SPY, DIA and five calm Dow members for each of the 12 distinct windows, 84 records in all. Returns between consecutive five-minute bars exclude the open and the overnight move and are divided by a diurnal clock, the slot's mean squared return over the name's prior year (or over the record itself for 7 recently listed names). With β the exponent of variance in the price, three estimates were made on each record. Within the day, a Gaussian quasi-likelihood with a free variance for every day estimates β from moves within each session alone, removing all variation across days. Across days, the log of each day's deseasonalized variance is regressed on the log opening price, controlling SPY's variance on the same day and a smooth volatility

state, an exponentially weighted average of past days with a half-life of 20 sessions. And an in-sample check compares $\beta \in [-20, 20]$ with $\beta \in [-20, 0]$, a null that allows leverage, with variance proportional to the state, the clock and S^{β}.

Group	**Records**	**Within-day** β	$t > 2$, $t < -2$	**Across-day** β	**Check above** $\log 20$
meme stocks, 2020–21	4	2.35	4, 0	0.65	3 or 4
growth names, 2019–22	4	−3.59	0, 4	−0.01	1
narrative names, 2023–26	20	1.48	12, 4	−0.04	7
SPY and DIA	24	−20	0, 24	−1.56	0
calm Dow members	60	−5.94	0, 28	−0.23	13

Scope. Medians of the estimated variance exponent per group, and counts of records. The t statistics are model-based, from the quasi-likelihood's curvature; the across-day errors are Newey–West. Within-day estimates for 23 of the 24 index records stop at the search bound of −20, which does not show that the exponent lies below it. The meme row's count in the last column is 4 in the original run and 3 in an independent reimplementation, which puts GameStop's 2020–21 statistic at 2.0 rather than 3.02. Declared bar closes, a median 50% of them sub-cent for the positives and 30% for the controls; not a rounding-exact likelihood.

Within the day, the meme and narrative run-ups show the sign a strict local martingale needs. Variance rises with the price in all four meme stocks (t about 5 to 10) and in 13 of the 20 narrative names, while calm names and the indices show the usual intraday leverage effect, variance rising as the price falls, down to the search bound for the index funds. Each positive's estimate exceeds the mean of its own window's controls in 26 of 28 records, a median gap of 11, understated where the controls are censored. The growth names, and Nvidia, Palantir and MicroStrategy among the narrative names, show leverage instead. Across days, with the stock's own volatility state controlled, the elasticity is much smaller, with medians from −0.46 (controls) to 0.65 (meme stocks), and the across-day t exceeds 2 in only 4 of 28 positives against 11 of 84 controls. The in-sample check exceeds $\log 20$ in 11 or 12 of the 28 positives (39 to 43%, depending on conventions that the design leaves open) against 13 of 84 controls and no index record; the smooth state absorbs most of the slow co-movement of level and variance, so the check is weaker than the within-day estimate. Section 6.11 qualifies both readings: the own-state estimators understate a simulated level effect, and a return–volatility correlation alone produces a large within-day estimate. Measured from onset to peak, the run-up segments alone are weaker still (median within-day β of 0.2 to 0.6 over 19 segments, sensitive to where each segment starts).

A positive within-day elasticity does not identify a level-dependent diffusion coefficient. An inverse leverage effect in attention-driven trading, volatility rising on up-moves through order flow, would produce the same sign, and this design does not separate the two. Tick discreteness is argued against rather than tested: bid–ask bounce adds variance in proportion to the squared tick-to-price ratio, which pushes β negative, not positive. A strict local martingale can also arise through the correlation of price and volatility in a stochastic-volatility model rather than through the level ([75]), which a level elasticity does not measure.

Scope. Descriptive and in-sample, not a test: the check uses each record's own data, and no false-alarm rate is estimated. Consensus and press labels; survivors and current listings only; five-minute closes, which see the diurnal pattern only through a fitted clock. The sub-cent shares were computed on split-adjusted closes, which can create sub-cent values before a split; the group medians are the same on raw closes.

6.11 Checking the estimators

Two features of the estimators could explain the weak signals of Sections 6.8 and 6.10: controlling the stock's own smooth volatility state may absorb a slow level effect, and the daily estimators of Section 6.8 fit one power law across run-up and crash. Both were checked on simulated paths with a known answer, using the same estimator code. Each path has two years of 78 five-minute bars a day, the first used only for the diurnal clock, which is U-shaped; an overnight move carrying a fifth of a day's variance; and log-normal stochastic volatility, mean-reverting with a half-life of 20 sessions, an annual level of 0.40 and a stationary standard deviation of 0.5 in log variance, multiplied by $(S/S_0)^{\beta}$. Its shocks have correlation ρ with the price's. There are seven cells of 20 paths each; a path whose log price leaves $[-5, 5]$ is kept as a failure (three in the $\beta = -2$ cell, none elsewhere).

True β, ρ	Within-day	Across-day, own state	Check, own state	Daily fit (Section 6.8)
−2, 0	−2.52	−1.24	−1.03	−1.84
0, 0	−0.09	0.43	0.37	0.32
0, −0.7	−4.14	−1.07	−0.38	−1.03
1, 0	1.10	1.02	0.73	0.64
2, 0	2.74	1.08	1.02	2.01
2, 0, drift 0.8	1.89	1.36	0.79	1.78

Scope. Median recovered β over 20 simulated paths per cell (17 in the first row); with 20 paths a median has a standard error of about 0.2 to 0.4. An independent reimplementation with 100 paths gives medians of −0.01 to 0.14 at $\beta = 0$ (standard errors 0.08 to 0.19) and, at $\beta = 2$, 1.95 within the day, 1.36 and 0.83 for the two own-state estimators and 1.87 for the daily fit (standard errors 0.13 to 0.21). One volatility model and one clock; Gaussian overnight moves.

The estimators recover a true level effect in sign and roughly in size. The daily fit of Section 6.8 recovers $\beta = 2$ (1.87 to 2.01), so its pooling across a whole record is not what hid the signal there. The estimators that control the stock's own state come out below the truth at $\beta = 2$, at about two-thirds across days and two-fifths for the check; no version without the state was simulated alongside, so the cause of the shortfall is not isolated. At $\beta = 0$ the estimates are centred near zero but spread widely, so a single record's sign is weak evidence. The within-day estimate is not specific to the level: with $\beta = 0$ and a return–volatility correlation of −0.7 it is negative on every path, with median −4.1. The within-day contrast of Section 6.10, calm names strongly negative and the meme run-ups positive, is therefore equally consistent with a difference in the sign of that correlation.

On the real records of Section 6.10, the same quantities were recomputed controlling only SPY's same-day variance and the diurnal clock, not the stock's own state. SPY's same-day variance is a descriptive covariate, not a forecast available in real time.

Group	Records	Across-day β	Check β	Check above log 20	Daily correlation
meme stocks, 2020–21	4	0.78	1.31	4	0.26
growth names, 2019–22	4	−0.02	−0.02	2	−0.04
narrative names, 2023–26	20	−0.02	0.24	10	0.21
run-up segments	19	0.51	0.75	15	0.11
calm controls	84	−0.47	−0.75	21	−0.14
SPY and DIA	24	−0.67	−1.12	3	−0.17

Scope. Medians over records and counts of records. The last column is the median correlation, per record, between a session's open-to-close return and the change in its deseasonalized log realized volatility from the previous session, over 68 to 858 session pairs per record, with no per-record interval. Run-up segments run from the record's low to its peak, chosen with hindsight. The no-state estimators were not checked in simulation.

Without the own state, the across-day association is positive in the meme stocks and in 84% of the run-up segments, and negative in calm controls. But the meme median was already 0.65 with the state controlled, and only 2 of the 4 meme slopes rose; the check also changes how SPY enters, and no matched simulation of the no-state estimators isolates what the state removed, so the attenuation is shown in simulation, not on these records. The check exceeds $\log 20$ in 21 of 84 controls, a reminder that it is in-sample, not a test; its null fit reports a bound hit in 64 of 131 fits, at an end the output does not record. At the daily scale, the meme and most narrative run-ups have a positive correlation between returns and changes in estimated volatility, and calm names a negative one (positive in 19 of 28 positives against 18 of 84 controls), in the ranges that simulated level effects (0.05 to 0.17) and leverage (-0.18) produce; the five-minute and hourly versions have small medians in every group, and no rejection rule was set at any scale.

None of this identifies a mechanism. The measured correlation is between finite-interval returns and estimated volatility changes, not the instantaneous correlation of drivers in Sin's stochastic-volatility model ([75]), whose strictness condition is specific to that model; and by Proposition 20, one physical law of prices can admit both a pricing measure under which the price is a martingale and one under which it is strict, so no measurement of the physical law alone settles the label. The measurements are consistent with a level-dependent diffusion, with positively correlated stochastic volatility, and with other mechanisms such as order flow, or the high uncertainty about future profitability that Pástor and Veronesi ([65]) argue could explain both the valuations and the high return volatility of Nasdaq stocks in the late 1990s.

Scope. Simulation with one volatility model, a discretized hybrid of a level effect and log-normal stochastic volatility that is not Sin's model and not a certified strict-local-martingale sample; design-used episodes and hindsight segments; five-minute closes with sub-penny prints; descriptive diagnostics, not tests.

6.12 A real-time monitor on five-minute data

Sections 6.10 and 6.11 suggest a monitor that looks for the level effect within each session against a volatility state that moves only slowly. One was built from fixed choices and replayed on the five-minute bars, as if run live with a one-minute delay. At each monthly launch it fits the nuisance on the 252 prior sessions alone, under $\beta = 0$: the drift, the initial variance state and the degrees of freedom of Student-t innovations. The model has a prior-year diurnal clock and a day-level variance state that updates once per completed session with a fixed half-life of 20 sessions. It then freezes these and scores the next 21 sessions, 1,617 five-minute checkpoints. The level is measured from each session's first bar, so the variance at a checkpoint is the clock times the state times $(S/s_d)^{\beta}$, with s_d that session's first price. The evidence at each checkpoint is the log ratio of an equally weighted mixture over eight values of β from 1/8 to 16, accumulated over all sessions so far, to the numerically profiled maximum over $\beta \le 0$ on the same prefix. A missing bar ends the launch's supported prefix. The alarm is the first checkpoint whose transformed evidence strictly exceeds a cutoff set by the conservative rank rule of Proposition 45, from the path maxima of 499 calm-start references in 2017–18, at most one per name per quarter; the cutoff corresponds to a log ratio of about 10. All targets start in February 2019 or later, after every reference has finished. GSADF was run in monitoring form on daily closes, over the 252 prior sessions and the sessions so far, with its alarm at the first session end above its own reference cutoff at the same rank.

Launches	Number	Monitor	Calibrated GSADF
run-up records, all	449	19.6	40.5
meme stocks, 2020–21	52	46.2	67.3
narrative names, 2023–26	233	21.0	45.1
growth names, 2019–22	164	9.1	25.6
same-date calm members	805	4.2	0.0
same-date SPY and DIA	322	0.6	0.0
unselected 100-name draw	8,600	5.0	5.4

Scope. Percentages of launches with an alarm within 21 sessions, at a reference cutoff of rank 475 of 499 ($\alpha = 1/20$). Launches that fail the prior fit stay in the denominators (109 of the run-up launches and 609 of the draw), every one for lack of 200 sessions with at least 70 bars in the prior year. Control rows reuse the same name and launch across episode windows: over the 610 distinct name-launches the monitor alarms on 30 (4.9%). Rates are empirical, not certified, because the premise of the rank rule, that the references and the targets share a law after selection, is not established; the run-ups were already used in Sections 6.10 and 6.11. Current listings only.

The monitor fires on real five-minute data. On the run-up records it alarms on a fifth of launches and on almost half in the meme stocks; on calm controls over the same dates its empirical rate is near the nominal 5% (4.2% of members, 0.6% of SPY and DIA), and on the unselected draw it is 5.0%, ranging by year from 2.3% (2022) to 9.8% (2020). As an alarm it is weaker than calibrated GSADF on the episodes. GSADF alarms on twice as many run-up launches and on no calm control. It alarms strictly before the record's peak in 20 of 29 records, with a median lead of 57.5 sessions, against 14 for the monitor, with a median lead of 142. But the two do not see the same evidence: 163 of GSADF's 182 run-up alarms come at the first monitored session, from the 252 prior closes, while the monitor's evidence starts at launch, so this is a comparison of procedures, not of speed on equal information. The monitor's early alarms are often far from the top: Tesla alarmed in January 2020 for a peak in November 2021, and in 7 records the first alarm precedes the run-up's onset. Nine records never alarm, seven of them because every launch failed the prior fit.

The monitor's score over a completed window carries modest information about drawdowns after it. On 8,298 launches of the unselected draw with forward data, its rank within the launch date has Driscoll–Kraay t of 4.55 and 3.47 for the maximum drawdown over the next 63 and 126 sessions, with out-of-sample R^2 of 0.39% and 0.31% on launches from 2023 (block-bootstrap standard errors 0.08 and 0.09). Together with GSADF, out of sample, it explains 1.39% and 1.74%, against 1.05% and 1.49% for GSADF alone. Those gains of 0.34 and 0.25 points have no paired uncertainty, and the full-sample t values are not out-of-sample tests. Neither score predicts forward returns. This concerns ranks of the whole window's score with outcomes after the window, not alarms. Ties in the rank (8.4% of scores) are broken in symbol order; with averaged ranks the t values are 4.48 and 3.39.

Scope. A counterfactual replay on split-adjusted closes, assuming that splits and the calendar were known in time. The score is a numerical likelihood ratio, not an e-value, and no sequential guarantee is claimed. The calibration is per launch, so it gives no control over repeated launches, a record or a panel. The run-ups are labelled by consensus and the press, and the same-date controls are descriptive.

6.13 The daily market-scale statistic on real data

The daily statistic of Section 5.10 was replayed on real adjusted daily closes from 1994 to 2026, unchanged. Its universe is the current S&P 500 constituents of Section 6.4, and it was also run on the consensus and collapsed-name episodes of Section 6.2 and the 2024–26 run-ups of Section 6.10. Each name is launched on the first session of every month, with 20 sessions to start the market scale, 252 prior sessions for the fit and 252 scored sessions. Calibration uses only the past. For each target month, the references are up to 499 universe launches whose scored year ended strictly before the target's launch and that pass the calm-start predicate of Section 6.9 on their training returns; they are taken in a fixed hash order, at most one per name per quarter. Their scores give the strict rank cutoff at $\alpha = 1/20$, separately for the batch score, the monitor's path maximum and GSADF's path maximum. The first year has no completed reference, so its 3,197 launches cannot flag; from 1996 the cutoffs are stable (batch 0.94 to 0.96, GSADF 2.88 to 3.25).

Launch period	Daily statistic, all	Calm starts	GSADF, all	Calm starts
1994–99	6.3	8.9	8.2	6.4
2000–02	1.8	0.0	7.1	8.8
2003–07	4.3	4.9	4.5	3.4
2008–09	0.9	0.2	5.9	3.9
2010–19	2.0	2.8	6.2	5.2
2020–21	1.4	1.5	14.7	5.3
2022–26	2.4	2.3	4.9	3.3
all, 153,716 launches	2.9	3.8	6.7	4.8

Scope. Percentages of launches flagged by the batch score and by GSADF at the reference cutoff (the monitor's rates are within 0.8 points of the batch score's). Periods classify launch dates, and the following year is scored. All launches share overlapping references, so flags are dependent and no binomial interval is given; there are no null labels, so these are flag rates, not false-alarm rates. Current constituents; a retrospective replay.

The statistic flags fewer launches than GSADF overall, and far fewer in turbulent years (0.9 to 1.9% of launches in 2000–02, 2008–09 and 2020–21, where GSADF reaches 14.7%). Its rate is highest for launches in the 1990s, 8.9% of calm starts, so calm references from earlier years do not give it a uniform rate even on calm starts, which are themselves references for later months. Without null labels, neither the low rates in turbulent years nor the high ones in the 1990s show which flags are false, or that the statistic is specific to bubbles; Section 5.10 found that its estimate responds to correlated stochastic volatility.

Its rank within the launch date has a small in-sample association with deeper drawdowns over the next 63 and 126 sessions, for the monitor (Driscoll–Kraay t of 3.4 and 3.4) more than for the batch score (t of 2.7 and 2.3, or 2.0 and 1.8 when the two-thirds of batch scores that are exactly zero share an averaged rank instead of being ordered by row). Out of sample, on launches whose prior year starts in 2011 or later, its R^2 is 0.03 to 0.16%, within two bootstrap standard errors of zero, against 0.93 to 0.98% for GSADF. Together the two explain 1.02% and 0.98% against GSADF's 0.98% and 0.93%, gains with no paired uncertainty. Neither predicts returns. On the 9,840 windows of Section 6.9 where both can be computed, its batch score has a rank correlation of 0.34 with the repaired score; it is significant for later drawdowns in sample (t about 2.1 to 2.2) where the repaired score is not, though that is not a test of the difference.

On the episodes it often alarms early, sometimes years early, and long leads are not by themselves useful timing. Of the 30 consensus positives with a US calendar and a complete prior year, the monitor alarms

before the peak on 15, a median of 377 sessions before, and GSADF on 14, a median of 142. The monitor alarms before the peak on the Nasdaq indices from 1995, on all four large homebuilders 377 to 411 sessions before, and on Lehman and Countrywide among the collapsed names. It alarms only after the peak on the 2007 financials of the consensus list. Of the 18 run-ups of 2024–26 with a daily prior year, it alarms before the peak on 7 and GSADF on 10. On the consensus negatives it alarms on 4.8% of 3,556 launches (GSADF 3.9%), and on at least one launch for 35 of 91 records. The two do not see the same evidence: GSADF's path includes the prior year, and 315 of its 650 alarmed episode launches alarm on the first scored day, against none for this monitor. Against the five-minute monitor of Section 6.12, on the 29 records both cover, both alarm on 14, where the daily monitor's first alarm is earlier on one and the five-minute monitor's on 13; the daily monitor alone alarms on 5, the five-minute monitor alone on 6, and neither on 4. Peaks are defined differently for three records, and the two use different data, horizons and information.

Scope. A retrospective replay on current constituents; calibration is empirical, and stable cutoffs from calm references do not establish the premise of the rank rule, nor control over repeated launches or combined scores. The episodes were already used in earlier sections, three consensus records (two on non-US calendars, one before SPY's history) and four collapsed names could not be scored, and one market scale (SPY) was used.

6.14 Summary

On real prices, no procedure tested both keeps false alarms down and finds the consensus bubbles. Most of the flags that the tests do raise reflect volatility rising through the window rather than jumps (Section 6.6). Scaled by the implied-volatility index, the score is silent on SPX and NDX, including 1999–2001 and 2020–21, while GSADF flags the dot-com SPX record (Section 6.7). The drift-robust test's silence comes from its construction, a training-chosen forecast that is often itself a null law, and a forecast scale fixed on training data against a null refitted on evaluation data, but even in sample a level-elasticity alternative fires on the consensus bubbles no more often than on calm controls, 12.1% against 11.7% (Section 6.8). A repaired score that shares the nuisance model on both sides fires at about the nominal rate on real data, but shows no useful prediction of drawdowns and flags calm controls more often than consensus bubbles (Sections 5.9 and 6.9). At five-minute resolution, variance rises with the price within the day in the meme run-ups and most press-labelled run-ups of 2023–26, the opposite of the leverage seen in calm names and the indices (Section 6.10); the estimators recover a known level effect in simulation, but the within-day estimate also responds to the correlation of returns with volatility, and at the daily scale that correlation is positive in these run-ups and negative in calm names, which identifies no mechanism (Section 6.11). A real-time monitor built on that within-day effect fires on real five-minute data at near-nominal empirical rates on calm controls, and its window score adds a little to GSADF's prediction of drawdowns, but as an alarm it is weaker than calibrated GSADF on the run-ups (Section 6.12). A daily statistic that takes its volatility scale from the market flags fewer launches than GSADF, especially in turbulent years, and often alarms years before consensus peaks, but its drawdown association is small and does not survive out of sample (Sections 5.10 and 6.13). The drift-robust test flags no calm control on daily data, and on daily, collapsed-name and hourly data it flags only the most extreme short episodes, where a jump explanation cannot be excluded; on hourly data, with a calendar clock that ignores the intraday pattern, it also flagged one calm name; this is the behaviour Section 5.2 predicts for daily data, not a finding about the pricing laws of these episodes. The existing tests flag more positives but also flag calm controls at the rates measured above: GSADF 13 to 19% of calm daily names and 13 of 98 calm hourly ones, and JKP's Method 2 most names whatever their label. On an unselected universe the selected separation of Section 6.2 does not carry over: the drift-robust test fires on about 0.1 to 0.2% of windows and its ranking does not predict drawdowns,

while GSADF fires on 12 to 52% of windows by period and its ranking predicts about 1% of the variation in forward drawdowns out of sample. With critical values set from the calm windows (Section 6.5), GSADF's held-out false-alarm rate on calm records is about nominal, and it still flags 59 to 76% of the consensus positives; on the unselected universe that threshold flags 6 to 7% of shorter windows in calm years but about 15% of 72-month ones, and its flag carries the same small drawdown signal. What the simulations and the data leave open is the volatility that changes within a window, which the null either ignores, and false-alarms, or absorbs, and loses its power.

7. Related work

Martingale criteria for stochastic exponentials. The general criterion we apply is [60], Theorem 2.1. Their Proposition 4.2 writes the discounted price as $s_0 Z$ with $b(x) = \sigma(x)/x$, the case $\pi = 1$ of our representation, and their Corollary 4.3 is the JKP criterion. That criterion has several earlier proofs, which [60] (p. 14) lists; we cite it in their form, which needs no local boundedness of σ or $1/\sigma$. The explosion test behind both is Feller's. We use it in the endpoint form recalled in [60], equations (20)–(23), and the global form of [45], Appendix A. Karatzas and Ruf [45] give the Feller test function for the drift family $s\,b$ with $b(x) = a\,s(x)/x$ in closed form (their equation (7.4)), which is exactly the auxiliary diffusion here with $a = \pi$; their Proposition 7.1 records the case $a = 1$ (our $\pi = 1$) and the exit of S at zero used in Corollary 2. Neither source states the classification of the constant-proportion exponential across π, or the zero-endpoint correction. For uniform integrability over an infinite horizon, [60] (Theorem 2.3) gives the one-dimensional criterion, and Ruf ([73], Proposition 3.1 and Theorem 3.3) characterizes both the martingale property and uniform integrability of stochastic exponentials of functionals of a possibly multidimensional, time-dependent equation, through an energy integral under an auxiliary measure that always exists; Propositions 4 and 5 apply the two to constant and to time-varying exposure.

Strict local martingales and bubbles. The bubble definition is that of [44]. Kardaras, Kreher and Nikeghbali [47], Section 3.1, use the same auxiliary drift $\mu+\sigma b$ and point back to the general criterion; their discussion of positive diffusions imposes a divergent integral at the lower endpoint, which is consistent with our finding that zero must be checked separately. The literature on investment when an asset has a bubble, for example [34] and [35], may contain special cases of Theorem 1; we have not yet compared it. Stochastic portfolio theory studies rebalanced portfolios in markets where the deflator, and with it the discounted prices, can be strict local martingales ([28], Section 9; [46]). Its setting is relative arbitrage without a common pricing measure under which every price is a local martingale, so its results do not transfer to Theorem 1 or to Section 4.2 without an argument, and none is given here.

Bubble detection and identification. Jarrow, Kchia and Protter [41] test for bubbles by estimating σ nonparametrically on the visited range and extrapolating its tail; Proposition 7 makes exact the obstruction they identify, and their extrapolation methods are the kind of structural assumption that Proposition 7 shows to be necessary. The nonparametric estimators such tests rely on recover σ only where the price has been: the precision of the estimator of [29] at a level is governed by the local time there, and the kernel rates of [39] hold pointwise and on compact sets. Bandi and Phillips [10] build their estimators on the occupation formula, which they state for a general continuous semimartingale (their Lemma 3); Proposition 23 uses only that formula, so it needs none of the recurrence assumptions of their estimation results, and its denominator is the semimartingale local time rather than their chronological local time. Nothing in that literature reaches the tail that decides the JKP integral (the visited part is identified exactly through local time, Proposition 23), which is the observation Propositions 7 and 17 formalize. Tests for an explosive autoregressive root [66] target a different notion of bubble, in which the price grows explosively relative

to fundamentals; a strict local martingale need not have an explosive root, and Theorem 18 says nothing about those tests. When volatility varies over time, the limiting distribution of these tests depends on the variance function, and Kurozumi, Skrobotov and Tsarev ([49]) restore a pivotal limit by deforming the time domain; this fits the over-rejection of GSADF under level-dependent volatility in Section 5.2. Greenwood, Shleifer and You ([32]) study bubbles as observed run-ups and crashes: for US industry portfolios from 1926 and international sectors, a sharp run-up does not on average predict low returns, but it does predict a higher probability of a crash, a drawdown of 40% within two years, and the run-up's volatility, turnover, issuance and price path help forecast it. Obayashi, Protter and Yang ([63]) date bubbles and their lifetimes by smoothing a daily JKP signal from intraday prices with a hidden Markov model. Section 5 compares these four procedures with the tests of Section 3.5 in simulation, and Section 6 on real prices. Pooling a tail across names has precedent only for the tail index of returns ([48]) and for repeated paths of one process ([25]); neither supplies the common-exponent model of Proposition 26. For merging e-values across names, Vovk and Wang ([80], Proposition 3.1) show that under arbitrary dependence the arithmetic mean essentially dominates every symmetric merging function, so the evidence does not accumulate with the number of names, while the product of independent e-values is an e-value (their Section 4). This is consistent with the negative pooling result after Proposition 41, whose null allows any correlation across names. In the panel bubble literature, Liu, Phillips and Yu ([53]) gain power by aggregating clusters of similar series, and note that a factor-based panel approach allowing cross-section dependence does not enhance discriminatory power; their tests concern an explosive root, not a strict local martingale. We use the term identified set as in the partial-identification literature (for example [19]), but define it directly as the image of the compatible class; no estimation or inference result from that literature is used, and we are not aware of an application of identified sets, or of a testing impossibility result, to bubbles in the strict-local-martingale sense. Classical results on the nonexistence of nonparametric tests ([8]) may anticipate the abstract form of Theorem 18; we have not yet checked their hypotheses against this problem. The bounded-stopping and supermartingale facts behind Theorem 18 and Proposition 19 are as recalled in [38], Section 2.

Calibrated likelihood inference under misspecification. Propositions 27–42 and 50 apply split-likelihood universal inference ([82]) and e-values with e-BH ([81]), whose guarantees assume the model is correct. Park, Balakrishnan and Wasserman ([64], Section 3.1) show that when it is not, the split likelihood-ratio set can fail to cover even the Kullback–Leibler projection of the true law, because likelihood ratios are unbounded; their robust construction targets other projections, which need not carry a bubble label. This is consistent with the nearly vacuous bounds of Proposition 43, and we found no source linking a projection's exponent to the true law's bubble label. Approximate likelihoods for discretely sampled diffusions ([1]) and for stochastic volatility ([2], which replaces the latent volatility by the implied volatility of a short-dated at-the-money option) give accuracy, not the finite-sample validity that Proposition 27 needs. Microstructure noise at high frequency ([5]) and jumps, for which a test exists ([7]), are further departures from the catalogues' noiseless diffusion observations. Jumps also change what a bubble criterion must look at: Hulley and Ruf ([38], Example 3.2) construct a nonnegative pure-jump martingale that is not uniformly integrable because its positive jumps grow combinatorially, and their conditions involve the jumps at exit times, not only a volatility function, so the diffusion criteria behind our labels do not cover jump models. A null enlarged by jumps or regime switches has a mixture likelihood that can be unbounded; the split-likelihood test needs only an upper bound on the null maximum, and its authors suggest powered likelihoods for robustness and smoothed likelihoods when the likelihood is unbounded ([82], Section 6). We found no source giving a likelihood test with a finite-sample guarantee for a jump-diffusion or regime-switching null for a bubble. Lions and Musiela ([52]) give necessary and sufficient conditions, through parameters such as the correlation, for classical stochastic-volatility models to have integrable solutions. Dandapani and Protter ([68]) note that such results depend at one key stage on Feller's test and so apply

only in one dimension, and use Lyapunov functions, in the manner of Khasminskii's test for explosions, to give sufficient conditions for components of a multidimensional system with stochastic volatility to be strict local martingales or martingales.

Intraday structure and observation noise. The high-frequency literature treats microstructure noise and rounding as problems for estimating integrated variance, not as threats to the size of a test. Aït-Sahalia, Mykland and Zhang ([3]) show that modelling the noise explicitly restores the benefit of sampling as often as possible, and that the volatility estimator keeps the same variance even if the noise distribution is misspecified. Li and Mykland ([51]) model the contamination as a Markov kernel on the latent price, including additive error and rounding, and find that the robustness of the two-scales estimator deteriorates as the kernel becomes less smooth, as with rounding. On magnitudes, Aït-Sahalia and Yu ([4], Table 6) estimate an average noise standard deviation of 5 basis points across NYSE stocks from 1995 to 2005, from transaction data. The two-point error with a persistent sign of Proposition 62 is the bid–ask structure written by Diebold and Strasser ([23]) as the efficient price plus half the spread times the trade direction, and Jacod, Li and Zheng ([40]) find real noise moderately positively autocorrelated, which is why persistence belongs in such a declaration. Lou, Polk and Skouras ([55]) find that overnight and intraday returns behave as different components, in their means, so an overnight gap is not just a longer intraday step. We found no source giving a test with a finite-sample guarantee under declared noise, a deterministic time-of-day pattern or overnight gaps. The exact cell-probability correction of Proposition 55 is the forward, or filtering, recursion of a hidden Markov model with interval observations. Particle filters exist for diffusions observed at discrete times, with an Euler discretization ([61]) or without time discretization ([27]), but they approximate conditional expectations of the state under observation densities. We found no certified bound on the likelihood of interval observations of a diffusion, which is what the denominators here would need. On clocks shared across names, which Proposition 36 assumes, Campbell, Lettau, Malkiel and Xu ([17]) find that firm-level volatility rose relative to market volatility over 1962–97, so that correlations among individual stocks fell. Herskovic, Kelly, Lustig and Van Nieuwerburgh ([36]) find that a single factor explains 35% of the time variation in firm-level idiosyncratic volatility. So a market clock leaves a large firm-level remainder, whose volatilities share a further common factor; both bear on that premise. Nonparametric estimators of a time-varying volatility assume it constant over an interval of time homogeneity ([58]) or a smooth path observed with noise ([37]); neither lets the volatility also depend on the price level, so neither addresses the near-confounding of time variation and level dependence found after Proposition 48 and in Section 5.3. Tests of the form of a volatility function exist. Dette and Podolskij ([22]) test a parametric null for $\sigma(t, X_t)$, such as constant volatility, from discrete observations, and detect local alternatives at rate $n^{-1/2}$ in a fixed direction; Bull ([16]) detects deviations of volatility-like processes from a parametric null at the near-optimal rate $n^{-1/4}$, up to logarithms, when the deviation is itself a semimartingale. Neither has a nonparametric time effect as a nuisance, which is the problem of Propositions 49 and 50. Spot-variance estimators converge pointwise at about $n^{-1/4}$ ([62]), slower than the $n^{-1/2}$ that the array bound of Proposition 49 needs on the grid where the bracket is computed.

Sequential testing and real-time monitoring. The monitored score of Proposition 69 is a filtered, conditional version of the running sequential likelihood ratio of Wasserman, Ramdas and Balakrishnan ([82], Section 8, Theorem 9). That ratio has a non-anticipating plug-in numerator and a null maximum in the denominator, with a type-I guarantee uniform over time; they note that with nuisance parameters or conditional likelihoods the underlying process may be only a supermartingale. Ramdas, Grünwald, Vovk and Shafer ([71], Section 3.2.1) trace the plug-in method to Wald and note that unsmoothed maximum-likelihood plug-ins can assign zero probability to what happens. They also recall (Section 2.9) that e-values and e-processes may be averaged with weights fixed before the data. Proposition 71 instead funds

new tests, launched at stopping times from one budget, each scoring only the data after its launch; with fixed dates and weights it reduces to such an average. Ramdas, Ruf, Larsson and Koolen ([70]) show that admissible anytime-valid inference must rest on nonnegative martingales. In the bubble literature, Whitehouse, Harvey and Leybourne ([83]) state the multiple-testing problem of real-time monitoring, in which repeating a test as each observation arrives inflates its size. The real-time detectors of Phillips, Shi and Yu ([67]) compare a scan statistic at each date with a critical value sequence and control false positives asymptotically, by letting that sequence grow, rather than bounding the probability of ever crossing in finite samples as Propositions 69–71 do. For the daily scales of Propositions 37–39, Pérez-Ortiz, Lardy, de Heide and Grünwald ([69]) show that the likelihood ratio of the statistic invariant under a group is a growth-rate-optimal e-statistic, a Bayes factor with a right Haar prior. With the scaling group this removes an unknown scale without estimating it; it applies to scale shifts of a fixed within-day structure, not to volatility that depends on the price level within a day. Shin, Ramdas and Rinaldo ([74]) build e-detectors, sums of e-processes started at consecutive times, with nonasymptotic bounds on the average run length between false alarms and, in some cases, on the delay after a change; the restarts of Propositions 70 and 71 are e-processes started at declared or stopping times, with a bound on ever crossing over a fixed horizon rather than a run length. In change-point detection, Lorden and Pollak ([54]) apply the same Robbins–Siegmund idea as the plug-in forecasts here, estimating the unknown post-change parameter from past data only inside CUSUM and Shiryaev–Roberts rules, which keeps the martingale property under no change; Mei ([57]) treats unknown parameters in the pre-change law, minimizing the frequency of false alarms uniformly over them for a required delay, the detection-side counterpart of the composite nulls of Section 3.5.

Empirically calibrated thresholds. The calibrated thresholds of Proposition 45 and of Section 6.5 are conformal in form. Without exchangeability, Barber, Candès, Ramdas and Tibshirani ([11]) bound the loss of coverage by a weighted total-variation distance between the data vector and its versions with the test point swapped with each calibration point, with no assumption on the joint law; no value of that distance is known for overlapping windows of one price path. For counterfactual inference with time series, Chernozhukov, Wüthrich and Zhu ([20]) obtain approximate validity under stationary weakly dependent errors by permuting blocks, and Gibbs and Candès ([30]) give up the guarantee for each new window in exchange for a long-run frequency of errors that holds under arbitrary distribution shift. For many test points, Bates, Candès, Lei, Romano and Sesia ([13]) show that conformal p-values are positively regression dependent on the inliers when the test points are independent of each other and of the calibration data, so the Benjamini–Hochberg rule applies; overlapping windows do not meet that premise, which shows only that the theorem does not apply. None of these gives a guarantee for calm windows selected from the realized path.

Stochastic volatility and incomplete markets. Sin [75] shows that in a class of stochastic-volatility models the price can be a strict local martingale under one pricing measure while an equivalent measure makes it a martingale; In that model the sign of $a \cdot \gamma$, which decides strictness, is the sign of the correlation between price and volatility shocks. Proposition 20 applies Sin's construction with the physical law held fixed. Sin treats the price itself, not constant-proportion wealth, so it does not anticipate Theorem 1, but it is the nearest source for the stochastic-volatility question that Theorem 1 leaves open. The measure-dependence of bubbles in incomplete markets is discussed by Jarrow, Protter and Shimbo ([44]; [43], unread), and their treatment may already contain the qualitative point.

Option prices under bubbles. Which option prices are observed matters once the underlying can be strict. Under expectation pricing, put-call parity fails by exactly the defect, $E_Q[(X_T-K)^+]-E_Q[(K-X_T)^+] = E_Q[X_T]-K$ ([47], Example 4.8), and this residual is what Proposition 72 uses. Carr, Fisher and Ruf price by

minimal superreplication instead, which adds the defect to the call and restores parity ([18], Corollaries 3.7 and 3.8); under that operator the residual vanishes and the test has no power. American calls differ again: parity holds with the American call in place of the European one, and early exercise can be optimal ([14], Section 2.2), and European prices need not preserve convexity in the presence of bubbles ([24]). Every option-based statement in this paper is therefore made under a declared operator: European expectation pricing in Section 4.1, the American exercise value in Section 4.3. The restriction of Proposition 77, a pricing kernel that depends on the index level alone, is the kind of object Rosenberg and Engle ([72]) estimate as an empirical pricing kernel, the state price per unit probability, from a daily cross-section of S&P 500 option premia and a GARCH-type physical model; that is an estimate under a parametric physical law, not a test of the restriction, and we know of no estimator for the killed-weight version beyond the synthetic one of Proposition 78.

Leveraged funds. A continuously rebalanced leveraged fund with leverage β is the exponential $\mathcal{E}(\beta \int dS/S)$, and its options are priced in [50] under the assumption that S is a strictly positive martingale. Theorem 1 treats $\beta = \pi > 0$ for local volatility without either assumption, but only in this continuous-rebalancing idealization. Funds traded on exchanges rebalance daily and pay fees and financing costs; [6] gives the exact daily relation between fund and underlying returns with these terms. Section 3.3 uses it. The threshold $1/2$ and the lower-endpoint case of Theorem 1 hold at most as continuous-rebalancing limits and should not be read as properties of such funds (Propositions 11 and 12). For a strictly positive price, the finite-grid funds do converge to the continuously rebalanced wealth (Proposition 13), without assuming the price is a martingale, which [50] assumes for the continuous model; for prices that can reach zero no limit is known. Jarrow [42] analyses the risk of leveraged funds; we have not read it, and we do not claim that the discrete-rebalancing results of Section 3.3 are new.

8. Outlook

What is not done:

- **The wealth identification (W).** We classify the stopped exponential $v_0 Z^\pi$. We do not prove that it is the wealth of an admissible constant-proportion strategy, and we do not define holdings after the price reaches zero. A different convention for wealth at default is outside Theorem 1.
- **Exact endpoint values.** The Feller test functions of the auxiliary diffusion satisfy $\tilde{v}_\pi(\infty) = I_\infty/(2\pi - 1)$ for $\pi > 1/2$ and $\tilde{v}_\pi(0) = I_0/(1 - 2\pi)$ for $0 < \pi < 1/2$, by monotone convergence from (4) and (5). The proof of Theorem 1 uses only the equivalent integrability tests, and these identities are not recorded as results here; the formal proof computes the limits as integrals but does not state the identities.
- **Other exposures.** For Theorem 1, $\pi \leq 0$ (short and inverse positions) and time-dependent or random π are not covered; Proposition 5 gives sufficient conditions for bounded predictable exposure and a classification only for geometric Brownian motion with a deterministic schedule. Two independent assets, each with its own endpoint test, are treated in Proposition 6; correlated assets and liquidation through default are open. Section 3.3 treats one holding interval for every real π and finite grids over a martingale price for $0 \leq \pi \leq 1$. The uniform integrability needed for bubble sizes to converge in Proposition 13 is proved when the limit is a martingale (Proposition 14). The limit for prices that can reach zero, convergence of bubble sizes when the continuous limit is strict ($1/2 < \pi < 1$), exposures above one on a grid without a liquidation rule, inverse funds whose floor is monitored discretely or whose price gaps, random grids, and a test across exposures that reaches inverse funds, or leveraged

funds without the continuously monitored floor, or that does not assume jointly covering intervals for expectation prices, are open.

- **Uniform integrability.** Theorem 1 is about the martingale property on finite horizons only. Applied to constant exposure, the criterion of Mijatović and Urusov [60] (Theorem 2.3) gives a uniformly integrable martingale on $[0, \infty)$ only at $\pi = 0$ (Proposition 4), and for geometric Brownian motion with a deterministic exposure schedule Ruf's criterion [73] gives one exactly when $\int_0^\infty \pi(t)^2 \, dt < \infty$ (Proposition 5); both are machine-checked, conditional on the respective criteria, and the weak-tail conditions of Hulley and Ruf [38] are not applied.
- **Stochastic volatility.** The argument is one-dimensional: σ depends on the level of S alone. Whether the threshold $1/2$ and the zero-endpoint effect survive when volatility has its own dynamics is open. Any such statement must start from Sin [75], where the price alone already changes its martingale property with the pricing measure.
- **Tests with power.** By Theorem 22, a test with power above size needs a model class that excludes the linear-tail completions σ_n, for example a stated restriction on the tail of σ that they violate. Restricting the physical dynamics alone does not suffice, since the theorem holds with $P = Q$. The declared power tail of Proposition 25, and its common-exponent version across names in Proposition 26, are such restrictions; the band rules need jointly covering confidence bands, constructed in Proposition 27 only for a correctly specified finite catalogue of likelihoods, which price data supply only under the strong assumptions of Proposition 28, and the threshold tests are only pointwise. Other restrictions, and a comparison with existing tests, are not treated. Any test with power above size must also exclude the hidden-measure nulls of Proposition 20, which lie outside local volatility.
- **A usable test.** Section 3.5 gives ingredients, not yet a usable test. Still to be done: a realistic model class (a grid of elasticity exponents, stochastic volatility, and jumps if feasible), since the misspecification bounds of Proposition 43 are nearly vacuous at high sampling frequency and a valid likelihood for rounded prices at high frequency needs heavy computation, and a contamination budget for jumps is uninformative over realistic windows, volatility that changes within the evaluation window has no informative budget, and nulls that allow one switch or Poisson jumps are valid but nearly powerless on daily data, the first because time variation and level dependence of volatility are nearly confounded there and the second because computable bounds on the jump family's supremum remain loose, with the false-alarm rate and power measured when the true model lies outside it; a better way to declare a volatility interval for a real asset in advance, since a known interval recovers much of the power that a free scale costs, but an honest declaration from an earlier window recovers little and assumes the volatility does not change between windows; power on realistic samples, where pooling across constituents was tried and did not beat per-name testing, even with a forecast that models the dependence across names, and where the simulations of Section 5 find useful power only with long daily records or with intraday data under a declared clock; and a procedure that, on real prices, both keeps false alarms down and flags the consensus episodes, which none of the procedures compared in Section 6 does.
- **What Section 3.4 does not give.** Bubble values conditional on nontrivial information; the supremum of the set in Proposition 20; a characterization of all feasible defect curves in Proposition 19; a confidence procedure; observations with option prices or an unbounded horizon.
- **Estimating the visited integral.** Proposition 23 is exact for a continuously observed path, and the sampled-price estimator of Proposition 24 covers only a fixed interval with a known floor on σ^2/x^2, and is machine-checked only conditionally on Jacod's kernel estimation theorem. An estimator over the

random visited range, its behaviour near the maximum of the path, where the local time is small, and confidence intervals are not treated.

- **Baskets.** Section 4.2 covers fixed positive holdings without dividends or rebalancing, and its recovery formula for single-name bubbles assumes a terminal density depending on the index alone, not shown to be testable, and idealized population inputs; Proposition 78 estimates the contributions and tests their necessary inequalities only in a synthetic experiment with repeated terminal panels and a supplied pricing law, where on a realistic basket its fixed-bin bias is as large as the contribution and its test has no power (Section 5.8), and estimating that law from option panels, or the physical law from one historical path of constituent prices, is not treated.
- **Certifying the absence of bubbles.** Section 4.3 bounds bubbles above, up to a tolerance, and an upper budget never singles out a bubbly name; a positive monetary lower bound for single names is not available; exact certification of no bubble from positive option prices is not available, and the upper bound on the American value and the tracking bound under Q are not supplied from data.
- **Which names are bubbly.** The discovery list of Proposition 40 controls the false-discovery rate only within an exact price-law catalogue under $P = Q$; it gives no guarantee conditional on a screen, for realized lists, or across repeated testing dates.
- **Prior art.** The investment-with-bubbles literature cited in Section 7 has not been checked for special cases of Theorem 1.

Appendix A. Machine verification

The formalization is in Lean 4 with Mathlib. Mathlib has no stochastic calculus for diffusions, so the cited stochastic results enter as hypothesis structures, and what is proved is the deterministic analysis built on them. The statement of Theorem 1 is `lean/Standalone/D1TwoEndpoint.lean`, readable without the proofs; the proofs are in `lean/Novel/D1TwoEndpointProof.lean`. There is no `sorry`, and no axiom beyond Lean's standard ones.

- **Theorem 1, deterministic core (machine-checked, unconditional).** For every σ satisfying standing assumption 1, every $\pi > 0$ and every $c > 0$: the endpoint condition of the martingale criterion for b_π, written out with the source's scale density, scale function, Feller test function and goodness conditions, holds if and only if the right-hand side of (2) fails. Both endpoints are proved bad, the auxiliary scale (3) is computed in closed form, and the finiteness of I_0 and I_∞ is proved independent of c. The formal proof differs from the proof in Section 3: it evaluates the endpoint limits of the auxiliary Feller test function directly, by dominated convergence in one direction and Fatou's lemma in the other, so it does not use Feller's test.
- **Theorem 1, wealth classification (machine-checked, conditional).** The standalone statement proves (2) and the table for $v_0 Z^\pi$ from the conclusion of the martingale criterion at this instance, taken as a premise. The proof file also derives it from hypothesis structures for the martingale criterion and the JKP criterion (`wealth_classification`, `counterexample`), and checks that the criterion's condition as stated there is the one restated in the standalone file. These structures currently have only vacuous instances, because the pinned Mathlib cannot construct a Brownian motion. The hypotheses of these conditional theorems have therefore never been shown jointly satisfiable in Lean, and the theorems are exactly as conditional as the prose: on the martingale criterion, the JKP criterion and (W). They add

no evidence that those inputs hold. Assumption (W) enters as the definition $V^\pi = \nu_0 Z^\pi$. The usual conditions of standing assumption 2 are not needed.

- **Corollary 2.** Its last sentence is machine-checked with $I_0 < \infty$ in place of "S can reach zero" and $I_\infty < \infty$ in place of "S is strict": the conjecture so read holds for all $\pi > 0$ when $I_0 = \infty$, and fails for every $0 < \pi < 1/2$ when $I_0 < \infty$. The equivalence of $I_0 < \infty$ with reaching zero, an application of Feller's test, is not formalized.

- **Example 3.** Part (a) is machine-checked: $I_0 = 1/2$, $I_\infty = \infty$, and, conditionally as above, S is a martingale and $V^{1/4}$ is strict on every finite horizon. Part (b) is not formalized; its numerical values come from a registered experiment (`experiments/001-d1-threshold.md`) and are evidence about the cases computed, not a proof.

- **Proposition 4 (machine-checked, conditional).** The endpoint computation is proved unconditionally: for $\pi \geq 0$, the criterion's four alternatives hold exactly when $\pi = 0$, with the vanishing alternative read as almost everywhere on $(0, \infty)$. The uniform-integrability criterion is a hypothesis structure with a vacuous instance, and Theorem 1's lemmas (both endpoints bad) are reused; the conclusion is kept as the conjunction of the martingale property and uniform integrability. That $V \equiv \nu_0$ at $\pi = 0$ is not encoded (the zero row comes from the vanishing alternative), and the strictness in the table is Theorem 1's. The statement is in `lean/Standalone/D33ConstantExposureUI.lean`, and the proof in `lean/Novel/D33ConstantExposureUIProof.lean`.

- **Proposition 5 (machine-checked, conditional).** Ruf's criterion is restated in Lean for this model, with the canonical space, the localization, the energies, the stopped σ-field and the auxiliary law's density identities defined as in the source, and the stochastic exponential and the shifted martingale problem carried as fields, since Mathlib has no stochastic integral; it has a degenerate instance and no bridge to an Upstream structure yet. Proved: the energy-density identity and pathwise bounds of (a); in (b) the exhaustion, the identity of the lifetimes, the nonexit under every auxiliary law, the criterion's integral and the three schedules, with $(1 + t)^{-\gamma}$ for every real γ; and (c) from Theorem 1's classification. Of the geometric law, only that its lifetime is infinite is used. The statement is in `lean/Standalone/D33TimeExposure.lean`, and the proof in `lean/Novel/D33TimeExposureProof.lean`.

- **Proposition 6 (machine-checked, conditional).** The supermartingale property is the restatement used for Proposition 8, and each factor's scalar inputs (the finite-horizon test of Theorem 1, the uniform-integrability condition of Proposition 4, and $Z_i \equiv 1$ at zero exposure) are supplied as premises; the independence of the full systems and the joint local-martingale property are explicit premises, and no trading through default is constructed. The tail bound used in (b) is proved; its converse, from the factors to the product, is not needed and not formalized. The statement is in `lean/Standalone/D33IndependentAbsorbed.lean`, and the proof in `lean/Novel/D33IndependentAbsorbedProof.lean`.

- **Proposition 7 (machine-checked, unconditional).** The identified sets (7) are proved as equalities of subsets of $[0, \infty]$, together with the strict lower bound $F_c(g) > J_{cR}$, the admissibility of the completions (9), the values $F_c(g_1) = \infty$ and (10), and their positivity and continuity when σ is positive and continuous. No hypothesis is added. The reading of θ as the bubble label is proved from the hypothesis structure for the JKP criterion, which has only a vacuous instance, so that reading is exactly as conditional as the prose. The statement is in `lean/Standalone/D8TailCompletions.lean`, and the proof in `lean/Novel/D8TailCompletionsProof.lean`.

- **Propositions 8, 9 and 11 (machine-checked, conditional).** Every part is proved from bounded stopping and the supermartingale property stated as hypotheses; the proof also derives these from hypothesis structures whose instances are degenerate, a constant process on a one-point space, so they give no evidence that the inputs hold for a non-constant price. The daily cost accounting (13) and its specializations are proved by algebra. The examples of Propositions 9 and 11 are conditional corollaries, the second also on the martingale criterion and (W); they take the processes in a version with every path continuous, while the cited results give continuity only almost surely. The statement is in `lean/Standalone/D9HoldingPeriod.lean`, and the proof in `lean/Novel/D9HoldingPeriodProof.lean`.
- **Proposition 10 (machine-checked, conditional).** Proved from bounded stopping, finite trading, the stopping-time property of exit times and the stopping property of stochastic integrals stated as hypotheses, whose structures have degenerate instances; the bounds are proved path by path. The positivity of $Q(\tau_a < T)$ is proved by comparison with the signed contract, without the supermartingale property. That the stopped process is the self-financing wealth closed into cash is its definition here; the stopped-integral identity that interprets it is not formalized. The statement is in `lean/Standalone/D9InverseLiquidation.lean`, and the proof in `lean/Novel/D9InverseLiquidationProof.lean`.
- **Proposition 12 (machine-checked, conditional).** Proved from the supermartingale property and the finite-trading fact stated as hypotheses; the hypothesis structure for finite trading has a degenerate instance, a constant price, so it gives no evidence for a non-constant one. The pull-out of the holdings is proved directly, using the supermartingale property for integrability. The diffusion classification and the $\sigma(x) = x^2$ comparison are conditional corollaries through hypothesis structures for the JKP criterion and the martingale criterion, whose instances are vacuous. The statement is in `lean/Standalone/D9FiniteGrid.lean`, and the proof in `lean/Novel/D9FiniteGridProof.lean`.
- **Proposition 13 (machine-checked, conditional).** Proved from Émery's stability theorem, the stochastic-integral calculus, the stopping-time property of exit times and the supermartingale property, all stated as hypotheses; the hypothesis structures for the first two have degenerate instances, so they give no evidence that the inputs hold for a non-constant price. The stochastic integral and its bracket are relations with the cited properties, and their existence is proved from those properties. Uniform integrability in the expectation statement is Mathlib's notion, assumed. The identification with Theorem 1's wealth is with the explicit exponent of the stochastic exponential; its link to the hypothesis structure for the martingale criterion is not formalized. The statement is in `lean/Standalone/D9Refinement.lean`, and the proof in `lean/Novel/D9RefinementProof.lean`.
- **Proposition 14 (machine-checked, conditional).** Conditional on the inputs of Propositions 12 and 13; the convergence-in-probability forms of Fatou's lemma and dominated convergence are proved from the Lean library (via a subsequence argument) rather than taken as cited inputs, and uniform integrability is Mathlib's notion. The diffusion regimes take the martingale criterion at this instance, in the form of Theorem 1's statement, as a premise, so they are conditional on it and on (W). The statement is in `lean/Standalone/D9TrueLimit.lean`, and the proof in `lean/Novel/D9TrueLimitProof.lean`.
- **Numerical evidence after Proposition 14.** The values for $\pi = 3/4$ come from a registered experiment (`experiments/012-d9-ui.md`); they are evidence about the cases computed, not a proof.
- **Propositions 15 and 73 (machine-checked, conditional).** Proved from the inputs of Propositions 10 and 12 stated as hypotheses, with degenerate instances; the companion process R is shown to be a

local martingale by a lemma on linear combinations of continuous local martingales, localized at joint exit times. The stopped process is the liquidated wealth by definition; the closing-trade interpretation is not formalized. The elimination in (b) is a separate unconditional statement. The statements are in `lean/Standalone/D9PositiveLiquidation.lean` and `lean/Standalone/OuterSetLev.lean`. The proofs are in the corresponding files under `lean/Novel/`, with `Proof` appended to the name.

- **Proposition 16 (machine-checked; (b) and (c) conditional).** Parts (a) and (d) are proved path by path with no probability; (d) uses explicit tent paths. Parts (b) and (c) are proved from the inputs of Propositions 10 and 15; dense monitoring means every time in $[0,T]$ is eventually within any ε of the monitoring set, and (c) uses eventual detection of an open excursion and dominated convergence. That d_F can be negative is not formalized. The statement is in `lean/Standalone/D9FloorMonitoring.lean`, and the proof in `lean/Novel/D9FloorMonitoringProof.lean`.
- **Proposition 17 (machine-checked, unconditional).** Both tails range independently over $(0,\infty] \times (0,\infty]$, with the known middle integrals added; the lower tail is (19) and the upper tail reuses the completions of Proposition 7.
- **Theorem 18, Propositions 19–20, Corollary 21 (machine-checked, conditional).** Proved from bounded stopping, the supermartingale property and, for the exactness of (22), the stopping-time property of exit times, all stated as hypotheses; the hypothesis structures for these inputs have degenerate instances, so they give no evidence that the inputs hold for a non-constant price. Tests are measurable functions of k paths on $[0,T]$, which is equivalent to the Borel tests of Section 3.4. Convexity in Proposition 19 is proved by mixing two whole models on their disjoint union, which avoids hitting-time theory there. Proposition 20 is proved from a configuration derived from a hypothesis structure for Sin's model whose instance is vacuous, because the pinned Mathlib cannot construct a Brownian motion; it is exactly as conditional as the prose. The statement is in `lean/Standalone/D8NoGo.lean`, and the proof in `lean/Novel/D8NoGoProof.lean`.
- **Theorem 22 (machine-checked, conditional).** Proved from stopped uniqueness in law stated as a hypothesis, whose structure has a degenerate instance (every law the point mass at a constant path), so it gives no evidence for genuine diffusion laws. The pricing laws themselves are a premise: a law for each admissible coefficient, a probability on paths that start at s_0, stay nonnegative and are absorbed at zero; weak existence is not formalized. The completions σ_n are those of Proposition 7. All four families (positive continuous coefficients or not; $P = Q$ or any equivalent law) are covered, and the bound, its sharpness and the asymptotic statement are proved. The statement is in `lean/Standalone/D3LocalVolNoGo.lean`, and the proof in `lean/Novel/D3LocalVolNoGoProof.lean`.
- **Proposition 23 (machine-checked, conditional).** Proved from the occupation formula, the bracket formula, local-time positivity and Feller's test stated as hypotheses; the first two structures have degenerate instances and the positivity structure a vacuous one, so they give no evidence that the inputs hold for a genuine diffusion. The solution is supplied, with its stochastic-integral representation, and the occupation formula is used in the form of a continuous local martingale whose bracket has a density in time. The deterministic step, dividing out a density where it is positive, is a separate unconditional statement, `lean/Standalone/DensityCancellation.lean`. The statement is in `lean/Standalone/D3VisitedRange.lean`, and the proof in `lean/Novel/D3VisitedRangeProof.lean`.
- **Proposition 24 (machine-checked, conditional).** Proved from Jacod's kernel estimation theorem stated as a hypothesis structure with a vacuous instance, with the representation of $\log S$,

the equivalence with the driftless equation and the local time as premises. The change of variables, the bounds, the finite sum and the clipping bound (b) are proved without probability, in `lean/Standalone/CompactClipping.lean`. The rate in (c) is stated in the bounded-in-probability form taken as a premise by Proposition 25(c). The reciprocal-local-time estimator mentioned after it is that of a registered experiment (`experiments/013-d3-visited-integral.md`). The statement is in `lean/Standalone/D3CompactEstimator.lean`, and the proof in `lean/Novel/D3CompactEstimatorProof.lean`.

- **Proposition 25 (machine-checked; (c) and the test in (d) conditional).** Parts (a) and (b), and the coefficient statement of (d), are proved unconditionally, with the tail by closed-form power integrals and the attaining coefficients from Proposition 7's completions; the size in (b) assumes measurable bands with the stated joint coverage. Part (c) takes the rate of the estimator of Proposition 24 and the convergence of its occupation events as premises; Proposition 24 proves them conditionally on Jacod's theorem. The test statement of (d) is conditional on the pricing-law family and stopped uniqueness, as in Theorem 22. The bubble reading of the label rests on the JKP criterion, whose instance is vacuous. The statement is in `lean/Standalone/D3PowerTail.lean`, and the proof in `lean/Novel/D3PowerTailProof.lean`.
- **Proposition 26 (machine-checked; (d) conditional).** Parts (a)–(c) are proved as finite algebra with Proposition 25's constructions, using no hypothesis structure; (c) assumes measurable rectangles with the stated simultaneous coverage, and the two-name example is an explicit finite space. Part (d) takes the premises of Proposition 25(c) for every name under one joint law. The statement is in `lean/Standalone/D3PooledTail.lean`, and the proof in `lean/Novel/D3PooledTailProof.lean`.
- **Proposition 27 (machine-checked, conditional).** Proved from the split-likelihood bound, stated in its unconditional moment form for every finite catalogue with standard Borel observation spaces, measurable blocks, strictly positive normalized conditional densities and a training-only selector; no hypothesis structure or instance exists for it yet, so this input is not covered by the structure-and-instance route. The tie of each likelihood to its pricing coefficient appears in no formal statement. The statement is in `lean/Standalone/D3CatalogueCalibration.lean`, and the proof in `lean/Novel/D3CatalogueCalibrationProof.lean`.
- **Proposition 28 (machine-checked, conditional).** The lognormal and Bessel transition laws enter as hypothesis structures with vacuous instances, because the pinned Mathlib cannot construct a Brownian motion. The laws of the prices on a grid and their conditional densities are supplied as premises, as finite-grid iterates of the cited kernels; one-step normalization is cited and the finite-product normalization is proved. Measurability of the observed prices at fixed times is a disclosed premise. The deterministic chain densities are a separate statement. The statement is in `lean/Standalone/D3PriceCatalogue.lean`, and the proof in `lean/Novel/D3PriceCatalogueProof.lean`.
- **Proposition 29 (machine-checked, conditional).** The constant-elasticity and lognormal kernels, the JKP criterion and the e-BH theorem enter as hypothesis structures with vacuous instances; the support-aware split-likelihood bound is proved. The grid laws are supplied as premises, as for Proposition 28, and the lognormal conditional law is transferred to the reference measure with the atom. The finite-array support argument is a separate deterministic statement. Measurability of the observed prices is a disclosed premise. The statement is in `lean/Standalone/D13CEVCatalogue.lean`, and the proof in `lean/Novel/D13CEVCatalogueProof.lean`.

- **Proposition 30 (machine-checked; (c) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance. The composite-null split-likelihood bound is proved from Mathlib and applied, the closed-form supremum and its measurability are proved for every input array, and the normalization of the Gaussian family is proved by change of variables and the Gaussian integral; the constant-elasticity kernels enter as for Proposition 29. The pricing reading of a Gaussian null, including the existence of the pricing law, is not formalized. The statement is in `lean/Standalone/D13DriftNull.lean`, and the proof in `lean/Novel/D13DriftNullProof.lean`.
- **Proposition 31 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. The composite-null split-likelihood bound, the closed-form supremum with its exceptional case, its measurability and the ordering and limits in (c) are proved. The pricing reading and the remarks on interval choice and power are not formalized. The statement is in `lean/Standalone/D13BoundedDrift.lean`, and the proof in `lean/Novel/D13BoundedDriftProof.lean`.
- **Proposition 32 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. The composite-null split-likelihood bound, the normalization of the Gaussian family, the exact five-candidate maximum, with no optimizer, grid or concavity assumption, and the ordering and limits in (c) are proved. The bounded-drift score enters through its definition; its closed form is proved separately, in Proposition 31. The pricing reading of a Gaussian null is not formalized. The statement is in `lean/Standalone/D13DriftRect.lean`, and the proof in `lean/Novel/D13DriftRectProof.lean`.
- **Proposition 33 (machine-checked; false-discovery bounds conditional).** The false-discovery bounds, including the bound conditional on the earlier record, are conditional on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Proved: the conditional moments on the coverage event, in set-integral form; the corrected size, threshold, safe-score moment and rejection condition; the measurability of the declared score; the earlier-window rectangle, its roots, its containment of the likelihood set, its coverage bound and the null fallback event; and the three obstructions in (d), the counterexample with no cited input. The remarks on which dynamics supply the conditional-law premise, and the pricing link, are not formalized. The statement is in `lean/Standalone/D13Declaration.lean`, and the proof in `lean/Novel/D13DeclarationProof.lean`.
- **Proposition 34 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the clock laws enter only as components of the null mixture, so their normalization is not used. Proved: the closed-form supremum and its unique maximizer, by an exact gap identity; the singular set and its measure zero; measurability; the scale invariance and the reduction to Proposition 30; validity under the mixture; and the orderings. The declaration and coverage of clocks are premises. The statement is in `lean/Standalone/D17ClockDrift.lean`, and the proof in `lean/Novel/D17ClockDriftProof.lean`.
- **Proposition 35 (machine-checked).** The product rule for conditional e-values is the theorem proved from Mathlib, and the statistics and profile lemmas of Proposition 34 are reused; no new hypothesis structure is added. The null is stated as a conditional law given $\mathcal{F}_{t-1}$ for every event and Borel set, and the step from it to the conditional mean of each oracle factor is proved. Proved: the clipped profile, its attainment, finiteness and measurability; the conditional stopped-mean and

crossing bounds given $\mathcal{F}_0$, and the same for smaller numerators and larger denominators; and the tracked local-lognormal law as a null. That the local-lognormal recursion keeps prices positive is a hypothesis; the remarks that E^{IV} need not be a supermartingale and on the scope of the null are not formalized. The statement is in `lean/Standalone/D28PredictableIV.lean`, and the proof in `lean/Novel/D28PredictableIVProof.lean`.

- **Proposition 36 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the clock profiles of Proposition 34 are reused. Proved: the clock set and the measurability of retention; the coverage bound with no union over the catalogue; the two-stage conditioning that gives the raw and safe bounds; the comparisons, ceiling and boundary cases; and a nonzero model satisfying the premises, on the calendar clock. The statement is in `lean/Standalone/D20ClockConfidence.lean`, and the proof in `lean/Novel/D20ClockConfidenceProof.lean`.
- **Proposition 37 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the profile of Proposition 34 is reused for each day. Proved: the separation of the supremum, the singular days and their null set, measurability and scale invariance; validity under the history-dependent mixture law; the comparisons with grouped days and with full-record clocks; and the calendar identity with its equality case, through the unique optimizer of Proposition 34. That the optional constant-elasticity forecast has no atom at zero is not formalized here. The statement is in `lean/Standalone/D23DailyProfile.lean`, and the proof in `lean/Novel/D23DailyProfileProof.lean`.
- **Proposition 38 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the daily statistics of Proposition 34, the daily profile of Proposition 37 and the partition machinery of Proposition 39 are reused. Proved: the run profile, its attainment and its singular criterion; the partition maximum, its infinity criterion, attainment, measurability and the recursion; validity under the path mixture, also for a finite catalogue of full-record shapes; the nesting and its reversal for scores; and invariance under a common shape multiplier. The operation counts and the remarks on what does not hold in general are not formalized. The statement is in `lean/Standalone/D23ScaleSwitches.lean`, and the proof in `lean/Novel/D23ScaleSwitchesProof.lean`.
- **Proposition 39 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the profiles of Propositions 34 and 37 are reused on concatenated runs. Proved: the pooled run residual and its singular criterion; the partition maximum, its exchange with products, infinity and attainment, and the recursion; measurability on positive prices; validity under the mixture, also for a finite catalogue of full-record shapes; the nesting including the free-daily-drift denominator; the identity with the common profile; and the calendar cost and its additivity. The operation count and the diagnostic code are not formalized. The statement is in `lean/Standalone/D23JointSwitches.lean`, and the proof in `lean/Novel/D23JointSwitchesProof.lean`.
- **Proposition 40 (machine-checked, conditional).** Proved from the e-BH theorem stated as a hypothesis, whose structure has a degenerate instance, and from the split-likelihood moment bound, which is proved; the rule is encoded without sorting and shown to report exactly the first k^* names. The bubble interpretation is commentary. The statement is in `lean/Standalone/D12CatalogueFDR.lean`, and the proof in `lean/Novel/D12CatalogueFDRProof.lean`.

- **Proposition 41 (machine-checked).** The Gaussian maximum, including the unbounded singular case, is proved from the spectral theorem and the determinant identities; so are the rank bound, the null set of singular arrays when $n \geq m + 1$, validity for any normalized forecast, the forecast's normalization and the separation of its training maximization, and the correlation identity (c). The forecast's constant-elasticity kernels enter as hypothesis structures with vacuous instances, as for Proposition 29; the Gaussian formula enters only as a component of the null mixture, so its normalization is not used. The operation counts and the comparisons with per-name testing are not formalized. The statement is in `lean/Standalone/D16GaussianPool.lean`, and the proof in `lean/Novel/D16GaussianPoolProof.lean`.

- **Proposition 42 (machine-checked).** The Gaussian copula construction enters as a hypothesis on each step's normalization and marginals, and the constant-elasticity kernels as for Proposition 29. Proved: finiteness of the normal scores, the training-only measurable correlation rule with its bounds, normalization of the whole-path copula and mixture forecasts by a chain of normalized steps, the validity conclusions through Proposition 41, the mixture floor, the exact gain and the reduction at $R = I$. The statement is in `lean/Standalone/D16CopulaForecast.lean`, and the proof in `lean/Novel/D16CopulaForecastProof.lean`.

- **Proposition 43 (machine-checked; (b) conditional).** Only the false-discovery bound in (b) is conditional, on the e-BH theorem with a degenerate instance; disintegration of the actual law, the split-likelihood bound, Markov's inequality, the bounded-function comparison and the Gaussian affinity and its limits are proved. The band rule's containment in the score rule is proved for any truth. The Gaussian part is a hypothesis on the law of log returns; the decimal values after the proposition are numerical evaluations of the formalized formulas, not formalized themselves. The statement is in `lean/Standalone/D14Misspecification.lean`, and the proofs in `lean/Novel/D14MisspecificationProof.lean` and `lean/Novel/D14GaussianProof.lean`.

- **Proposition 44 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity and lognormal kernels enter as for Proposition 29. The support-restricted split-likelihood step, Markov's inequality, the comparison on the full observation space and the absorption bound are proved. The observation space is the reals with one error symbol, as a standard Borel space; the result is proved for any finite catalogue of normalized densities and then instantiated at the objects of Proposition 29, and the screen may take values in $[0, 1]$. The statement is in `lean/Standalone/D14SupportRobustness.lean`, and the proof in `lean/Novel/D14SupportRobustnessProof.lean`.

- **Proposition 45 (machine-checked, conditional).** The exchangeable quantile lemma and the training-conditional guarantee enter at the reference laws used, as hypothesis structures bridged from interface structures whose only instances are degenerate (a shared fair bit and a point-mass score law), so they are not discharged for arbitrary reference laws. Proved: measurability of the rank threshold and the flag event; that a common measurable map preserves exchangeability and that independent laws are exchangeable; the flag probability as the mean of the realized risk; (a), (b) and (c), with the full calibration record on a standard Borel space and the kernel premise simultaneous over events on one full-measure set; $r = n - k$; and (d) exactly, with its total variations. (b) is proved through the flag event, not through a bound on the total variation of the whole joint law, and the budgets may be any nonnegative numbers. The scope remarks after the proof are not encoded. The statement is in `lean/Standalone/D34ScreenCalibration.lean`,

and the proof in `lean/Novel/D34ScreenCalibrationProof.lean`, with `D34ScreenCalibrationFlag`, `D34ScreenCalibrationExample` and `D34ScreenCalibrationBridge`.

- **Proposition 46 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Proved: the free-drift score's reference moment at every training record, including impossible ones; the size bounds, the adjusted threshold, the capped moment and the rejection condition, for any score with such a moment; that the no-jump law is the Gaussian law of Proposition 30, with the Bernoulli construction realizing every $p \in (0,1)$; and the total-variation bound for dependent jumps. The mixture weight and jump law need not depend measurably on the training record. The remarks that finer sampling does not reduce the budget and that the mixture form can fail for dependent jumps are commentary. The statement is in `lean/Standalone/D14JumpRobustness.lean`, and the proof in `lean/Novel/D14JumpRobustnessProof.lean`.
- **Proposition 47 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Proved: the affinity (a), on log returns and on prices through the log-normal change of variables, with its measurability; the averaged size, threshold, clipped-moment and zero-budget conclusions for any latent kernel; the reference moment for both scores at every training record; and the envelope (c). The optional constant-elasticity reference with power one and zero drift for the rectangle score is not encoded. The statement is in `lean/Standalone/D14VolPaths.lean`, and the proof in `lean/Novel/D14VolPathsProof.lean`.
- **Proposition 48 (machine-checked; (b) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Proved: the factorization over segments, the finiteness criterion, the collapse at a one-step segment, that the exceptional set has measure zero, the normalization of the switch laws, the moments under the latent mixture, the ordering and the absorption cases. In the example showing that the score is not identically zero, positivity of the fitted likelihood on positive arrays is taken as a hypothesis. The interpretation of the switch null is not formalized. The statement is in `lean/Standalone/D15Switch.lean`, and the proof in `lean/Novel/D15SwitchProof.lean`.
- **Proposition 49 (machine-checked, conditional).** Part (a) is proved from the quadratic variation of a Brownian integral plus a continuous finite-variation term, in its realized form (sums of squared increments along partitions with mesh tending to zero, converging in probability), stated as a hypothesis structure with a vacuous instance; the finite-variation-only case is proved by uniform continuity, not assumed. The stochastic integral is a relation quantified over, and existence of the representation is a premise. Parts (b) and (c), including sharpness, the size and power bounds for measurable arrays and the sampled-path constraint, are proved without hypothesis structures. The statement is in `lean/Standalone/D30BVIdentification.lean`, and the proof in `lean/Novel/D30BVIdentificationProof.lean`, with the bridge in `lean/Novel/D30BVIdentificationBridge.lean`.
- **Proposition 50 (machine-checked).** The split-likelihood bound enters through its proved domination inequality, and the grid law of the time-changed price is derived from Mathlib's pre-Brownian motion; no hypothesis structure is added. The certificate values are extended reals and the scores lie in $[0,\infty]$. Proved: the interpretation in both directions, by the mean value of the clock rate on each step; measurability, positivity, finiteness, the unique maximizer, the zero-return case, monotonicity and the closed form at $V = 0$; the scalar suprema in all cases, the certificate bounds with the nonnegative gap de-

composition, existence of a finite certificate and of a zero-gap one at $V = 0$; and validity for the score, the certified score and every score below it, also for mixtures of null laws. The example forecast from constant-elasticity transitions, and the remark that jumps of the clock rate do not make the price jump, are not formalized. The statement is in `lean/Standalone/D30VariationProfile.lean`, and the proof in `lean/Novel/D30VariationProfileProof.lean`.

- **Proposition 51 (machine-checked, conditional).** The reciprocal-Bessel kernel's normalization is a hypothesis structure bridged from the cited Bessel kernel, the convergence theorem enters through the lab's mean-one L^1 lemma, and the geometric Brownian law in coordinates is proved by induction with one-dimensional change of variables and Tonelli; the price laws are the normalized chains of Proposition 28 with $P = Q$. The consequences for larger classes and for the time-and-level models are remarks, not encoded. The statement is in `lean/Standalone/D30SmallScale.lean`, and the proof in `lean/Novel/D30SmallScaleProof.lean`.
- **Proposition 52 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29. Proved: the domination of every jump law by the envelope, for the series formula as written, without using its normalization or its interpretation as Merton's law; the envelope's finiteness, measurability, monotonicity and limits; the truncation bound; that the infinite-denominator set has measure zero; the moments under the latent mixture; and the orderings. In the example of a positive score, positivity of the fitted likelihood is taken as a hypothesis. The statement is in `lean/Standalone/D15JumpEnvelope.lean`, and the proof in `lean/Novel/D15JumpEnvelopeProof.lean`.
- **Proposition 53 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the conclusions of Proposition 52(b) are proved for any valid envelope. Proved: the domination and the comparison with Proposition 52's envelope; measurability, through a supremum over rational parameters; the enclosing rectangle and attainment of the maximum; the box bounds, their monotonicity under subdivision and their convergence; and validity for every measurable cover rule. No outward-rounding certificate is given for floating-point evaluation. The statement is in `lean/Standalone/D15SharedJump.lean`, and the proof in `lean/Novel/D15SharedJumpProof.lean`.
- **Proposition 54 (machine-checked).** From Mathlib's measure maps, restrictions, Dirac masses and the Gaussian density and integral; the split-likelihood step is proved with the supplied densities, and no hypothesis structure is used. Means are lower integrals of the nonnegative scores, and the profiled denominator in (b) is written directly as the supremum over the drift. The positivity of the prices $1, e^R$ and $1, e^Y$ is not encoded. The statement is in `lean/Standalone/D31FilterScope.lean`, and the proof in `lean/Novel/D31FilterScopeProof.lean`.
- **Proposition 55 (machine-checked, conditional).** The price kernels are hypothesis structures with vacuous instances and the grid laws are supplied as premises, as for Proposition 28; the false-discovery bound is conditional on the e-BH theorem with a degenerate instance. Proved outright: (a), for every law on the reals and, for vectors, every absolutely continuous reference; the cell probability and the divergence of the plug-in mean in (b), and the bounds at $h = 1/2$, $\kappa = 1/100$; the conditional-density facts in (c), for an arbitrary selector; and (d), for every $h > 0$, by a bound on the log density ratio. The decimal 0.175 and the remark that the weights use the whole training prefix are not formalized. Measurability of the observed prices is a disclosed premise. The deterministic parts are stated in `lean/Standalone/RoundedCells.lean` and proved in `lean/Novel/RoundedCellsProof.lean`.

The statement is in `lean/Standalone/D14RoundedObservations.lean`, and the proof in `lean/Novel/D14RoundedObservationsProof.lean`.

- **Proposition 56 (machine-checked; (c) conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance. Proved: the restarted laws as images of the whole block, their positivity and normalization; the first-return shift and its affinity; the budget at almost every training record, for any conditional law of the exact price in its cell; the size, threshold and clipped-moment consequences; and the tolerance and limits in (d). The remarks on which dynamics supply the premise are not formalized. The statement is in `lean/Standalone/D14RoundedStart.lean`, and the proof in `lean/Novel/D14RoundedStartProof.lean`.
- **Proposition 57 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance. Proved: the normal distribution function's properties, the clipped-midpoint supremum and its range; the domination along the path, by induction, for any law of the exact price in its cell; the moments and size; the row sums and total mass in (c); and the counterexample. The operation count is not a mathematical statement, and no floating-point evaluation is certified. The statement is in `lean/Standalone/D14Cell.lean`, and the proof in `lean/Novel/D14CellProof.lean`.
- **Proposition 58 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance. Proved: the interval maximum for subcells of any width; the backward bounds from every exact state in a subcell, through a general partition bound; that level zero equals the envelope of Proposition 57; monotonicity in the level; and validity for any level chosen after seeing the evaluation cells. The operation and storage counts are not mathematical statements, and no floating-point evaluation is certified. The statement is in `lean/Standalone/D14CellSubdivision.lean`, and the proof in `lean/Novel/D14CellSubdivisionProof.lean`.
- **Proposition 59 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the Gaussian cell lemmas of Proposition 57 are reused. Proved: the three-case step supremum and its range, including the trapped zero; the mixture over the exact last price with an atom at zero, which is derived rather than assumed; the direct envelope and the subcell recursion with its ordering; validity at data-selected levels; the normalized prototype forecast; the log-width asymptotics; the positive probability of leaving the zero cell from a quarter tick, and zero from an exact zero; and the all-zero boundary. The operation count is not a mathematical statement. The statement is in `lean/Standalone/D14FixedTick.lean`, and the proof in `lean/Novel/D14FixedTickProof.lean`.
- **Proposition 60 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; no constant-elasticity kernel is used. Proved: the component bound and averaging of Proposition 47, the domination of the cent-tick envelope from every exact starting price, the measurable choice of the closest catalogue law, the comparison (a), the size, threshold, clipped-moment and rejection conclusions, the zero-budget case and the uniform envelope. The remarks on the ceiling, refinement and numerical approximation are not formalized. The statement is in `lean/Standalone/D14RoundedModel.lean`, and the proof in `lean/Novel/D14RoundedModelProof.lean`.
- **Proposition 61 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the cent-tick interval supre-

mum and forecast are reused, and nothing else is cited. Proved: the widened cells, the explicit uniform emission and its subcell supremum, the domination under each contract by a single backward induction, validity at data-selected levels, the orderings and the reduction at $a = 0$, the all-zero bounds and the misuse counterexample. The operation count and the reading of zero quotes are commentary. The statement is in `lean/Standalone/D18BoundedNoise.lean`, and the proof in `lean/Novel/D18BoundedNoiseProof.lean`.

- **Proposition 62 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the cent-tick transition bound and forecast, and the support bound of Proposition 61, are reused, and nothing else is cited. Proved: the declared joint law of returns and signs and the sign-transition bounds; the domination of both recursions under an arbitrary joint posterior; validity at data-selected levels; monotonicity, the comparison with the support score and the three reductions; and the counterexample, including its arithmetic. The operation count is commentary. The statement is in `lean/Standalone/D18SignNoise.lean`, and the proof in `lean/Novel/D18SignNoiseProof.lean`.
- **Proposition 63 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the cent-tick model and forecast are reused, and nothing else is cited. Proved: the rectangle bound, its range and its reduction for a single parameter, with the scale optimum by a derivative-sign argument; the recursion's range and monotonicity under box bisection and cell subdivision; validity at data-selected levels for any normalized forecast; and the boundary cases. The operation count, and that the bound is only an upper bound for the coupled model, are commentary. The statement is in `lean/Standalone/D17ClockCells.lean`, and the proof in `lean/Novel/D17ClockCellsProof.lean`.
- **Proposition 64 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the constant-elasticity kernels enter as for Proposition 29, and the denominators of Propositions 34 and 63 are reused unchanged. Proved: normalization of both forecasts and their mixtures, the positivity and normalization of the cell rows, the training-only first-maximizer selection, the validity conclusions in both domains, the support check that the exact-price forecast is positive only where the denominator is, the mixture floor, the boundary cases and the counterexample (d). The statement is in `lean/Standalone/D17ClockForecast.lean`, and the proof in `lean/Novel/D17ClockForecastProof.lean`.
- **Proposition 65 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the forecast's normalization uses the constant-elasticity kernels as for Proposition 29, while the denominator and the validity for a supplied forecast use none. The bounds of Propositions 61–63 and the forecast of Proposition 64 are reused. Proved: the range and monotonicity of both recursions and their minimum; domination for every cover and every data-selected level; positivity and normalization of the sign-chain forecast and its mixtures, and its reduction to Proposition 64 with no noise; validity and the mixture floor; and the three reductions in (d). The operation counts are commentary. The statement is in `lean/Standalone/D18ClockNoise.lean`, and the proof in `lean/Novel/D18ClockNoiseProof.lean`.
- **Proposition 66 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; no stochastic upstream input is used, and the cent-tick cells and the box bound of Proposition 63 are reused. Proved: positivity of every string's mass and the forward recursion; normalization, continuity and attainment of the maximum; validity for any denominator between the filtered maximum and one; the

relative quadrature enclosure, by induction along the string; the cover bounds from both sides and their convergence; and the comparison (e). The outward-arithmetic evaluator, the operation counts and the absence of an order between the certified bound and the box bound are not formalized. The statement is in `lean/Standalone/D19FilteredLikelihood.lean`, and the proof in `lean/Novel/D19FilteredLikelihoodProof.lean`.

- **Proposition 67 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the model, statement and validity of Proposition 66 are reused, and nothing new is cited. Proved: the oscillation and deviation bounds on the path statistics, the parameter-displacement bounds, the second-order evaluation bound, the cancellation of the training infimum without any posterior approximation, the ordering $D_{\text{filter}} \leq D_{\text{cond}} \leq D_{\text{cert}}$ with the validity of every variant, and the limits. The operation counts are not formalized. The statement is in `lean/Standalone/D19CondSensitivity.lean`, and the proof in `lean/Novel/D19CondSensitivityProof.lean`.
- **Proposition 68 (machine-checked; false-discovery bound conditional).** Only the false-discovery bound is conditional, on the e-BH theorem with a degenerate instance; the cosh bound is Mathlib's, and the model, quadrature and certificates of Propositions 66 and 67 are reused. Proved: the unitriangular change of variables, the tensor subcell decomposition, the reflection lower bound, the factorized cosh upper bound with coefficients by summation by parts, the summation back to the quadrature of Proposition 66, the chain of bounds, validity, the exact M^{-2} scaling and the limits. The operation counts, the outward-arithmetic contract and the symmetric allowance used by the evaluator are not formalized. The statement is in `lean/Standalone/D19CenteredQuadrature.lean`, and the proof in `lean/Novel/D19CenteredQuadratureProof.lean`.
- **Proposition 69 (machine-checked).** Nothing in it is conditional on the product rule for conditional e-values, which is proved from Mathlib and applied; the false-discovery bound for panels is conditional on the e-BH theorem. The filtered likelihood of Proposition 66 is reused. Proved: the profile facts; the stopped-mean and anytime bounds for every adapted score between zero and E^{seq}, conditionally on the training cells and unconditionally, through the actual mixture's conditional probability; the identification of the denominator with that of Proposition 66, so that its certificates qualify; the supermartingale property of the step process and the ordering; and the smoothed forecast. Freezing the score after the horizon is a remark. The statement is in `lean/Standalone/D22FilteredSequential.lean`, and the proof in `lean/Novel/D22FilteredSequentialProof.lean`.
- **Proposition 70 (machine-checked).** The model, null, prefix masses and oracle bounds of Proposition 69 are reused, and nothing else is cited; no panel conclusion is stated. Proved: the restart denominators and their range, adaptedness, the domination $E_t \geq \lambda_r E_{r,t}$, the stopped-mean and crossing bounds conditionally on the training cells and unconditionally, and the same for lower scores. That the certificate of Proposition 66 at training length k_r qualifies as a restart denominator, and the scope remarks, are not formalized. The statement is in `lean/Standalone/D22DeclaredRestarts.lean`, and the proof in `lean/Novel/D22DeclaredRestartsProof.lean`.
- **Proposition 71 (machine-checked).** The model, null, prefix masses and forecasts of Propositions 69 and 70 are reused, with the product rule for conditional e-values proved from Mathlib. Proved: the restart denominators at random launches, the total evidence, the stopped-mean and crossing bounds through a mass-weighted evidence whose one-step sums are preserved, the slot domination, the identity with Proposition 70's restart scores and the domination of its scan, and the lower-score version. That the total need not be a supermartingale, and the link to the certificates of Proposition

66, are remarks. The statement is in `lean/Standalone/D22FundedRestarts.lean`, and the proof in `lean/Novel/D22FundedRestartsProof.lean`.

- **Section 4 and its conventions.** The machine-checked statements of Section 4 take the option-pricing convention they name as part of their premises: Lean checks the implications from the convention, not the convention itself, which is a modelling assumption about how option quotes relate to the bubble.
- **Proposition 72 (machine-checked, conditional).** The set computation in (a) is a separate statement proved with no probability, for arbitrary, possibly inconsistent, intervals. The application is proved from the inputs of Proposition 12 stated as hypotheses, with the call-put identity and the interval bounds as proved algebra; coverage and convergence are on a separate observation space, not under Q. The statements are in `lean/Standalone/OuterSet.lean` and `lean/Standalone/D9CrossPi.lean`, and the proofs in `lean/Novel/OuterSetProof.lean` and `lean/Novel/D9CrossPiProof.lean`.
- **Proposition 74 (machine-checked).** Parts (a)–(d) are proved as finite inequalities, integrals over terminal laws and one event inclusion, with no hypothesis structure; terminal laws are not path laws, and no embedding is asserted. The application to δ_T uses the supermartingale property, and the fund containment enters as part of the coverage event, from Proposition 73. The statement is in `lean/Standalone/D9IncompleteCalls.lean`, and the proof in `lean/Novel/D9IncompleteCallsProof.lean`.
- **Propositions 75 and 76 (machine-checked, conditional).** Proved from the supermartingale property and the existence of the share probability stated as hypotheses; the share-probability structure has a degenerate instance on a one-point system, so it gives no evidence for a genuine basket. Prices are taken continuous and nonnegative on every path, as the supermartingale interface requires, which narrows the raw-space setting with almost-sure versions. The index lifetime is a supplied stopping time, a premise the prose derives. The finite-sum and derivative algebra is a separate module. The statement is in `lean/Standalone/D4Basket.lean`, and the proof in `lean/Novel/D4BasketProof.lean`.
- **Proposition 77 (machine-checked, conditional).** Proved in the setting of Propositions 75 and 76, with their narrowings; the restriction and the conditional means are stated as an equality of measures on $\mathscr{F}_T$ and as conditional expectations given the terminal index, and the conditioned share probability is never replaced by the full one. No compatible model is constructed. The statement is in `lean/Standalone/D4Recovery.lean`, and the proof in `lean/Novel/D4RecoveryProof.lean`.
- **Proposition 78 (machine-checked).** From Mathlib only, with no hypothesis structure; the reading of θ as a bubble contribution enters through Proposition 77. The coverage in (b) is proved for the actual independent sampling experiment, by conditioning on which draws fall in each cell. The tilt in (d) is stated on a generic probability space, with the true-martingale identities as premises, and is not linked to the process model of Proposition 77; the histogram is a valid fit in the sense of (a) only almost surely; and the per-cell-radius variant is not part of the statement. The statement is in `lean/Standalone/D32RecoveryEstimator.lean`, and the proof in `lean/Novel/D32RecoveryEstimatorProof.lean`.
- **Propositions 79 and 80 (machine-checked, conditional).** Proved from bounded stopping, the supermartingale property and the stopping-time property of exit times, stated as hypotheses with degenerate instances; the bounds U and η are premises. The American value is the supremum over all stopping times of the expected payoff, valued in $[0,\infty]$; the dividend process need not be càdlàg. The proof bounds the exercise payoff using D_T rather than D at the random time, which avoids progressive measurability. The remarks on quotes, exercise calendars and the interpretation of flags are

not formalized. The statement is in `lean/Standalone/D10Certification.lean`, and the proof in `lean/Novel/D10CertificationProof.lean`.

- **Proposition 81 (machine-checked).** Parts (a)–(c) are finite real algebra and one event inclusion, with no hypothesis structure; the coverage in (c) is on a separate observation space with measurable endpoints. The application to gain defects is conditional on the supermartingale property. The statement is in `lean/Standalone/D4DispersionBudget.lean`, and the proof in `lean/Novel/D4DispersionBudgetProof.lean`.
- **Sections 5 and 6.** No simulation or real-data figure is machine-checked. The numerical checks of Section 5.7 are evidence for the cases computed, and every figure in Sections 5 and 6 comes from a design fixed before it was run and reproduced by an independent implementation.
- **Not formalized.** The exact endpoint values of the Feller test functions (Section 8), the identification (W) as a statement about trading strategies, and the remark in Section 2 that wealth is absorbed at zero when S is.

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